

Complete Publication List

Oliver Gutsche
June 01, 2026

1. Search for light scalar particles produced in Higgs boson decays in exclusive final states with two muons and two hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.27875](#) | 2026-05-27

2. Search for a new heavy scalar resonance decaying to a pair of Z bosons in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.26462](#) | 2026-05-26

3. Search for low-mass resonances decaying to $\tau\tau$ and measurement of the $Y \rightarrow \tau\tau$ decay in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.25103](#) | 2026-05-24

4. Studies of $Z \rightarrow 4\ell$ decays in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Tumasyan, Armen et al.

[arXiv:2605.25158](#) | 2026-05-24

5. Search for Higgs boson decays into two neutral scalars with unequal masses in final states with b quarks and tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.19144](#) | 2026-05-18

6. Search for the nonresonant production of a pair of additional Higgs bosons in the Type-X two-Higgs-doublet model in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.16655](#) | 2026-05-15

7. Search for new physics in triple boson production in proton-proton collisions at $\sqrt{s} = 13$ TeV using the effective field theory approach

Hayrapetyan, Aram et al.

[arXiv:2605.15023](#) | 2026-05-14

8. Constraints on anomalous Higgs boson couplings to vector bosons and fermions using the $\gamma\gamma$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.14245](#) | 2026-05-14

9. First measurements of vector boson scattering in $W^{\pm}W^{\pm}$ and WZ production in all-leptonic final states at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.15396](#) | 2026-05-14

10. Search for pair production of additional neutral scalars within the Inert Doublet Model in a final state with two electrons or two muons in proton-proton collisions at $\sqrt{s} = 13$ TeV and 13.6 TeV

Gevorgyan, Arzunik et al.

[arXiv:2605.13614](#) | 2026-05-13

11. Search for single vector-like quark production in opposite-sign dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.13649](#) | 2026-05-13

12. Search for a new heavy scalar resonance decaying into the Higgs boson and a new scalar particle in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2605.02848](#) | 2026-05-04

13. Observation of the rare decay $\eta \rightarrow \mu^+ \mu^- e^+ e^-$

Hayrapetyan, Aram et al.

[arXiv:2605.00615](#) | 2026-05-01

14. Search for light charged Higgs bosons decaying to charm and strange quarks in $t\bar{t}$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.27795](#) | 2026-05-01

15. Improved results on Higgs boson pair production in the $4b$ final state

Hayrapetyan, Aram et al.

[arXiv:2604.27044](#) | 2026-04-29

16. Search for electroweakinos in compressed-spectrum scenarios with low-momentum isolated tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.25604](#) | 2026-04-28

17. Simultaneous measurements of N-subjettiness observables in jets from gluons and light-flavour quarks, and in decays of boosted W bosons and top quarks

Hayrapetyan, Aram et al.

[arXiv:2604.25538](#) | 2026-04-28

18. Measurement of the $Z \rightarrow \mu^+ \mu^-$ angular coefficients in pp collisions at $\sqrt{s} = 13$ TeV as functions of transverse momentum and rapidity

Gevorgyan, Arzunik et al.

[arXiv:2604.25678](#) | 2026-04-28

19. Search for associated production of a Higgs boson and two vector bosons via vector boson scattering at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.24531](#) | 2026-04-27

20. Search for dark matter produced in association with a dark Higgs boson decaying into a bottom quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.20999](#) | 2026-04-22

21. Search for the single production of vector-like quarks decaying into a W boson and a b quark using single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.17564](#) | 2026-04-19

22. Highly boosted dielectron identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.13320](#) | 2026-04-14

23. Search for heavy resonances decaying into four-lepton final states via light bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.14236](#) | 2026-04-14

24. A search for microscopic black holes, string balls, and sphalerons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10732](#) | 2026-04-12

25. Search for a new heavy resonance decaying to a top quark and a neutral scalar boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.10729](#) | 2026-04-12

26. Particle transformers for identifying Lorentz-boosted Higgs bosons decaying to a pair of W bosons

Hayrapetyan, Aram et al.

[arXiv:2604.09809](#) | 2026-04-10

27. Observation of nuclear suppression in coherent $\Upsilon(1S)$ photoproduction off heavy nuclei at the LHC

Belyaev, Andrey et al.

[arXiv:2604.05814](#) | 2026-04-07

28. Search for soft unclustered energy patterns produced in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.05996](#) | 2026-04-07

29. Evidence of $ZZ\gamma$ production and observation of $4\ell\gamma$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2604.02594](#) | 2026-04-03

30. Search for Higgs boson pair production in the $b\bar{b}WW$ decay channel with two leptons in the final state using proton-proton collision data at $\sqrt{s} = 13.6$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2604.02127](#) | 2026-04-02

31. Measurement of dijet angular distributions and search for beyond the standard model physics in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.25458](#) | 2026-03-26

32. Search for new particles decaying into top quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Gevorgyan, Arzunik et al.

[arXiv:2603.23454](#) | 2026-03-24

33. Search for low-mass vector and scalar resonances decaying into a quark-antiquark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2603.21965](#) | 2026-03-23

34. Search for Higgs boson production at high transverse momentum in the WW decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[arXiv:2603.22233](#) | 2026-03-23

35. Measurement of the jet mass in hadronic decays of boosted W bosons at 13 TeV and extraction of the W boson mass

Hayrapetyan, Aram et al.
[arXiv:2603.19963](#) | 2026-03-20

36. Jet peak shapes based on two-particle angular correlations in lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.
[arXiv:2603.14385](#) | 2026-03-15

37. Measurement of the t-channel single top quark cross section in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.13592](#) | 2026-03-13

38. Search for Z' bosons decaying into charginos in final states with two oppositely charged leptons and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.11035](#) | 2026-03-11

39. Measurement of angular correlations inside jets induced by gluon polarization in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.
[arXiv:2603.03689](#) | 2026-03-04

40. Search for a narrow resonance with a mass between 10 and 70 GeV decaying to a pair of photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[arXiv:2603.03250](#) | 2026-03-03

41. Combination of ATLAS and CMS searches for Higgs boson pair production at $\sqrt{s} = 13$ TeV

Aad, Georges et al.
[arXiv:2602.23991](#) | 2026-02-27

42. Search for pair production of heavy resonances in final states with a photon and large-radius jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.
[arXiv:2602.20477](#) | 2026-02-24

43. System-size dependence of charged-particle suppression in ultrarelativistic nucleus-nucleus collisions

Belyaev, Andrey et al.
[arXiv:2602.21325](#) | 2026-02-24

44. Observation of the jet diffusion wake using dijets in heavy ion collisions

Hayrapetyan, Aram et al.
[arXiv:2602.19431](#) | 2026-02-23

45. First measurement of jet axis decorrelation with photon-tagged jets in pp and PbPb collisions at 5.02 TeV

Hayrapetyan, Aram et al.
[arXiv:2602.18279](#) | 2026-02-20

46. Search for a new resonance decaying to a Higgs boson and a scalar boson in events with two b jets and two Z bosons in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.18223](#) | 2026-02-20

47. Combined measurements and interpretations of Higgs boson production and decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[arXiv:2602.18611](#) | 2026-02-20

48. Measurement of event shape variables using charged particles inside jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[arXiv:2602.17509](#) | 2026-02-19

49. Building an AI-native Research Ecosystem for Experimental Particle Physics: A Community Vision

Aarrestad, Thea Klæboe et al.

[arXiv:2602.17582](#) | 2026-02-19

50. Interpretable Models for Workflow Differentiation in High-Performance Scientific Networks

Giannakou, Anna et al.

(2026) 349 | DOI:10.1109/ICNC68183.2026.11416957 | 2026

51. Influence of strangeness on the anisotropic flow of prompt $D^{\pm}_s$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Belyaev, Andrey et al.

[arXiv:2602.14221](#) | 2026-02-15

52. High-level hadronic tau lepton triggers of the CMS experiment in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

JINST 21 (2026) P04002 | [arXiv:2602.11359](#) | DOI:10.1088/1748-0221/21/04/P04002 | 2026-02-11

53. Measurement of B meson production fraction ratios in proton-proton collisions at $\sqrt{s} = 13$ TeV using open-charm and charmonium decays

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[arXiv:2602.10270](#) | 2026-02-10

54. Probing early parton emissions in heavy ion collisions using the Lund jet plane

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[arXiv:2602.09271](#) | 2026-02-09

55. Observation of a family of all-charm tetraquarks

Hayrapetyan, Aram et al.

[arXiv:2602.02252](#) | 2026-02-02

56. Observation of $\Upsilon(1S) + Z$ associated production and measurement of the effective double-parton scattering cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.02770](#) | 2026-02-02

57. Search for heavy scalar resonances decaying to Lorentz-boosted Higgs and Higgs-like bosons in the $b\bar{b}4q$ final state at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2602.00273](#) | 2026-01-30

58. Measurement and effective field theory interpretation of the photon-fusion production cross section of a pair of W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.21574](#) | 2026-01-29

59. Search for heavy long-lived charged particles with level-1 trigger scouting data from proton-proton collisions at $\sqrt{s} = 13.6$ TeV

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Phys.Lett.B 878 (2026) 140498 | [arXiv:2601.20063](#) | DOI:10.1016/j.physletb.2026.140498 | 2026-01-27

60. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s} = 13.6$ TeV

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[arXiv:2601.20023](#) | 2026-01-27

61. Search for heavy resonances decaying into two Higgs bosons in the $b\bar{b}\tau^+\tau^-$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[arXiv:2601.20011](#) | 2026-01-27

62. Evaluating Application Characteristics for GPU Portability Layer Selection

Atif, Mohammad et al.

[arXiv:2601.17526](#) | 2026-01-24

63. Search for the pair production of long-lived supersymmetric partners of the tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17576](#) | 2026-01-24

64. Full event interpretation with machine-learning-based particle-flow reconstruction in the CMS detector

Hayrapetyan, Aram et al.

[arXiv:2601.17554](#) | 2026-01-24

65. Strategy and performance of the CMS long-lived particle trigger program in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.17544](#) | 2026-01-24

66. Measurement of the $Z\gamma$ production cross section and search for anomalous neutral triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.14102](#) | 2026-01-20

67. Combination of searches for heavy vector boson resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.12583](#) | 2026-01-18

68. Search for dark matter produced in association with a Higgs boson decaying to bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

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Phys.Rev.D 113 (2026) 092016 | [arXiv:2601.11330](#) | DOI:10.1103/2m87-3dlr | 2026-01-16

69. Measurement of the Higgs boson total decay width using the $H \rightarrow WW \rightarrow e\nu\mu\nu$ decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 092014](#) | [arXiv:2601.05168](#) | [DOI:10.1103/rny9-zby1](#) | 2026-01-08

70. Search for a boosted Higgs boson decaying to bottom quark pairs in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.05362](#) | 2026-01-08

71. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A}$ with $\mathcal{A}\mathcal{A} \rightarrow \gamma\gamma$ in events with a semi-merged topology in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2601.00183](#) | 2026-01-01

72. Search for charged Higgs bosons decaying into top and bottom quarks in lepton+jets final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2512.24471](#) | 2025-12-30

73. Machine-learning techniques for model-independent searches in dijet final states

Hayrapetyan, Aram et al.

[arXiv:2512.20395](#) | [DOI:10.5281/zenodo.16656501](#) | 2025-12-23

74. Characterization of the quantum state of top quark pairs produced in proton-proton collisions at $\sqrt{s}=13\text{TeV}$ using the beam and helicity bases

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[Phys.Rev.D 113 \(2026\) L051101](#) | [arXiv:2512.17557](#) | [DOI:10.1103/hkgw-dztk](#) | 2025-12-19

75. Measurement of $Z\gamma$ production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV and constraints on neutral triple gauge couplings

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[arXiv:2512.08582](#) | 2025-12-09

76. Measurements of electroweak production of a photon in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 05 \(2026\) 020](#) | [arXiv:2512.00502](#) | [DOI:10.1007/JHEP05\(2026\)020](#) | 2025-11-29

77. Search for Light Pseudoscalar Bosons, Pair-Produced in Higgs Boson Decays in the Four-Electron Final State in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

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[Phys.Rev.Lett. 136 \(2026\) 181807](#) | [arXiv:2511.19563](#) | [DOI:10.1103/r53x-8fqp](#) | 2025-11-24

78. Search for new physics in the final state with a single photon and large missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 072020](#) | [arXiv:2511.17310](#) | [DOI:10.1103/f9r8-styj](#) | 2025-11-21

79. Search for Higgsinos in final states with low-momentum lepton-track pairs at 13 TeV

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[Phys.Rev.D 113 \(2026\) 092009](#) | [arXiv:2511.16394](#) | [DOI:10.1103/w3b5-92ls](#) | 2025-11-20

80. Search for heavy $H\gamma$ and $Z\gamma$ resonances with a bottom quark-antiquark pair in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2511.14583](#) | 2025-11-18

81. Search for low-mass hidden-valley dark showers with non-prompt muon pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 03 \(2026\) 189](#) | [arXiv:2511.11888](#) | [DOI:10.1007/JHEP03\(2026\)189](#) | 2025-11-14

82. Search for a new neutral gauge boson produced in association with one or two b jets and decaying into a pair of muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

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[arXiv:2511.11853](#) | 2025-11-14

83. Search for long-lived particles using displaced vertices with low-momentum tracks in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 05 \(2026\) 191](#) | [arXiv:2511.08212](#) | [DOI:10.1007/JHEP05\(2026\)191](#) | 2025-11-11

84. Identification of tau leptons using a convolutional neural network with domain adaptation

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[JINST 20 \(2025\) P12032](#) | [arXiv:2511.05468](#) | [DOI:10.1088/1748-0221/20/12/P12032](#) | 2025-11-07

85. Jet fragmentation function and groomed substructure of bottom quark jets in proton-proton collisions at 5.02 TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 147](#) | [arXiv:2511.10666](#) | [DOI:10.1007/JHEP04\(2026\)147](#) | 2025-11-06

86. Inclusive and differential measurements of the $t\bar{t}\gamma$ cross section and the $t\bar{t}\gamma / t\bar{t}$ cross section ratio in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 086](#) | [arXiv:2511.01995](#) | [DOI:10.1007/JHEP04\(2026\)086](#) | 2025-11-03

87. Characterizing the initial state and dynamical evolution in XeXe and PbPb collisions using multiparticle cumulants

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[Phys.Lett.B 876 \(2026\) 140359](#) | [arXiv:2510.26766](#) | [DOI:10.1016/j.physletb.2026.140359](#) | 2025-10-30

88. Search for single production of a vector-like T quark decaying to a top quark and a neutral scalar boson in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2510.25874](#) | 2025-10-29

89. Measurement of the ratio of the $B_c^+ \rightarrow J/\psi \tau^+ \nu_\tau$ and $B_c^+ \rightarrow J/\psi \mu^+ \nu_\mu$ branching fractions using three-prong τ lepton decays

Chekhovsky, Vladimir et al.

[arXiv:2510.21559](#) | 2025-10-24

90. Search for dijet resonances with data scouting in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2026\) 133](#) | [arXiv:2510.21641](#) | [DOI:10.1007/JHEP04\(2026\)133](#) | 2025-10-24

91. Observation of tWZ Production at the CMS Experiment

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 081802](#) | [arXiv:2510.19080](#) | [DOI:10.1103/rk6w-1pcl](#) | 2025-10-21

92. Search for dark matter production in association with bottom quarks and a lepton pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

[arXiv:2510.12396](#) | 2025-10-14

93. Performance of heavy-flavour jet identification in Lorentz-boosted topologies in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P11006](#) | [arXiv:2510.10228](#) | [DOI:10.1088/1748-0221/20/11/P11006](#) | 2025-10-11

94. Observation of Suppressed Charged-Particle Production in Ultrarelativistic Oxygen-Oxygen Collisions

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[Phys.Rev.Lett. 136 \(2026\) 162301](#) | [arXiv:2510.09864](#) | [DOI:10.1103/89sf-9t1x](#) | 2025-10-10

95. Combination of searches for nonresonant Higgs boson pair production in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2510.07527](#) | 2025-10-08

96. Wasserstein normalized autoencoder for anomaly detection

Hayrapetyan, Aram et al.

[Mach.Learn.Sci.Tech. 7 \(2026\) 035030](#) | [arXiv:2510.02168](#) | [DOI:10.1088/2632-2153/ae6168](#) | 2025-10-02

97. Observation of long-range collective flow in OO and NeNe collisions and implications for nuclear structure studies

Hayrapetyan, Aram et al.

[arXiv:2510.02580](#) | 2025-10-02

98. Vector boson scattering and anomalous quartic couplings in final states with $\ell\nu qq$ or $\ell\ell qq$ plus jets using proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 022](#) | [arXiv:2510.00118](#) | [DOI:10.1007/JHEP03\(2026\)022](#) | 2025-09-30

99. Measurement of the dineutrino system kinematic variables in dileptonic top quark pair production in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 05 \(2026\) 175](#) | [arXiv:2510.00160](#) | [DOI:10.1007/JHEP05\(2026\)175](#) | 2025-09-30

100. Simultaneous Probe of the Charm and Bottom Quark Yukawa Couplings Using $t\bar{t}H$ Events

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 011801](#) | [arXiv:2509.22535](#) | [DOI:10.1103/9nwb-splk](#) | 2025-09-26

101. Search for resonances decaying to an anomalous jet and a Higgs boson in proton-proton collisions at $\sqrt{s}=13$ TeV

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[Eur.Phys.J.C 86 \(2026\) 137](#) | [arXiv:2509.13635](#) | [DOI:10.1140/epjc/s10052-025-15176-9](#) | 2025-09-17

102. Measurements of $t\bar{t}W$ differential cross sections and the leptonic charge asymmetry at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 083](#) | [arXiv:2509.13512](#) | [DOI:10.1007/JHEP03\(2026\)083](#) | 2025-09-16

103. Improving missing transverse momentum estimation with a deep neural network

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[Phys.Rev.D 113 \(2026\) 072010](#) | [arXiv:2509.12012](#) | [DOI:10.1103/c4z7-tqvc](#) | 2025-09-15

104. Measurement of D^0 Meson Photoproduction in Ultraperipheral Heavy Ion Collisions

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 136 \(2026\) 122303](#) | [arXiv:2509.08626](#) | [DOI:10.1103/lckg-sdh9](#) | 2025-09-10

105. Model-independent measurement of the Higgs boson associated production with two jets and decaying to a pair of W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 03 \(2026\) 073](#) | [arXiv:2509.07958](#) | [DOI:10.1007/JHEP03\(2026\)073](#) | 2025-09-09

106. Search for a Higgs boson produced in association with a charm quark and decaying to a W boson pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2508.14988](#) | 2025-08-20

107. General search for supersymmetric particles in scenarios with compressed mass spectra using proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.D 112 \(2025\) 112023](#) | [arXiv:2508.13900](#) | [DOI:10.1103/b26z-zmpy](#) | 2025-08-19

108. Search for a new scalar resonance decaying to a Higgs boson and another new scalar particle in the final state with two bottom quarks and two photons in proton-proton collisions at $\sqrt{s}=13$ TeV

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[JHEP 12 \(2025\) 178](#) | [arXiv:2508.11494](#) | [DOI:10.1007/JHEP12\(2025\)178](#) | 2025-08-15

109. Search for charged lepton flavor violating Z and Z' boson decays in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 112 \(2025\) 112011](#) | [arXiv:2508.07512](#) | [DOI:10.1103/8lw4-6lyk](#) | 2025-08-11

110. Search for light pseudoscalar boson pairs produced from Higgs boson decays using the 4τ and $2\mu 2\tau$ final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 132](#) | [arXiv:2508.06947](#) | [DOI:10.1007/JHEP04\(2026\)132](#) | 2025-08-09

111. Search for b-hadron decays to long-lived particles in the CMS endcap muon detectors

Hayrapetyan, Aram et al.

[Phys.Rev.D 113 \(2026\) 012009](#) | [arXiv:2508.06363](#) | [DOI:10.1103/7ldn-snzn](#) | 2025-08-08

112. First Exclusive Reconstruction of the B^{*+} , B^{*0} , and B_s^{*0} Mesons and Precise Measurement of Their Masses

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 031902](#) | [arXiv:2508.05820](#) | [DOI:10.1103/njf9-4zfv](#) | 2025-08-07

113. Probing the flavour structure of dimension-6 EFT operators in multilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2026\) 140](#) | [arXiv:2507.17498](#) | [DOI:10.1007/JHEP03\(2026\)140](#) | 2025-07-23

114. Search for resonant production of pairs of dijet resonances through broad mediators in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[arXiv:2507.17884](#) | 2025-07-23

115. Exploring small-angle emissions in charm quark jets in proton-proton collisions at $\sqrt{s}=5.02$ TeV

Chekhovsky, Vladimir et al.

[JHEP 05 \(2026\) 176](#) | [arXiv:2507.13469](#) | [DOI:10.1007/JHEP05\(2026\)176](#) | 2025-07-17

116. Evidence of medium response to hard probes using correlations of Z bosons with hadrons in heavy ion collisions

Chekhovsky, Vladimir et al.

[Phys.Lett.B 874 \(2026\) 140120](#) | [arXiv:2507.09307](#) | [DOI:10.1016/j.physletb.2025.140120](#) | 2025-07-12

117. A method for correcting the substructure of multiprong jets using the Lund jet plane

Hayrapetyan, Aram et al.

[JHEP 11 \(2025\) 038](#) | [arXiv:2507.07775](#) | [DOI:10.1007/JHEP11\(2025\)038](#) | 2025-07-10

118. Search for heavy pseudoscalar and scalar bosons decaying to a top quark pair in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 88 \(2025\) 127801](#) | [arXiv:2507.05119](#) | [DOI:10.1088/1361-6633/ae2207](#) | 2025-07-07

119. Search for the nonresonant and resonant production of a Higgs boson in association with an additional scalar boson in the $\gamma\gamma\tau\tau$ final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2026\) 186](#) | [arXiv:2506.23012](#) | [DOI:10.1007/JHEP05\(2026\)186](#) | 2025-06-28

120. Operation and performance of the CMS silicon strip tracker with proton-proton collisions at the CERN LHC

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P08027](#) | [arXiv:2506.17195](#) | [DOI:10.1088/1748-0221/20/08/P08027](#) | 2025-06-20

121. Search for nonresonant new physics signals in high-mass dilepton events produced in association with b-tagged jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 02 \(2026\) 141](#) | [arXiv:2506.13565](#) | [DOI:10.1007/JHEP02\(2026\)141](#) | 2025-06-16

122. Search for top squarks in final states with many light-flavor jets and 0, 1, or 2 charged leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 10 \(2025\) 236](#) | [arXiv:2506.08825](#) | [DOI:10.1007/JHEP10\(2025\)236](#) | 2025-06-10

123. Machine learning method for enforcing variable independence in background estimation with LHC data: ABCDisCoTEC

Hayrapetyan, Aram et al.

[arXiv:2506.08826](#) | 2025-06-10

124. Determination of the spin and parity of all-charm tetraquarks

Hayrapetyan, Aram et al.

[Nature 648 \(2025\) 58](#) | [arXiv:2506.07944](#) | [DOI:10.1038/s41586-025-09711-7](#) | 2025-06-09

125. Search for the Rare Decay $D^0 \rightarrow \mu^+ \mu^-$ in Proton-Proton Collisions at $\sqrt{s}=13.6$ TeV

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 135 \(2025\) 151803](#) | [arXiv:2506.06152](#) | [DOI:10.1103/zc76-rgcp](#) | 2025-06-06

126. Search for dark matter produced in association with a Higgs boson decaying to a τ lepton pair in proton-proton collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

[JHEP 10 \(2025\) 170](#) | [arXiv:2506.04431](#) | [DOI:10.1007/JHEP10\(2025\)170](#) | 2025-06-04

127. Search for CP violation in events with top quarks and Z bosons at $\sqrt{s} = 13$ and 13.6 TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 869 \(2025\) 139857](#) | [arXiv:2505.21206](#) | [DOI:10.1016/j.physletb.2025.139857](#) | 2025-05-27

128. Measurement of WWZ and ZH Production Cross Sections at $\sqrt{s}=13$ and 13.6 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 135 \(2025\) 091802](#) | [arXiv:2505.20483](#) | [DOI:10.1103/6z3d-zjw4](#) | 2025-05-26

129. Measurement of event shapes in minimum-bias events from proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Chekhovsky, Vladimir et al.

[Phys.Rev.D 112 \(2025\) 112006](#) | [arXiv:2505.17850](#) | [DOI:10.1103/rqvb-gfb7](#) | 2025-05-23

130. Search for dark matter produced in association with one or two top quarks in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Chekhovsky, Vladimir et al.

[JHEP 08 \(2025\) 085](#) | [arXiv:2505.05300](#) | [DOI:10.1007/JHEP08\(2025\)085](#) | 2025-05-08

131. Measurements of inclusive and differential Higgs boson production cross sections at $\sqrt{s}=13.6\text{ TeV}$ in the $H \rightarrow \gamma\gamma$ decay channel

Hayrapetyan, Aram et al.

[JHEP 09 \(2025\) 070](#) | [arXiv:2504.17755](#) | [DOI:10.1007/JHEP09\(2025\)070](#) | 2025-04-24

132. Combination and interpretation of differential Higgs boson production cross sections in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Chekhovsky, Vladimir et al.

[JHEP 04 \(2026\) 093](#) | [arXiv:2504.13081](#) | [DOI:10.1007/JHEP04\(2026\)093](#) | 2025-04-17

133. Search for charged-lepton flavour violation in top quark interactions with an up-type quark, a muon, and a τ lepton in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 12 \(2025\) 065](#) | [arXiv:2504.08532](#) | [DOI:10.1007/JHEP12\(2025\)065](#) | 2025-04-11

134. Search for jet quenching with dijets from high-multiplicity pPb collisions at $\sqrt{s_{\text{NN}}}=8.16\text{ TeV}$

Chekhovsky, Vladimir et al.

[JHEP 07 \(2025\) 118](#) | [arXiv:2504.08507](#) | [DOI:10.1007/JHEP07\(2025\)118](#) | 2025-04-11

135. Observation of Coherent $\phi(1020)$ Meson Photoproduction in Ultraperipheral PbPb Collisions at $\sqrt{s_{\text{NN}}}=5.36\text{TeV}$

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 135 \(2025\) 262301](#) | [arXiv:2504.05193](#) | [DOI:10.1103/2ssw-wwyy](#) | 2025-04-07

136. Combined effective field theory interpretation of Higgs boson, electroweak vector boson, top quark, and multijet measurements

Chekhovsky, Vladimir et al.

[Eur.Phys.J.C 86 \(2026\) 331](#) | [arXiv:2504.02958](#) | [DOI:10.1140/epjc/s10052-025-14997-y](#) | 2025-04-03

137. Highlights of the HL-LHC physics projections by ATLAS and CMS

Aad, Georges et al.

[arXiv:2504.00672](#) | 2025-04-01

138. Observation of a pseudoscalar excess at the top quark pair production threshold

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 88 \(2025\) 087801](#) | [arXiv:2503.22382](#) | [DOI:10.1088/1361-6633/adf7d3](#) | 2025-03-28

139. Observation of $WZ\gamma$ production and constraints on new physics scenarios in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Chekhovsky, V. et al.

[Phys.Rev.D 112 \(2025\) 012009](#) | [arXiv:2503.21977](#) | [DOI:10.1103/cm24-665b](#) | 2025-03-27

140. Observation of nuclear modification of energy-energy correlators inside jets in heavy ion collisions

Chekhovsky, Vladimir et al.

Phys.Lett.B 866 (2025) 139556 | [arXiv:2503.19993](#) | [DOI:10.1016/j.physletb.2025.139556](#) | 2025-03-25

141. Search for t-channel scalar and vector leptoquark exchange in the high-mass dimuon and dielectron spectra in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, A. et al.

JHEP 12 (2025) 052 | [arXiv:2503.20023](#) | [DOI:10.1007/JHEP12\(2025\)052](#) | 2025-03-25

142. Search for dark matter production in association with a single top quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

JHEP 09 (2025) 141 | [arXiv:2503.20033](#) | [DOI:10.1007/JHEP09\(2025\)141](#) | 2025-03-25

143. Search for vector-like leptons with long-lived particle decays in the CMS muon system in proton-proton collisions at $\sqrt{s} = 13$ TeV

Chekhovsky, Vladimir et al.

JHEP 08 (2025) 156 | [arXiv:2503.16699](#) | [DOI:10.1007/JHEP08\(2025\)156](#) | 2025-03-20

144. Measurements of the inclusive W and Z boson production cross sections and their ratios in proton-proton collisions at $\sqrt{s}=13.6$ TeV

Chekhovsky, Vladimir et al.

JHEP 01 (2026) 047 | [arXiv:2503.09742](#) | [DOI:10.1007/JHEP01\(2026\)047](#) | 2025-03-12

145. Probing Gluon Fluctuations in Nuclei with the First Energy-Dependent Measurement of Incoherent J/ψ Photoproduction in Ultraperipheral PbPb Collisions

Chekhovsky, Vladimir et al.

Phys.Rev.Lett. 135 (2025) 112301 | [arXiv:2503.08903](#) | [DOI:10.1103/w9kp-f8xr](#) | 2025-03-11

146. Search for a cH signal in the associated production of at least one charm quark with a Higgs boson in the diphoton decay channel in pp collisions at $\sqrt{s}=13$ TeV

Chekhovsky, Vladimir et al.

JHEP 11 (2025) 060 | [arXiv:2503.08797](#) | [DOI:10.1007/JHEP11\(2025\)060](#) | 2025-03-11

147. Search for New Physics in Jet Multiplicity Patterns of Multilepton Events at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

Phys.Rev.Lett. 135 (2025) 231804 | [arXiv:2503.06726](#) | [DOI:10.1103/51fw-klz3](#) | 2025-03-09

148. Luminosity measurement for lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV in 2015 and 2018 at CMS

Chekhovsky, Vladimir et al.

[arXiv:2503.03946](#) | 2025-03-05

149. Observation of the Charged-Particle Multiplicity Dependence of $\sigma_{\psi(2S)}/\sigma_{J/\psi}$ in p-Pb Collisions at 8.16 TeV

Chekhovsky, Vladimir et al.

Phys.Rev.Lett. 135 (2025) 092301 | [arXiv:2503.02139](#) | [DOI:10.1103/c9wp-5tq3](#) | 2025-03-03

150. Search for medium effects using jet axis decorrelation in inclusive jets from PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Chekhovsky, Vladimir et al.

JHEP 06 (2025) 120 | [arXiv:2502.13020](#) | [DOI:10.1007/JHEP06\(2025\)120](#) | 2025-02-18

151. Development of systematic uncertainty-aware neural network trainings for binned-likelihood analyses at the LHC

Chekhovsky, Vladimir et al.

Eur.Phys.J.C 85 (2025) 1360 | [arXiv:2502.13047](#) | [DOI:10.1140/epjc/s10052-025-14713-w](#) | 2025-02-18

152. Evidence for Similar Collectivity of High Transverse-Momentum Particles in p-Pb and Pb-Pb Collisions

Chekhovsky, Vladimir et al.

[Phys.Rev.Lett. 135 \(2025\) 071903](#) | [arXiv:2502.07525](#) | [DOI:10.1103/t5kp-vsv7](#) | 2025-02-11

153. Observation of Λ Hyperon Local Polarization in p-Pb Collisions at $\sqrt{s_{NN}}=8.16\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 135 \(2025\) 132301](#) | [arXiv:2502.07898](#) | [DOI:10.1103/6ywq-gm61](#) | 2025-02-11

154. Search for bosons of an extended Higgs sector in b quark final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Chekhovsky, Vladimir et al.

[JHEP 06 \(2025\) 144](#) | [arXiv:2502.06568](#) | [DOI:10.1007/JHEP06\(2025\)144](#) | 2025-02-10

155. Search for γH production and constraints on the Yukawa couplings of light quarks to the Higgs boson

Chekhovsky, Vladimir et al.

[Phys.Rev.D 112 \(2025\) 112001](#) | [arXiv:2502.05665](#) | [DOI:10.1103/6s8n-pvmy](#) | 2025-02-08

156. Measurements of the Higgs boson production cross section in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13.6\text{ TeV}$

Chekhovsky, Vladimir et al.

[JHEP 05 \(2025\) 079](#) | [arXiv:2501.14849](#) | [DOI:10.1007/JHEP05\(2025\)079](#) | 2025-01-24

157. Search for a heavy pseudoscalar Higgs boson decaying to a 125 GeV Higgs boson and a Z boson in final states with two tau and two light leptons in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Chekhovsky, Vladimir et al.

[JHEP 10 \(2025\) 074](#) | [arXiv:2501.14825](#) | [DOI:10.1007/JHEP10\(2025\)074](#) | 2025-01-21

158. Evidence for CP violation and measurement of CP-violating parameters in $B^0_s \rightarrow J/\psi \phi(1020)$ decays in pp collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 136 \(2026\) 181801](#) | [arXiv:2412.19952](#) | [DOI:10.1103/ctqx-vxrj](#) | 2024-12-27

159. Search for a Neutral Gauge Boson with Nonuniversal Fermion Couplings in Vector Boson Fusion Processes in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 135 \(2025\) 061803](#) | [arXiv:2412.19261](#) | [DOI:10.1103/srvm-f1h3](#) | 2024-12-26

160. Identification of low-momentum muons in the CMS detector using multivariate techniques in proton-proton collisions at $\sqrt{s} = 13.6\text{ TeV}$

Chekhovsky, Vladimir et al.

[JINST 20 \(2025\) P04021](#) | [arXiv:2412.17590](#) | [DOI:10.1088/1748-0221/20/04/P04021](#) | 2024-12-23

161. Determination of the strong coupling and its running from measurements of inclusive jet production

Chekhovsky, Vladimir et al.

[Phys.Lett.B 868 \(2025\) 139651](#) | [arXiv:2412.16665](#) | [DOI:10.1016/j.physletb.2025.139651](#) | 2024-12-21

162. Measurement of the W boson decay branching fraction ratio $\mathcal{B}(W \rightarrow cq)/\mathcal{B}(W \rightarrow q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 868 \(2025\) 139754](#) | [arXiv:2412.16296](#) | [DOI:10.1016/j.physletb.2025.139754](#) | 2024-12-20

163. Measurement of light-by-light scattering and the Breit-Wheeler process, and search for axion-like particles in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 08 \(2025\) 006](#) | [arXiv:2412.15413](#) | [DOI:10.1007/JHEP08\(2025\)006](#) | 2024-12-19

164. High-precision measurement of the W boson mass with the CMS experiment

Chekhovsky, Vladimir et al.

[Nature 652 \(2026\) 321](#) | [arXiv:2412.13872](#) | [DOI:10.1038/s41586-026-10168-5](#) | 2024-12-18

165. Search for heavy neutral resonances decaying to tau lepton pairs in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 112004](#) | [arXiv:2412.04357](#) | [DOI:10.1103/PhysRevD.111.112004](#) | 2024-12-05

166. Model-agnostic search for dijet resonances with anomalous jet substructure in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Chekhovsky, Vladimir et al.

[Rept.Prog.Phys. 88 \(2025\) 067802](#) | [arXiv:2412.03747](#) | [DOI:10.1088/1361-6633/add762](#) | 2024-12-04

167. Measurement of the inclusive WZ production cross section in pp collisions at $\sqrt{s} = 13.6 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 04 \(2025\) 115](#) | [arXiv:2412.02477](#) | [DOI:10.1007/JHEP04\(2025\)115](#) | 2024-12-03

168. Search for heavy neutral Higgs bosons A and H in the $t\bar{t}Z$ channel in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 866 \(2025\) 139568](#) | [arXiv:2412.00570](#) | [DOI:10.1016/j.physletb.2025.139568](#) | 2024-11-30

169. Proton reconstruction with the TOTEM Roman pot detectors for high- β^* LHC data

Hayrapetyan, Aram et al.

[JINST 20 \(2025\) P04012](#) | [arXiv:2411.19749](#) | [DOI:10.1088/1748-0221/20/04/P04012](#) | 2024-11-29

170. Constraints on standard model effective field theory for a Higgs boson produced in association with W or Z bosons in the $H \rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Chekhovsky, Vladimir et al.

[JHEP 03 \(2025\) 114](#) | [arXiv:2411.16907](#) | [DOI:10.1007/JHEP03\(2025\)114](#) | 2024-11-25

171. Search for rare decays of the Z and Higgs bosons to a J/ψ or $\psi(2S)$ meson and a photon in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 865 \(2025\) 139462](#) | [arXiv:2411.15000](#) | [DOI:10.1016/j.physletb.2025.139462](#) | 2024-11-22

172. Angular analysis of the $B^0 \rightarrow K^{*(892)^0} \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 864 \(2025\) 139406](#) | [arXiv:2411.11820](#) | [DOI:10.1016/j.physletb.2025.139406](#) | 2024-11-18

173. Reweighting simulated events using machine-learning techniques in the CMS experiment

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 495](#) | [arXiv:2411.03023](#) | [DOI:10.1140/epjc/s10052-025-14097-x](#) | 2024-11-05

174. Search for a heavy resonance decaying into a Z and a Higgs boson in events with an energetic jet and two electrons, two muons, or missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 089](#) | [arXiv:2411.00202](#) | [DOI:10.1007/JHEP02\(2025\)089](#) | 2024-10-31

175. Measurements of inclusive and differential cross sections for top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 177](#) | [arXiv:2410.23475](#) | [DOI:10.1007/JHEP02\(2025\)177](#) | 2024-10-30

176. Measurement of the inclusive $t\bar{t}$ cross section in final states with at least one lepton and additional jets with 302 pb^{-1} of pp collisions at $\sqrt{s} = 5.02$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2025\) 099](#) | [arXiv:2410.21631](#) | [DOI:10.1007/JHEP04\(2025\)099](#) | 2024-10-28

177. Search for excited tau leptons in the $\tau\tau\gamma$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2025\) 006](#) | [arXiv:2410.21137](#) | [DOI:10.1007/JHEP06\(2025\)006](#) | 2024-10-28

178. Search for pair production of heavy particles decaying to a top quark and a gluon in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 342](#) | [arXiv:2410.20601](#) | [DOI:10.1140/epjc/s10052-024-13729-y](#) | 2024-10-27

179. Search for the Higgs boson decays to a ρ^0 , ϕ , or K^*_0 meson and a photon in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 862 \(2025\) 139296](#) | [arXiv:2410.18289](#) | [DOI:10.1016/j.physletb.2025.139296](#) | 2024-10-23

180. Search for high-mass resonances in a final state comprising a gluon and two hadronically decaying W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 199](#) | [arXiv:2410.17303](#) | [DOI:10.1007/JHEP02\(2025\)199](#) | 2024-10-22

181. Performance of the CMS high-level trigger during LHC Run 2

Hayrapetyan, A. et al.

[JINST 19 \(2024\) P11021](#) | [arXiv:2410.17038](#) | [DOI:10.1088/1748-0221/19/11/P11021](#) | 2024-10-22

182. Search for heavy long-lived charged particles with large ionization energy loss in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2025\) 109](#) | [arXiv:2410.09164](#) | [DOI:10.1007/JHEP04\(2025\)109](#) | 2024-10-11

183. Correlations between azimuthal anisotropy and mean transverse momentum in pp, pPb, and peripheral PbPb collisions

Tumasyan, Armen et al.

[arXiv:2410.04578](#) | 2024-10-06

184. Study of same-sign W boson scattering and anomalous couplings in events with one tau lepton from pp collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2025\) 219](#) | [arXiv:2410.04210](#) | [DOI:10.1007/JHEP10\(2025\)219](#) | 2024-10-05

185. Energy-scaling behavior of intrinsic transverse-momentum parameters in Drell-Yan simulation

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 072003](#) | [arXiv:2409.17770](#) | [DOI:10.1103/PhysRevD.111.072003](#) | 2024-09-26

186. Measurement of the Higgs boson mass and width using the four-lepton final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 092014](#) | [arXiv:2409.13663](#) | [DOI:10.1103/PhysRevD.111.092014](#) | 2024-09-20

187. Measurements of polarization and spin correlation and observation of entanglement in top quark pairs using lepton+jets events from proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 112016](#) | [arXiv:2409.11067](#) | [DOI:10.1103/PhysRevD.110.112016](#) | 2024-09-17

188. Search for light long-lived particles decaying to displaced jets in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 88 \(2025\) 037801](#) | [arXiv:2409.10806](#) | [DOI:10.1088/1361-6633/adaa13](#) | 2024-09-16

189. Bottom quark energy loss and hadronization with B^+ and B_s^0 nuclear modification factors using pp and PbPb collisions at $\sqrt{s_{NN}} = 5.02 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 195](#) | [arXiv:2409.07258](#) | [DOI:10.1007/JHEP02\(2025\)195](#) | 2024-09-11

190. Measurement of inclusive and differential cross sections of single top quark production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 01 \(2025\) 107](#) | [arXiv:2409.06444](#) | [DOI:10.1007/JHEP01\(2025\)107](#) | 2024-09-10

191. Pseudorapidity distributions of charged hadrons in lead-lead collisions at $\sqrt{s_{NN}}=5.36 \text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 861 \(2025\) 139279](#) | [arXiv:2409.00838](#) | [DOI:10.1016/j.physletb.2025.139279](#) | 2024-09-01

192. Search for dark matter produced in association with a pair of bottom quarks in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 050](#) | [arXiv:2408.17336](#) | [DOI:10.1007/JHEP02\(2025\)050](#) | 2024-08-30

193. Measurement of the Drell-Yan forward-backward asymmetry and of the effective leptonic weak mixing angle in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 866 \(2025\) 139526](#) | [arXiv:2408.07622](#) | [DOI:10.1016/j.physletb.2025.139526](#) | 2024-08-14

194. Measurement of the inclusive cross sections for W and Z boson production in proton-proton collisions at $\sqrt{s} = 5.02$ and 13 TeV

Hayrapetyan, Aram et al.

[JHEP 04 \(2025\) 162](#) | [arXiv:2408.03744](#) | [DOI:10.1007/JHEP04\(2025\)162](#) | 2024-08-07

195. Search for bottom quark associated production of the standard model Higgs boson in final states with leptons in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 860 \(2025\) 139173](#) | [arXiv:2408.01344](#) | [DOI:10.1016/j.physletb.2024.139173](#) | 2024-08-02

196. Test of lepton flavor universality in semileptonic B_c^+ meson decays in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) L051102](#) | [arXiv:2408.00678](#) | [DOI:10.1103/PhysRevD.111.L051102](#) | 2024-08-01

197. Model-independent search for pair production of new bosons decaying into muons in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 12 \(2024\) 172](#) | [arXiv:2407.20425](#) | [DOI:10.1007/JHEP12\(2024\)172](#) | 2024-07-29

198. Search for flavor-changing neutral current interactions of the top quark mediated by a Higgs boson in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 112 \(2025\) 032008](#) | [arXiv:2407.15172](#) | [DOI:10.1103/95q6-vvlp](#) | 2024-07-21

199. Multiplicity dependence of charm baryon and charm meson production in pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 868 \(2025\) 139672](#) | [arXiv:2407.13615](#) | [DOI:10.1016/j.physletb.2025.139672](#) | 2024-07-18

200. Constraints on the Higgs boson self-coupling from the combination of single and double Higgs boson production in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 861 \(2025\) 139210](#) | [arXiv:2407.13554](#) | [DOI:10.1016/j.physletb.2024.139210](#) | 2024-07-18

201. Measurement of the $B^0_s \rightarrow J/\psi K^0_S$ effective lifetime from proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2024\) 247](#) | [arXiv:2407.13441](#) | [DOI:10.1007/JHEP10\(2024\)247](#) | 2024-07-18

202. Search for resonant pair production of Higgs bosons in the bbbb final state using large-area jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2025\) 040](#) | [arXiv:2407.13872](#) | [DOI:10.1007/JHEP02\(2025\)040](#) | 2024-07-18

203. Search for long-lived heavy neutral leptons in proton-proton collision events with a lepton-jet pair associated with a secondary vertex at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 036](#) | [arXiv:2407.10717](#) | [DOI:10.1007/JHEP02\(2025\)036](#) | 2024-07-15

204. Measurement of the ttH and tH production rates in the $H \rightarrow b\bar{b}$ decay channel using proton-proton collision data at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 02 \(2025\) 097](#) | [arXiv:2407.10896](#) | [DOI:10.1007/JHEP02\(2025\)097](#) | 2024-07-15

205. Measurement of boosted Higgs bosons produced via vector boson fusion or gluon fusion in the $H \rightarrow b\bar{b}$ decay mode using LHC proton-proton collision data at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 12 \(2024\) 035](#) | [arXiv:2407.08012](#) | [DOI:10.1007/JHEP12\(2024\)035](#) | 2024-07-10

206. Search for Nuclear Modifications of $B^+ +$ Meson Production in p-Pb Collisions at $\sqrt{s_{NN}}=8.16$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 134 \(2025\) 111903](#) | [arXiv:2407.05402](#) | [DOI:10.1103/PhysRevLett.134.111903](#) | 2024-07-07

207. Observation of double J/ψ meson production in pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 092002](#) | [arXiv:2407.03223](#) | [DOI:10.1103/PhysRevD.110.092002](#) | 2024-07-03

208. Measurement of the polarizations of prompt and non-prompt J/ψ and $\psi(2S)$ mesons produced in pp collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 858 \(2024\) 139044](#) | [arXiv:2406.14409](#) | [DOI:10.1016/j.physletb.2024.139044](#) | 2024-06-20

209. Search for a resonance decaying to a W boson and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV using leptonic W boson decays

Hayrapetyan, Aram et al.

210. Measurement of inclusive and differential cross sections for W^+W^- production in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 861 \(2025\) 139231](#) | [arXiv:2406.05101](#) | [DOI:10.1016/j.physletb.2024.139231](#) | 2024-06-07

211. Observation of quantum entanglement in top quark pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 117801](#) | [arXiv:2406.03976](#) | [DOI:10.1088/1361-6633/ad7e4d](#) | 2024-06-06

212. Observation of $\gamma\gamma \rightarrow \tau\tau$ in proton-proton collisions and limits on the anomalous electromagnetic moments of the τ lepton

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 107801](#) | [arXiv:2406.03975](#) | [DOI:10.1088/1361-6633/ad6fcb](#) | 2024-06-06

213. Search for a standard model-like Higgs boson in the mass range between 70 and 110 GeV in the diphoton final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 860 \(2025\) 139067](#) | [arXiv:2405.18149](#) | [DOI:10.1016/j.physletb.2024.139067](#) | 2024-05-28

214. Stairway to discovery: A report on the CMS programme of cross section measurements from millibarns to femtobarns

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 3](#) | [arXiv:2405.18661](#) | [DOI:10.1016/j.physrep.2024.11.005](#) | 2024-05-28

215. Review of searches for vector-like quarks, vector-like leptons, and heavy neutral leptons in proton-proton collisions at $\sqrt{s}=13$ TeV at the CMS experiment

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2350](#) | [arXiv:2405.17605](#) | [DOI:10.1016/j.physrep.2024.09.012](#) | 2024-05-27

216. Study of WH production through vector boson scattering and extraction of the relative sign of the W and Z couplings to the Higgs boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 860 \(2025\) 139202](#) | [arXiv:2405.16566](#) | [DOI:10.1016/j.physletb.2024.139202](#) | 2024-05-26

217. Searches for violation of Lorentz invariance in top quark pair production using dilepton events in 13 TeV proton-proton collisions

Hayrapetyan, Aram et al.

[Phys.Lett.B 857 \(2024\) 138979](#) | [arXiv:2405.14757](#) | [DOI:10.1016/j.physletb.2024.138979](#) | 2024-05-23

218. Dark sector searches with the CMS experiment

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2351](#) | [arXiv:2405.13778](#) | [DOI:10.1016/j.physrep.2024.09.013](#) | 2024-05-22

219. Search for CP violation in $D^0 \rightarrow K^0_{\text{SK}} S$ decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 1264](#) | [arXiv:2405.11606](#) | [DOI:10.1140/epjc/s10052-024-13244-0](#) | 2024-05-19

220. Overview of high-density QCD studies with the CMS experiment at the LHC

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2367](#) | [arXiv:2405.10785](#) | [DOI:10.1016/j.physrep.2024.11.007](#) | 2024-05-17

221. Search for new physics in high-mass diphoton events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2024\) 215](#) | [arXiv:2405.09320](#) | [DOI:10.1007/JHEP08\(2024\)215](#) | 2024-05-15

222. Search for production of a single vectorlike quark decaying to tH or tZ in the all-hadronic final state in pp collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 072012](#) | [arXiv:2405.05071](#) | [DOI:10.1103/PhysRevD.110.072012](#) | 2024-05-08

223. Girth and groomed radius of jets recoiling against isolated photons in lead-lead and proton-proton collisions at $\sqrt{s_{NN}}=5.02\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 861 \(2025\) 139088](#) | [arXiv:2405.02737](#) | [DOI:10.1016/j.physletb.2024.139088](#) | 2024-05-04

224. Search for New Resonances Decaying to Pairs of Merged Diphotons in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 134 \(2025\) 041801](#) | [arXiv:2405.00834](#) | [DOI:10.1103/PhysRevLett.134.041801](#) | 2024-05-01

225. Search for the Z Boson Decay to $\tau\tau\mu\mu$ in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 161805](#) | [arXiv:2404.18298](#) | [DOI:10.1103/PhysRevLett.133.161805](#) | 2024-04-28

226. Performance of CMS muon reconstruction from proton-proton to heavy ion collisions

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P09012](#) | [arXiv:2404.17377](#) | [DOI:10.1088/1748-0221/19/09/P09012](#) | 2024-04-26

227. Measurement of multijet azimuthal correlations and determination of the strong coupling in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 842](#) | [arXiv:2404.16082](#) | [DOI:10.1140/epjc/s10052-024-13116-7](#) | 2024-04-24

228. Search for Higgs boson pair production with one associated vector boson in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 10 \(2024\) 061](#) | [arXiv:2404.08462](#) | [DOI:10.1007/JHEP10\(2024\)061](#) | 2024-04-12

229. The CMS Statistical Analysis and Combination Tool: Combine

Hayrapetyan, Aram et al.

[Comput.Softw.Big Sci. 8 \(2024\) 19](#) | [arXiv:2404.06614](#) | [DOI:10.1007/s41781-024-00121-4](#) | 2024-04-09

230. Measurement of differential $ZZ + \text{jets}$ production cross sections in pp collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 10 \(2024\) 209](#) | [arXiv:2404.02711](#) | [DOI:10.1007/JHEP10\(2024\)209](#) | 2024-04-03

231. Searches for Pair-Produced Multijet Resonances Using Data Scouting in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 201803](#) | [arXiv:2404.02992](#) | [DOI:10.1103/PhysRevLett.133.201803](#) | 2024-04-03

232. Search for ZZ and ZH production in the $b\bar{b}b\bar{b}$ final state using proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 712](#) | [arXiv:2403.20241](#) | [DOI:10.1140/epjc/s10052-024-13021-z](#) | 2024-03-29

233. Measurement of the production cross section of a Higgs boson with large transverse momentum in its decays to a pair of τ leptons in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 857 \(2024\) 138964](#) | [arXiv:2403.20201](#) | [DOI:10.1016/j.physletb.2024.138964](#) | 2024-03-29

234. Searches for Higgs boson production through decays of heavy resonances

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2342](#) | [arXiv:2403.16926](#) | [DOI:10.1016/j.physrep.2024.09.004](#) | 2024-03-25

235. Enriching the physics program of the CMS experiment via data scouting and data parking

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2344](#) | [arXiv:2403.16134](#) | [DOI:10.1016/j.physrep.2024.09.006](#) | 2024-03-24

236. Performance of the CMS electromagnetic calorimeter in pp collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P09004](#) | [arXiv:2403.15518](#) | [DOI:10.1088/1748-0221/19/09/P09004](#) | 2024-03-22

237. Observation of the $J/\psi \rightarrow \mu^+ \mu^- \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) L111101](#) | [arXiv:2403.11352](#) | [DOI:10.1103/PhysRevD.109.L111101](#) | 2024-03-17

238. Search for the decay of the Higgs boson to a pair of light pseudoscalar bosons in the final state with four bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 097](#) | [arXiv:2403.10341](#) | [DOI:10.1007/JHEP06\(2024\)097](#) | 2024-03-15

239. Search for Higgs boson pair production in the bbW^+W^- decay mode in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2024\) 293](#) | [arXiv:2403.09430](#) | [DOI:10.1007/JHEP07\(2024\)293](#) | 2024-03-14

240. Search for Soft Unclustered Energy Patterns in Proton-Proton Collisions at 13 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 191902](#) | [arXiv:2403.05311](#) | [DOI:10.1103/PhysRevLett.133.191902](#) | 2024-03-08

241. Search for long-lived heavy neutrinos in the decays of B mesons produced in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 183](#) | [arXiv:2403.04584](#) | [DOI:10.1007/JHEP06\(2024\)183](#) | 2024-03-07

242. Search for dark QCD with emerging jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 07 \(2024\) 142](#) | [arXiv:2403.01556](#) | [DOI:10.1007/JHEP07\(2024\)142](#) | 2024-03-03

243. Review of top quark mass measurements in CMS

Hayrapetyan, Aram et al.

[Phys.Rept. 1115 \(2025\) 2369](#) | [arXiv:2403.01313](#) | [DOI:10.1016/j.physrep.2024.12.002](#) | 2024-03-02

244. Constraints on anomalous Higgs boson couplings from its production and decay using the WW channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 779](#) | [arXiv:2403.00657](#) | [DOI:10.1140/epjc/s10052-024-12925-0](#) | 2024-03-01

245. Search for heavy neutral leptons in final states with electrons, muons, and hadronically decaying tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 06 \(2024\) 123](#) | [arXiv:2403.00100](#) | [DOI:10.1007/JHEP06\(2024\)123](#) | 2024-02-29

246. Search for baryon number violation in top quark production and decay using proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 241802](#) | [arXiv:2402.18461](#) | [DOI:10.1103/PhysRevLett.132.241802](#) | 2024-02-28

247. Search for long-lived heavy neutral leptons decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012004](#) | [arXiv:2402.18658](#) | [DOI:10.1103/PhysRevD.110.012004](#) | 2024-02-28

248. Observation of the $\Xi_b^- \rightarrow \psi(2S)\Xi^-$ decay and studies of the $\Xi_b(5945)^0$ baryon in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 012002](#) | [arXiv:2402.17738](#) | [DOI:10.1103/PhysRevD.110.012002](#) | 2024-02-27

249. Search for long-lived particles using displaced vertices and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112005](#) | [arXiv:2402.15804](#) | [DOI:10.1103/PhysRevD.109.112005](#) | 2024-02-24

250. Portable Acceleration of CMS Computing Workflows with Coprocessors as a Service

Hayrapetyan, Aram et al.

[Comput.Softw.Big Sci. 8 \(2024\) 17](#) | [arXiv:2402.15366](#) | [DOI:10.1007/s41781-024-00124-1](#) | 2024-02-23

251. Search for long-lived particles decaying to final states with a pair of muons in proton-proton collisions at $\sqrt{s} = 13.6 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 047](#) | [arXiv:2402.14491](#) | [DOI:10.1007/JHEP05\(2024\)047](#) | 2024-02-22

252. Measurement of Energy Correlators inside Jets and Determination of the Strong Coupling $\alpha_S(m_Z)$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 071903](#) | [arXiv:2402.13864](#) | [DOI:10.1103/PhysRevLett.133.071903](#) | 2024-02-21

253. Search for bottom-type vectorlike quark pair production in dileptonic and fully hadronic final states in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 052004](#) | [arXiv:2402.13808](#) | [DOI:10.1103/PhysRevD.110.052004](#) | 2024-02-21

254. Search for exotic decays of the Higgs boson to a pair of pseudoscalars in the $\mu\mu b\bar{b}$ and $\tau\tau b\bar{b}$ final states

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 493](#) | [arXiv:2402.13358](#) | [DOI:10.1140/epjc/s10052-024-12727-4](#) | 2024-02-20

255. Search for a scalar or pseudoscalar dilepton resonance produced in association with a massive vector boson or top quark-antiquark pair in multilepton events at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012013](#) | [arXiv:2402.11098](#) | [DOI:10.1103/PhysRevD.110.012013](#) | 2024-02-16

256. Search for Fractionally Charged Particles in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 134 \(2025\) 131802](#) | [arXiv:2402.09932](#) | [DOI:10.1103/PhysRevLett.134.131802](#) | 2024-02-15

257. Differential cross section measurements for the production of top quark pairs and of additional jets using dilepton events from pp collisions at $\sqrt{s} = 13 \text{ TeV}$

Tumasyan, Armen et al.

[JHEP 02 \(2025\) 064](#) | [arXiv:2402.08486](#) | [DOI:10.1007/JHEP02\(2025\)064](#) | 2024-02-13

258. Combination of Measurements of the Top Quark Mass from Data Collected by the ATLAS and CMS Experiments at $\sqrt{s}=7$ and 8 TeV

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 261902](#) | [arXiv:2402.08713](#) | [DOI:10.1103/PhysRevLett.132.261902](#) | 2024-02-13

259. Search for pair production of scalar and vector leptoquarks decaying to muons and bottom quarks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112003](#) | [arXiv:2402.08668](#) | [DOI:10.1103/PhysRevD.109.112003](#) | 2024-02-09

260. Combined search for electroweak production of winos, binos, higgsinos, and sleptons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112001](#) | [arXiv:2402.01888](#) | [DOI:10.1103/PhysRevD.109.112001](#) | 2024-02-02

261. Search for long-lived particles decaying in the CMS muon detectors in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 110 \(2024\) 032007](#) | [arXiv:2402.01898](#) | [DOI:10.1103/PhysRevD.110.032007](#) | 2024-02-02

262. Observation of the $\Lambda_b^0 \rightarrow J/\psi \Xi^- K^+$ decay

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 1062](#) | [arXiv:2401.16303](#) | [DOI:10.1140/epjc/s10052-024-13114-9](#) | 2024-01-29

263. Application of performance portability solutions for GPUs and many-core CPUs to track reconstruction kernels

Kwok, Ka Hei Martin et al.

[EPJ Web Conf. 295 \(2024\) 11003](#) | [arXiv:2401.14221](#) | [DOI:10.1051/epjconf/202429511003](#) | 2024-01-25

264. Nonresonant central exclusive production of charged-hadron pairs in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112013](#) | [arXiv:2401.14494](#) | [DOI:10.1103/PhysRevD.109.112013](#) | 2024-01-25

265. Measurement of the double-differential inclusive jet cross section in proton-proton collisions at $\sqrt{s} = 5.02 \text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 01 \(2025\) 011](#) | [arXiv:2401.11355](#) | [DOI:10.1007/JHEP01\(2025\)011](#) | 2024-01-20

266. Test of lepton flavor universality in $B^{\pm} \rightarrow K^{\pm} \mu^+ \mu^-$ and $B^{\pm} \rightarrow K^{\pm} e^+ e^-$ decays in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 077802](#) | [arXiv:2401.07090](#) | [DOI:10.1088/1361-6633/ad4e65](#) | 2024-01-13

267. Extracting the speed of sound in quark-gluon plasma with ultrarelativistic lead-lead collisions at the LHC

Hayrapetyan, Aram et al.

[Rept.Prog.Phys. 87 \(2024\) 077801](#) | [arXiv:2401.06896](#) | [DOI:10.1088/1361-6633/ad4b9b](#) | 2024-01-12

268. Observation of Enhanced Long-Range Elliptic Anisotropies Inside High-Multiplicity Jets in pp Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 142301](#) | [arXiv:2312.17103](#) | [DOI:10.1103/PhysRevLett.133.142301](#) | 2023-12-28

269. Elliptic anisotropy measurement of the $f_0(980)$ hadron in proton-lead collisions and evidence for its quark-antiquark composition

Hayrapetyan, Aram et al.

[Nature Commun. 16 \(2025\) 7990](#) | [arXiv:2312.17092](#) | [DOI:10.1038/s41467-025-56200-6](#) | 2023-12-28

270. Measurement of multidifferential cross sections for dijet production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 85 \(2025\) 72](#) | [arXiv:2312.16669](#) | [DOI:10.1140/epjc/s10052-024-13606-8](#) | 2023-12-27

271. Measurement of the primary Lund jet plane density in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 116](#) | [arXiv:2312.16343](#) | [DOI:10.1007/JHEP05\(2024\)116](#) | 2023-12-26

272. Evidence for tWZ production in proton-proton collisions at $\sqrt{s}=13$ TeV in multilepton final states

Hayrapetyan, Aram et al.

[Phys.Lett.B 855 \(2024\) 138815](#) | [arXiv:2312.11668](#) | [DOI:10.1016/j.physletb.2024.138815](#) | 2023-12-18

273. Search for Long-Lived Heavy Neutral Leptons with Lepton Flavour Conserving or Violating Decays to a Jet and a Charged Lepton

Hayrapetyan, Aram et al.

[JHEP 03 \(2024\) 105](#) | [arXiv:2312.07484](#) | [DOI:10.1007/JHEP03\(2024\)105](#) | 2023-12-12

274. Search for flavor changing neutral current interactions of the top quark in final states with a photon and additional jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072004](#) | [arXiv:2312.08229](#) | [DOI:10.1103/PhysRevD.109.072004](#) | 2023-12-11

275. Measurement of simplified template cross sections of the Higgs boson produced in association with W or Z bosons in the $H \rightarrow b\bar{b}$ decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 109 \(2024\) 092011](#) | [arXiv:2312.07562](#) | [DOI:10.1103/PhysRevD.109.092011](#) | 2023-12-09

276. Search for charged-lepton flavor violation in the production and decay of top quarks using trilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 111 \(2025\) 012009](#) | [arXiv:2312.03199](#) | [DOI:10.1103/PhysRevD.111.012009](#) | 2023-12-05

277. Search for the lepton flavor violating $\tau \rightarrow 3\mu$ decay in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 853 \(2024\) 138633](#) | [arXiv:2312.02371](#) | [DOI:10.1016/j.physletb.2024.138633](#) | 2023-12-04

278. The U.S. CMS HL-LHC R D Strategic Plan

Gutsche, Oliver et al.

[EPJ Web Conf. 295 \(2024\) 04050](#) | [arXiv:2312.00772](#) | [DOI:10.1051/epjconf/202429504050](#) | 2023-12-01

279. A Ceph S3 Object Data Store for HEP

Smith, Nick et al.

[EPJ Web Conf. 295 \(2024\) 01003](#) | [arXiv:2311.16321](#) | [DOI:10.1051/epjconf/202429501003](#) | 2023-11-27

280. Higher-order moments of the elliptic flow distribution in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 2024 \(2024\) 106](#) | [arXiv:2311.11370](#) | [DOI:10.1007/JHEP02\(2024\)106](#) | 2023-11-19

281. Search for new Higgs bosons via same-sign top quark pair production in association with a jet in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 850 \(2024\) 138478](#) | [arXiv:2311.03261](#) | [DOI:10.1016/j.physletb.2024.138478](#) | 2023-11-06

282. Search for high-mass exclusive diphoton production with tagged protons in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.D 110 \(2024\) 012010](#) | [arXiv:2311.02725](#) | [DOI:10.1103/PhysRevD.110.012010](#) | 2023-11-05

283. Search for an exotic decay of the Higgs boson into a Z boson and a pseudoscalar particle in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Lett.B 852 \(2024\) 138582](#) | [arXiv:2311.00130](#) | [DOI:10.1016/j.physletb.2024.138582](#) | 2023-10-31

284. Search for W' bosons decaying to a top and a bottom quark in leptonic final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 046](#) | [arXiv:2310.19893](#) | [DOI:10.1007/JHEP05\(2024\)046](#) | 2023-10-30

285. Search for Narrow Trijet Resonances in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 133 \(2024\) 011801](#) | [arXiv:2310.14023](#) | [DOI:10.1103/PhysRevLett.133.011801](#) | 2023-10-21

286. Search for central exclusive production of top quark pairs in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$ with tagged protons

Tumasyan, Armen et al.

[JHEP 06 \(2024\) 187](#) | [arXiv:2310.11231](#) | [DOI:10.1007/JHEP06\(2024\)187](#) | 2023-10-17

287. Observation of $WW\gamma$ production and search for $H\gamma$ production in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 121901](#) | [arXiv:2310.05164](#) | [DOI:10.1103/PhysRevLett.132.121901](#) | 2023-10-08

288. Muon identification using multivariate techniques in the CMS experiment in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P02031](#) | [arXiv:2310.03844](#) | [DOI:10.1088/1748-0221/19/02/P02031](#) | 2023-10-05

289. Search for stealth supersymmetry in final states with two photons, jets, and low missing transverse momentum in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 112009](#) | [arXiv:2310.03154](#) | [DOI:10.1103/PhysRevD.109.112009](#) | 2023-10-04

290. Study of azimuthal anisotropy of $\phi(1S)$ mesons in pPb collisions at $\sqrt{s_{NN}} = 8.16\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138518](#) | [arXiv:2310.03233](#) | [DOI:10.1016/j.physletb.2024.138518](#) | 2023-10-04

291. Search for a new resonance decaying into two spin-0 bosons in a final state with two photons and two bottom quarks in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2024\) 316](#) | [arXiv:2310.01643](#) | [DOI:10.1007/JHEP05\(2024\)316](#) | 2023-10-02

292. Search for supersymmetry in final states with disappearing tracks in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.D 109 \(2024\) 072007](#) | [arXiv:2309.16823](#) | [DOI:10.1103/PhysRevD.109.072007](#) | 2023-09-28

293. Search for direct production of GeV-scale resonances decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 070](#) | [arXiv:2309.16003](#) | [DOI:10.1007/JHEP12\(2023\)070](#) | 2023-09-27

294. Inclusive and differential cross section measurements of ttbb production in the lepton+jets channel at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 042](#) | [arXiv:2309.14442](#) | [DOI:10.1007/JHEP05\(2024\)042](#) | 2023-09-25

295. Measurement of the τ lepton polarization in Z boson decays in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 101](#) | [arXiv:2309.12408](#) | [DOI:10.1007/JHEP01\(2024\)101](#) | 2023-09-21

296. Development of the CMS detector for the CERN LHC Run 3

Hayrapetyan, Aram et al.

[JINST 19 \(2024\) P05064](#) | [arXiv:2309.05466](#) | [DOI:10.1088/1748-0221/19/05/P05064](#) | 2023-09-11

297. Evidence for the Higgs Boson Decay to a Z Boson and a Photon at the LHC

Aad, Georges et al.

[Phys.Rev.Lett. 132 \(2024\) 021803](#) | [arXiv:2309.03501](#) | [DOI:10.1103/PhysRevLett.132.021803](#) | 2023-09-07

298. Luminosity determination using Z boson production at the CMS experiment

Hayrapetyan, Aram et al.

[Eur.Phys.J.C 84 \(2024\) 26](#) | [arXiv:2309.01008](#) | [DOI:10.1140/epjc/s10052-023-12268-2](#) | 2023-09-02

299. Search for a third-generation leptoquark coupled to a τ lepton and a b quark through single, pair, and nonresonant production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 05 \(2024\) 311](#) | [arXiv:2308.07826](#) | [DOI:10.1007/JHEP05\(2024\)311](#) | 2023-08-15

300. Search for Scalar Leptoquarks Produced via τ -Lepton-Quark Scattering in pp Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 061801](#) | [arXiv:2308.06143](#) | [DOI:10.1103/PhysRevLett.132.061801](#) | 2023-08-11

301. Measurement of the production cross section for a W boson in association with a charm quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 84 \(2024\) 27](#) | [arXiv:2308.02285](#) | [DOI:10.1140/epjc/s10052-023-12258-4](#) | 2023-08-04

302. Measurement of the Higgs boson production via vector boson fusion and its decay into bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 01 \(2024\) 173](#) | [arXiv:2308.01253](#) | [DOI:10.1007/JHEP01\(2024\)173](#) | 2023-08-02

303. Search for new physics in multijet events with at least one photon and large missing transverse momentum in proton-proton collisions at 13 TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 046](#) | [arXiv:2307.16216](#) | [DOI:10.1007/JHEP10\(2023\)046](#) | 2023-07-30

304. Search for physics beyond the standard model in top quark production with additional leptons in the context of effective field theory

Hayrapetyan, Aram et al.

[JHEP 12 \(2023\) 068](#) | [arXiv:2307.15761](#) | [DOI:10.1007/JHEP12\(2023\)068](#) | 2023-07-28

305. Multiplicity and transverse momentum dependence of charge-balance functions in pPb and PbPb collisions at LHC energies

Tumasyan, Armen et al.

[JHEP 08 \(2024\) 148](#) | [arXiv:2307.11185](#) | [DOI:10.1007/JHEP08\(2024\)148](#) | 2023-07-20

306. Study of charm hadronization with prompt Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2024\) 128](#) | [arXiv:2307.11186](#) | [DOI:10.1007/JHEP01\(2024\)128](#) | 2023-07-20

307. Search for a high-mass dimuon resonance produced in association with b quark jets at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 10 \(2023\) 043](#) | [arXiv:2307.08708](#) | [DOI:10.1007/JHEP10\(2023\)043](#) | 2023-07-14

308. Search for Z' bosons decaying to pairs of heavy Majorana neutrinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2023\) 181](#) | [arXiv:2307.06959](#) | [DOI:10.1007/JHEP11\(2023\)181](#) | 2023-07-12

309. Evaluating Portable Parallelization Strategies for Heterogeneous Architectures in High Energy Physics

Atif, Mohammad et al.

[arXiv:2306.15869](#) | 2023-06-27

310. Detector R D needs for the next generation e^+e^- collider

Apresyan, A. et al.

[arXiv:2306.13567](#) | 2023-06-23

311. Two-particle Bose-Einstein correlations and their Lévy parameters in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.C 109 \(2024\) 024914](#) | [arXiv:2306.11574](#) | [DOI:10.1103/PhysRevC.109.024914](#) | 2023-06-20

312. Performance of the local reconstruction algorithms for the CMS hadron calorimeter with Run 2 data

Tumasyan, Armen et al.

[JINST 18 \(2023\) P11017](#) | [arXiv:2306.10355](#) | [DOI:10.1088/1748-0221/18/11/P11017](#) | 2023-06-17

313. New Structures in the $J/\psi J/\psi$ Mass Spectrum in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 111901](#) | [arXiv:2306.07164](#) | [DOI:10.1103/PhysRevLett.132.111901](#) | 2023-06-12

314. Search for the lepton-flavor violating decay of the Higgs boson and additional Higgs bosons in the $e\mu$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[Phys.Rev.D 108 \(2023\) 072004](#) | [arXiv:2305.18106](#) | [DOI:10.1103/PhysRevD.108.072004](#) | 2023-05-29

315. Measurements of the azimuthal anisotropy of prompt and nonprompt charmonia in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2023\) 115](#) | [arXiv:2305.16928](#) | [DOI:10.1007/JHEP10\(2023\)115](#) | 2023-05-26

316. Observation of four top quark production in proton-proton collisions at $\sqrt{s}=13$ TeV

Hayrapetyan, Aram et al.

[Phys.Lett.B 847 \(2023\) 138290](#) | [arXiv:2305.13439](#) | [DOI:10.1016/j.physletb.2023.138290](#) | 2023-05-22

317. Search for Inelastic Dark Matter in Events with Two Displaced Muons and Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 132 \(2024\) 041802](#) | [arXiv:2305.11649](#) | [DOI:10.1103/PhysRevLett.132.041802](#) | 2023-05-19

318. Search for resonances in events with photon and jet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2023\) 189](#) | [arXiv:2305.07998](#) | [DOI:10.1007/JHEP12\(2023\)189](#) | 2023-05-13

319. Measurements of inclusive and differential cross sections for the Higgs boson production and decay to four-leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 08 \(2023\) 040](#) | [arXiv:2305.07532](#) | [DOI:10.1007/JHEP08\(2023\)040](#) | 2023-05-12

320. Automated Network Services for Exascale Data Movement

Balcas, Justas et al.

[EPJ Web Conf. 295 \(2024\) 01009](#) | [DOI:10.1051/epjconf/202429501009](#) | 2024

321. Observation of the rare decay of the η meson to four muons

Hayrapetyan, Aram et al.

[Phys.Rev.Lett. 131 \(2023\) 091903](#) | [arXiv:2305.04904](#) | [DOI:10.1103/PhysRevLett.131.091903](#) | 2023-05-08

322. Search for top squark pair production in a final state with at least one hadronically decaying tau lepton in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 110](#) | [arXiv:2304.07174](#) | [DOI:10.1007/JHEP07\(2023\)110](#) | 2023-04-14

323. Observation of the $\chi(3S)$ Meson and Suppression of χ States in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 133 \(2024\) 022302](#) | [arXiv:2303.17026](#) | [DOI:10.1103/PhysRevLett.133.022302](#) | 2023-03-29

324. Probing Small Bjorken-x Nuclear Gluonic Structure via Coherent J/ψ Photoproduction in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 262301](#) | [arXiv:2303.16984](#) | [DOI:10.1103/PhysRevLett.131.262301](#) | 2023-03-29

325. First measurement of the top quark pair production cross section in proton-proton collisions at $\sqrt{s} = 13.6$ TeV

Tumasyan, Armen et al.

[JHEP 08 \(2023\) 204](#) | [arXiv:2303.10680](#) | [DOI:10.1007/JHEP08\(2023\)204](#) | 2023-03-19

326. A search for new physics in central exclusive production using the missing mass technique with the CMS detector and the CMS-TOTEM precision proton spectrometer

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 827](#) | [arXiv:2303.04596](#) | [DOI:10.1140/epjc/s10052-023-11687-5](#) | 2023-03-08

327. Evidence for Four-Top Quark Production in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 138076](#) | [arXiv:2303.03864](#) | [DOI:10.1016/j.physletb.2023.138076](#) | 2023-03-07

328. A search for decays of the Higgs boson to invisible particles in events with a top-antitop quark pair or a vector boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 933](#) | [arXiv:2303.01214](#) | [DOI:10.1140/epjc/s10052-023-11952-7](#) | 2023-03-02

329. Search for dark matter particles in W^+W^- events with transverse momentum imbalance in proton-proton collisions at $\sqrt{s} = 13$ TeV

Hayrapetyan, Aram et al.

[JHEP 03 \(2024\) 134](#) | [arXiv:2310.12229](#) | [DOI:10.1007/JHEP03\(2024\)134](#) | 2023-03

330. Search for a vector-like quark $T \rightarrow tH$ via the diphoton decay mode of the Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 057](#) | [arXiv:2302.12802](#) | [DOI:10.1007/JHEP09\(2023\)057](#) | 2023-02-24

331. Measurement of the top quark mass using a profile likelihood approach with the lepton + jets final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 963](#) | [arXiv:2302.01967](#) | [DOI:10.1140/epjc/s10052-023-12050-4](#) | 2023-02-03

332. IRIS-HEP Strategic Plan for the Next Phase of Software Upgrades for HL-LHC Physics

Bockelman, Brian et al.

[arXiv:2302.01317](#) | 2023-02-02

333. Search for top squarks in the four-body decay mode with single lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 060](#) | [arXiv:2301.08096](#) | [DOI:10.1007/JHEP06\(2023\)060](#) | 2023-01-19

334. First Measurement of the Forward Rapidity Gap Distribution in pPb Collisions at $\sqrt{s_{NN}}=8.16$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 092004](#) | [arXiv:2301.07630](#) | [DOI:10.1103/PhysRevD.108.092004](#) | 2023-01-18

335. K_S^0 and $\Lambda(\Lambda^0)$ two-particle femtoscopic correlations in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 857 \(2024\) 138936](#) | [arXiv:2301.05290](#) | [DOI:10.1016/j.physletb.2024.138936](#) | 2023-01-12

336. Measurement of the electroweak production of $W\gamma$ in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032017](#) | [arXiv:2212.12592](#) | [DOI:10.1103/PhysRevD.108.032017](#) | 2022-12-23

337. Search for new physics in the τ lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[JHEP 09 \(2023\) 051](#) | [arXiv:2212.12604](#) | [DOI:10.1007/JHEP09\(2023\)051](#) | 2022-12-23

338. Measurement of the $B^0_S \rightarrow \mu^+ \mu^-$ decay properties and search for the $B^0 \rightarrow \mu^+ \mu^-$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137955](#) | [arXiv:2212.10311](#) | [DOI:10.1016/j.physletb.2023.137955](#) | 2022-12-20

339. Search for long-lived particles using out-of-time trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 210](#) | [arXiv:2212.06695](#) | [DOI:10.1007/JHEP07\(2023\)210](#) | 2022-12-13

340. Measurement of the dependence of the hadron production fraction ratio f_s / f_u and f_d / f_u on B meson kinematic variables in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 121901](#) | [arXiv:2212.02309](#) | [DOI:10.1103/PhysRevLett.131.121901](#) | 2022-12-05

341. Measurements of azimuthal anisotropy of nonprompt D^0 mesons in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 850 \(2024\) 138389](#) | [arXiv:2212.01636](#) | [DOI:10.1016/j.physletb.2023.138389](#) | 2022-12-03

342. Search for high-mass exclusive $\gamma\gamma \rightarrow WW$ and $\gamma\gamma \rightarrow ZZ$ production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 229](#) | [arXiv:2211.16320](#) | [DOI:10.1007/JHEP07\(2023\)229](#) | 2022-11-29

343. Search for Higgs Boson and Observation of Z Boson through their Decay into a Charm Quark-Antiquark Pair in Boosted Topologies in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041801](#) | [arXiv:2211.14181](#) | [DOI:10.1103/PhysRevLett.131.041801](#) | 2022-11-25

344. Search for supersymmetry in final states with a single electron or muon using angular correlations and heavy-object identification in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 149](#) | [arXiv:2211.08476](#) | [DOI:10.1007/JHEP09\(2023\)149](#) | 2022-11-15

345. Measurement of the differential $t\bar{t}$ production cross section as a function of the jet mass and extraction of the top quark mass in hadronic decays of boosted top quarks

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 560](#) | [arXiv:2211.01456](#) | [DOI:10.1140/epjc/s10052-023-11587-8](#) | 2022-11-02

346. Azimuthal correlations in Z +jets events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 722](#) | [arXiv:2210.16139](#) | [DOI:10.1140/epjc/s10052-023-11833-z](#) | 2022-10-28

347. Application of Portable Parallelization Strategies for GPUs on track reconstruction kernels

Kwok, Martin et al.

[J.Phys.Conf.Ser. 3206 \(2026\) 012052](#) | [DOI:10.1088/1742-6596/3206/1/012052](#) | 2026

348. Measurements of jet multiplicity and jet transverse momentum in multijet events in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 742](#) | [arXiv:2210.13557](#) | [DOI:10.1140/epjc/s10052-023-11753-y](#) | 2022-10-24

349. Search for medium effects using jets from bottom quarks in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137849](#) | [arXiv:2210.08547](#) | [DOI:10.1016/j.physletb.2023.137849](#) | 2022-10-16

350. Azimuthal anisotropy of dijet events in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 139](#) | [arXiv:2210.08325](#) | [DOI:10.1007/JHEP07\(2023\)139](#) | 2022-10-15

351. The Future of High Energy Physics Software and Computing

Elvira, V. Daniel et al.

[arXiv:2210.05822](#) | [DOI:10.2172/1898754](#) | 2022-10-11

352. Proton reconstruction with the CMS-TOTEM Precision Proton Spectrometer

Tumasyan, Armen et al.

[JINST 18 \(2023\) P09009](#) | [arXiv:2210.05854](#) | [DOI:10.1088/1748-0221/18/09/P09009](#) | 2022-10-11

353. Search for a heavy composite Majorana neutrino in events with dilepton signatures from proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 843 \(2023\) 137803](#) | [arXiv:2210.03082](#) | [DOI:10.1016/j.physletb.2023.137803](#) | 2022-10-06

354. Search for new heavy resonances decaying to WW, WZ, ZZ, WH, or ZH boson pairs in the all-jets final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137813](#) | [arXiv:2210.00043](#) | [DOI:10.1016/j.physletb.2023.137813](#) | 2022-09-30

355. Snowmass Computational Frontier: Topical Group Report on Experimental Algorithm Parallelization

Cerati, G. et al.

[arXiv:2209.07356](#) | 2022-09-15

356. Search for pair production of vector-like quarks in leptonic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 020](#) | [arXiv:2209.07327](#) | [DOI:10.1007/JHEP07\(2023\)020](#) | 2022-09-15

357. Search for exotic Higgs boson decays $H \rightarrow \mathcal{A}\mathcal{A} \rightarrow 4\gamma$ with events containing two merged diphotons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 101801](#) | [arXiv:2209.06197](#) | [DOI:10.1103/PhysRevLett.131.101801](#) | 2022-09-13

358. Search for new physics using effective field theory in 13 TeV pp collision events that contain a top quark pair and a boosted Z or Higgs boson

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032008](#) | [arXiv:2208.12837](#) | [DOI:10.1103/PhysRevD.108.032008](#) | 2022-08-26

359. Measurement of the Higgs boson inclusive and differential fiducial production cross sections in the diphoton decay channel with pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 091](#) | [arXiv:2208.12279](#) | [DOI:10.1007/JHEP07\(2023\)091](#) | 2022-08-25

360. Search for pair-produced vector-like leptons in final states with third-generation leptons and at least three b quark jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137713](#) | [arXiv:2208.09700](#) | [DOI:10.1016/j.physletb.2023.137713](#) | 2022-08-20

361. Measurement of the cross section of top quark-antiquark pair production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 219](#) | [arXiv:2208.06485](#) | [DOI:10.1007/JHEP07\(2023\)219](#) | 2022-08-12

362. Measurement of the $t\bar{t}$ charge asymmetry in events with highly Lorentz-boosted top quarks in pp collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137703](#) | [arXiv:2208.02751](#) | [DOI:10.1016/j.physletb.2023.137703](#) | 2022-08-04

363. Searches for additional Higgs bosons and for vector leptoquarks in $\tau\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 073](#) | [arXiv:2208.02717](#) | [DOI:10.1007/JHEP07\(2023\)073](#) | 2022-08-04

364. Search for CP violation in $t\bar{t}H$ and tH production in multilepton channels in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 092](#) | [arXiv:2208.02686](#) | [DOI:10.1007/JHEP07\(2023\)092](#) | 2022-08-04

365. Search for the exotic decay of the Higgs boson into two light pseudoscalars with four photons in the final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 148](#) | [arXiv:2208.01469](#) | [DOI:10.1007/JHEP07\(2023\)148](#) | 2022-08-02

366. Measurement of inclusive and differential cross sections for single top quark production in association with a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 046](#) | [arXiv:2208.00924](#) | [DOI:10.1007/JHEP07\(2023\)046](#) | 2022-08-01

367. Search for the Higgs boson decay to a pair of electrons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 846 \(2023\) 137783](#) | [arXiv:2208.00265](#) | [DOI:10.1016/j.physletb.2023.137783](#) | 2022-07-30

368. Search for new physics in the lepton plus missing transverse momentum final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 067](#) | [arXiv:2202.06075](#) | [DOI:10.1007/JHEP07\(2022\)067](#) | 2022-07-11

369. Measurement of the top quark pole mass using $t\bar{t}$ +jet events in the dilepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 077](#) | [arXiv:2207.02270](#) | [DOI:10.1007/JHEP07\(2023\)077](#) | 2022-07-05

370. Search for direct pair production of supersymmetric partners of τ leptons in the final state with two hadronically decaying τ leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012011](#) | [arXiv:2207.02254](#) | [DOI:10.1103/PhysRevD.108.012011](#) | 2022-07-05

371. Search for a charged Higgs boson decaying into a heavy neutral Higgs boson and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2023\) 032](#) | [arXiv:2207.01046](#) | [DOI:10.1007/JHEP09\(2023\)032](#) | 2022-07-03

372. A portrait of the Higgs boson by the CMS experiment ten years after the discovery.

Tumasyan, Armen et al.

[Nature 607 \(2022\) 60](#) | [arXiv:2207.00043](#) | [DOI:10.1038/s41586-022-04892-x](#) | 2022-06-30

373. Search for Higgs boson pairs decaying to WW^*WW^* , $WW^*\tau\tau$, and $\tau\tau\tau\tau$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 095](#) | [arXiv:2206.10268](#) | [DOI:10.1007/JHEP07\(2023\)095](#) | 2022-06-21

374. Search for nonresonant Higgs boson pair production in the four leptons plus two jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 130](#) | [arXiv:2206.10657](#) | [DOI:10.1007/JHEP06\(2023\)130](#) | 2022-06-21

375. Measurements of the Higgs boson production cross section and couplings in the W boson pair decay channel in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 667](#) | [arXiv:2206.09466](#) | [DOI:10.1140/epjc/s10052-023-11632-6](#) | 2022-06-19

376. Search for nonresonant Higgs boson pair production in final state with two bottom quarks and two tau leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137531](#) | [arXiv:2206.09401](#) | [DOI:10.1016/j.physletb.2022.137531](#) | 2022-06-19

377. Probing Heavy Majorana Neutrinos and the Weinberg Operator through Vector Boson Fusion Processes in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 011803](#) | [arXiv:2206.08956](#) | [DOI:10.1103/PhysRevLett.131.011803](#) | 2022-06-17

378. Precision measurement of the Z boson invisible width in pp collisions at $\sqrt{s}=13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137563](#) | [arXiv:2206.07110](#) | [DOI:10.1016/j.physletb.2022.137563](#) | 2022-06-14

379. Observation of τ lepton pair production in ultraperipheral lead-lead collisions at $\sqrt{s_{\text{NN}}} = 5.02\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 151803](#) | [arXiv:2206.05192](#) | [DOI:10.1103/PhysRevLett.131.151803](#) | 2022-06-10

380. Search for Higgs boson decays into Z and J/ψ and for Higgs and Z boson decays into J/ψ or Y pairs in pp collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137534](#) | [arXiv:2206.03525](#) | [DOI:10.1016/j.physletb.2022.137534](#) | 2022-06-07

381. Observation of same-sign WW production from double parton scattering in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 091803](#) | [arXiv:2206.02681](#) | [DOI:10.1103/PhysRevLett.131.091803](#) | 2022-06-06

382. Combination of inclusive top-quark pair production cross-section measurements using ATLAS and CMS data at $\sqrt{s} = 7$ and 8 TeV

Aad, G. et al.

[JHEP 07 \(2023\) 213](#) | [arXiv:2205.13830](#) | [DOI:10.1007/JHEP07\(2023\)213](#) | 2022-05-27

383. Search for electroweak production of charginos and neutralinos at $\sqrt{s}=13\text{ TeV}$ in final states containing hadronic decays of WW, WZ, or WH and missing transverse momentum

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137460](#) | [arXiv:2205.09597](#) | [DOI:10.1016/j.physletb.2022.137460](#) | 2022-05-19

384. Search for long-lived particles decaying to a pair of muons in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 228](#) | [arXiv:2205.08582](#) | [DOI:10.1007/JHEP05\(2023\)228](#) | 2022-05-17

385. Search for CP violating top quark couplings in pp collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 07 \(2023\) 023](#) | [arXiv:2205.07434](#) | [DOI:10.1007/JHEP07\(2023\)023](#) | 2022-05-15

386. Search for heavy resonances and quantum black holes in $e\mu$, $e\tau$, and $\mu\tau$ final states in proton-proton collisions at $\sqrt{s} = 13\text{ TeV}$

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 227](#) | [arXiv:2205.06709](#) | [DOI:10.1007/JHEP05\(2023\)227](#) | 2022-05-13

387. Search for Nonresonant Pair Production of Highly Energetic Higgs Bosons Decaying to Bottom Quarks

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 041803](#) | [arXiv:2205.06667](#) | [DOI:10.1103/PhysRevLett.131.041803](#) | 2022-05-13

388. Search for Higgs Boson Decay to a Charm Quark-Antiquark Pair in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 061801](#) | [arXiv:2205.05550](#) | [DOI:10.1103/PhysRevLett.131.061801](#) | 2022-05-11

389. Observation of electroweak $W+W^-$ pair production in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 841 \(2023\) 137495](#) | [arXiv:2205.05711](#) | [DOI:10.1016/j.physletb.2022.137495](#) | 2022-05-11

390. Measurement of the mass dependence of the transverse momentum of lepton pairs in Drell-Yan production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 628](#) | [arXiv:2205.04897](#) | [DOI:10.1140/epjc/s10052-023-11631-7](#) | 2022-05-10

391. Constraints on anomalous Higgs boson couplings to vector bosons and fermions from the production of Higgs bosons using the $\tau\tau$ final state

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 032013](#) | [arXiv:2205.05120](#) | [DOI:10.1103/PhysRevD.108.032013](#) | 2022-05-10

392. Measurement of differential cross sections for the production of a Z boson in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 052004](#) | [arXiv:2205.02872](#) | [DOI:10.1103/PhysRevD.108.052004](#) | 2022-05-05

393. CMS pythia 8 colour reconnection tunes based on underlying-event data

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 587](#) | [arXiv:2205.02905](#) | [DOI:10.1140/epjc/s10052-023-11630-8](#) | 2022-05-05

394. Search for CP violation using $t\bar{t}$ events in the lepton+jets channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2023\) 081](#) | [arXiv:2205.02314](#) | [DOI:10.1007/JHEP06\(2023\)081](#) | 2022-05-04

395. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 012009](#) | [arXiv:2205.01835](#) | [DOI:10.1103/PhysRevD.108.012009](#) | 2022-05-03

396. Azimuthal Correlations within Exclusive Dijets with Large Momentum Transfer in Photon-Lead Collisions

Tumasyan, Armen et al.

[Phys.Rev.Lett. 131 \(2023\) 051901](#) | [arXiv:2205.00045](#) | [DOI:10.1103/PhysRevLett.131.051901](#) | 2022-04-29

397. Strange hadron collectivity in pPb and PbPb collisions

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 007](#) | [arXiv:2205.00080](#) | [DOI:10.1007/JHEP05\(2023\)007](#) | 2022-04-29

398. Two-particle azimuthal correlations in γp interactions using pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 844 \(2023\) 137905](#) | [arXiv:2204.13486](#) | [DOI:10.1016/j.physletb.2023.137905](#) | 2022-04-28

399. Search for light Higgs bosons from supersymmetric cascade decays in pp collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 571](#) | [arXiv:2204.13532](#) | [DOI:10.1140/epjc/s10052-023-11581-0](#) | 2022-04-28

400. Search for Higgs boson decays to a Z boson and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2023\) 233](#) | [arXiv:2204.12945](#) | [DOI:10.1007/JHEP05\(2023\)233](#) | 2022-04-27

401. Measurements of Higgs boson production in the decay channel with a pair of τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 83 \(2023\) 562](#) | [arXiv:2204.12957](#) | [DOI:10.1140/epjc/s10052-023-11452-8](#) | 2022-04-27

402. Reconstruction of decays to merged photons using end-to-end deep learning with domain continuation in the CMS detector

Tumasyan, Armen et al.

[Phys.Rev.D 108 \(2023\) 052002](#) | [arXiv:2204.12313](#) | [DOI:10.1103/PhysRevD.108.052002](#) | 2022-04-26

403. Search for a massive scalar resonance decaying to a light scalar and a Higgs boson in the four b quarks final state with boosted topology

Tumasyan, Armen et al.

[Phys.Lett.B 842 \(2023\) 137392](#) | [arXiv:2204.12413](#) | [DOI:10.1016/j.physletb.2022.137392](#) | 2022-04-26

404. Portability: A Necessary Approach for Future Scientific Software

Bhattacharya, Meghna et al.

[arXiv:2203.09945](#) | 2022-03-15

405. Search for new particles in an extended Higgs sector with four b quarks in the final state at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 835 \(2022\) 137566](#) | [arXiv:2203.00480](#) | [DOI:10.1016/j.physletb.2022.137566](#) | 2022-03-01

406. Search for a W' boson decaying to a vector-like quark and a top or bottom quark in the all-jets final state at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 09 \(2022\) 088](#) | [arXiv:2202.12988](#) | [DOI:10.1007/JHEP09\(2022\)088](#) | 2022-02-25

407. Measurement of the Drell-Yan forward-backward asymmetry at high dilepton masses in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 2022 \(2022\) 063](#) | [arXiv:2202.12327](#) | [DOI:10.1007/JHEP08\(2022\)063](#) | 2022-02-24

408. Nuclear modification of Υ states in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 835 \(2022\) 137397](#) | [arXiv:2202.11807](#) | [DOI:10.1016/j.physletb.2022.137397](#) | 2022-02-23

409. Search for Higgs Boson Pair Production in the Four b Quark Final State in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 081802](#) | [arXiv:2202.09617](#) | [DOI:10.1103/PhysRevLett.129.081802](#) | 2022-02-19

410. Inclusive nonresonant multilepton probes of new phenomena at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 112007](#) | [arXiv:2202.08676](#) | [DOI:10.1103/PhysRevD.105.112007](#) | 2022-02-17

411. Measurement of the Higgs boson width and evidence of its off-shell contributions to ZZ production

Tumasyan, Armen et al.

[Nature Phys. 18 \(2022\) 1329](#) | [arXiv:2202.06923](#) | [DOI:10.1038/s41567-022-01682-0](#) | 2022-02-14

412. Search for invisible decays of the Higgs boson produced via vector boson fusion in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092007](#) | [arXiv:2201.11585](#) | [DOI:10.1103/PhysRevD.105.092007](#) | 2022-01-27

413. Observation of $B^0 \rightarrow \psi(2S) K^0_S \pi^+ \pi^-$ and $B^0_S \rightarrow \psi(2S) K^0_S$ decays

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 499](#) | [arXiv:2201.09131](#) | [DOI:10.1140/epjc/s10052-022-10315-y](#) | 2022-01-22

414. Search for resonances decaying to three W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 021802](#) | [arXiv:2201.08476](#) | [DOI:10.1103/PhysRevLett.129.021802](#) | 2022-01-20

415. Identification of hadronic tau lepton decays using a deep neural network

Tumasyan, Armen et al.

[JINST 17 \(2022\) P07023](#) | [arXiv:2201.08458](#) | [DOI:10.1088/1748-0221/17/07/P07023](#) | 2022-01-20

416. Search for charged-lepton flavor violation in top quark production and decay in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 082](#) | [arXiv:2201.07859](#) | [DOI:10.1007/JHEP06\(2022\)082](#) | 2022-01-19

417. Precision measurement of the W boson decay branching fractions in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 072008](#) | [arXiv:2201.07861](#) | [DOI:10.1103/PhysRevD.105.072008](#) | 2022-01-19

418. Measurement of the inclusive and differential $t\bar{t}\gamma$ cross sections in the dilepton channel and effective field theory interpretation in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 091](#) | [arXiv:2201.07301](#) | [DOI:10.1007/JHEP05\(2022\)091](#) | 2022-01-18

419. Search for long-lived heavy neutral leptons with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 081](#) | [arXiv:2201.05578](#) | [DOI:10.1007/JHEP07\(2022\)081](#) | 2022-01-14

420. Search for higgsinos decaying to two Higgs bosons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 014](#) | [arXiv:2201.04206](#) | [DOI:10.1007/JHEP05\(2022\)014](#) | 2022-01-11

421. Observation of the B_c^+ Meson in Pb-Pb and pp Collisions at $\sqrt{s_{NN}} = 5.02$ TeV and Measurement of its Nuclear Modification Factor

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 252301](#) | [arXiv:2201.02659](#) | [DOI:10.1103/PhysRevLett.128.252301](#) | 2022-01-07

422. Search for high-mass resonances decaying to a jet and a Lorentz-boosted resonance in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 832 \(2022\) 137263](#) | [arXiv:2201.02140](#) | [DOI:10.1016/j.physletb.2022.137263](#) | 2022-01-06

423. Search for single production of a vector-like T quark decaying to a top quark and a Z boson in the final state with jets and missing transverse momentum at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 093](#) | [arXiv:2201.02227](#) | [DOI:10.1007/JHEP05\(2022\)093](#) | 2022-01-06

424. Search for long-lived particles decaying into muon pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV collected with a dedicated high-rate data stream

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 062](#) | [arXiv:2112.13769](#) | [DOI:10.1007/JHEP04\(2022\)062](#) | 2021-12-27

425. Search for resonances decaying to three W bosons in the hadronic final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012002](#) | [arXiv:2112.13090](#) | [DOI:10.1103/PhysRevD.106.012002](#) | 2021-12-24

426. Probing Charm Quark Dynamics via Multiparticle Correlations in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 022001](#) | [arXiv:2112.12236](#) | [DOI:10.1103/PhysRevLett.129.022001](#) | 2021-12-22

427. Search for resonant production of strongly coupled dark matter in proton-proton collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 156](#) | [arXiv:2112.11125](#) | [DOI:10.1007/JHEP06\(2022\)156](#) | 2021-12-21

428. Measurement of the production cross section for Z+b jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 092014](#) | [arXiv:2112.09659](#) | [DOI:10.1103/PhysRevD.105.092014](#) | 2021-12-17

429. Search for flavor-changing neutral current interactions of the top quark and the Higgs boson decaying to a bottom quark-antiquark pair at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 169](#) | [arXiv:2112.09734](#) | [DOI:10.1007/JHEP02\(2022\)169](#) | 2021-12-17

430. Measurement of the inclusive tt production cross section in proton-proton collisions at $\sqrt{s}=5.02$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 144](#) | [arXiv:2112.09114](#) | [DOI:10.1007/JHEP04\(2022\)144](#) | 2021-12-16

431. Evidence for WW/WZ vector boson scattering in the decay channel $\ell\nu qq$ produced in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 834 \(2022\) 137438](#) | [arXiv:2112.05259](#) | [DOI:10.1016/j.physletb.2022.137438](#) | 2021-12-09

432. Search for a right-handed W boson and a heavy neutrino in proton-proton collisions at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 047](#) | [arXiv:2112.03949](#) | [DOI:10.1007/JHEP04\(2022\)047](#) | 2021-12-07

433. Search for heavy resonances decaying to a pair of Lorentz-boosted Higgs bosons in final states with leptons and a bottom quark pair at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 05 \(2022\) 005](#) | [arXiv:2112.03161](#) | [DOI:10.1007/JHEP05\(2022\)005](#) | 2021-12-06

434. Measurements of the associated production of a W boson and a charm quark in proton-proton collisions at $\sqrt{s}=8$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 1094](#) | [arXiv:2112.00895](#) | [DOI:10.1140/epjc/s10052-022-10897-7](#) | 2021-12-01

435. Measurement of $W^{\pm}\gamma$ differential cross sections in proton-proton collisions at $\sqrt{s}=13$ TeV and effective field theory constraints

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 052003](#) | [arXiv:2111.13948](#) | [DOI:10.1103/PhysRevD.105.052003](#) | 2021-11-27

436. Search for heavy resonances decaying to ZZ or ZW and axion-like particles mediating nonresonant ZZ or ZH production at $\sqrt{s}=13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 087](#) | [arXiv:2111.13669](#) | [DOI:10.1007/JHEP04\(2022\)087](#) | 2021-11-26

437. Search for a heavy resonance decaying into a top quark and a W boson in the lepton+jets final state at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[JHEP 04 \(2022\) 048](#) | [arXiv:2111.10216](#) | [DOI:10.1007/JHEP04\(2022\)048](#) | 2021-11-19

438. Measurement and QCD analysis of double-differential inclusive jet cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 142](#) | [arXiv:2111.10431](#) | [DOI:10.1007/JHEP02\(2022\)142](#) | 2021-11-19

439. Strategies and performance of the CMS silicon tracker alignment during LHC Run 2

Tumasyan, Armen et al.

[Nucl.Instrum.Meth.A 1037 \(2022\) 166795](#) | [arXiv:2111.08757](#) | [DOI:10.1016/j.nima.2022.166795](#) | 2021-11-16

440. Search for supersymmetry in final states with two or three soft leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 091](#) | [arXiv:2111.06296](#) | [DOI:10.1007/JHEP04\(2022\)091](#) | 2021-11-11

441. Observation of triple J/ψ meson production in proton-proton collisions

Tumasyan, Armen et al.

[Nature Phys. 19 \(2023\) 338](#) | [arXiv:2111.05370](#) | [DOI:10.1038/s41567-022-01838-y](#) | 2021-11-09

442. Study of dijet events with large rapidity separation in proton-proton collisions at $\sqrt{s} = 2.76$ TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 189](#) | [arXiv:2111.04605](#) | [DOI:10.1007/JHEP03\(2022\)189](#) | 2021-11-08

443. Inclusive and differential cross section measurements of single top quark production in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 02 \(2022\) 107](#) | [arXiv:2111.02860](#) | [DOI:10.1007/JHEP02\(2022\)107](#) | 2021-11-04

444. A new calibration method for charm jet identification validated with proton-proton collision events at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JINST 17 \(2022\) P03014](#) | [arXiv:2111.03027](#) | [DOI:10.1088/1748-0221/17/03/P03014](#) | 2021-11-04

445. Search for Flavor-Changing Neutral Current Interactions of the Top Quark and Higgs Boson in Final States with Two Photons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[Phys.Rev.Lett. 129 \(2022\) 032001](#) | [arXiv:2111.02219](#) | [DOI:10.1103/PhysRevLett.129.032001](#) | 2021-11-03

446. Search for low-mass dilepton resonances in Higgs boson decays to four-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, A. et al.

[Eur.Phys.J.C 82 \(2022\) 290](#) | [arXiv:2111.01299](#) | [DOI:10.1140/epjc/s10052-022-10127-0](#) | 2021-11-01

447. Search for long-lived particles produced in association with a Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 03 \(2022\) 160](#) | [arXiv:2110.13218](#) | [DOI:10.1007/JHEP03\(2022\)160](#) | 2021-10-25

448. Measurement of the inclusive and differential WZ production cross sections, polarization angles, and triple gauge couplings in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 07 \(2022\) 032](#) | [arXiv:2110.11231](#) | [DOI:10.1007/JHEP07\(2022\)032](#) | 2021-10-21

449. First Search for Exclusive Diphoton Production at High Mass with Tagged Protons in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 129 \(2022\) 011801](#) | [arXiv:2110.05916](#) | [DOI:10.1103/PhysRevLett.129.011801](#) | 2021-10-12

450. Analysis of the CP structure of the Yukawa coupling between the Higgs boson and τ leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 06 \(2022\) 012](#) | [arXiv:2110.04836](#) | [DOI:10.1007/JHEP06\(2022\)012](#) | 2021-10-10

451. Search for long-lived particles decaying to leptons with large impact parameter in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 153](#) | [arXiv:2110.04809](#) | [DOI:10.1140/epjc/s10052-022-10027-3](#) | 2021-10-10

452. Measurement of double-parton scattering in inclusive production of four jets with low transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 177](#) | [arXiv:2109.13822](#) | [DOI:10.1007/JHEP01\(2022\)177](#) | 2021-09-28

453. Search for heavy resonances decaying to $Z(\nu\bar{\nu})V(q\bar{q}')$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 106 \(2022\) 012004](#) | [arXiv:2109.08268](#) | [DOI:10.1103/PhysRevD.106.012004](#) | 2021-09-17

454. Search for heavy resonances decaying to WW, WZ, or WH boson pairs in the lepton plus merged jet final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 105 \(2022\) 032008](#) | [arXiv:2109.06055](#) | [DOI:10.1103/PhysRevD.105.032008](#) | 2021-09-13

455. Study of quark and gluon jet substructure in Z+jet and dijet events from pp collisions

Tumasyan, Armen et al.

[JHEP 01 \(2022\) 188](#) | [arXiv:2109.03340](#) | [DOI:10.1007/JHEP01\(2022\)188](#) | 2021-09-07

456. Observation of B^0_s mesons and measurement of the B^0_s/B^+ yield ratio in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 829 \(2022\) 137062](#) | [arXiv:2109.01908](#) | [DOI:10.1016/j.physletb.2022.137062](#) | 2021-09-04

457. Observation of $t\bar{W}$ production in the single-lepton channel in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 111](#) | [arXiv:2109.01706](#) | [DOI:10.1007/JHEP11\(2021\)111](#) | 2021-09-03

458. Measurement of the top quark mass using events with a single reconstructed top quark in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 161](#) | [arXiv:2108.10407](#) | [DOI:10.1007/JHEP12\(2021\)161](#) | 2021-08-23

459. Measurement of differential $t\bar{t}$ production cross sections in the full kinematic range using lepton+jets events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 092013](#) | [arXiv:2108.02803](#) | [DOI:10.1103/PhysRevD.104.092013](#) | 2021-08-05

460. Probing effective field theory operators in the associated production of top quarks with a Z boson in multilepton final states at $\sqrt{s} = 13$ TeV

Lee, Kyeongpil et al.

[JHEP 12 \(2021\) 083](#) | [arXiv:2107.13896](#) | [DOI:10.1007/JHEP12\(2021\)083](#) | 2021-07-29

461. Search for new particles in events with energetic jets and large missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 153](#) | [arXiv:2107.13021](#) | [DOI:10.1007/JHEP11\(2021\)153](#) | 2021-07-27

462. Search for chargino-neutralino production in events with Higgs and W bosons using 137 fb^{-1} of proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 045](#) | [arXiv:2107.12553](#) | [DOI:10.1007/JHEP10\(2021\)045](#) | 2021-07-27

463. Measurement of the inclusive and differential Higgs boson production cross sections in the decay mode to a pair of τ leptons in pp collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 128 \(2022\) 081805](#) | [arXiv:2107.11486](#) | [DOI:10.1103/PhysRevLett.128.081805](#) | 2021-07-23

464. Combined searches for the production of supersymmetric top quark partners in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 81 \(2021\) 970](#) | [arXiv:2107.10892](#) | [DOI:10.1140/epjc/s10052-021-09721-5](#) | 2021-07-22

465. Search for Long-Lived Particles Decaying in the CMS End Cap Muon Detectors in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 261804](#) | [arXiv:2107.04838](#) | [DOI:10.1103/PhysRevLett.127.261804](#) | 2021-07-10

466. Measurement of prompt open-charm production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 225](#) | [arXiv:2107.01476](#) | [DOI:10.1007/JHEP11\(2021\)225](#) | 2021-07-03

467. Measurement of the inclusive and differential tt cross sections in the single-lepton channel and EFT interpretation at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 12 \(2021\) 180](#) | [arXiv:2107.01508](#) | [DOI:10.1007/JHEP12\(2021\)180](#) | 2021-07-03

468. Measurements of the electroweak diboson production cross sections in proton-proton collisions at $\sqrt{s} = 5.02$ TeV using leptonic decays

Tumasyan, Armen et al.

[Phys.Rev.Lett. 127 \(2021\) 191801](#) | [arXiv:2107.01137](#) | [DOI:10.1103/PhysRevLett.127.191801](#) | 2021-07-02

469. Search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[JHEP 04 \(2022\) 147](#) | [arXiv:2106.14246](#) | [DOI:10.1007/JHEP04\(2022\)147](#) | 2021-06-27

470. Fragmentation of jets containing a prompt J/ψ meson in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Tumasyan, Armen et al.

[Phys.Lett.B 825 \(2022\) 136842](#) | [arXiv:2106.13235](#) | [DOI:10.1016/j.physletb.2021.136842](#) | 2021-06-23

471. Measurement of the electroweak production of $Z\gamma$ and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[Phys.Rev.D 104 \(2021\) 072001](#) | [arXiv:2106.11082](#) | [DOI:10.1103/PhysRevD.104.072001](#) | 2021-06-21

472. Search for $W\gamma$ resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV using hadronic decays of Lorentz-boosted W bosons

Tumasyan, Armen et al.

[Phys.Lett.B 826 \(2022\) 136888](#) | [arXiv:2106.10509](#) | [DOI:10.1016/j.physletb.2022.136888](#) | 2021-06-19

473. Search for a heavy Higgs boson decaying into two lighter Higgs bosons in the $\tau\tau b\bar{b}$ final state at 13 TeV

Tumasyan, Armen et al.

[JHEP 11 \(2021\) 057](#) | [arXiv:2106.10361](#) | [DOI:10.1007/JHEP11\(2021\)057](#) | 2021-06-18

474. Study of Z boson plus jets events using variables sensitive to double-parton scattering in pp collisions at 13 TeV

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 176](#) | [arXiv:2105.14511](#) | [DOI:10.1007/JHEP10\(2021\)176](#) | 2021-05-30

475. Measurements of the $pp \rightarrow W^\pm \gamma \gamma$ and $pp \rightarrow Z \gamma \gamma$ cross sections at $\sqrt{s} = 13$ TeV and limits on anomalous quartic gauge couplings

Tumasyan, Armen et al.

[JHEP 10 \(2021\) 174](#) | [arXiv:2105.12780](#) | [DOI:10.1007/JHEP10\(2021\)174](#) | 2021-05-26

476. Search for strongly interacting massive particles generating trackless jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Tumasyan, Armen et al.

[Eur.Phys.J.C 82 \(2022\) 213](#) | [arXiv:2105.09178](#) | [DOI:10.1140/epjc/s10052-022-10095-5](#) | 2021-05-19

477. Search for lepton-flavor violating decays of the Higgs boson in the $\mu\tau$ and $e\tau$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 032013](#) | [arXiv:2105.03007](#) | [DOI:10.1103/PhysRevD.104.032013](#) | 2021-05-06

478. Search for long-lived particles decaying to jets with displaced vertices in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Rev.D 104 \(2021\) 052011](#) | [arXiv:2104.13474](#) | [DOI:10.1103/PhysRevD.104.052011](#) | 2021-04-27

479. Search for a heavy resonance decaying to a top quark and a W boson at $\sqrt{s} = 13$ TeV in the fully hadronic final state

Sirunyan, Albert M et al.

[JHEP 12 \(2021\) 106](#) | [arXiv:2104.12853](#) | [DOI:10.1007/JHEP12\(2021\)106](#) | 2021-04-26

480. Constraints on anomalous Higgs boson couplings to vector bosons and fermions in its production and decay using the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052004](#) | [arXiv:2104.12152](#) | [DOI:10.1103/PhysRevD.104.052004](#) | 2021-04-25

481. Search for W' bosons decaying to a top and a bottom quark at $\sqrt{s} = 13$ TeV in the hadronic final state

Sirunyan, Albert M et al.

[Phys.Lett.B 820 \(2021\) 136535](#) | [arXiv:2104.04831](#) | [DOI:10.1016/j.physletb.2021.136535](#) | 2021-04-10

482. Search for charged Higgs bosons produced in vector boson fusion processes and decaying into vector boson pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 723](#) | [arXiv:2104.04762](#) | [DOI:10.1140/epjc/s10052-021-09472-3](#) | 2021-04-10

483. Precision luminosity measurement in proton-proton collisions at $\sqrt{s} = 13$ TeV in 2015 and 2016 at CMS

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 800](#) | [arXiv:2104.01927](#) | [DOI:10.1140/epjc/s10052-021-09538-2](#) | 2021-04-05

484. Constraints on the Initial State of Pb-Pb Collisions via Measurements of Z-Boson Yields and Azimuthal Anisotropy at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 102002](#) | [arXiv:2103.14089](#) | [DOI:10.1103/PhysRevLett.127.102002](#) | 2021-03-25

485. Measurements of Higgs boson production cross sections and couplings in the diphoton decay channel at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 027](#) | [arXiv:2103.06956](#) | [DOI:10.1007/JHEP07\(2021\)027](#) | 2021-03-11

486. Measurements of production cross sections of the Higgs boson in the four-lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 488](#) | [arXiv:2103.04956](#) | [DOI:10.1140/epjc/s10052-021-09200-x](#) | 2021-03-08

487. Using Z Boson Events to Study Parton-Medium Interactions in Pb-Pb Collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 128 \(2022\) 122301](#) | [arXiv:2103.04377](#) | [DOI:10.1103/PhysRevLett.128.122301](#) | 2021-03-07

488. Search for resonant and nonresonant new phenomena in high-mass dilepton final states at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2021\) 208](#) | [arXiv:2103.02708](#) | [DOI:10.1007/JHEP07\(2021\)208](#) | 2021-03-03

489. Search for top squark production in fully-hadronic final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 052001](#) | [arXiv:2103.01290](#) | [DOI:10.1103/PhysRevD.104.052001](#) | 2021-03-01

490. Study of Drell-Yan dimuon production in proton-lead collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 182](#) | [arXiv:2102.13648](#) | [DOI:10.1007/JHEP05\(2021\)182](#) | 2021-02-26

491. Evidence for X(3872) in Pb-Pb Collisions and Studies of its Prompt Production at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 128 \(2022\) 032001](#) | [arXiv:2102.13048](#) | [DOI:10.1103/PhysRevLett.128.032001](#) | 2021-02-25

492. First measurement of large area jet transverse momentum spectra in heavy-ion collisions

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 284](#) | [arXiv:2102.13080](#) | [DOI:10.1007/JHEP05\(2021\)284](#) | 2021-02-25

493. Measurements of angular distance and momentum ratio distributions in three-jet and Z + two-jet final states in pp collisions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 852](#) | [arXiv:2102.08816](#) | [DOI:10.1140/epjc/s10052-021-09570-2](#) | 2021-02-17

494. Search for a heavy vector resonance decaying to a Z^0 boson and a Higgs boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 688](#) | [arXiv:2102.08198](#) | [DOI:10.1140/epjc/s10052-021-09348-6](#) | 2021-02-16

495. Hard color-singlet exchange in dijet events in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[Phys.Rev.D 104 \(2021\) 032009](#) | [arXiv:2102.06945](#) | [DOI:10.1103/PhysRevD.104.032009](#) | 2021-02-13
496. Search for top squarks in final states with two top quarks and several light-flavor jets in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[Phys.Rev.D 104 \(2021\) 032006](#) | [arXiv:2102.06976](#) | [DOI:10.1103/PhysRevD.104.032006](#) | 2021-02-13
497. Performance of the CMS muon trigger system in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[JINST 16 \(2021\) P07001](#) | [arXiv:2102.04790](#) | [DOI:10.1088/1748-0221/16/07/P07001](#) | 2021-02-09
498. Observation of a New Excited Beauty Strange Baryon Decaying to $\Xi^-_b \pi^+ \pi^-$
 Sirunyan, Albert M et al.
[Phys.Rev.Lett. 126 \(2021\) 252003](#) | [arXiv:2102.04524](#) | [DOI:10.1103/PhysRevLett.126.252003](#) | 2021-02-08
499. Measurements of the differential cross sections of the production of Z + jets and γ + jets and of Z boson emission collinear with a jet in pp collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[JHEP 05 \(2021\) 285](#) | [arXiv:2102.02238](#) | [DOI:10.1007/JHEP05\(2021\)285](#) | 2021-02-03
500. Measurement of the $W\gamma$ Production Cross Section in Proton-Proton Collisions at $\sqrt{s}=13$ TeV and Constraints on Effective Field Theory Coefficients
 Sirunyan, Albert M et al.
[Phys.Rev.Lett. 126 \(2021\) 252002](#) | [arXiv:2102.02283](#) | [DOI:10.1103/PhysRevLett.126.252002](#) | 2021-02-03
501. In-medium modification of dijets in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV
 Sirunyan, Albert M et al.
[JHEP 05 \(2021\) 116](#) | [arXiv:2101.04720](#) | [DOI:10.1007/JHEP05\(2021\)116](#) | 2021-01-12
502. First measurement of the cross section for top quark pair production with additional charm jets using dileptonic final states in pp collisions at $s=13\text{TeV}$
 Sirunyan, Albert M et al.
[Phys.Lett.B 820 \(2021\) 136565](#) | [arXiv:2012.09225](#) | [DOI:10.1016/j.physletb.2021.136565](#) | 2020-12-16
503. Measurement of the Z boson differential production cross section using its invisible decay mode ($Z\nu\bar{\nu}$) in proton-proton collisions at $\sqrt{s}=13$ TeV
 Sirunyan, Albert M et al.
[JHEP 05 \(2021\) 205](#) | [arXiv:2012.09254](#) | [DOI:10.1007/JHEP05\(2021\)205](#) | 2020-12-16
504. Search for supersymmetry in final states with two oppositely charged same-flavor leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[JHEP 04 \(2021\) 123](#) | [arXiv:2012.08600](#) | [DOI:10.1007/JHEP04\(2021\)123](#) | 2020-12-15
505. Electron and photon reconstruction and identification with the CMS experiment at the CERN LHC
 Sirunyan, Albert M et al.
[JINST 16 \(2021\) P05014](#) | [arXiv:2012.06888](#) | [DOI:10.1088/1748-0221/16/05/P05014](#) | 2020-12-12
506. Measurement of differential cross sections for Z bosons produced in association with charm jets in pp collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[JHEP 04 \(2021\) 109](#) | [arXiv:2012.04119](#) | [DOI:10.1007/JHEP04\(2021\)109](#) | 2020-12-07

507. Search for singly and pair-produced leptoquarks coupling to third-generation fermions in proton-proton collisions at $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136446](#) | [arXiv:2012.04178](#) | [DOI:10.1016/j.physletb.2021.136446](#) | 2020-12-07

508. Search for new physics in top quark production with additional leptons in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$ using effective field theory

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 095](#) | [arXiv:2012.04120](#) | [DOI:10.1007/JHEP03\(2021\)095](#) | 2020-12-07

509. Search for long-lived particles using displaced jets in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.D 104 \(2021\) 012015](#) | [arXiv:2012.01581](#) | [DOI:10.1103/PhysRevD.104.012015](#) | 2020-12-02

510. Search for nonresonant Higgs boson pair production in final states with two bottom quarks and two photons in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 03 \(2021\) 257](#) | [arXiv:2011.12373](#) | [DOI:10.1007/JHEP03\(2021\)257](#) | 2020-11-24

511. Search for the rare decay of the W boson into a pion and a photon in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136409](#) | [arXiv:2011.06028](#) | [DOI:10.1016/j.physletb.2021.136409](#) | 2020-11-11

512. Observation of Forward Neutron Multiplicity Dependence of Dimuon Acoplanarity in Ultraperipheral Pb-Pb Collisions at $\sqrt{s_{\text{NN}}}=5.02 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 127 \(2021\) 122001](#) | [arXiv:2011.05239](#) | [DOI:10.1103/PhysRevLett.127.122001](#) | 2020-11-10

513. Development and validation of HERWIG 7 tunes from CMS underlying-event measurements

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 312](#) | [arXiv:2011.03422](#) | [DOI:10.1140/epjc/s10052-021-08949-5](#) | 2020-11-06

514. Measurement of the Higgs boson production rate in association with top quarks in final states with electrons, muons, and hadronically decaying tau leptons at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 378](#) | [arXiv:2011.03652](#) | [DOI:10.1140/epjc/s10052-021-09014-x](#) | 2020-11-06

515. The very forward CASTOR calorimeter of the CMS experiment

Khachatryan, Vardan et al.

[JINST 16 \(2021\) P02010](#) | [arXiv:2011.01185](#) | [DOI:10.1088/1748-0221/16/02/P02010](#) | 2020-11-02

516. Angular analysis of the decay $B^+ \rightarrow K^{*(892)^+} \mu^+ \mu^-$ in proton-proton collisions at $\sqrt{s} = 8 \text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 04 \(2021\) 124](#) | [arXiv:2010.13968](#) | [DOI:10.1007/JHEP04\(2021\)124](#) | 2020-10-26

517. MUSiC: a model-unspecific search for new physics in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 629](#) | [arXiv:2010.02984](#) | [DOI:10.1140/epjc/s10052-021-09236-z](#) | 2020-10-06

518. Search for dark photons in Higgs boson production via vector boson fusion in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

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519. Measurement of prompt D^0 and $\bar{D}^0$ meson azimuthal anisotropy and search for strong electric fields in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 816 \(2021\) 136253](#) | [arXiv:2009.12628](#) | [DOI:10.1016/j.physletb.2021.136253](#) | 2020-09-26

520. Measurements of production cross sections of polarized same-sign W boson pairs in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 812 \(2021\) 136018](#) | [arXiv:2009.09429](#) | [DOI:10.1016/j.physletb.2020.136018](#) | 2020-09-20

521. Studies of charm and beauty hadron long-range correlations in pp and pPb collisions at LHC energies

Sirunyan, Albert M et al.

[Phys.Lett.B 813 \(2021\) 136036](#) | [arXiv:2009.07065](#) | [DOI:10.1016/j.physletb.2020.136036](#) | 2020-09-15

522. Measurement of the top quark Yukawa coupling from $t\bar{t}$ kinematic distributions in the dilepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 092013](#) | [arXiv:2009.07123](#) | [DOI:10.1103/PhysRevD.102.092013](#) | 2020-09-15

523. Evidence for Higgs boson decay to a pair of muons

Sirunyan, Albert M et al.

[JHEP 01 \(2021\) 148](#) | [arXiv:2009.04363](#) | [DOI:10.1007/JHEP01\(2021\)148](#) | 2020-09-09

524. Measurements of $pp \rightarrow ZZ$ production cross sections and constraints on anomalous triple gauge couplings at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 81 \(2021\) 200](#) | [arXiv:2009.01186](#) | [DOI:10.1140/epjc/s10052-020-08817-8](#) | 2020-09-02

525. W^+W^- boson pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 092001](#) | [arXiv:2009.00119](#) | [DOI:10.1103/PhysRevD.102.092001](#) | 2020-08-31

526. Coffea: Columnar Object Framework For Effective Analysis

Smith, Nicholas et al.

[EPJ Web Conf. 245 \(2020\) 06012](#) | [arXiv:2008.12712](#) | [DOI:10.1051/epjconf/202024506012](#) | 2020-08-28

527. Observation of electroweak production of $W\gamma$ with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 811 \(2020\) 135988](#) | [arXiv:2008.10521](#) | [DOI:10.1016/j.physletb.2020.135988](#) | 2020-08-24

528. A search for bottom-type, vector-like quark pair production in a fully hadronic final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 112004](#) | [arXiv:2008.09835](#) | [DOI:10.1103/PhysRevD.102.112004](#) | 2020-08-22

529. Measurement of $B_c(2S)^+$ and $B_c^{*+}(2S)^+$ cross section ratios in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 092007](#) | [arXiv:2008.08629](#) | [DOI:10.1103/PhysRevD.102.092007](#) | 2020-08-19

530. Measurement of differential $t\bar{t}$ production cross sections using top quarks at large transverse momenta in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 103 \(2021\) 052008](#) | [arXiv:2008.07860](#) | [DOI:10.1103/PhysRevD.103.052008](#) | 2020-08-18

531. Evidence for electroweak production of four charged leptons and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Lett.B 812 \(2021\) 135992](#) | [arXiv:2008.07013](#) | [DOI:10.1016/j.physletb.2020.135992](#) | 2020-08-16
532. Search for top squark pair production using dilepton final states in pp collision data collected at $\sqrt{s}=13$ TeV
Sirunyan, Albert M et al.
[Eur.Phys.J.C 81 \(2021\) 3](#) | [arXiv:2008.05936](#) | [DOI:10.1140/epjc/s10052-020-08701-5](#) | 2020-08-13
533. Search for dark matter produced in association with a leptonically decaying Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Eur.Phys.J.C 81 \(2021\) 13](#) | [arXiv:2008.04735](#) | [DOI:10.1140/epjc/s10052-020-08739-5](#) | 2020-08-11
534. Measurements of the W boson rapidity, helicity, double-differential cross sections, and charge asymmetry in pp collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[Phys.Rev.D 102 \(2020\) 092012](#) | [arXiv:2008.04174](#) | [DOI:10.1103/PhysRevD.102.092012](#) | 2020-08-10
535. Search for supersymmetry in proton-proton collisions at $\sqrt{s} = 13$ TeV in events with high-momentum Z bosons and missing transverse momentum
Sirunyan, Albert M et al.
[JHEP 09 \(2020\) 149](#) | [arXiv:2008.04422](#) | [DOI:10.1007/JHEP09\(2020\)149](#) | 2020-08-10
536. Search for decays of the 125 GeV Higgs boson into a Z boson and a ρ or ϕ meson
Sirunyan, Albert M et al.
[JHEP 11 \(2020\) 039](#) | [arXiv:2007.05122](#) | [DOI:10.1007/JHEP11\(2020\)039](#) | 2020-07-10
537. Search for the lepton flavor violating decay $\tau \rightarrow 3\mu$ in proton-proton collisions at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[JHEP 01 \(2021\) 163](#) | [arXiv:2007.05658](#) | [DOI:10.1007/JHEP01\(2021\)163](#) | 2020-07-10
538. Investigation into the event-activity dependence of $Y(nS)$ relative production in proton-proton collisions at $\sqrt{s} = 7$ TeV
Sirunyan, Albert M et al.
[JHEP 11 \(2020\) 001](#) | [arXiv:2007.04277](#) | [DOI:10.1007/JHEP11\(2020\)001](#) | 2020-07-09
539. Measurement of the CP-violating phase ϕ_s in the $B_s^0 \rightarrow J/\psi \phi(1020) \rightarrow \mu^+ \mu^- K^+ K^-$ channel in proton-proton collisions at $\sqrt{s}=13$ TeV
Sirunyan, Albert M et al.
[Phys.Lett.B 816 \(2021\) 136188](#) | [arXiv:2007.02434](#) | [DOI:10.1016/j.physletb.2021.136188](#) | 2020-07-05
540. Measurement of the inclusive and differential Higgs boson production cross sections in the leptonic WW decay mode at $\sqrt{s} = 13$ TeV
Sirunyan, Albert M et al.
[JHEP 03 \(2021\) 003](#) | [arXiv:2007.01984](#) | [DOI:10.1007/JHEP03\(2021\)003](#) | 2020-07-03
541. Search for resonant and nonresonant production of pairs of dijet resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV
Tumasyan, Armen et al.
[JHEP 07 \(2023\) 161](#) | [arXiv:2206.09997](#) | [DOI:10.1007/JHEP07\(2023\)161](#) | 2020
542. Reconstruction of signal amplitudes in the CMS electromagnetic calorimeter in the presence of overlapping proton-proton interactions

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P10002](#) | [arXiv:2006.14359](#) | [DOI:10.1088/1748-0221/15/10/P10002](#) | 2020-06-25

543. Inclusive search for highly boosted Higgs bosons decaying to bottom quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2020\) 085](#) | [arXiv:2006.13251](#) | [DOI:10.1007/JHEP12\(2020\)085](#) | 2020-06-23

544. Evidence for Top Quark Production in Nucleus-Nucleus Collisions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 222001](#) | [arXiv:2006.11110](#) | [DOI:10.1103/PhysRevLett.125.222001](#) | 2020-06-19

545. Observation of the Production of Three Massive Gauge Bosons at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 151802](#) | [arXiv:2006.11191](#) | [DOI:10.1103/PhysRevLett.125.151802](#) | 2020-06-19

546. Performance of the CMS Level-1 trigger in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P10017](#) | [arXiv:2006.10165](#) | [DOI:10.1088/1748-0221/15/10/P10017](#) | 2020-06-17

547. Measurement of the azimuthal anisotropy of $\Upsilon(1S)$ and $\Upsilon(2S)$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 819 \(2021\) 136385](#) | [arXiv:2006.07707](#) | [DOI:10.1016/j.physletb.2021.136385](#) | 2020-06-13

548. Search for resonant pair production of Higgs bosons in the $bbZZ$ channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 032003](#) | [arXiv:2006.06391](#) | [DOI:10.1103/PhysRevD.102.032003](#) | 2020-06-11

549. Measurement of b jet shapes in proton-proton collisions at $\sqrt{s} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2021\) 054](#) | [arXiv:2005.14219](#) | [DOI:10.1007/JHEP05\(2021\)054](#) | 2020-05-28

550. Search for a light charged Higgs boson in the $H^{\pm} \rightarrow cs$ channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 102 \(2020\) 072001](#) | [arXiv:2005.08900](#) | [DOI:10.1103/PhysRevD.102.072001](#) | 2020-05-18

551. Search for a light pseudoscalar Higgs boson in the boosted $\mu\mu\tau\tau$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2020\) 139](#) | [arXiv:2005.08694](#) | [DOI:10.1007/JHEP08\(2020\)139](#) | 2020-05-18

552. Dependence of inclusive jet production on the anti- k_T distance parameter in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2020\) 082](#) | [arXiv:2005.05159](#) | [DOI:10.1007/JHEP12\(2020\)082](#) | 2020-05-11

553. Observation of the $B^0_s \rightarrow X(3872)\phi$ decay

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 152001](#) | [arXiv:2005.04764](#) | [DOI:10.1103/PhysRevLett.125.152001](#) | 2020-05-10

554. Combination of the W boson polarization measurements in top quark decays using ATLAS and CMS data at $\sqrt{s} = 8$ TeV

Aad, Georges et al.

[JHEP 08 \(2020\) 051](#) | [arXiv:2005.03799](#) | [DOI:10.1007/JHEP08\(2020\)051](#) | 2020-05-07

555. Measurements of production cross sections of WZ and same-sign WW boson pairs in association with two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 809 \(2020\) 135710](#) | [arXiv:2005.01173](#) | [DOI:10.1016/j.physletb.2020.135710](#) | 2020-05-03

556. Measurement of CKM matrix elements in single top quark t-channel production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 808 \(2020\) 135609](#) | [arXiv:2004.12181](#) | [DOI:10.1016/j.physletb.2020.135609](#) | 2020-04-25

557. Identification of heavy, energetic, hadronically decaying particles using machine-learning techniques

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P06005](#) | [arXiv:2004.08262](#) | [DOI:10.1088/1748-0221/15/06/P06005](#) | 2020-04-17

558. Search for disappearing tracks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 806 \(2020\) 135502](#) | [arXiv:2004.05153](#) | [DOI:10.1016/j.physletb.2020.135502](#) | 2020-04-10

559. Measurement of quark- and gluon-like jet fractions using jet charge in PbPb and pp collisions at 5.02 TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 115](#) | [arXiv:2004.00602](#) | [DOI:10.1007/JHEP07\(2020\)115](#) | 2020-04-01

560. The production of isolated photons in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 116](#) | [arXiv:2003.12797](#) | [DOI:10.1007/JHEP07\(2020\)116](#) | 2020-03-28

561. Search for supersymmetry in pp collisions at $\sqrt{s} = 13$ TeV with 137 fb^{-1} in final states with a single lepton using the sum of masses of large-radius jets

Sirunyan, Albert M et al.

[Phys.Rev.D 101 \(2020\) 052010](#) | [arXiv:1911.07558](#) | [DOI:10.1103/PhysRevD.101.052010](#) | 2020-03-25

562. Measurements of $t\bar{t}H$ Production and the CP Structure of the Yukawa Interaction between the Higgs Boson and Top Quark in the Diphoton Decay Channel

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 061801](#) | [arXiv:2003.10866](#) | [DOI:10.1103/PhysRevLett.125.061801](#) | 2020-03-24

563. Measurement of the cross section for $t\bar{t}$ production with additional jets and b jets in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2020\) 125](#) | [arXiv:2003.06467](#) | [DOI:10.1007/JHEP07\(2020\)125](#) | 2020-03-13

564. Study of central exclusive $\pi^+ \pi^-$ production in proton-proton collisions at $\sqrt{s} = 5.02$ and 13 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 718](#) | [arXiv:2003.02811](#) | [DOI:10.1140/epjc/s10052-020-8166-5](#) | 2020-03-05

565. Pileup mitigation at CMS in 13 TeV data

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P09018](#) | [arXiv:2003.00503](#) | [DOI:10.1088/1748-0221/15/09/P09018](#) | 2020-03-01

566. Measurement of single-diffractive dijet production in proton-proton collisions at $\sqrt{s} = 8$ TeV with the CMS and TOTEM experiments

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 1164](#) | [arXiv:2002.12146](#) | [DOI:10.1140/epjc/s10052-020-08562-y](#) | 2020-02-27

567. Measurement of the cross section for electroweak production of a Z boson, a photon and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on anomalous quartic couplings

Sirunyan, Albert M et al.

JHEP 06 (2020) 076 | [arXiv:2002.09902](#) | [DOI:10.1007/JHEP06\(2020\)076](#) | 2020-02-23

568. Measurement of the $\Upsilon(1S)$ pair production cross section and search for resonances decaying to $\Upsilon(1S)\mu^+\mu^-$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

Phys.Lett.B 808 (2020) 135578 | [arXiv:2002.06393](#) | [DOI:10.1016/j.physletb.2020.135578](#) | 2020-02-15

569. A measurement of the Higgs boson mass in the diphoton decay channel

Sirunyan, Albert M et al.

Phys.Lett.B 805 (2020) 135425 | [arXiv:2002.06398](#) | [DOI:10.1016/j.physletb.2020.135425](#) | 2020-02-15

570. Search for physics beyond the standard model in events with jets and two same-sign or at least three charged leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

Eur.Phys.J.C 80 (2020) 752 | [arXiv:2001.10086](#) | [DOI:10.1140/epjc/s10052-020-8168-3](#) | 2020-01-27

571. Search for charged Higgs bosons decaying into a top and a bottom quark in the all-jet final state of pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

JHEP 07 (2020) 126 | [arXiv:2001.07763](#) | [DOI:10.1007/JHEP07\(2020\)126](#) | 2020-01-21

572. Measurement of the associated production of a Z boson with charm or bottom quark jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

Phys.Rev.D 102 (2020) 032007 | [arXiv:2001.06899](#) | [DOI:10.1103/PhysRevD.102.032007](#) | 2020-01-19

573. Study of excited Λ_b^0 states decaying to $\Lambda_b^0\pi^+\pi^-$ in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

Phys.Lett.B 803 (2020) 135345 | [arXiv:2001.06533](#) | [DOI:10.1016/j.physletb.2020.135345](#) | 2020-01-17

574. Measurements with silicon photomultipliers of dose-rate effects in the radiation damage of plastic scintillator tiles in the CMS hadron endcap calorimeter

Sirunyan, Albert M et al.

JINST 15 (2020) P06009 | [arXiv:2001.06553](#) | [DOI:10.1088/1748-0221/15/06/P06009](#) | 2020-01-17

575. Search for an excited lepton that decays via a contact interaction to a lepton and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

JHEP 05 (2020) 052 | [arXiv:2001.04521](#) | [DOI:10.1007/JHEP05\(2020\)052](#) | 2020-01-13

576. A deep neural network to search for new long-lived particles decaying to jets

Sirunyan, Albert M et al.

Mach.Learn.Sci.Tech. 1 (2020) 035012 | [arXiv:1912.12238](#) | [DOI:10.1088/2632-2153/ab9023](#) | 2019-12-27

577. Measurement of the top quark forward-backward production asymmetry and the anomalous chromoelectric and chromomagnetic moments in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

JHEP 06 (2020) 146 | [arXiv:1912.09540](#) | [DOI:10.1007/JHEP06\(2020\)146](#) | 2019-12-19

578. Search for direct top squark pair production in events with one lepton, jets, and missing transverse momentum at 13 TeV with the CMS experiment

Sirunyan, Albert M et al.

JHEP 05 (2020) 032 | [arXiv:1912.08887](#) | [DOI:10.1007/JHEP05\(2020\)032](#) | 2019-12-18

579. Constraints on the χ_{c1} versus χ_{c2} Polarizations in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 124 \(2020\) 162002](#) | [arXiv:1912.07706](#) | [DOI:10.1103/PhysRevLett.124.162002](#) | 2019-12-16

580. A Deep Neural Network for Simultaneous Estimation of b Jet Energy and Resolution

Sirunyan, Albert M et al.

[Comput.Softw.Big Sci. 4 \(2020\) 10](#) | [arXiv:1912.06046](#) | [DOI:10.1007/s41781-020-00041-z](#) | 2019-12-12

581. Search for a Narrow Resonance Lighter than 200 GeV Decaying to a Pair of Muons in Proton-Proton Collisions at $\sqrt{s} = \text{TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 124 \(2020\) 131802](#) | [arXiv:1912.04776](#) | [DOI:10.1103/PhysRevLett.124.131802](#) | 2019-12-10

582. Determination of the strong coupling constant $\alpha_S(m_Z)$ from measurements of inclusive $W^\pm$ and Z boson production cross sections in proton-proton collisions at $\sqrt{s} = 7$ and 8 TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2020\) 018](#) | [arXiv:1912.04387](#) | [DOI:10.1007/JHEP06\(2020\)018](#) | 2019-12-09

583. Performance of the reconstruction and identification of high-momentum muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P02027](#) | [arXiv:1912.03516](#) | [DOI:10.1088/1748-0221/15/02/P02027](#) | 2019-12-07

584. Search for a heavy Higgs boson decaying to a pair of W bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 034](#) | [arXiv:1912.01594](#) | [DOI:10.1007/JHEP03\(2020\)034](#) | 2019-12-03

585. A search for the standard model Higgs boson decaying to charm quarks

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 131](#) | [arXiv:1912.01662](#) | [DOI:10.1007/JHEP03\(2020\)131](#) | 2019-12-03

586. Measurement of the top quark pair production cross section in dilepton final states containing one τ lepton in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2020\) 191](#) | [arXiv:1911.13204](#) | [DOI:10.1007/JHEP02\(2020\)191](#) | 2019-11-29

587. Search for lepton flavour violating decays of a neutral heavy Higgs boson to $\mu\tau$ and $e\tau$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 103](#) | [arXiv:1911.10267](#) | [DOI:10.1007/JHEP03\(2020\)103](#) | 2019-11-22

588. Search for physics beyond the standard model in multilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 051](#) | [arXiv:1911.04968](#) | [DOI:10.1007/JHEP03\(2020\)051](#) | 2019-11-12

589. Search for high mass dijet resonances with a new background prediction method in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2020\) 033](#) | [arXiv:1911.03947](#) | [DOI:10.1007/JHEP05\(2020\)033](#) | 2019-11-10

590. Measurement of the Jet Mass Distribution and Top Quark Mass in Hadronic Decays of Boosted Top Quarks in pp Collisions at $\sqrt{s} = \text{TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 124 \(2020\) 202001](#) | [arXiv:1911.03800](#) | [DOI:10.1103/PhysRevLett.124.202001](#) | 2019-11-09

591. Observation of the $\Lambda_b^0 \rightarrow J/\psi \Lambda \phi$ decay in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 802 \(2020\) 135203](#) | [arXiv:1911.03789](#) | [DOI:10.1016/j.physletb.2020.135203](#) | 2019-11-09

592. Search for new neutral Higgs bosons through the $H \rightarrow ZA \rightarrow \ell^+ \ell^- b \bar{b}$ process in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 055](#) | [arXiv:1911.03781](#) | [DOI:10.1007/JHEP03\(2020\)055](#) | 2019-11-09

593. Search for dijet resonances using events with three jets in proton-proton collisions at $s=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 805 \(2020\) 135448](#) | [arXiv:1911.03761](#) | [DOI:10.1016/j.physletb.2020.135448](#) | 2019-11-09

594. Striped Data Analysis Framework

Gutsche, Oliver et al.

[EPJ Web Conf. 245 \(2020\) 06042](#) | [DOI:10.1051/epjconf/202024506042](#) | 2020

595. Studies of charm quark diffusion inside jets using PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 125 \(2020\) 102001](#) | [arXiv:1911.01461](#) | [DOI:10.1103/PhysRevLett.125.102001](#) | 2019-11-04

596. Response to NITRD, NCO, NSF Request for Information on "Update to the 2016 National Artificial Intelligence Research and Development Strategic Plan"

Amundson, J. et al.

[arXiv:1911.05796](#) | [DOI:10.2172/1592156](#) | 2019-11-04

597. Search for top squark pair production in a final state with two tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2020\) 015](#) | [arXiv:1910.12932](#) | [DOI:10.1007/JHEP02\(2020\)015](#) | 2019-10-28

598. Measurement of properties of $B^0_s \rightarrow \mu^+ \mu^-$ decays and search for $B^0 \rightarrow \mu^+ \mu^-$ with the CMS experiment

Sirunyan, Albert M et al.

[JHEP 04 \(2020\) 188](#) | [arXiv:1910.12127](#) | [DOI:10.1007/JHEP04\(2020\)188](#) | 2019-10-26

599. Search for a heavy pseudoscalar Higgs boson decaying into a 125 GeV Higgs boson and a Z boson in final states with two tau and two light leptons at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 065](#) | [arXiv:1910.11634](#) | [DOI:10.1007/JHEP03\(2020\)065](#) | 2019-10-25

600. Mixed higher-order anisotropic flow and nonlinear response coefficients of charged particles in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ and 5.02 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 534](#) | [arXiv:1910.08789](#) | [DOI:10.1140/epjc/s10052-020-7834-9](#) | 2019-10-19

601. Bose-Einstein correlations of charged hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 014](#) | [arXiv:1910.08815](#) | [DOI:10.1007/JHEP03\(2020\)014](#) | 2019-10-19

602. Strange hadron production in pp and pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 101 \(2020\) 064906](#) | [arXiv:1910.04812](#) | [DOI:10.1103/PhysRevC.101.064906](#) | 2019-10-10

603. Measurement of the top quark polarization and $t\bar{t}$ spin correlations using dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 072002](#) | [arXiv:1907.03729](#) | [DOI:10.1103/PhysRevD.100.072002](#) | 2019-10-09

604. Study of J/ψ meson production inside jets in pp collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 804 \(2020\) 135409](#) | [arXiv:1910.01686](#) | [DOI:10.1016/j.physletb.2020.135409](#) | 2019-10-03

605. Search for Supersymmetry with a Compressed Mass Spectrum in Events with a Soft τ Lepton, a Highly Energetic Jet, and Large Missing Transverse Momentum in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 124 \(2020\) 041803](#) | [arXiv:1910.01185](#) | [DOI:10.1103/PhysRevLett.124.041803](#) | 2019-10-02

606. Calibration of the CMS hadron calorimeters using proton-proton collision data at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JINST 15 \(2020\) P05002](#) | [arXiv:1910.00079](#) | [DOI:10.1088/1748-0221/15/05/P05002](#) | 2019-09-30

607. Running of the top quark mass from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 803 \(2020\) 135263](#) | [arXiv:1909.09193](#) | [DOI:10.1016/j.physletb.2020.135263](#) | 2019-09-19

608. CMS Computing Resources: Meeting the Demands of the High-Luminosity LHC Physics Program

Lange, David et al.

[EPJ Web Conf. 214 \(2019\) 03055](#) | [DOI:10.1051/epjconf/201921403055](#) | 2019-09-17

609. Evidence for WW production from double-parton interactions in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 41](#) | [arXiv:1909.06265](#) | [DOI:10.1140/epjc/s10052-019-7541-6](#) | 2019-09-13

610. Search for long-lived particles using delayed photons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 112003](#) | [arXiv:1909.06166](#) | [DOI:10.1103/PhysRevD.100.112003](#) | 2019-09-13

611. Measurement of the $t\bar{t}b\bar{b}$ production cross section in the all-jet final state in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 803 \(2020\) 135285](#) | [arXiv:1909.05306](#) | [DOI:10.1016/j.physletb.2020.135285](#) | 2019-09-11

612. Search for electroweak production of a vector-like T quark using fully hadronic final states

Sirunyan, Albert M et al.

[JHEP 01 \(2020\) 036](#) | [arXiv:1909.04721](#) | [DOI:10.1007/JHEP01\(2020\)036](#) | 2019-09-10

613. Search for low mass vector resonances decaying into quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 112007](#) | [arXiv:1909.04114](#) | [DOI:10.1103/PhysRevD.100.112007](#) | 2019-09-09

614. Measurements of differential Z boson production cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2019\) 061](#) | [arXiv:1909.04133](#) | [DOI:10.1007/JHEP12\(2019\)061](#) | 2019-09-09

615. Searches for physics beyond the standard model with the M_{T2} variable in hadronic final states with and without disappearing tracks in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 3](#) | [arXiv:1909.03460](#) | [DOI:10.1140/epjc/s10052-019-7493-x](#) | 2019-09-08

616. Search for a charged Higgs boson decaying into top and bottom quarks in events with electrons or muons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2020\) 096](#) | [arXiv:1908.09206](#) | [DOI:10.1007/JHEP01\(2020\)096](#) | 2019-08-24

617. Search for supersymmetry using Higgs boson to diphoton decays at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2019\) 109](#) | [arXiv:1908.08500](#) | [DOI:10.1007/JHEP11\(2019\)109](#) | 2019-08-22

618. Search for production of four top quarks in final states with same-sign or multiple leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 75](#) | [arXiv:1908.06463](#) | [DOI:10.1140/epjc/s10052-019-7593-7](#) | 2019-08-18

619. Search for supersymmetry in proton-proton collisions at 13 TeV in final states with jets and missing transverse momentum

Collaboration, The Cms et al.

[JHEP 10 \(2019\) 244](#) | [arXiv:1908.04722](#) | [DOI:10.1007/JHEP10\(2019\)244](#) | 2019-08-13

620. Search for dark photons in decays of Higgs bosons produced in association with Z bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2019\) 139](#) | [arXiv:1908.02699](#) | [DOI:10.1007/JHEP10\(2019\)139](#) | 2019-08-07

621. Measurement of the average very forward energy as a function of the track multiplicity at central pseudorapidities in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 893](#) | [arXiv:1908.01750](#) | [DOI:10.1140/epjc/s10052-019-7402-3](#) | 2019-08-05

622. Search for dark matter particles produced in association with a Higgs boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 025](#) | [arXiv:1908.01713](#) | [DOI:10.1007/JHEP03\(2020\)025](#) | 2019-08-05

623. Search for heavy Higgs bosons decaying to a top quark pair in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2020\) 171](#) | [arXiv:1908.01115](#) | [DOI:10.1007/JHEP04\(2020\)171](#) | 2019-08-02

624. New Technologies for Discovery

Ahmed, Z. et al.

[arXiv:1908.00194](#) | 2019-07-31

625. Search for direct pair production of supersymmetric partners to the τ lepton in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 189](#) | [arXiv:1907.13179](#) | [DOI:10.1140/epjc/s10052-020-7739-7](#) | 2019-07-30

626. Measurement of top quark pair production in association with a Z boson in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2020\) 056](#) | [arXiv:1907.11270](#) | [DOI:10.1007/JHEP03\(2020\)056](#) | 2019-07-25

627. Measurements of triple-differential cross sections for inclusive isolated-photon+jet events in pp collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 969](#) | [arXiv:1907.08155](#) | [DOI:10.1140/epjc/s10052-019-7451-7](#) | 2019-07-18

628. Measurement of differential cross sections and charge ratios for t-channel single top quark production in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 370](#) | [arXiv:1907.08330](#) | [DOI:10.1140/epjc/s10052-020-7858-1](#) | 2019-07-18

629. Search for anomalous triple gauge couplings in WW and WZ production in lepton + jet events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2019\) 062](#) | [arXiv:1907.08354](#) | [DOI:10.1007/JHEP12\(2019\)062](#) | 2019-07-18

630. Search for light pseudoscalar boson pairs produced from decays of the 125 GeV Higgs boson in final states with two muons and two nearby tracks in pp collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 800 \(2020\) 135087](#) | [arXiv:1907.07235](#) | [DOI:10.1016/j.physletb.2019.135087](#) | 2019-07-16

631. Search for Physics beyond the Standard Model in Events with Overlapping Photons and Jets

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 123 \(2019\) 241801](#) | [arXiv:1907.06275](#) | [DOI:10.1103/PhysRevLett.123.241801](#) | 2019-07-14

632. Study of the $B^+ \rightarrow J/\psi \Lambda p$ decay in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2019\) 100](#) | [arXiv:1907.05461](#) | [DOI:10.1007/JHEP12\(2019\)100](#) | 2019-07-11

633. Search for MSSM Higgs bosons decaying to $\mu + \mu -$ in proton-proton collisions at $s=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 798 \(2019\) 134992](#) | [arXiv:1907.03152](#) | [DOI:10.1016/j.physletb.2019.134992](#) | 2019-07-06

634. Measurement of the top quark Yukawa coupling from $t\bar{t}$ kinematic distributions in the lepton+jets final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 072007](#) | [arXiv:1907.01590](#) | [DOI:10.1103/PhysRevD.100.072007](#) | 2019-07-02

635. Combined search for supersymmetry with photons in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 801 \(2020\) 135183](#) | [arXiv:1907.00857](#) | [DOI:10.1016/j.physletb.2019.135183](#) | 2019-07-01

636. Search for pair production of vectorlike quarks in the fully hadronic final state

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 072001](#) | [arXiv:1906.11903](#) | [DOI:10.1103/PhysRevD.100.072001](#) | 2019-06-27

637. Search for long-lived particles using nonprompt jets and missing transverse momentum with proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 797 \(2019\) 134876](#) | [arXiv:1906.06441](#) | [DOI:10.1016/j.physletb.2019.134876](#) | 2019-06-14

638. A multi-dimensional search for new heavy resonances decaying to boosted WW, WZ, or ZZ boson pairs in the dijet final state at 13 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 237](#) | [arXiv:1906.05977](#) | [DOI:10.1140/epjc/s10052-020-7773-5](#) | 2019-06-13

639. Production of Λ_c^+ baryons in proton-proton and lead-lead collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 803 \(2020\) 135328](#) | [arXiv:1906.03322](#) | [DOI:10.1016/j.physletb.2020.135328](#) | 2019-06-07

640. Search for the production of four top quarks in the single-lepton and opposite-sign dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2019\) 082](#) | [arXiv:1906.02805](#) | [DOI:10.1007/JHEP11\(2019\)082](#) | 2019-06-06

641. Combination of CMS searches for heavy resonances decaying to pairs of bosons or leptons

Sirunyan, Albert M et al.

[Phys.Lett.B 798 \(2019\) 134952](#) | [arXiv:1906.00057](#) | [DOI:10.1016/j.physletb.2019.134952](#) | 2019-05-31

642. Search for supersymmetry with a compressed mass spectrum in the vector boson fusion topology with 1-lepton and 0-lepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2019\) 150](#) | [arXiv:1905.13059](#) | [DOI:10.1007/JHEP08\(2019\)150](#) | 2019-05-30

643. Search for vector-like leptons in multilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 052003](#) | [arXiv:1905.10853](#) | [DOI:10.1103/PhysRevD.100.052003](#) | 2019-05-26

644. Search for Low-Mass Quark-Antiquark Resonances Produced in Association with a Photon at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 123 \(2019\) 231803](#) | [arXiv:1905.10331](#) | [DOI:10.1103/PhysRevLett.123.231803](#) | 2019-05-24

645. Search for Higgs and Z boson decays to J/ψ or Y pairs in the four-muon final state in proton-proton collisions at $s=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 797 \(2019\) 134811](#) | [arXiv:1905.10408](#) | [DOI:10.1016/j.physletb.2019.134811](#) | 2019-05-24

646. Correlations of azimuthal anisotropy Fourier harmonics with subevent cumulants in pPb collisions at $\sqrt{s_{NN}}=8.16$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 103 \(2021\) 014902](#) | [arXiv:1905.09935](#) | [DOI:10.1103/PhysRevC.103.014902](#) | 2019-05-23

647. Search for a light charged Higgs boson decaying to a W boson and a CP-odd Higgs boson in final states with $e\mu\mu$ or $\mu\mu\mu$ in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 123 \(2019\) 131802](#) | [arXiv:1905.07453](#) | [DOI:10.1103/PhysRevLett.123.131802](#) | 2019-05-17

648. Search for anomalous electroweak production of vector boson pairs in association with two jets in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 798 \(2019\) 134985](#) | [arXiv:1905.07445](#) | [DOI:10.1016/j.physletb.2019.134985](#) | 2019-05-17

649. Search for the production of $W^\pm W^\pm W^\mp$ events at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 012004](#) | [arXiv:1905.04246](#) | [DOI:10.1103/PhysRevD.100.012004](#) | 2019-05-10

650. Observation of nuclear modifications in $W^{\pm}$ boson production in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 800 \(2020\) 135048](#) | [arXiv:1905.01486](#) | [DOI:10.1016/j.physletb.2019.135048](#) | 2019-05-04

651. Multiparticle correlation studies in pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 101 \(2020\) 014912](#) | [arXiv:1904.11519](#) | [DOI:10.1103/PhysRevC.101.014912](#) | 2019-04-25

652. Measurement of $t\bar{t}$ normalised multi-differential cross sections in pp collisions at $\sqrt{s}=13$ TeV, and simultaneous determination of the strong coupling strength, top quark pole mass, and parton distribution functions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 658](#) | [arXiv:1904.05237](#) | [DOI:10.1140/epjc/s10052-020-7917-7](#) | 2019-04-10

653. Search for resonances decaying to a pair of Higgs bosons in the $b\bar{b}q\bar{q}'\ell\nu$ final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2019\) 125](#) | [arXiv:1904.04193](#) | [DOI:10.1007/JHEP10\(2019\)125](#) | 2019-04-08

654. Extraction and validation of a new set of CMS PYTHIA8 tunes from underlying-event measurements

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 4](#) | [arXiv:1903.12179](#) | [DOI:10.1140/epjc/s10052-019-7499-4](#) | 2019-03-28

655. Search for new physics in top quark production in dilepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 886](#) | [arXiv:1903.11144](#) | [DOI:10.1140/epjc/s10052-019-7387-y](#) | 2019-03-26

656. Search for a low-mass $\tau^+\tau^-$ resonance in association with a bottom quark in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2019\) 210](#) | [arXiv:1903.10228](#) | [DOI:10.1007/JHEP05\(2019\)210](#) | 2019-03-25

657. Search for supersymmetry in final states with photons and missing transverse momentum in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2019\) 143](#) | [arXiv:1903.07070](#) | [DOI:10.1007/JHEP06\(2019\)143](#) | 2019-03-17

658. Constraints on anomalous HVV couplings from the production of Higgs bosons decaying to τ lepton pairs

Sirunyan, Albert M et al.

[Phys.Rev.D 100 \(2019\) 112002](#) | [arXiv:1903.06973](#) | [DOI:10.1103/PhysRevD.100.112002](#) | 2019-03-16

659. Performance of missing transverse momentum reconstruction in proton-proton collisions at $\sqrt{s} = 13$ TeV using the CMS detector

Sirunyan, Albert M et al.

[JINST 14 \(2019\) P07004](#) | [arXiv:1903.06078](#) | [DOI:10.1088/1748-0221/14/07/P07004](#) | 2019-03-14

660. Search for charged Higgs bosons in the $H^{\pm} \rightarrow \tau^{\pm}\nu_{\tau}$ decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2019\) 142](#) | [arXiv:1903.04560](#) | [DOI:10.1007/JHEP07\(2019\)142](#) | 2019-03-11

661. Measurement of electroweak production of a W boson in association with two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 80 \(2020\) 43](#) | [arXiv:1903.04040](#) | [DOI:10.1140/epjc/s10052-019-7585-7](#) | 2019-03-10

662. An embedding technique to determine $\tau\tau$ backgrounds in proton-proton collision data

Sirunyan, Albert M et al.

[JINST 14 \(2019\) P06032](#) | [arXiv:1903.01216](#) | [DOI:10.1088/1748-0221/14/06/P06032](#) | 2019-03-04

663. Search for a heavy pseudoscalar boson decaying to a Z and a Higgs boson at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 564](#) | [arXiv:1903.00941](#) | [DOI:10.1140/epjc/s10052-019-7058-z](#) | 2019-03-03

664. Combinations of single-top-quark production cross-section measurements and $|f_{LVV_{tb}}|$ determinations at $\sqrt{s} = 7$ and 8 TeV with the ATLAS and CMS experiments

Aaboud, Morad et al.

[JHEP 05 \(2019\) 088](#) | [arXiv:1902.07158](#) | [DOI:10.1007/JHEP05\(2019\)088](#) | 2019-02-19

665. Azimuthal separation in nearly back-to-back jet topologies in inclusive 2- and 3-jet events in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 773](#) | [arXiv:1902.04374](#) | [DOI:10.1140/epjc/s10052-019-7276-4](#) | 2019-02-12

666. Pseudorapidity distributions of charged hadrons in xenon-xenon collisions at $\sqrt{s_{NN}} = 5.44$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 799 \(2019\) 135049](#) | [arXiv:1902.03603](#) | [DOI:10.1016/j.physletb.2019.135049](#) | 2019-02-10

667. Measurement of exclusive $\rho(770)^0$ photoproduction in ultraperipheral pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 702](#) | [arXiv:1902.01339](#) | [DOI:10.1140/epjc/s10052-019-7202-9](#) | 2019-02-04

668. Observation of Two Excited B^+_c States and Measurement of the $B^+_c(2S)$ Mass in pp Collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 132001](#) | [arXiv:1902.00571](#) | [DOI:10.1103/PhysRevLett.122.132001](#) | 2019-02-01

669. Search for W boson decays to three charged pions

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 151802](#) | [arXiv:1901.11201](#) | [DOI:10.1103/PhysRevLett.122.151802](#) | 2019-01-30

670. Charged-particle angular correlations in XeXe collisions at $\sqrt{s_{NN}} = 5.44$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 100 \(2019\) 044902](#) | [arXiv:1901.07997](#) | [DOI:10.1103/PhysRevC.100.044902](#) | 2019-01-23

671. Using Big Data Technologies for HEP Analysis

Cremonesi, Matteo et al.

[EPJ Web Conf. 214 \(2019\) 06030](#) | [arXiv:1901.07143](#) | [DOI:10.1051/epjconf/201921406030](#) | 2019-01-21

672. Search for supersymmetry in events with a photon, jets, b-jets, and missing transverse momentum in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 444](#) | [arXiv:1901.06726](#) | [DOI:10.1140/epjc/s10052-019-6926-x](#) | 2019-01-20

673. Measurement of electroweak WZ boson production and search for new physics in WZ + two jets events in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 795 \(2019\) 281](#) | [arXiv:1901.04060](#) | [DOI:10.1016/j.physletb.2019.05.042](#) | 2019-01-13

674. Measurements of the $pp \rightarrow WZ$ inclusive and differential production cross section and constraints on charged anomalous triple gauge couplings at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 122](#) | [arXiv:1901.03428](#) | [DOI:10.1007/JHEP04\(2019\)122](#) | 2019-01-10

675. Search for dark matter produced in association with a single top quark or a top quark pair in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 141](#) | [arXiv:1901.01553](#) | [DOI:10.1007/JHEP03\(2019\)141](#) | 2019-01-06

676. Search for the pair production of light top squarks in the $e^{\pm}\mu^{\mp}$ final state in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 101](#) | [arXiv:1901.01288](#) | [DOI:10.1007/JHEP03\(2019\)101](#) | 2019-01-04

677. Measurements of the Higgs boson width and anomalous HVV couplings from on-shell and off-shell production in the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 112003](#) | [arXiv:1901.00174](#) | [DOI:10.1103/PhysRevD.99.112003](#) | 2019-01-01

678. Search for contact interactions and large extra dimensions in the dilepton mass spectra from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 114](#) | [arXiv:1812.10443](#) | [DOI:10.1007/JHEP04\(2019\)114](#) | 2018-12-26

679. Measurement of the differential Drell-Yan cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2019\) 059](#) | [arXiv:1812.10529](#) | [DOI:10.1007/JHEP12\(2019\)059](#) | 2018-12-26

680. Measurement of the $t\bar{t}$ production cross section, the top quark mass, and the strong coupling constant using dilepton events in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 368](#) | [arXiv:1812.10505](#) | [DOI:10.1140/epjc/s10052-019-6863-8](#) | 2018-12-26

681. Measurement of the top quark mass in the all-jets final state at $\sqrt{s} = 13$ TeV and combination with the lepton+jets channel

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 313](#) | [arXiv:1812.10534](#) | [DOI:10.1140/epjc/s10052-019-6788-2](#) | 2018-12-26

682. Measurement of the single top quark and antiquark production cross sections in the t channel and their ratio in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 800 \(2020\) 135042](#) | [arXiv:1812.10514](#) | [DOI:10.1016/j.physletb.2019.135042](#) | 2018-12-26

683. Search for vector-like quarks in events with two oppositely charged leptons and jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 364](#) | [arXiv:1812.09768](#) | [DOI:10.1140/epjc/s10052-019-6855-8](#) | 2018-12-23

684. Measurement and interpretation of differential cross sections for Higgs boson production at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 792 \(2019\) 369](#) | [arXiv:1812.06504](#) | [DOI:10.1016/j.physletb.2019.03.059](#) | 2018-12-16

685. Search for a heavy resonance decaying to a top quark and a vector-like top quark in the lepton+jets final state in pp collisions at $\sqrt{s} = 13$ TeV

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[Eur.Phys.J.C 79 \(2019\) 208](#) | [arXiv:1812.06489](#) | [DOI:10.1140/epjc/s10052-019-6688-5](#) | 2018-12-16

686. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two muons and two b quarks in pp collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 795 \(2019\) 398](#) | [arXiv:1812.06359](#) | [DOI:10.1016/j.physletb.2019.06.021](#) | 2018-12-15

687. Inclusive search for supersymmetry in pp collisions at $\sqrt{s}=13$ TeV using razor variables and boosted object identification in zero and one lepton final states

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 031](#) | [arXiv:1812.06302](#) | [DOI:10.1007/JHEP03\(2019\)031](#) | 2018-12-15

688. Observation of Single Top Quark Production in Association with a Z Boson in Proton-Proton Collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 132003](#) | [arXiv:1812.05900](#) | [DOI:10.1103/PhysRevLett.122.132003](#) | 2018-12-14

689. Search for supersymmetry in events with a photon, a lepton, and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 154](#) | [arXiv:1812.04066](#) | [DOI:10.1007/JHEP01\(2019\)154](#) | 2018-12-10

690. Measurement of the energy density as a function of pseudorapidity in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 391](#) | [arXiv:1812.04095](#) | [DOI:10.1140/epjc/s10052-019-6861-x](#) | 2018-12-10

691. Measurement of inclusive very forward jet cross sections in proton-lead collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2019\) 043](#) | [arXiv:1812.01691](#) | [DOI:10.1007/JHEP05\(2019\)043](#) | 2018-12-04

692. A search for pair production of new light bosons decaying into muons in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 796 \(2019\) 131](#) | [arXiv:1812.00380](#) | [DOI:10.1016/j.physletb.2019.07.013](#) | 2018-12-02

693. HEP Software Foundation Community White Paper Working Group – Data Organization, Management and Access (DOMA)

Berzano, Dario et al.

[arXiv:1812.00761](#) | 2018-11-30

694. HEP Software Foundation Community White Paper Working Group – Visualization

Bellis, Matthew et al.

[arXiv:1811.10309](#) | 2018-11-26

695. Search for dark matter in events with a leptoquark and missing transverse momentum in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 795 \(2019\) 76](#) | [arXiv:1811.10151](#) | [DOI:10.1016/j.physletb.2019.05.046](#) | 2018-11-25

696. Measurement of associated production of a W boson and a charm quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 269](#) | [arXiv:1811.10021](#) | [DOI:10.1140/epjc/s10052-019-6752-1](#) | 2018-11-25

697. Combination of searches for Higgs boson pair production in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 121803](#) | [arXiv:1811.09689](#) | [DOI:10.1103/PhysRevLett.122.121803](#) | 2018-11-23

698. Search for associated production of a Higgs boson and a single top quark in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 092005](#) | [arXiv:1811.09696](#) | [DOI:10.1103/PhysRevD.99.092005](#) | 2018-11-23

699. Search for resonant production of second-generation sleptons with same-sign dimuon events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 305](#) | [arXiv:1811.09760](#) | [DOI:10.1140/epjc/s10052-019-6800-x](#) | 2018-11-23

700. Search for a standard model-like Higgs boson in the mass range between 70 and 110 GeV in the diphoton final state in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 793 \(2019\) 320](#) | [arXiv:1811.08459](#) | [DOI:10.1016/j.physletb.2019.03.064](#) | 2018-11-20

701. Search for long-lived particles decaying into displaced jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 032011](#) | [arXiv:1811.07991](#) | [DOI:10.1103/PhysRevD.99.032011](#) | 2018-11-19

702. Search for a W' boson decaying to a vector-like quark and a top or bottom quark in the all-jets final state

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 127](#) | [arXiv:1811.07010](#) | [DOI:10.1007/JHEP03\(2019\)127](#) | 2018-11-16

703. Search for dark matter produced in association with a Higgs boson decaying to a pair of bottom quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 280](#) | [arXiv:1811.06562](#) | [DOI:10.1140/epjc/s10052-019-6730-7](#) | 2018-11-15

704. Measurements of $t\bar{t}$ differential cross sections in proton-proton collisions at $\sqrt{s} = 13$ TeV using events containing two leptons

Sirunyan, Albert M et al.

[JHEP 02 \(2019\) 149](#) | [arXiv:1811.06625](#) | [DOI:10.1007/JHEP02\(2019\)149](#) | 2018-11-15

705. Search for excited leptons in $\ell\ell\gamma$ final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 015](#) | [arXiv:1811.03052](#) | [DOI:10.1007/JHEP04\(2019\)015](#) | 2018-11-07

706. Search for pair production of first-generation scalar leptoquarks at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 052002](#) | [arXiv:1811.01197](#) | [DOI:10.1103/PhysRevD.99.052002](#) | 2018-11-03

707. Search for heavy neutrinos and third-generation leptoquarks in hadronic states of two τ leptons and two jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 170](#) | [arXiv:1811.00806](#) | [DOI:10.1007/JHEP03\(2019\)170](#) | 2018-11-02

708. Event shape variables measured using multijet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2018\) 117](#) | [arXiv:1811.00588](#) | [DOI:10.1007/JHEP12\(2018\)117](#) | 2018-11-01

709. Search for low-mass resonances decaying into bottom quark-antiquark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 012005](#) | [arXiv:1810.11822](#) | [DOI:10.1103/PhysRevD.99.012005](#) | 2018-10-28

710. Search for nonresonant Higgs boson pair production in the bbbb final state at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 112](#) | [arXiv:1810.11854](#) | [DOI:10.1007/JHEP04\(2019\)112](#) | 2018-10-28

711. Studies of Beauty Suppression via Nonprompt D^0 Mesons in Pb-Pb Collisions at $Q^2 = 4 \text{ GeV}^2$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 123 \(2019\) 022001](#) | [arXiv:1810.11102](#) | [DOI:10.1103/PhysRevLett.123.022001](#) | 2018-10-25

712. Search for pair-produced three-jet resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 012010](#) | [arXiv:1810.10092](#) | [DOI:10.1103/PhysRevD.99.012010](#) | 2018-10-23

713. Search for new particles decaying to a jet and an emerging jet

Sirunyan, Albert M et al.

[JHEP 02 \(2019\) 179](#) | [arXiv:1810.10069](#) | [DOI:10.1007/JHEP02\(2019\)179](#) | 2018-10-23

714. Search for rare decays of Z and Higgs bosons to J/ψ and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 94](#) | [arXiv:1810.10056](#) | [DOI:10.1140/epjc/s10052-019-6562-5](#) | 2018-10-23

715. HPC resource integration into CMS Computing via HEPCloud

Hufnagel, Dirk et al.

[EPJ Web Conf. 214 \(2019\) 03031](#) | [DOI:10.1051/epjconf/201921403031](#) | 2018-10-16

716. Search for resonant $t\bar{t}$ production in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2019\) 031](#) | [arXiv:1810.05905](#) | [DOI:10.1007/JHEP04\(2019\)031](#) | 2018-10-13

717. Centrality and pseudorapidity dependence of the transverse energy density in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 100 \(2019\) 024902](#) | [arXiv:1810.05745](#) | [DOI:10.1103/PhysRevC.100.024902](#) | 2018-10-12

718. Evidence for light-by-light scattering and searches for axion-like particles in ultraperipheral PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 797 \(2019\) 134826](#) | [arXiv:1810.04602](#) | [DOI:10.1016/j.physletb.2019.134826](#) | 2018-10-10

719. Search for top quark partners with charge $5/3$ in the same-sign dilepton and single-lepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 082](#) | [arXiv:1810.03188](#) | [DOI:10.1007/JHEP03\(2019\)082](#) | 2018-10-07

720. Measurement of B^0_s meson production in pp and PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 796 \(2019\) 168](#) | [arXiv:1810.03022](#) | [DOI:10.1016/j.physletb.2019.07.014](#) | 2018-10-06

721. Observation of prompt J/ψ meson elliptic flow in high-multiplicity pPb collisions at $\sqrt{s_{NN}} = 8.16$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 791 \(2019\) 172](#) | [arXiv:1810.01473](#) | [DOI:10.1016/j.physletb.2019.02.018](#) | 2018-10-02

722. Search for new physics in final states with a single photon and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2019\) 074](#) | [arXiv:1810.00196](#) | [DOI:10.1007/JHEP02\(2019\)074](#) | 2018-09-29

723. Measurement of exclusive Υ photoproduction from protons in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 79 \(2019\) 277](#) | [arXiv:1809.11080](#) | [DOI:10.1140/epjc/s10052-019-6774-8](#) | 2018-09-28

724. Combined measurements of Higgs boson couplings in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 421](#) | [arXiv:1809.10733](#) | [DOI:10.1140/epjc/s10052-019-6909-y](#) | 2018-09-27

725. Jet Shapes of Isolated Photon-Tagged Jets in Pb-Pb and pp Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 152001](#) | [arXiv:1809.08602](#) | [DOI:10.1103/PhysRevLett.122.152001](#) | 2018-09-23

726. Search for single production of vector-like quarks decaying to a top quark and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 79 \(2019\) 90](#) | [arXiv:1809.08597](#) | [DOI:10.1140/epjc/s10052-019-6556-3](#) | 2018-09-23

727. Search for invisible decays of a Higgs boson produced through vector boson fusion in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 793 \(2019\) 520](#) | [arXiv:1809.05937](#) | [DOI:10.1016/j.physletb.2019.04.025](#) | 2018-09-16

728. Search for leptoquarks coupled to third-generation quarks in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 241802](#) | [arXiv:1809.05558](#) | [DOI:10.1103/PhysRevLett.121.241802](#) | 2018-09-14

729. Studies of $B^*_s(5840)^0$ and $B_s(5830)^0$ mesons including the observation of the $B^*_s(5840)^0 \rightarrow B^0 K^0_S$ decay in proton-proton collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 939](#) | [arXiv:1809.03578](#) | [DOI:10.1140/epjc/s10052-018-6390-z](#) | 2018-09-10

730. Search for the associated production of the Higgs boson and a vector boson in proton-proton collisions at $\sqrt{s}= 13$ TeV via Higgs boson decays to τ leptons

Sirunyan, Albert M et al.

[JHEP 06 \(2019\) 093](#) | [arXiv:1809.03590](#) | [DOI:10.1007/JHEP06\(2019\)093](#) | 2018-09-10

731. Performance of reconstruction and identification of τ leptons decaying to hadrons and ν_τ in pp collisions at $\sqrt{s}= 13$ TeV

Sirunyan, A.M. et al.

[JINST 13 \(2018\) P10005](#) | [arXiv:1809.02816](#) | [DOI:10.1088/1748-0221/13/10/P10005](#) | 2018-09-08

732. Search for physics beyond the standard model in high-mass diphoton events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.D 98 \(2018\) 092001](#) | [arXiv:1809.00327](#) | [DOI:10.1103/PhysRevD.98.092001](#) | 2018-09-02

733. Charged-particle nuclear modification factors in XeXe collisions at $\sqrt{s_{NN}} = 5.44$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 138](#) | [arXiv:1809.00201](#) | [DOI:10.1007/JHEP10\(2018\)138](#) | 2018-09-01

734. Observation of Higgs boson decay to bottom quarks

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 121801](#) | [arXiv:1808.08242](#) | [DOI:10.1103/PhysRevLett.121.121801](#) | 2018-08-24

735. Measurement of jet substructure observables in tt events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.D 98 \(2018\) 092014](#) | [arXiv:1808.07340](#) | [DOI:10.1103/PhysRevD.98.092014](#) | 2018-08-22

736. Search for a charged Higgs boson decaying to charm and bottom quarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 115](#) | [arXiv:1808.06575](#) | [DOI:10.1007/JHEP11\(2018\)115](#) | 2018-08-20

737. Search for pair production of second-generation leptoquarks at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 99 \(2019\) 032014](#) | [arXiv:1808.05082](#) | [DOI:10.1103/PhysRevD.99.032014](#) | 2018-08-15

738. Search for an L_μ - L_τ gauge boson using $Z \rightarrow 4\mu$ events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 792 \(2019\) 345](#) | [arXiv:1808.03684](#) | [DOI:10.1016/j.physletb.2019.01.072](#) | 2018-08-10

739. Search for pair-produced resonances decaying to quark pairs in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 112014](#) | [arXiv:1808.03124](#) | [DOI:10.1103/PhysRevD.98.112014](#) | 2018-08-09

740. Search for long-lived particles with displaced vertices in multijet events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 092011](#) | [arXiv:1808.03078](#) | [DOI:10.1103/PhysRevD.98.092011](#) | 2018-08-09

741. Evidence for the associated production of a single top quark and a photon in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 221802](#) | [arXiv:1808.02913](#) | [DOI:10.1103/PhysRevLett.121.221802](#) | 2018-08-08

742. Search for resonances in the mass spectrum of muon pairs produced in association with b quark jets in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Sirunyan, A.M. et al.

[JHEP 11 \(2018\) 161](#) | [arXiv:1808.01890](#) | [DOI:10.1007/JHEP11\(2018\)161](#) | 2018-08-06

743. Search for production of Higgs boson pairs in the four b quark final state using large-area jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 040](#) | [arXiv:1808.01473](#) | [DOI:10.1007/JHEP01\(2019\)040](#) | 2018-08-04

744. Search for narrow H_γ resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 122 \(2019\) 081804](#) | [arXiv:1808.01257](#) | [DOI:10.1103/PhysRevLett.122.081804](#) | 2018-08-03

745. Search for heavy resonances decaying into two Higgs bosons or into a Higgs boson and a W or Z boson in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2019\) 051](#) | [arXiv:1808.01365](#) | [DOI:10.1007/JHEP01\(2019\)051](#) | 2018-08-03

746. Search for a W' boson decaying to a τ lepton and a neutrino in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 792 \(2019\) 107](#) | [arXiv:1807.11421](#) | [DOI:10.1016/j.physletb.2019.01.069](#) | 2018-07-30

747. Searches for pair production of charginos and top squarks in final states with two oppositely charged leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 079](#) | [arXiv:1807.07799](#) | [DOI:10.1007/JHEP11\(2018\)079](#) | 2018-07-20

748. Search for the Higgs boson decaying to two muons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 122 \(2019\) 021801](#) | [arXiv:1807.06325](#) | [DOI:10.1103/PhysRevLett.122.021801](#) | 2018-07-17

749. Search for dark matter particles produced in association with a top quark pair at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 122 \(2019\) 011803](#) | [arXiv:1807.06522](#) | [DOI:10.1103/PhysRevLett.122.011803](#) | 2018-07-17

750. Measurements of the differential jet cross section as a function of the jet mass in dijet events from proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 113](#) | [arXiv:1807.05974](#) | [DOI:10.1007/JHEP11\(2018\)113](#) | 2018-07-16

751. Measurement of inclusive and differential Higgs boson production cross sections in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 01 \(2019\) 183](#) | [arXiv:1807.03825](#) | [DOI:10.1007/JHEP01\(2019\)183](#) | 2018-07-10

752. Precision measurement of the structure of the CMS inner tracking system using nuclear interactions

Sirunyan, Albert M et al.

[JINST 13 \(2018\) P10034](#) | [arXiv:1807.03289](#) | [DOI:10.1088/1748-0221/13/10/P10034](#) | 2018-07-09

753. Study of the underlying event in top quark pair production in pp collisions at 13 TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 79 \(2019\) 123](#) | [arXiv:1807.02810](#) | [DOI:10.1140/epjc/s10052-019-6620-z](#) | 2018-07-08

754. Search for heavy resonances decaying into a vector boson and a Higgs boson in final states with charged leptons, neutrinos and b quarks at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 172](#) | [arXiv:1807.02826](#) | [DOI:10.1007/JHEP11\(2018\)172](#) | 2018-07-08

755. Search for supersymmetry in events with a τ lepton pair and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 11 \(2018\) 151](#) | [arXiv:1807.02048](#) | [DOI:10.1007/JHEP11\(2018\)151](#) | 2018-07-05

756. Measurement of differential cross sections for inclusive isolated-photon and photon+jets production in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[Eur.Phys.J.C 79 \(2019\) 20](#) | [arXiv:1807.00782](#) | [DOI:10.1140/epjc/s10052-018-6482-9](#) | 2018-07-02
757. Measurement of charged particle spectra in minimum-bias events from proton-proton collisions at $\sqrt{s}=13$ TeV
 Sirunyan, Albert M. et al.
[Eur.Phys.J.C 78 \(2018\) 697](#) | [arXiv:1806.11245](#) | [DOI:10.1140/epjc/s10052-018-6144-y](#) | 2018-06-29
758. Search for heavy Majorana neutrinos in same-sign dilepton channels in proton-proton collisions at $\sqrt{s}=13$ TeV
 Sirunyan, Albert M et al.
[JHEP 01 \(2019\) 122](#) | [arXiv:1806.10905](#) | [DOI:10.1007/JHEP01\(2019\)122](#) | 2018-06-28
759. Measurement of differential cross sections for Z boson pair production in association with jets at $\sqrt{s} = 8$ and 13 TeV
 Sirunyan, Albert M. et al.
[Phys.Lett.B 789 \(2019\) 19](#) | [arXiv:1806.11073](#) | [DOI:10.1016/j.physletb.2018.11.007](#) | 2018-06-28
760. Search for the decay of a Higgs boson in the $\ell\ell\gamma$ channel in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M. et al.
[JHEP 11 \(2018\) 152](#) | [arXiv:1806.05996](#) | [DOI:10.1007/JHEP11\(2018\)152](#) | 2018-06-15
761. Measurements of properties of the Higgs boson decaying to a W boson pair in pp collisions at $\sqrt{s}= 13$ TeV
 Sirunyan, Albert M. et al.
[Phys.Lett.B 791 \(2019\) 96](#) | [arXiv:1806.05246](#) | [DOI:10.1016/j.physletb.2018.12.073](#) | 2018-06-13
762. Search for supersymmetric partners of electrons and muons in proton-proton collisions at $\sqrt{s}= 13$ TeV
 Sirunyan, Albert M. et al.
[Phys.Lett.B 790 \(2019\) 140](#) | [arXiv:1806.05264](#) | [DOI:10.1016/j.physletb.2019.01.005](#) | 2018-06-13
763. Search for dark matter produced in association with a Higgs boson decaying to $\gamma\gamma$ or $\tau^+\tau^-$ at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M. et al.
[JHEP 09 \(2018\) 046](#) | [arXiv:1806.04771](#) | [DOI:10.1007/JHEP09\(2018\)046](#) | 2018-06-12
764. Observation of the $Z\rightarrow\psi\ell^++\ell^-$ decay in pp collisions at $\sqrt{s}= 13$ TeV
 Sirunyan, Albert M. et al.
[Phys.Rev.Lett. 121 \(2018\) 141801](#) | [arXiv:1806.04213](#) | [DOI:10.1103/PhysRevLett.121.141801](#) | 2018-06-11
765. Search for a singly produced third-generation scalar leptoquark decaying to a τ lepton and a bottom quark in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M. et al.
[JHEP 07 \(2018\) 115](#) | [arXiv:1806.03472](#) | [DOI:10.1007/JHEP07\(2018\)115](#) | 2018-06-09
766. Search for resonant pair production of Higgs bosons decaying to bottom quark-antiquark pairs in proton-proton collisions at 13 TeV
 Sirunyan, Albert M. et al.
[JHEP 08 \(2018\) 152](#) | [arXiv:1806.03548](#) | [DOI:10.1007/JHEP08\(2018\)152](#) | 2018-06-09
767. Search for pair-produced resonances each decaying into at least four quarks in proton-proton collisions at $\sqrt{s}= 13$ TeV
 Sirunyan, Albert M. et al.

[Phys.Rev.Lett. 121 \(2018\) 141802](#) | [arXiv:1806.01058](#) | [DOI:10.1103/PhysRevLett.121.141802](#) | 2018-06-04

768. Search for narrow and broad dijet resonances in proton-proton collisions at $\sqrt{s}=13$ TeV and constraints on dark matter mediators and other new particles

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 130](#) | [arXiv:1806.00843](#) | [DOI:10.1007/JHEP08\(2018\)130](#) | 2018-06-03

769. Measurement of the weak mixing angle using the forward-backward asymmetry of Drell-Yan events in pp collisions at 8 TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 701](#) | [arXiv:1806.00863](#) | [DOI:10.1140/epjc/s10052-018-6148-7](#) | 2018-06-03

770. Angular analysis of the decay $B^+ \rightarrow K^+ \mu^+ \mu^-$ in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 112011](#) | [arXiv:1806.00636](#) | [DOI:10.1103/PhysRevD.98.112011](#) | 2018-06-02

771. Search for Higgs boson pair production in the $\gamma\gamma b\bar{b}$ final state in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 788 \(2019\) 7](#) | [arXiv:1806.00408](#) | [DOI:10.1016/j.physletb.2018.10.056](#) | 2018-06-01

772. Search for beyond the standard model Higgs bosons decaying into a $b\bar{b}$ pair in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 113](#) | [arXiv:1805.12191](#) | [DOI:10.1007/JHEP08\(2018\)113](#) | 2018-05-30

773. Observation of the $\chi_{b1}(3P)$ and $\chi_{b2}(3P)$ and measurement of their masses

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 092002](#) | [arXiv:1805.11192](#) | [DOI:10.1103/PhysRevLett.121.092002](#) | 2018-05-28

774. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state with two b quarks and two τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 785 \(2018\) 462](#) | [arXiv:1805.10191](#) | [DOI:10.1016/j.physletb.2018.08.057](#) | 2018-05-25

775. Constraints on models of scalar and vector leptoquarks decaying to a quark and a neutrino at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 98 \(2018\) 032005](#) | [arXiv:1805.10228](#) | [DOI:10.1103/PhysRevD.98.032005](#) | 2018-05-25

776. Measurement of nuclear modification factors of $Y(1S)$, $Y(2S)$, and $Y(3S)$ mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 270](#) | [arXiv:1805.09215](#) | [DOI:10.1016/j.physletb.2019.01.006](#) | 2018-05-23

777. Measurement of the production cross section for single top quarks in association with W bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 117](#) | [arXiv:1805.07399](#) | [DOI:10.1007/JHEP10\(2018\)117](#) | 2018-05-18

778. Search for top squarks decaying via four-body or chargino-mediated modes in single-lepton final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[JHEP 09 \(2018\) 065](#) | [arXiv:1805.05784](#) | [DOI:10.1007/JHEP09\(2018\)065](#) | 2018-05-15

779. Search for black holes and sphalerons in high-multiplicity final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 042](#) | [arXiv:1805.06013](#) | [DOI:10.1007/JHEP11\(2018\)042](#) | 2018-05-15

780. Measurement of the groomed jet mass in PbPb and pp collisions at $\sqrt{s_{\text{NN}}}=5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2018\) 161](#) | [arXiv:1805.05145](#) | [DOI:10.1007/JHEP10\(2018\)161](#) | 2018-05-14

781. Search for an exotic decay of the Higgs boson to a pair of light pseudoscalars in the final state of two muons and two τ leptons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2018\) 018](#) | [arXiv:1805.04865](#) | [DOI:10.1007/JHEP11\(2018\)018](#) | 2018-05-13

782. Constraining gluon distributions in nuclei using dijets in proton-proton and proton-lead collisions at $\sqrt{s_{\text{NN}}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 062002](#) | [arXiv:1805.04736](#) | [DOI:10.1103/PhysRevLett.121.062002](#) | 2018-05-12

783. Search for vector-like T and B quark pairs in final states with leptons at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 177](#) | [arXiv:1805.04758](#) | [DOI:10.1007/JHEP08\(2018\)177](#) | 2018-05-12

784. Measurement of prompt $\psi(2S)$ production cross sections in proton-lead and proton-proton collisions at $\sqrt{s_{\text{NN}}}= 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 790 \(2019\) 509](#) | [arXiv:1805.02248](#) | [DOI:10.1016/j.physletb.2019.01.058](#) | 2018-05-06

785. Measurement of the top quark mass with lepton+jets final states using p p collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 891](#) | [arXiv:1805.01428](#) | [DOI:10.1140/epjc/s10052-018-6332-9](#) | 2018-05-03

786. Elliptic flow of charm and strange hadrons in high-multiplicity pPb collisions at $\sqrt{s_{\text{NN}}} = 8.16$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 121 \(2018\) 082301](#) | [arXiv:1804.09767](#) | [DOI:10.1103/PhysRevLett.121.082301](#) | 2018-04-25

787. Search for disappearing tracks as a signature of new long-lived particles in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 016](#) | [arXiv:1804.07321](#) | [DOI:10.1007/JHEP08\(2018\)016](#) | 2018-04-19

788. Measurement of differential cross sections for Z boson production in association with jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 965](#) | [arXiv:1804.05252](#) | [DOI:10.1140/epjc/s10052-018-6373-0](#) | 2018-04-14

789. Performance of the CMS muon detector and muon reconstruction with proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, A.M. et al.

[JINST 13 \(2018\) P06015](#) | [arXiv:1804.04528](#) | [DOI:10.1088/1748-0221/13/06/P06015](#) | 2018-04-12

790. Search for ttH production in the $H \rightarrow b\bar{b}$ decay channel with leptonic tt decays in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2019\) 026](#) | [arXiv:1804.03682](#) | [DOI:10.1007/JHEP03\(2019\)026](#) | 2018-04-10

791. HEP Software Foundation Community White Paper Working Group - Data Analysis and Interpretation

Bauerdick, Lothar et al.

[arXiv:1804.03983](#) | 2018-04-09

792. Measurements of Higgs boson properties in the diphoton decay channel in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[JHEP 11 \(2018\) 185](#) | [arXiv:1804.02716](#) | [DOI:10.1007/JHEP11\(2018\)185](#) | 2018-04-08

793. Observation of ttH production

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 231801](#) | [arXiv:1804.02610](#) | [DOI:10.1103/PhysRevLett.120.231801](#) | 2018-04-07

794. Search for a new scalar resonance decaying to a pair of Z bosons in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 127](#) | [arXiv:1804.01939](#) | [DOI:10.1007/JHEP06\(2018\)127](#) | 2018-04-05

795. Search for a heavy right-handed W boson and a heavy neutrino in events with two same-flavor leptons and two jets at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 148](#) | [arXiv:1803.11116](#) | [DOI:10.1007/JHEP05\(2018\)148](#) | 2018-03-29

796. Search for high-mass resonances in final states with a lepton and missing transverse momentum at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 128](#) | [arXiv:1803.11133](#) | [DOI:10.1007/JHEP06\(2018\)128](#) | 2018-03-29

797. Search for a heavy resonance decaying into a Z boson and a Z or W boson in $2\ell 2q$ final states at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 101](#) | [arXiv:1803.10093](#) | [DOI:10.1007/JHEP09\(2018\)101](#) | 2018-03-27

798. Measurement of differential cross sections for the production of top quark pairs and of additional jets in lepton+jets events from pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 112003](#) | [arXiv:1803.08856](#) | [DOI:10.1103/PhysRevD.97.112003](#) | 2018-03-23

799. Search for new physics in dijet angular distributions using proton-proton collisions at $\sqrt{s}= 13$ TeV and constraints on dark matter and other models

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 789](#) | [arXiv:1803.08030](#) | [DOI:10.1140/epjc/s10052-018-6242-x](#) | 2018-03-21

800. Search for ttH production in the all-jet final state in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 101](#) | [arXiv:1803.06986](#) | [DOI:10.1007/JHEP06\(2018\)101](#) | 2018-03-19

801. Search for additional neutral MSSM Higgs bosons in the $\tau\tau$ final state in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 007](#) | [arXiv:1803.06553](#) | [DOI:10.1007/JHEP09\(2018\)007](#) | 2018-03-17

802. Search for high-mass resonances in dilepton final states in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 120](#) | [arXiv:1803.06292](#) | [DOI:10.1007/JHEP06\(2018\)120](#) | 2018-03-16

803. Evidence for associated production of a Higgs boson with a top quark pair in final states with electrons, muons, and hadronically decaying τ leptons at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 066](#) | [arXiv:1803.05485](#) | [DOI:10.1007/JHEP08\(2018\)066](#) | 2018-03-14

804. Observation of proton-tagged, central (semi)exclusive production of high-mass lepton pairs in pp collisions at 13 TeV with the CMS-TOTEM precision proton spectrometer

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 153](#) | [arXiv:1803.04496](#) | [DOI:10.1007/JHEP07\(2018\)153](#) | 2018-03-12

805. Measurements of differential cross sections of top quark pair production as a function of kinematic event variables in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 06 \(2018\) 002](#) | [arXiv:1803.03991](#) | [DOI:10.1007/JHEP06\(2018\)002](#) | 2018-03-11

806. Search for a heavy resonance decaying into a Z boson and a vector boson in the $\nu\nu qq$ final state

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 075](#) | [arXiv:1803.03838](#) | [DOI:10.1007/JHEP07\(2018\)075](#) | 2018-03-10

807. Search for third-generation scalar leptoquarks decaying to a top quark and a τ lepton at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 707](#) | [arXiv:1803.02864](#) | [DOI:10.1140/epjc/s10052-018-6143-z](#) | 2018-03-07

808. Jet properties in PbPb and pp collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 006](#) | [arXiv:1803.00042](#) | [DOI:10.1007/JHEP05\(2018\)006](#) | 2018-02-28

809. Search for a heavy resonance decaying to a pair of vector bosons in the lepton plus merged jet final state at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 088](#) | [arXiv:1802.09407](#) | [DOI:10.1007/JHEP05\(2018\)088](#) | 2018-02-26

810. Search for narrow resonances in the b-tagged dijet mass spectrum in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 120 \(2018\) 201801](#) | [arXiv:1802.06149](#) | [DOI:10.1103/PhysRevLett.120.201801](#) | 2018-02-16

811. Measurement of the Λ_b polarization and angular parameters in $\Lambda_b \rightarrow J/\psi \Lambda$ decays from pp collisions at $\sqrt{s}=7$ and 8 TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 072010](#) | [arXiv:1802.04867](#) | [DOI:10.1103/PhysRevD.97.072010](#) | 2018-02-13

812. Search for heavy neutral leptons in events with three charged leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 221801](#) | [arXiv:1802.02965](#) | [DOI:10.1103/PhysRevLett.120.221801](#) | 2018-02-08

813. Measurement of the inelastic proton-proton cross section at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2018\) 161](#) | [arXiv:1802.02613](#) | [DOI:10.1007/JHEP07\(2018\)161](#) | 2018-02-07

814. Search for natural and split supersymmetry in proton-proton collisions at $\sqrt{s}=13$ TeV in final states with jets and missing transverse momentum

Sirunyan, Albert M et al.

[JHEP 05 \(2018\) 025](#) | [arXiv:1802.02110](#) | [DOI:10.1007/JHEP05\(2018\)025](#) | 2018-02-06

815. Search for single production of vector-like quarks decaying to a b quark and a Higgs boson

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 031](#) | [arXiv:1802.01486](#) | [DOI:10.1007/JHEP06\(2018\)031](#) | 2018-02-05

816. Search for lepton-flavor violating decays of heavy resonances and quantum black holes to $e\mu$ final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 04 \(2018\) 073](#) | [arXiv:1802.01122](#) | [DOI:10.1007/JHEP04\(2018\)073](#) | 2018-02-04

817. Comparing transverse momentum balance of b jet pairs in pp and PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2018\) 181](#) | [arXiv:1802.00707](#) | [DOI:10.1007/JHEP03\(2018\)181](#) | 2018-02-02

818. Search for dark matter in events with energetic, hadronically decaying top quarks and missing transverse momentum at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 027](#) | [arXiv:1801.08427](#) | [DOI:10.1007/JHEP06\(2018\)027](#) | 2018-01-25

819. Observation of Medium-Induced Modifications of Jet Fragmentation in Pb-Pb Collisions at $\sqrt{s_{NN}}= 5.02$ TeV Using Isolated Photon-Tagged Jets

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 121 \(2018\) 242301](#) | [arXiv:1801.04895](#) | [DOI:10.1103/PhysRevLett.121.242301](#) | 2018-01-15

820. Combined search for electroweak production of charginos and neutralinos in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 160](#) | [arXiv:1801.03957](#) | [DOI:10.1007/JHEP03\(2018\)160](#) | 2018-01-11

821. Measurement of the $Z\gamma^* \rightarrow \tau\tau$ cross section in pp collisions at $\sqrt{s} = 13$ TeV and validation of τ lepton analysis techniques

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 708](#) | [arXiv:1801.03535](#) | [DOI:10.1140/epjc/s10052-018-6146-9](#) | 2018-01-10

822. Search for new physics in events with two soft oppositely charged leptons and missing transverse momentum in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 782 \(2018\) 440](#) | [arXiv:1801.01846](#) | [DOI:10.1016/j.physletb.2018.05.062](#) | 2018-01-05

823. Search for decays of stopped exotic long-lived particles produced in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, A.M. et al.

[JHEP 05 \(2018\) 127](#) | [arXiv:1801.00359](#) | [DOI:10.1007/JHEP05\(2018\)127](#) | 2017-12-31

824. Electroweak production of two jets in association with a Z boson in proton-proton collisions at $\sqrt{s}= 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 589](#) | [arXiv:1712.09814](#) | [DOI:10.1140/epjc/s10052-018-6049-9](#) | 2017-12-28

825. Search for R-parity violating supersymmetry in pp collisions at $\sqrt{s} = 13$ TeV using b jets in a final state with a single lepton, many jets, and high sum of large-radius jet masses

Sirunyan, Albert M et al.

[Phys.Lett.B 783 \(2018\) 114](#) | [arXiv:1712.08920](#) | [DOI:10.1016/j.physletb.2018.06.028](#) | 2017-12-24

826. Measurement of prompt and nonprompt charmonium suppression in PbPb collisions at 5.02 TeV

Sirunyan, Albert M. et al.

[Eur.Phys.J.C 78 \(2018\) 509](#) | [arXiv:1712.08959](#) | [DOI:10.1140/epjc/s10052-018-5950-6](#) | 2017-12-24

827. Search for Physics Beyond the Standard Model in Events with High-Momentum Higgs Bosons and Missing Transverse Momentum in Proton-Proton Collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 241801](#) | [arXiv:1712.08501](#) | [DOI:10.1103/PhysRevLett.120.241801](#) | 2017-12-22

828. Identification of heavy-flavour jets with the CMS detector in pp collisions at 13 TeV

Sirunyan, A.M. et al.

[JINST 13 \(2018\) P05011](#) | [arXiv:1712.07158](#) | [DOI:10.1088/1748-0221/13/05/P05011](#) | 2017-12-19

829. Search for lepton flavour violating decays of the Higgs boson to $\mu\tau$ and $e\tau$ in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 001](#) | [arXiv:1712.07173](#) | [DOI:10.1007/JHEP06\(2018\)001](#) | 2017-12-19

830. Bose-Einstein correlations in pp, pPb, and PbPb collisions at $\sqrt{s_{NN}}=0.9-7$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.C 97 \(2018\) 064912](#) | [arXiv:1712.07198](#) | [DOI:10.1103/PhysRevC.97.064912](#) | 2017-12-19

831. A Roadmap for HEP Software and Computing R D for the 2020s

Albrecht, Johannes et al.

[Comput.Softw.Big Sci. 3 \(2019\) 7](#) | [arXiv:1712.06982](#) | [DOI:10.1007/s41781-018-0018-8](#) | 2017-12-18

832. Search for the $X(5568)$ state decaying into $B^0_{s1}\pi^\pm$ in proton-proton collisions at $\sqrt{s}=8$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.Lett. 120 \(2018\) 202005](#) | [arXiv:1712.06144](#) | [DOI:10.1103/PhysRevLett.120.202005](#) | 2017-12-17

833. Azimuthal correlations for inclusive 2-jet, 3-jet, and 4-jet events in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 566](#) | [arXiv:1712.05471](#) | [DOI:10.1140/epjc/s10052-018-6033-4](#) | 2017-12-14

834. Search for $Z\gamma$ resonances using leptonic and hadronic final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2018\) 148](#) | [arXiv:1712.03143](#) | [DOI:10.1007/JHEP09\(2018\)148](#) | 2017-12-08

835. Measurement of the associated production of a single top quark and a Z boson in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 779 \(2018\) 358](#) | [arXiv:1712.02825](#) | [DOI:10.1016/j.physletb.2018.02.025](#) | 2017-12-07

836. Constraints on the double-parton scattering cross section from same-sign W boson pair production in proton-proton collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2018\) 032](#) | [arXiv:1712.02280](#) | [DOI:10.1007/JHEP02\(2018\)032](#) | 2017-12-06

837. Search for new physics in final states with an energetic jet or a hadronically decaying W or Z boson and transverse momentum imbalance at $\sqrt{s}=13$ TeV

Sirunyan, A. M. et al.

[Phys.Rev.D 97 \(2018\) 092005](#) | [arXiv:1712.02345](#) | [DOI:10.1103/PhysRevD.97.092005](#) | 2017-12-06

838. Search for the flavor-changing neutral current interactions of the top quark and the Higgs boson which decays into a pair of b quarks at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 06 \(2018\) 102](#) | [arXiv:1712.02399](#) | [DOI:10.1007/JHEP06\(2018\)102](#) | 2017-12-06

839. Search for pair production of excited top quarks in the lepton + jets final state

Sirunyan, Albert M et al.

[Phys.Lett.B 778 \(2018\) 349](#) | [arXiv:1711.10949](#) | [DOI:10.1016/j.physletb.2018.01.049](#) | 2017-11-29

840. Study of jet quenching with isolated-photon+jet correlations in PbPb and pp collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 785 \(2018\) 14](#) | [arXiv:1711.09738](#) | [DOI:10.1016/j.physletb.2018.07.061](#) | 2017-11-27

841. Search for new long-lived particles at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 780 \(2018\) 432](#) | [arXiv:1711.09120](#) | [DOI:10.1016/j.physletb.2018.03.019](#) | 2017-11-24

842. Experience in using commercial clouds in CMS

Bauerdick, L. et al.

[J.Phys.Conf.Ser. 898 \(2017\) 052019](#) | [DOI:10.1088/1742-6596/898/5/052019](#) | 2017-11-22

843. Search for gauge-mediated supersymmetry in events with at least one photon and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 780 \(2018\) 118](#) | [arXiv:1711.08008](#) | [DOI:10.1016/j.physletb.2018.02.045](#) | 2017-11-21

844. Non-Gaussian elliptic-flow fluctuations in PbPb collisions at $\sqrt{[b]s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 789 \(2019\) 643](#) | [arXiv:1711.05594](#) | [DOI:10.1016/j.physletb.2018.11.063](#) | 2017-11-15

845. Search for excited quarks of light and heavy flavor in γ + jet final states in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 781 \(2018\) 390](#) | [arXiv:1711.04652](#) | [DOI:10.1016/j.physletb.2018.04.007](#) | 2017-11-13

846. Measurement of the underlying event activity in inclusive Z boson production in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 07 \(2018\) 032](#) | [arXiv:1711.04299](#) | [DOI:10.1007/JHEP07\(2018\)032](#) | 2017-11-12

847. Search for ZZ resonances in the $2\ell 2\nu$ final state in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2018\) 003](#) | [arXiv:1711.04370](#) | [DOI:10.1007/JHEP03\(2018\)003](#) | 2017-11-12

848. Measurement of the inclusive tt cross section in pp collisions at $\sqrt{s}=5.02$ TeV using final states with at least one charged lepton

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 115](#) | [arXiv:1711.03143](#) | [DOI:10.1007/JHEP03\(2018\)115](#) | 2017-11-08

849. Measurement of the cross section for top quark pair production in association with a W or Z boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2018\) 011](#) | [arXiv:1711.02547](#) | [DOI:10.1007/JHEP08\(2018\)011](#) | 2017-11-07

850. Measurement of associated Z + charm production in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, A.M. et al.

[Eur.Phys.J.C 78 \(2018\) 287](#) | [arXiv:1711.02143](#) | [DOI:10.1140/epjc/s10052-018-5752-x](#) | 2017-11-06

851. Search for top squarks and dark matter particles in opposite-charge dilepton final states at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 032009](#) | [arXiv:1711.00752](#) | [DOI:10.1103/PhysRevD.97.032009](#) | 2017-11-02

852. Search for new physics in events with a leptonically decaying Z boson and a large transverse momentum imbalance in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[Eur.Phys.J.C 78 \(2018\) 291](#) | [arXiv:1711.00431](#) | [DOI:10.1140/epjc/s10052-018-5740-1](#) | 2017-11-01

853. CMS Analysis and Data Reduction with Apache Spark

Gutsche, Oliver et al.

[J.Phys.Conf.Ser. 1085 \(2018\) 042030](#) | [arXiv:1711.00375](#) | [DOI:10.1088/1742-6596/1085/4/042030](#) | 2017-10-31

854. Measurement of quarkonium production cross sections in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 780 \(2018\) 251](#) | [arXiv:1710.11002](#) | [DOI:10.1016/j.physletb.2018.02.033](#) | 2017-10-30

855. Search for supersymmetry in proton-proton collisions at 13 TeV using identified top quarks

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 012007](#) | [arXiv:1710.11188](#) | [DOI:10.1103/PhysRevD.97.012007](#) | 2017-10-30

856. Search for standard model production of four top quarks with same-sign and multilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 140](#) | [arXiv:1710.10614](#) | [DOI:10.1140/epjc/s10052-018-5607-5](#) | 2017-10-29

857. Search for supersymmetry in events with at least three electrons or muons, jets, and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 02 \(2018\) 067](#) | [arXiv:1710.09154](#) | [DOI:10.1007/JHEP02\(2018\)067](#) | 2017-10-25

858. Pseudorapidity distributions of charged hadrons in proton-lead collisions at $\sqrt{s_{NN}}=5.02$ and 8.16 TeV

Sirunyan, Albert M et al.

[JHEP 01 \(2018\) 045](#) | [arXiv:1710.09355](#) | [DOI:10.1007/JHEP01\(2018\)045](#) | 2017-10-25

859. Measurement of b hadron lifetimes in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 78 \(2018\) 457](#) | [arXiv:1710.08949](#) | [DOI:10.1140/epjc/s10052-018-5929-3](#) | 2017-10-24

860. Measurement of differential cross sections in the kinematic angular variable ϕ^* for inclusive Z boson production in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 172](#) | [arXiv:1710.07955](#) | [DOI:10.1007/JHEP03\(2018\)172](#) | 2017-10-22

861. Pseudorapidity and transverse momentum dependence of flow harmonics in pPb and PbPb collisions

Sirunyan, Albert M et al.

[Phys.Rev.C 98 \(2018\) 044902](#) | [arXiv:1710.07864](#) | [DOI:10.1103/PhysRevC.98.044902](#) | 2017-10-21

862. Search for a massive resonance decaying to a pair of Higgs bosons in the four b quark final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 781 \(2018\) 244](#) | [arXiv:1710.04960](#) | [DOI:10.1016/j.physletb.2018.03.084](#) | 2017-10-13

863. Measurement of angular parameters from the decay $B^0 \rightarrow K^*0 \mu^+ \mu^-$ in proton-proton collisions at $\sqrt{s} = 8$ TeV
 Sirunyan, Albert M et al.
[Phys.Lett.B 781 \(2018\) 517](#) | [arXiv:1710.02846](#) | [DOI:10.1016/j.physletb.2018.04.030](#) | 2017-10-08
864. Study of dijet events with a large rapidity gap between the two leading jets in pp collisions at $\sqrt{s} = 7$ TeV
 Sirunyan, Albert M et al.
[Eur.Phys.J.C 78 \(2018\) 242](#) | [arXiv:1710.02586](#) | [DOI:10.1140/epjc/s10052-018-5691-6](#) | 2017-10-06
865. Search for pair production of vector-like quarks in the $bWbW$ channel from proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[Phys.Lett.B 779 \(2018\) 82](#) | [arXiv:1710.01539](#) | [DOI:10.1016/j.physletb.2018.01.077](#) | 2017-10-04
866. Search for low mass vector resonances decaying into quark-antiquark pairs in proton-proton collisions at $\sqrt{s}=13$ TeV
 Sirunyan, Albert M et al.
[JHEP 01 \(2018\) 097](#) | [arXiv:1710.00159](#) | [DOI:10.1007/JHEP01\(2018\)097](#) | 2017-09-30
867. HEPCloud, a New Paradigm for HEP Facilities: CMS Amazon Web Services Investigation
 Holzman, Burt et al.
[Comput.Softw.Big Sci. 1 \(2017\) 1](#) | [arXiv:1710.00100](#) | [DOI:10.1007/s41781-017-0001-9](#) | 2017-09-29
868. Search for supersymmetry in events with one lepton and multiple jets exploiting the angular correlation between the lepton and the missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, A.M. et al.
[Phys.Lett.B 780 \(2018\) 384](#) | [arXiv:1709.09814](#) | [DOI:10.1016/j.physletb.2018.03.028](#) | 2017-09-28
869. Search for new phenomena in final states with two opposite-charge, same-flavor leptons, jets, and missing transverse momentum in pp collisions at $\sqrt{s}=13$ TeV
 Sirunyan, Albert M et al.
[JHEP 03 \(2018\) 076](#) | [arXiv:1709.08908](#) | [DOI:10.1007/JHEP03\(2018\)076](#) | 2017-09-26
870. Observation of Correlated Azimuthal Anisotropy Fourier Harmonics in pp and p+Pb Collisions at the LHC
 Sirunyan, Albert M et al.
[Phys.Rev.Lett. 120 \(2018\) 092301](#) | [arXiv:1709.09189](#) | [DOI:10.1103/PhysRevLett.120.092301](#) | 2017-09-26
871. Measurements of the $pp \rightarrow ZZ$ production cross section and the $Z \rightarrow 4\ell$ branching fraction, and constraints on anomalous triple gauge couplings at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.
[Eur.Phys.J.C 78 \(2018\) 165](#) | [arXiv:1709.08601](#) | [DOI:10.1140/epjc/s10052-018-5567-9](#) | 2017-09-25
872. Observation of top quark production in proton-nucleus collisions
 Sirunyan, Albert M et al.
[Phys.Rev.Lett. 119 \(2017\) 242001](#) | [arXiv:1709.07411](#) | [DOI:10.1103/PhysRevLett.119.242001](#) | 2017-09-21
873. Evidence for the Higgs boson decay to a bottom quark-antiquark pair
 Sirunyan, Albert M et al.
[Phys.Lett.B 780 \(2018\) 501](#) | [arXiv:1709.07497](#) | [DOI:10.1016/j.physletb.2018.02.050](#) | 2017-09-21
874. Observation of electroweak production of same-sign W boson pairs in the two jet and two same-sign lepton final state in proton-proton collisions at $\sqrt{s} = 13$ TeV
 Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 081801](#) | [arXiv:1709.05822](#) | [DOI:10.1103/PhysRevLett.120.081801](#) | 2017-09-18

875. Inclusive search for a highly boosted Higgs boson decaying to a bottom quark-antiquark pair

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 071802](#) | [arXiv:1709.05543](#) | [DOI:10.1103/PhysRevLett.120.071802](#) | 2017-09-16

876. Combination of inclusive and differential tt charge asymmetry measurements using ATLAS and CMS data at $\sqrt{s}=7$ and 8 TeV

Aaboud, Morad et al.

[JHEP 04 \(2018\) 033](#) | [arXiv:1709.05327](#) | [DOI:10.1007/JHEP04\(2018\)033](#) | 2017-09-15

877. Search for electroweak production of charginos and neutralinos in multilepton final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 166](#) | [arXiv:1709.05406](#) | [DOI:10.1007/JHEP03\(2018\)166](#) | 2017-09-15

878. Search for Higgsino pair production in pp collisions at $\sqrt{s}=13$ TeV in final states with large missing transverse momentum and two Higgs bosons decaying via $H \rightarrow b\bar{b}$

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 032007](#) | [arXiv:1709.04896](#) | [DOI:10.1103/PhysRevD.97.032007](#) | 2017-09-14

879. Search for supersymmetry with Higgs boson to diphoton decays using the razor variables at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 779 \(2018\) 166](#) | [arXiv:1709.00384](#) | [DOI:10.1016/j.physletb.2017.12.069](#) | 2017-09-01

880. Measurement of the Splitting Function in pp and Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 142302](#) | [arXiv:1708.09429](#) | [DOI:10.1103/PhysRevLett.120.142302](#) | 2017-08-30

881. Probing the chiral magnetic wave in pPb and PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV using charge-dependent azimuthal anisotropies

Sirunyan, Albert M et al.

[Phys.Rev.C 100 \(2019\) 064908](#) | [arXiv:1708.08901](#) | [DOI:10.1103/PhysRevC.100.064908](#) | 2017-08-29

882. Search for heavy resonances decaying to a top quark and a bottom quark in the lepton+jets final state in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 777 \(2018\) 39](#) | [arXiv:1708.08539](#) | [DOI:10.1016/j.physletb.2017.12.006](#) | 2017-08-28

883. Search for Evidence of the Type-III Seesaw Mechanism in Multilepton Final States in Proton-Proton Collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 221802](#) | [arXiv:1708.07962](#) | [DOI:10.1103/PhysRevLett.119.221802](#) | 2017-08-26

884. Measurement of normalized differential tt cross sections in the dilepton channel from pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 04 \(2018\) 060](#) | [arXiv:1708.07638](#) | [DOI:10.1007/JHEP04\(2018\)060](#) | 2017-08-25

885. Principal-component analysis of two-particle azimuthal correlations in PbPb and pPb collisions at CMS

Sirunyan, A.M. et al.

[Phys.Rev.C 96 \(2017\) 064902](#) | [arXiv:1708.07113](#) | [DOI:10.1103/PhysRevC.96.064902](#) | 2017-08-23

886. Striped Data Server for Scalable Parallel Data Analysis

Chang, Jin et al.

[J.Phys.Conf.Ser. 1085 \(2018\) 042035](#) | [DOI:10.1088/1742-6596/1085/4/042035](#) | 2017

887. Search for massive resonances decaying into WW, WZ, ZZ, qW, and qZ with dijet final states at $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.D 97 \(2018\) 072006](#) | [arXiv:1708.05379](#) | [DOI:10.1103/PhysRevD.97.072006](#) | 2017-08-17

888. Nuclear modification factor of D^0 mesons in PbPb collisions at $\sqrt{s_{NN}} = 5.02\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 782 \(2018\) 474](#) | [arXiv:1708.04962](#) | [DOI:10.1016/j.physletb.2018.05.074](#) | 2017-08-16

889. Search for resonant and nonresonant Higgs boson pair production in the $b\bar{b}\ell\nu\ell\nu$ final state in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Sirunyan, Albert M et al.

[JHEP 01 \(2018\) 054](#) | [arXiv:1708.04188](#) | [DOI:10.1007/JHEP01\(2018\)054](#) | 2017-08-14

890. Measurement of prompt D^0 meson azimuthal anisotropy in Pb-Pb collisions at $\sqrt{s_{NN}} = 5.02\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 202301](#) | [arXiv:1708.03497](#) | [DOI:10.1103/PhysRevLett.120.202301](#) | 2017-08-11

891. Measurement of vector boson scattering and constraints on anomalous quartic couplings from events with four leptons and two jets in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 774 \(2017\) 682](#) | [arXiv:1708.02812](#) | [DOI:10.1016/j.physletb.2017.10.020](#) | 2017-08-09

892. Search for vectorlike light-flavor quark partners in proton-proton collisions at $\sqrt{s}=8\text{ TeV}$

Sirunyan, A. M. et al.

[Phys.Rev.D 97 \(2018\) 072008](#) | [arXiv:1708.02510](#) | [DOI:10.1103/PhysRevD.97.072008](#) | 2017-08-08

893. Constraints on the chiral magnetic effect using charge-dependent azimuthal correlations in pPb and PbPb collisions at the CERN Large Hadron Collider

Sirunyan, Albert M et al.

[Phys.Rev.C 97 \(2018\) 044912](#) | [arXiv:1708.01602](#) | [DOI:10.1103/PhysRevC.97.044912](#) | 2017-08-04

894. Search for single production of a vector-like T quark decaying to a Z boson and a top quark in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Sirunyan, A.M. et al.

[Phys.Lett.B 781 \(2018\) 574](#) | [arXiv:1708.01062](#) | [DOI:10.1016/j.physletb.2018.04.036](#) | 2017-08-03

895. Observation of the Higgs boson decay to a pair of τ leptons with the CMS detector

Sirunyan, Albert M et al.

[Phys.Lett.B 779 \(2018\) 283](#) | [arXiv:1708.00373](#) | [DOI:10.1016/j.physletb.2018.02.004](#) | 2017-08-01

896. Search for the pair production of third-generation squarks with two-body decays to a bottom or charm quark and a neutralino in proton-proton collisions at $\sqrt{s}=13\text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 778 \(2018\) 263](#) | [arXiv:1707.07274](#) | [DOI:10.1016/j.physletb.2018.01.012](#) | 2017-07-23

897. Search for a light pseudoscalar Higgs boson produced in association with bottom quarks in pp collisions at $\sqrt{s}=8\text{ TeV}$

Sirunyan, A.M. et al.

[JHEP 11 \(2017\) 010](#) | [arXiv:1707.07283](#) | [DOI:10.1007/JHEP11\(2017\)010](#) | 2017-07-23

898. Measurement of the differential cross sections for the associated production of a W boson and jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 072005](#) | [arXiv:1707.05979](#) | [DOI:10.1103/PhysRevD.96.072005](#) | 2017-07-19

899. Search for supersymmetry in events with at least one photon, missing transverse momentum, and large transverse event activity in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 12 \(2017\) 142](#) | [arXiv:1707.06193](#) | [DOI:10.1007/JHEP12\(2017\)142](#) | 2017-07-19

900. Search for direct production of supersymmetric partners of the top quark in the all-jets final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 005](#) | [arXiv:1707.03316](#) | [DOI:10.1007/JHEP10\(2017\)005](#) | 2017-07-11

901. Search for natural supersymmetry in events with top quark pairs and photons in pp collisions at $\sqrt{s} = 8$ TeV

Sirunyan, A.M. et al.

[JHEP 03 \(2018\) 167](#) | [arXiv:1707.03325](#) | [DOI:10.1007/JHEP03\(2018\)167](#) | 2017-07-11

902. Search for Higgs boson pair production in events with two bottom quarks and two tau leptons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 778 \(2018\) 101](#) | [arXiv:1707.02909](#) | [DOI:10.1016/j.physletb.2018.01.001](#) | 2017-07-10

903. Search for heavy resonances that decay into a vector boson and a Higgs boson in hadronic final states at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 636](#) | [arXiv:1707.01303](#) | [DOI:10.1140/epjc/s10052-017-5192-z](#) | 2017-07-05

904. Constraints on anomalous Higgs boson couplings using production and decay information in the four-lepton final state

Sirunyan, Albert M et al.

[Phys.Lett.B 775 \(2017\) 1](#) | [arXiv:1707.00541](#) | [DOI:10.1016/j.physletb.2017.10.021](#) | 2017-07-03

905. Search for Higgs boson pair production in the $b\bar{b}\tau\tau$ final state in proton-proton collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 072004](#) | [arXiv:1707.00350](#) | [DOI:10.1103/PhysRevD.96.072004](#) | 2017-07-02

906. Measurement of charged pion, kaon, and proton production in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 112003](#) | [arXiv:1706.10194](#) | [DOI:10.1103/PhysRevD.96.112003](#) | 2017-06-30

907. The Phase-2 Upgrade of the CMS Tracker

Klein, K. et al.

[DOI:10.17181/CERN.QZ28.FLHW](#) | 2017-06-30

908. Search for electroweak production of charginos and neutralinos in WH events in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2017\) 029](#) | [arXiv:1706.09933](#) | [DOI:10.1007/JHEP11\(2017\)029](#) | 2017-06-29

909. Measurements of properties of the Higgs boson decaying into the four-lepton final state in pp collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 11 \(2017\) 047](#) | [arXiv:1706.09936](#) | [DOI:10.1007/JHEP11\(2017\)047](#) | 2017-06-29

910. Exclusive and semi-exclusive $\pi^+\pi^-$ production in proton-proton collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[arXiv:1706.08310](#) | 2017-06-26

911. Search for a heavy composite Majorana neutrino in the final state with two leptons and two quarks at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 775 \(2017\) 315](#) | [arXiv:1706.08578](#) | [DOI:10.1016/j.physletb.2017.11.001](#) | 2017-06-26

912. Measurement of the semileptonic $t\bar{t} + \gamma$ production cross section in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 006](#) | [arXiv:1706.08128](#) | [DOI:10.1007/JHEP10\(2017\)006](#) | 2017-06-25

913. Measurements of jet charge with dijet events in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 131](#) | [arXiv:1706.05868](#) | [DOI:10.1007/JHEP10\(2017\)131](#) | 2017-06-19

914. Suppression of Excited Υ States Relative to the Ground State in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 120 \(2018\) 142301](#) | [arXiv:1706.05984](#) | [DOI:10.1103/PhysRevLett.120.142301](#) | 2017-06-19

915. Particle-flow reconstruction and global event description with the CMS detector

Sirunyan, A.M. et al.

[JINST 12 \(2017\) P10003](#) | [arXiv:1706.04965](#) | [DOI:10.1088/1748-0221/12/10/P10003](#) | 2017-06-15

916. Search for top squark pair production in pp collisions at $\sqrt{s}=13$ TeV using single lepton events

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 019](#) | [arXiv:1706.04402](#) | [DOI:10.1007/JHEP10\(2017\)019](#) | 2017-06-14

917. Searches for W' bosons decaying to a top quark and a bottom quark in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2017\) 029](#) | [arXiv:1706.04260](#) | [DOI:10.1007/JHEP08\(2017\)029](#) | 2017-06-13

918. Search for new physics in the monophoton final state in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 073](#) | [arXiv:1706.03794](#) | [DOI:10.1007/JHEP10\(2017\)073](#) | 2017-06-12

919. Search for pair production of vector-like T and B quarks in single-lepton final states using boosted jet substructure in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[JHEP 11 \(2017\) 085](#) | [arXiv:1706.03408](#) | [DOI:10.1007/JHEP11\(2017\)085](#) | 2017-06-11

920. Search for dark matter produced in association with heavy-flavor quark pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, A.M. et al.

[Eur.Phys.J.C 77 \(2017\) 845](#) | [arXiv:1706.02581](#) | [DOI:10.1140/epjc/s10052-017-5317-4](#) | 2017-06-08

921. Search for top quark partners with charge $5/3$ in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

922. Search for Low Mass Vector Resonances Decaying to Quark-Antiquark Pairs in Proton-Proton Collisions at $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 111802](#) | [arXiv:1705.10532](#) | [DOI:10.1103/PhysRevLett.119.111802](#) | 2017-05-30

923. Measurements of $t\bar{t}$ cross sections in association with b jets and inclusive jets and their ratio using dilepton final states in pp collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 776 \(2018\) 355](#) | [arXiv:1705.10141](#) | [DOI:10.1016/j.physletb.2017.11.043](#) | 2017-05-29

924. Combination of searches for heavy resonances decaying to WW, WZ, ZZ, WH, and ZH boson pairs in proton-proton collisions at $\sqrt{s}=8$ and 13 TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 774 \(2017\) 533](#) | [arXiv:1705.09171](#) | [DOI:10.1016/j.physletb.2017.09.083](#) | 2017-05-25

925. Search for new phenomena with the M_{T2} variable in the all-hadronic final state produced in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 710](#) | [arXiv:1705.04650](#) | [DOI:10.1140/epjc/s10052-017-5267-x](#) | 2017-05-12

926. Search for Supersymmetry in pp Collisions at $\sqrt{s}=13\text{TeV}$ in the Single-Lepton Final State Using the Sum of Masses of Large-Radius Jets

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 151802](#) | [arXiv:1705.04673](#) | [DOI:10.1103/PhysRevLett.119.151802](#) | 2017-05-12

927. Measurement of the $B^{\pm}$ Meson Nuclear Modification Factor in Pb-Pb Collisions at $\sqrt{s_{NN}}=5.02\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 152301](#) | [arXiv:1705.04727](#) | [DOI:10.1103/PhysRevLett.119.152301](#) | 2017-05-12

928. Search for Charged Higgs Bosons Produced via Vector Boson Fusion and Decaying into a Pair of W and Z Bosons Using pp Collisions at $\sqrt{s}=13\text{TeV}$

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 141802](#) | [arXiv:1705.02942](#) | [DOI:10.1103/PhysRevLett.119.141802](#) | 2017-05-08

929. Measurement of the triple-differential dijet cross section in proton-proton collisions at $\sqrt{s}=8 \text{ TeV}$ and constraints on parton distribution functions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 746](#) | [arXiv:1705.02628](#) | [DOI:10.1140/epjc/s10052-017-5286-7](#) | 2017-05-07

930. Search for black holes in high-multiplicity final states in proton-proton collisions at $\sqrt{s}=13 \text{ TeV}$

Sirunyan, Albert M et al.

[Phys.Lett.B 774 \(2017\) 279](#) | [arXiv:1705.01403](#) | [DOI:10.1016/j.physletb.2017.09.053](#) | 2017-05-03

931. Search for supersymmetry in multijet events with missing transverse momentum in proton-proton collisions at 13 TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 032003](#) | [arXiv:1704.07781](#) | [DOI:10.1103/PhysRevD.96.032003](#) | 2017-04-25

932. Search for physics beyond the standard model in events with two leptons of same sign, missing transverse momentum, and jets in proton-proton collisions at $\sqrt{s} = 13 \text{ TeV}$

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 578](#) | [arXiv:1704.07323](#) | [DOI:10.1140/epjc/s10052-017-5079-z](#) | 2017-04-24

933. Measurement of the top quark mass in the dileptonic $t\bar{t}$ decay channel using the mass observables $M_{b\ell}$, M_{T2} , and $M_{b\ell\nu}$ in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.D 96 \(2017\) 032002](#) | [arXiv:1704.06142](#) | [DOI:10.1103/PhysRevD.96.032002](#) | 2017-04-20

934. Search for $t\bar{t}$ resonances in highly boosted lepton+jets and fully hadronic final states in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 001](#) | [arXiv:1704.03366](#) | [DOI:10.1007/JHEP07\(2017\)001](#) | 2017-04-11

935. Measurements of the $pp \rightarrow W\gamma\gamma$ and $pp \rightarrow Z\gamma\gamma$ cross sections and limits on anomalous quartic gauge couplings at $\sqrt{s}=8$ TeV

Sirunyan, A.M. et al.

[JHEP 10 \(2017\) 072](#) | [arXiv:1704.00366](#) | [DOI:10.1007/JHEP10\(2017\)072](#) | 2017-04-02

936. Search for new physics with dijet angular distributions in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 013](#) | [arXiv:1703.09986](#) | [DOI:10.1007/JHEP07\(2017\)013](#) | 2017-03-29

937. Measurement of the jet mass in highly boosted $t\bar{t}$ events from pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 467](#) | [arXiv:1703.06330](#) | [DOI:10.1140/epjc/s10052-017-5030-3](#) | 2017-03-18

938. Search for a heavy resonance decaying to a top quark and a vector-like top quark at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 09 \(2017\) 053](#) | [arXiv:1703.06352](#) | [DOI:10.1007/JHEP09\(2017\)053](#) | 2017-03-18

939. Search for anomalous couplings in boosted $WW/WZ \rightarrow \ell\nu q \bar{q}$ production in proton-proton collisions at $\sqrt{s} = 8$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 21](#) | [arXiv:1703.06095](#) | [DOI:10.1016/j.physletb.2017.06.009](#) | 2017-03-17

940. Search for associated production of dark matter with a Higgs boson decaying to $b\bar{b}$ or $\gamma\gamma$ at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 10 \(2017\) 180](#) | [arXiv:1703.05236](#) | [DOI:10.1007/JHEP10\(2017\)180](#) | 2017-03-15

941. Big Data in HEP: A comprehensive use case study

Gutsche, Oliver et al.

[J.Phys.Conf.Ser. 898 \(2017\) 072012](#) | [arXiv:1703.04171](#) | [DOI:10.1088/1742-6596/898/7/072012](#) | 2017-03-12

942. Search for third-generation scalar leptoquarks and heavy right-handed neutrinos in final states with two tau leptons and two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 121](#) | [arXiv:1703.03995](#) | [DOI:10.1007/JHEP07\(2017\)121](#) | 2017-03-11

943. Measurement of the top quark mass using single top quark events in proton-proton collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 354](#) | [arXiv:1703.02530](#) | [DOI:10.1140/epjc/s10052-017-4912-8](#) | 2017-03-07

944. Measurement of double-differential cross sections for top quark pair production in pp collisions at $\sqrt{s} = 8$ TeV and impact on parton distribution functions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 459](#) | [arXiv:1703.01630](#) | [DOI:10.1140/epjc/s10052-017-4984-5](#) | 2017-03-05

945. Search for dark matter produced with an energetic jet or a hadronically decaying W or Z boson at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 014](#) | [arXiv:1703.01651](#) | [DOI:10.1007/JHEP07\(2017\)014](#) | 2017-03-05

946. Search for standard model production of four top quarks in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 772 \(2017\) 336](#) | [arXiv:1702.06164](#) | [DOI:10.1016/j.physletb.2017.06.064](#) | 2017-02-20

947. Measurement of the cross section for electroweak production of $Z\gamma$ in association with two jets and constraints on anomalous quartic gauge couplings in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 770 \(2017\) 380](#) | [arXiv:1702.03025](#) | [DOI:10.1016/j.physletb.2017.04.071](#) | 2017-02-09

948. Search for associated production of a Z boson with a single top quark and for tZ flavour-changing interactions in pp collisions at $\sqrt{s}=8$ TeV

Sirunyan, Albert M et al.

[JHEP 07 \(2017\) 003](#) | [arXiv:1702.01404](#) | [DOI:10.1007/JHEP07\(2017\)003](#) | 2017-02-05

949. Measurement of prompt and nonprompt J/ψ production in pp and pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 269](#) | [arXiv:1702.01462](#) | [DOI:10.1140/epjc/s10052-017-4828-3](#) | 2017-02-05

950. Study of Jet Quenching with Z+jet Correlations in Pb-Pb and pp Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 119 \(2017\) 082301](#) | [arXiv:1702.01060](#) | [DOI:10.1103/PhysRevLett.119.082301](#) | 2017-02-03

951. Azimuthal anisotropy of charged particles with transverse momentum up to 100 GeV/ c in PbPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, A.M. et al.

[Phys.Lett.B 776 \(2018\) 195](#) | [arXiv:1702.00630](#) | [DOI:10.1016/j.physletb.2017.11.041](#) | 2017-02-02

952. Measurement of the inclusive energy spectrum in the very forward direction in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 08 \(2017\) 046](#) | [arXiv:1701.08695](#) | [DOI:10.1007/JHEP08\(2017\)046](#) | 2017-01-30

953. Search for single production of vector-like quarks decaying into a b quark and a W boson in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 634](#) | [arXiv:1701.08328](#) | [DOI:10.1016/j.physletb.2017.07.022](#) | 2017-01-28

954. Search for single production of vector-like quarks decaying to a Z boson and a top or a bottom quark in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 05 \(2017\) 029](#) | [arXiv:1701.07409](#) | [DOI:10.1007/JHEP05\(2017\)029](#) | 2017-01-25

955. Search for new phenomena with multiple charged leptons in proton-proton collisions at $\sqrt{s}= 13$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 635](#) | [arXiv:1701.06940](#) | [DOI:10.1140/epjc/s10052-017-5182-1](#) | 2017-01-24

956. Measurement of the $t\bar{t}$ production cross section using events with one lepton and at least one jet in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M. et al.

[JHEP 09 \(2017\) 051](#) | [arXiv:1701.06228](#) | [DOI:10.1007/JHEP09\(2017\)051](#) | 2017-01-22

957. Dark Matter and Super Symmetry: Exploring and Explaining the Universe with Simulations at the LHC

Gutsche, Oliver et al.

[\(2017\) 4](#) | [DOI:10.1109/WSC.2016.7822075](#) | 2017-01-19

958. Search for supersymmetry in the all-hadronic final state using top quark tagging in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 96 \(2017\) 012004](#) | [arXiv:1701.01954](#) | [DOI:10.1103/PhysRevD.96.012004](#) | 2017-01-08

959. Mechanical stability of the CMS strip tracker measured with a laser alignment system

Sirunyan, Albert M et al.

[JINST 12 \(2017\) P04023](#) | [arXiv:1701.02022](#) | [DOI:10.1088/1748-0221/12/04/P04023](#) | 2017-01-08

960. Search for light bosons in decays of the 125 GeV Higgs boson in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, V. et al.

[JHEP 10 \(2017\) 076](#) | [arXiv:1701.02032](#) | [DOI:10.1007/JHEP10\(2017\)076](#) | 2017-01-08

961. Search for dark matter and unparticles in events with a Z boson and missing transverse momentum in proton-proton collisions at $\sqrt{s}=13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2017\) 061](#) | [arXiv:1701.02042](#) | [DOI:10.1007/JHEP03\(2017\)061](#) | 2017-01-08

962. Search for leptophobic Z' bosons decaying into four-lepton final states in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, V. et al.

[Phys.Lett.B 773 \(2017\) 563](#) | [arXiv:1701.01345](#) | [DOI:10.1016/j.physletb.2017.08.069](#) | 2017-01-05

963. Search for high-mass $Z \gamma$ resonances in proton-proton collisions at $\sqrt{s}= 8$ and 13 TeV using jet substructure techniques

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 363](#) | [arXiv:1612.09516](#) | [DOI:10.1016/j.physletb.2017.06.062](#) | 2016-12-30

964. Search for massive resonances decaying into WW, WZ or ZZ bosons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[JHEP 03 \(2017\) 162](#) | [arXiv:1612.09159](#) | [DOI:10.1007/JHEP03\(2017\)162](#) | 2016-12-29

965. Measurement of electroweak-induced production of $W\gamma$ with two jets in pp collisions at $\sqrt{s}=8$ TeV and constraints on anomalous quartic gauge couplings

Khachatryan, Vardan et al.

[JHEP 06 \(2017\) 106](#) | [arXiv:1612.09256](#) | [DOI:10.1007/JHEP06\(2017\)106](#) | 2016-12-29

966. Search for heavy gauge W' boson in events with an energetic lepton and large missing transverse momentum at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 770 \(2017\) 278](#) | [arXiv:1612.09274](#) | [DOI:10.1016/j.physletb.2017.04.043](#) | 2016-12-29

967. Measurements of the charm jet cross section and nuclear modification factor in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 306](#) | [arXiv:1612.08972](#) | [DOI:10.1016/j.physletb.2017.06.053](#) | 2016-12-28

968. Search for electroweak production of a vector-like quark decaying to a top quark and a Higgs boson using boosted topologies in fully hadronic final states

Sirunyan, Albert M et al.

[JHEP 04 \(2017\) 136](#) | [arXiv:1612.05336](#) | [DOI:10.1007/JHEP04\(2017\)136](#) | 2016-12-15

969. Searches for pair production of third-generation squarks in $\sqrt{s}=13$ TeV pp collisions

Sirunyan, Albert M et al.

[Eur.Phys.J.C 77 \(2017\) 327](#) | [arXiv:1612.03877](#) | [DOI:10.1140/epjc/s10052-017-4853-2](#) | 2016-12-12

970. Search for heavy neutrinos or third-generation leptoquarks in final states with two hadronically decaying τ leptons and two jets in proton-proton collisions at $\sqrt{s}=13$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2017\) 077](#) | [arXiv:1612.01190](#) | [DOI:10.1007/JHEP03\(2017\)077](#) | 2016-12-04

971. Search for single production of a heavy vector-like T quark decaying to a Higgs boson and a top quark with a lepton and jets in the final state

Khachatryan, Vardan et al.

[Phys.Lett.B 771 \(2017\) 80](#) | [arXiv:1612.00999](#) | [DOI:10.1016/j.physletb.2017.05.019](#) | 2016-12-03

972. Search for CP violation in tt production and decay in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2017\) 101](#) | [arXiv:1611.08931](#) | [DOI:10.1007/JHEP03\(2017\)101](#) | 2016-11-27

973. Measurements of the associated production of a Z boson and b jets in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, V. et al.

[Eur.Phys.J.C 77 \(2017\) 751](#) | [arXiv:1611.06507](#) | [DOI:10.1140/epjc/s10052-017-5140-y](#) | 2016-11-20

974. Search for heavy resonances decaying to tau lepton pairs in proton-proton collisions at $\sqrt{s}=13$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 048](#) | [arXiv:1611.06594](#) | [DOI:10.1007/JHEP02\(2017\)048](#) | 2016-11-20

975. Search for supersymmetry in events with photons and missing transverse energy in pp collisions at 13 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 769 \(2017\) 391](#) | [arXiv:1611.06604](#) | [DOI:10.1016/j.physletb.2017.04.005](#) | 2016-11-20

976. Measurement of the $t\bar{t}$ production cross section using events in the $e\mu$ final state in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 172](#) | [arXiv:1611.04040](#) | [DOI:10.1140/epjc/s10052-017-4718-8](#) | 2016-11-12

977. Measurements of differential production cross sections for a Z boson in association with jets in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2017\) 022](#) | [arXiv:1611.03844](#) | [DOI:10.1007/JHEP04\(2017\)022](#) | 2016-11-11

978. Search for dijet resonances in proton-proton collisions at $\sqrt{s} = 13$ TeV and constraints on dark matter and other models

Sirunyan, Albert M et al.

[Phys.Lett.B 769 \(2017\) 520](#) | [arXiv:1611.03568](#) | [DOI:10.1016/j.physletb.2017.02.012](#) | 2016-11-10

979. Charged-particle nuclear modification factors in PbPb and pPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2017\) 039](#) | [arXiv:1611.01664](#) | [DOI:10.1007/JHEP04\(2017\)039](#) | 2016-11-05

980. Relative Modification of Prompt $\psi(2S)$ and J/ψ Yields from pp to PbPb Collisions at $\sqrt{s_{NN}}=5.02$ TeV

Sirunyan, Albert M et al.

[Phys.Rev.Lett. 118 \(2017\) 162301](#) | [arXiv:1611.01438](#) | [DOI:10.1103/PhysRevLett.118.162301](#) | 2016-11-04

981. Suppression of $Y(1S)$, $Y(2S)$ and $Y(3S)$ production in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 770 \(2017\) 357](#) | [arXiv:1611.01510](#) | [DOI:10.1016/j.physletb.2017.04.031](#) | 2016-11-04

982. A search for new phenomena in pp collisions at $\sqrt{s} = 13$ TeV in final states with missing transverse momentum and at least one jet using the α_T variable

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 294](#) | [arXiv:1611.00338](#) | [DOI:10.1140/epjc/s10052-017-4787-8](#) | 2016-11-01

983. Measurement of the mass difference between top quark and antiquark in pp collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 770 \(2017\) 50](#) | [arXiv:1610.09551](#) | [DOI:10.1016/j.physletb.2017.04.028](#) | 2016-10-29

984. Searches for invisible decays of the Higgs boson in pp collisions at $\sqrt{s} = 7, 8$, and 13 TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 135](#) | [arXiv:1610.09218](#) | [DOI:10.1007/JHEP02\(2017\)135](#) | 2016-10-28

985. Search for heavy resonances decaying into a vector boson and a Higgs boson in final states with charged leptons, neutrinos, and b quarks

Khachatryan, Vardan et al.

[Phys.Lett.B 768 \(2017\) 137](#) | [arXiv:1610.08066](#) | [DOI:10.1016/j.physletb.2017.02.040](#) | 2016-10-25

986. Observation of $Y(1S)$ pair production in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 05 \(2017\) 013](#) | [arXiv:1610.07095](#) | [DOI:10.1007/JHEP05\(2017\)013](#) | 2016-10-22

987. Search for R-parity violating supersymmetry with displaced vertices in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 012009](#) | [arXiv:1610.05133](#) | [DOI:10.1103/PhysRevD.95.012009](#) | 2016-10-17

988. Search for top quark decays via Higgs-boson-mediated flavor-changing neutral currents in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 079](#) | [arXiv:1610.04857](#) | [DOI:10.1007/JHEP02\(2017\)079](#) | 2016-10-16

989. Search for electroweak production of charginos in final states with two τ leptons in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2017\) 018](#) | [arXiv:1610.04870](#) | [DOI:10.1007/JHEP04\(2017\)018](#) | 2016-10-16

990. Measurement of differential cross sections for top quark pair production using the lepton+jets final state in proton-proton collisions at 13 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 092001](#) | [arXiv:1610.04191](#) | [DOI:10.1103/PhysRevD.95.092001](#) | 2016-10-13

991. Measurements of differential cross sections for associated production of a W boson and jets in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 052002](#) | [arXiv:1610.04222](#) | [DOI:10.1103/PhysRevD.95.052002](#) | 2016-10-13

992. Search for anomalous Wtb couplings and flavour-changing neutral currents in t-channel single top quark production in pp collisions at $\sqrt{s} = 7$ and 8 TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 028](#) | [arXiv:1610.03545](#) | [DOI:10.1007/JHEP02\(2017\)028](#) | 2016-10-11

993. Search for high-mass $Z\gamma$ resonances in $e^+e^-\gamma$ and $\mu^+\mu^-\gamma$ final states in proton-proton collisions at $\sqrt{s} = 8$ and 13 TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2017\) 076](#) | [arXiv:1610.02960](#) | [DOI:10.1007/JHEP01\(2017\)076](#) | 2016-10-10

994. Suppression and azimuthal anisotropy of prompt and nonprompt J/ψ production in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 252](#) | [arXiv:1610.00613](#) | [DOI:10.1140/epjc/s10052-017-4781-1](#) | 2016-10-03

995. Cross section measurement of t-channel single top quark production in pp collisions at $\sqrt{s} = 13$ TeV

Sirunyan, Albert M et al.

[Phys.Lett.B 772 \(2017\) 752](#) | [arXiv:1610.00678](#) | [DOI:10.1016/j.physletb.2017.07.047](#) | 2016-10-03

996. Observation of charge-dependent azimuthal correlations in p-Pb collisions and its implication for the search for the chiral magnetic effect

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 118 \(2017\) 122301](#) | [arXiv:1610.00263](#) | [DOI:10.1103/PhysRevLett.118.122301](#) | 2016-10-02

997. Search for supersymmetry in events with one lepton and multiple jets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 012011](#) | [arXiv:1609.09386](#) | [DOI:10.1103/PhysRevD.95.012011](#) | 2016-09-29

998. Search for long-lived charged particles in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 112004](#) | [arXiv:1609.08382](#) | [DOI:10.1103/PhysRevD.94.112004](#) | 2016-09-27

999. Inclusive search for supersymmetry using razor variables in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 95 \(2017\) 012003](#) | [arXiv:1609.07658](#) | [DOI:10.1103/PhysRevD.95.012003](#) | 2016-09-24

1000. Measurement of the WZ production cross section in pp collisions at $\sqrt{s} = 7$ and 8 TeV and search for anomalous triple gauge couplings at $\sqrt{s} = 8$ TeV

Khachatryan, V. et al.

[Eur.Phys.J.C 77 \(2017\) 236](#) | [arXiv:1609.05721](#) | [DOI:10.1140/epjc/s10052-017-4730-z](#) | 2016-09-19

1001. Measurement and QCD analysis of double-differential inclusive jet cross sections in pp collisions at $\sqrt{s} = 8$ TeV and cross section ratios to 2.76 and 7 TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2017\) 156](#) | [arXiv:1609.05331](#) | [DOI:10.1007/JHEP03\(2017\)156](#) | 2016-09-17

1002. Measurement of inclusive jet cross sections in pp and PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.C 96 \(2017\) 015202](#) | [arXiv:1609.05383](#) | [DOI:10.1103/PhysRevC.96.015202](#) | 2016-09-17

1003. Search for narrow resonances in dilepton mass spectra in proton-proton collisions at $\sqrt{s} = 13$ TeV and combination with 8 TeV data

Khachatryan, Vardan et al.

[Phys.Lett.B 768 \(2017\) 57](#) | [arXiv:1609.05391](#) | [DOI:10.1016/j.physletb.2017.02.010](#) | 2016-09-17

1004. Studies of inclusive four-jet production with two b-tagged jets in proton-proton collisions at 7 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 112005](#) | [arXiv:1609.03489](#) | [DOI:10.1103/PhysRevD.94.112005](#) | 2016-09-12

1005. The CMS trigger system

Khachatryan, Vardan et al.

[JINST 12 \(2017\) P01020](#) | [arXiv:1609.02366](#) | [DOI:10.1088/1748-0221/12/01/P01020](#) | 2016-09-08

1006. Decomposing transverse momentum balance contributions for quenched jets in PbPb collisions at $\sqrt{s_{NN}}=2.76$ TeV

Khachatryan, Vardan et al.

[JHEP 11 \(2016\) 055](#) | [arXiv:1609.02466](#) | [DOI:10.1007/JHEP11\(2016\)055](#) | 2016-09-08

1007. Search for high-mass diphoton resonances in proton-proton collisions at 13 TeV and combination with 8 TeV search

Khachatryan, Vardan et al.

[Phys.Lett.B 767 \(2017\) 147](#) | [arXiv:1609.02507](#) | [DOI:10.1016/j.physletb.2017.01.027](#) | 2016-09-08

1008. Measurement of the total and differential inclusive B^+ hadron cross sections in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 771 \(2017\) 435](#) | [arXiv:1609.00873](#) | [DOI:10.1016/j.physletb.2017.05.074](#) | 2016-09-03

1009. Measurement of the production cross section of a W boson in association with two b jets in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 92](#) | [arXiv:1608.07561](#) | [DOI:10.1140/epjc/s10052-016-4573-z](#) | 2016-08-26

1010. Measurement of the mass of the top quark in decays with a J/ψ meson in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[JHEP 12 \(2016\) 123](#) | [arXiv:1608.03560](#) | [DOI:10.1007/JHEP12\(2016\)123](#) | 2016-08-11

1011. Search for new phenomena in events with high jet multiplicity and low missing transverse momentum in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 770 \(2017\) 257](#) | [arXiv:1608.01224](#) | [DOI:10.1016/j.physletb.2017.01.073](#) | 2016-08-03

1012. Measurement of the ZZ production cross section and $Z \rightarrow \ell^+ \ell^- \ell'^+ \ell'^-$ branching fraction in pp collisions at $\sqrt{s}=13$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 763 \(2016\) 280](#) | [arXiv:1607.08834](#) | [DOI:10.1016/j.physletb.2016.10.054](#) | 2016-07-29

1013. Measurement of the WZ production cross section in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 766 \(2017\) 268](#) | [arXiv:1607.06943](#) | [DOI:10.1016/j.physletb.2017.01.011](#) | 2016-07-23

1014. Measurement of electroweak production of a W boson and two forward jets in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 11 \(2016\) 147](#) | [arXiv:1607.06975](#) | [DOI:10.1007/JHEP11\(2016\)147](#) | 2016-07-23

1015. Search for dark matter in proton-proton collisions at 8 TeV with missing transverse momentum and vector boson tagged jets

Khachatryan, Vardan et al.

[JHEP 12 \(2016\) 083](#) | [arXiv:1607.05764](#) | [DOI:10.1007/JHEP12\(2016\)083](#) | 2016-07-19

1016. Jet energy scale and resolution in the CMS experiment in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[JINST 12 \(2017\) P02014](#) | [arXiv:1607.03663](#) | [DOI:10.1088/1748-0221/12/02/P02014](#) | 2016-07-13

1017. Search for lepton flavour violating decays of the Higgs boson to $e\tau$ and $e\mu$ in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 763 \(2016\) 472](#) | [arXiv:1607.03561](#) | [DOI:10.1016/j.physletb.2016.09.062](#) | 2016-07-12

1018. Observation of the decay $B^+ \rightarrow \psi(2S) \phi(1020) K^+$ in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 764 \(2017\) 66](#) | [arXiv:1607.02638](#) | [DOI:10.1016/j.physletb.2016.11.001](#) | 2016-07-09

1019. Measurement of the differential cross sections for top quark pair production as a function of kinematic event variables in pp collisions at $\sqrt{s}=7$ and 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 052006](#) | [arXiv:1607.00837](#) | [DOI:10.1103/PhysRevD.94.052006](#) | 2016-07-04

1020. Search for new physics in final states with two opposite-sign, same-flavor leptons, jets, and missing transverse momentum in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[JHEP 12 \(2016\) 013](#) | [arXiv:1607.00915](#) | [DOI:10.1007/JHEP12\(2016\)013](#) | 2016-07-04

1021. Searches for R-parity-violating supersymmetry in ppcollisions at $\sqrt{s} = 8$ TeV in final states with 0-4 leptons

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 112009](#) | [arXiv:1606.08076](#) | [DOI:10.1103/PhysRevD.94.112009](#) | 2016-06-26

1022. Evidence for collectivity in pp collisions at the LHC

Khachatryan, Vardan et al.

[Phys.Lett.B 765 \(2017\) 193](#) | [arXiv:1606.06198](#) | [DOI:10.1016/j.physletb.2016.12.009](#) | 2016-06-20

1023. Measurement of the transverse momentum spectra of weak vector bosons produced in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2017\) 096](#) | [arXiv:1606.05864](#) | [DOI:10.1007/JHEP02\(2017\)096](#) | 2016-06-19

1024. Search for Resonant Production of High-Mass Photon Pairs in Proton-Proton Collisions at $\sqrt{s} = 8$ and 13 TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 117 \(2016\) 051802](#) | [arXiv:1606.04093](#) | [DOI:10.1103/PhysRevLett.117.051802](#) | 2016-06-13

1025. Phenomenological MSSM interpretation of CMS searches in pp collisions at $\sqrt{s} = 7$ and 8 TeV

Khachatryan, Vardan et al.

[JHEP 10 \(2016\) 129](#) | [arXiv:1606.03577](#) | [DOI:10.1007/JHEP10\(2016\)129](#) | 2016-06-11

1026. Measurements of the Higgs boson production and decay rates and constraints on its couplings from a combined ATLAS and CMS analysis of the LHC pp collision data at $\sqrt{s}=7$ and 8 TeV

Aad, Georges et al.

[JHEP 08 \(2016\) 045](#) | [arXiv:1606.02266](#) | [DOI:10.1007/JHEP08\(2016\)045](#) | 2016-06-07

1027. Measurement of the transverse momentum spectrum of the Higgs boson produced in pp collisions at $\sqrt{s}=8$ TeV using $H \rightarrow WW$ decays

Khachatryan, Vardan et al.

[JHEP 03 \(2017\) 032](#) | [arXiv:1606.01522](#) | [DOI:10.1007/JHEP03\(2017\)032](#) | 2016-06-05

1028. Search for Dark Matter and Supersymmetry with a Compressed Mass Spectrum in the Vector Boson Fusion Topology in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 118 \(2017\) 021802](#) | [arXiv:1605.09305](#) | [DOI:10.1103/PhysRevLett.118.021802](#) | 2016-05-30

1029. Search for top squark pair production in compressed-mass-spectrum scenarios in proton-proton collisions at $\sqrt{s} = 8$ TeV using the α_T variable

Khachatryan, Vardan et al.

[Phys.Lett.B 767 \(2017\) 403](#) | [arXiv:1605.08993](#) | [DOI:10.1016/j.physletb.2017.02.007](#) | 2016-05-29

1030. Measurement of the W boson helicity fractions in the decays of top quark pairs to lepton + jets final states produced in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 762 \(2016\) 512](#) | [arXiv:1605.09047](#) | [DOI:10.1016/j.physletb.2016.10.007](#) | 2016-05-29

1031. Coherent J/ψ photoproduction in ultra-peripheral PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV with the CMS experiment

Khachatryan, Vardan et al.

[Phys.Lett.B 772 \(2017\) 489](#) | [arXiv:1605.06966](#) | [DOI:10.1016/j.physletb.2017.07.001](#) | 2016-05-23

1032. Multiplicity and rapidity dependence of strange hadron production in pp, pPb, and PbPb collisions at the LHC

Khachatryan, Vardan et al.

[Phys.Lett.B 768 \(2017\) 103](#) | [arXiv:1605.06699](#) | [DOI:10.1016/j.physletb.2017.01.075](#) | 2016-05-21

1033. Search for supersymmetry in pp collisions at $\sqrt{s}=13$ TeV in the single-lepton final state using the sum of masses of large-radius jets

Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 122](#) | [arXiv:1605.04608](#) | [DOI:10.1007/JHEP08\(2016\)122](#) | 2016-05-15

1034. Measurement of the double-differential inclusive jet cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 451](#) | [arXiv:1605.04436](#) | [DOI:10.1140/epjc/s10052-016-4286-3](#) | 2016-05-14

1035. Search for new physics in same-sign dilepton events in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 439](#) | [arXiv:1605.03171](#) | [DOI:10.1140/epjc/s10052-016-4261-z](#) | 2016-05-10

1036. Search for Higgs boson off-shell production in proton-proton collisions at 7 and 8 TeV and derivation of constraints on its total decay width

Khachatryan, Vardan et al.

[JHEP 09 \(2016\) 051](#) | [arXiv:1605.02329](#) | [DOI:10.1007/JHEP09\(2016\)051](#) | 2016-05-08

1037. Measurement of the integrated and differential $t\bar{t}$ production cross sections for high- p_T top quarks in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 072002](#) | [arXiv:1605.00116](#) | [DOI:10.1103/PhysRevD.94.072002](#) | 2016-04-30

1038. Search for narrow resonances in dijet final states at $\sqrt{s}=8$ TeV with the novel CMS technique of data scouting

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 117 \(2016\) 031802](#) | [arXiv:1604.08907](#) | [DOI:10.1103/PhysRevLett.117.031802](#) | 2016-04-29

1039. Search for lepton flavour violating decays of heavy resonances and quantum black holes to an $e\mu$ pair in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 317](#) | [arXiv:1604.05239](#) | [DOI:10.1140/epjc/s10052-016-4149-y](#) | 2016-04-18

1040. Pseudorapidity dependence of long-range two-particle correlations in pPb collisions at $\sqrt{s_{NN}}=5.02$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.C 96 \(2017\) 014915](#) | [arXiv:1604.05347](#) | [DOI:10.1103/PhysRevC.96.014915](#) | 2016-04-18

1041. Evidence for exclusive $\gamma\gamma \rightarrow W^+W^-$ production and constraints on anomalous quartic gauge couplings in pp collisions at $\sqrt{s}=7$ and 8 TeV

Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 119](#) | [arXiv:1604.04464](#) | [DOI:10.1007/JHEP08\(2016\)119](#) | 2016-04-15

1042. ASCR/HEP Exascale Requirements Review Report

Habib, Salman et al.

[arXiv:1603.09303](#) | 2016-03-30

1043. Search for dark matter particles in proton-proton collisions at $\sqrt{s}=8$ TeV using the razor variables

Khachatryan, Vardan et al.

[JHEP 12 \(2016\) 088](#) | [arXiv:1603.08914](#) | [DOI:10.1007/JHEP12\(2016\)088](#) | 2016-03-29

1044. Search for two Higgs bosons in final states containing two photons and two bottom quarks in proton-proton collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 94 \(2016\) 052012](#) | [arXiv:1603.06896](#) | [DOI:10.1103/PhysRevD.94.052012](#) | 2016-03-22

1045. Measurement of the top quark mass using charged particles in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 092006](#) | [arXiv:1603.06536](#) | [DOI:10.1103/PhysRevD.93.092006](#) | 2016-03-21

1046. Measurements of $t\bar{t}$ charge asymmetry using dilepton final states in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 760 \(2016\) 365](#) | [arXiv:1603.06221](#) | [DOI:10.1016/j.physletb.2016.07.006](#) | 2016-03-20

1047. Search for new physics with the M_{T2} variable in all-jets final states produced in pp collisions at $\sqrt{s}=13$ TeV

Khachatryan, Vardan et al.

[JHEP 10 \(2016\) 006](#) | [arXiv:1603.04053](#) | [DOI:10.1007/JHEP10\(2016\)006](#) | 2016-03-13

1048. $Y(nS)$ polarizations versus particle multiplicity in pp collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 761 \(2016\) 31](#) | [arXiv:1603.02913](#) | [DOI:10.1016/j.physletb.2016.07.065](#) | 2016-03-09

1049. Search for neutral resonances decaying into a Z boson and a pair of b jets or τ leptons

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 369](#) | [arXiv:1603.02991](#) | [DOI:10.1016/j.physletb.2016.05.087](#) | 2016-03-09

1050. Search for s channel single top quark production in pp collisions at $\sqrt{s}=7$ and 8 TeV

Khachatryan, Vardan et al.

[JHEP 09 \(2016\) 027](#) | [arXiv:1603.02555](#) | [DOI:10.1007/JHEP09\(2016\)027](#) | 2016-03-08

1051. Search for heavy Majorana neutrinos in $e^+e^+ + \text{jets}$ and $e^+e^- + \mu^+ + \text{jets}$ events in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 169](#) | [arXiv:1603.02248](#) | [DOI:10.1007/JHEP04\(2016\)169](#) | 2016-03-07

1052. Measurement of the $t\text{-}\bar{t}$ production cross section in the $e\text{-}\mu$ channel in proton-proton collisions at $\sqrt{s} = 7$ and 8 TeV

Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 029](#) | [arXiv:1603.02303](#) | [DOI:10.1007/JHEP08\(2016\)029](#) | 2016-03-07

1053. Measurement of the differential cross section and charge asymmetry for inclusive $pp \rightarrow W^{\pm} + X$ production at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 469](#) | [arXiv:1603.01803](#) | [DOI:10.1140/epjc/s10052-016-4293-4](#) | 2016-03-06

1054. Search for direct pair production of supersymmetric top quarks decaying to all-hadronic final states in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 460](#) | [arXiv:1603.00765](#) | [DOI:10.1140/epjc/s10052-016-4292-5](#) | 2016-03-02

1055. Measurements of the $t\bar{t}$ production cross section in lepton+jets final states in pp collisions at 8 TeV and ratio of 8 to 7 TeV cross sections

Khachatryan, Vardan et al.

[Eur.Phys.J.C 77 \(2017\) 15](#) | [arXiv:1602.09024](#) | [DOI:10.1140/epjc/s10052-016-4504-z](#) | 2016-02-29

1056. Search for heavy resonances decaying to two Higgs bosons in final states containing four b quarks

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 371](#) | [arXiv:1602.08762](#) | [DOI:10.1140/epjc/s10052-016-4206-6](#) | 2016-02-28

1057. Search for supersymmetry in electroweak production with photons and large missing transverse energy in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 479](#) | [arXiv:1602.08772](#) | [DOI:10.1016/j.physletb.2016.05.088](#) | 2016-02-28

1058. Measurement of the $Z \gamma \rightarrow \nu \bar{\nu} \gamma$ production cross section in pp collisions at $\sqrt{s}=8$ TeV and limits on anomalous $ZZ\gamma$ and $Z\gamma\gamma$ trilinear gauge boson couplings

Khachatryan, Vardan et al.

[Phys.Lett.B 760 \(2016\) 448](#) | [arXiv:1602.07152](#) | [DOI:10.1016/j.physletb.2016.06.080](#) | 2016-02-23

1059. Search for supersymmetry in the multijet and missing transverse momentum final state in pp collisions at 13 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 152](#) | [arXiv:1602.06581](#) | [DOI:10.1016/j.physletb.2016.05.002](#) | 2016-02-21

1060. Combined search for anomalous pseudoscalar HVV couplings in $VH(H \rightarrow b\bar{b})$ production and $H \rightarrow VV$ decay

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 672](#) | [arXiv:1602.04305](#) | [DOI:10.1016/j.physletb.2016.06.004](#) | 2016-02-13

1061. Search for R-parity violating decays of a top squark in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 760 \(2016\) 178](#) | [arXiv:1602.04334](#) | [DOI:10.1016/j.physletb.2016.06.039](#) | 2016-02-13

1062. Measurement of dijet azimuthal decorrelation in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 536](#) | [arXiv:1602.04384](#) | [DOI:10.1140/epjc/s10052-016-4346-8](#) | 2016-02-13

1063. Search for supersymmetry in pp collisions at $\sqrt{s} = 8$ TeV in final states with boosted W bosons and b jets using razor variables

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 092009](#) | [arXiv:1602.02917](#) | [DOI:10.1103/PhysRevD.93.092009](#) | 2016-02-09

1064. Search for direct pair production of scalar top quarks in the single- and dilepton channels in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 07 \(2016\) 027](#) | [arXiv:1602.03169](#) | [DOI:10.1007/JHEP07\(2016\)027](#) | 2016-02-09

1065. Azimuthal decorrelation of jets widely separated in rapidity in pp collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[JHEP 08 \(2016\) 139](#) | [arXiv:1601.06713](#) | [DOI:10.1007/JHEP08\(2016\)139](#) | 2016-01-25

1066. Search for massive WH resonances decaying into the $\ell\nu b\bar{b}$ final state at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 237](#) | [arXiv:1601.06431](#) | [DOI:10.1140/epjc/s10052-016-4067-z](#) | 2016-01-24

1067. Forward-backward asymmetry of Drell-Yan lepton pairs in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 325](#) | [arXiv:1601.04768](#) | [DOI:10.1140/epjc/s10052-016-4156-z](#) | 2016-01-18

1068. Measurement of inclusive jet production and nuclear modifications in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 372](#) | [arXiv:1601.02001](#) | [DOI:10.1140/epjc/s10052-016-4205-7](#) | 2016-01-08

1069. Measurements of t t-bar spin correlations and top quark polarization using dilepton final states in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, V. et al.

[Phys.Rev.D 93 \(2016\) 052007](#) | [arXiv:1601.01107](#) | [DOI:10.1103/PhysRevD.93.052007](#) | 2016-01-06

1070. Correlations between jets and charged particles in PbPb and pp collisions at $\sqrt{s_{NN}}=2.76$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2016\) 156](#) | [arXiv:1601.00079](#) | [DOI:10.1007/JHEP02\(2016\)156](#) | 2016-01-01

1071. Measurement of differential and integrated fiducial cross sections for Higgs boson production in the four-lepton decay channel in pp collisions at $\sqrt{s}=7$ and 8 TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 005](#) | [arXiv:1512.08377](#) | [DOI:10.1007/JHEP04\(2016\)005](#) | 2015-12-28

1072. Search for supersymmetry in events with soft leptons, low jet multiplicity, and missing transverse energy in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 9](#) | [arXiv:1512.08002](#) | [DOI:10.1016/j.physletb.2016.05.033](#) | 2015-12-25

1073. Diversity in Computing Technologies and Strategies for Dynamic Resource Allocation

Garzoglio, G. et al.

[J.Phys.Conf.Ser. 664 \(2015\) 012001](#) | [DOI:10.1088/1742-6596/664/1/012001](#) | 2015-12-23

1074. Fermilab Computing at the Intensity Frontier

Group, Craig et al.

[J.Phys.Conf.Ser. 664 \(2015\) 032012](#) | [DOI:10.1088/1742-6596/664/3/032012](#) | 2015-12-23

1075. Pooling the resources of the CMS Tier-1 sites

Apyan, A. et al.

[J.Phys.Conf.Ser. 664 \(2015\) 042056](#) | [DOI:10.1088/1742-6596/664/4/042056](#) | 2015-12-23

1076. Pushing HTCondor and glideinWMS to 200K+ Jobs in a Global Pool for CMS before Run 2

Balcas, J. et al.

[J.Phys.Conf.Ser. 664 \(2015\) 062030](#) | [DOI:10.1088/1742-6596/664/6/062030](#) | 2015-12-23

1077. Using the GlideinWMS System as a Common Resource Provisioning Layer in CMS

Balcas, J. et al.

[J.Phys.Conf.Ser. 664 \(2015\) 062031](#) | [DOI:10.1088/1742-6596/664/6/062031](#) | 2015-12-23

1078. Study of Z boson production in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 36](#) | [arXiv:1512.06461](#) | [DOI:10.1016/j.physletb.2016.05.044](#) | 2015-12-20

1079. Measurement of the inclusive jet cross section in pp collisions at $\sqrt{s} = 2.76$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 265](#) | [arXiv:1512.06212](#) | [DOI:10.1140/epjc/s10052-016-4083-z](#) | 2015-12-19

1080. Search for narrow resonances decaying to dijets in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 071801](#) | [arXiv:1512.01224](#) | [DOI:10.1103/PhysRevLett.116.071801](#) | 2015-12-03

1081. Event generator tunes obtained from underlying event and multiparton scattering measurements

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 155](#) | [arXiv:1512.00815](#) | [DOI:10.1140/epjc/s10052-016-3988-x](#) | 2015-12-02

1082. Search for dark matter and unparticles produced in association with a Z boson in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 052011](#) | [arXiv:1511.09375](#) | [DOI:10.1103/PhysRevD.93.052011](#) | 2015-11-30

1083. Measurement of Spin Correlations in $t\bar{t}$ Production using the Matrix Element Method in the Muon+Jets Final State in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 321](#) | [arXiv:1511.06170](#) | [DOI:10.1016/j.physletb.2016.05.005](#) | 2015-11-19

1084. Search for Anomalous Single Top Quark Production in Association with a Photon in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 035](#) | [arXiv:1511.03951](#) | [DOI:10.1007/JHEP04\(2016\)035](#) | 2015-11-12

1085. Search for a Low-Mass Pseudoscalar Higgs Boson Produced in Association with a $b\bar{b}$ Pair in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 758 \(2016\) 296](#) | [arXiv:1511.03610](#) | [DOI:10.1016/j.physletb.2016.05.003](#) | 2015-11-11

1086. Measurement of Top Quark Polarisation in T-Channel Single Top Quark Production

Khachatryan, Vardan et al.

[JHEP 04 \(2016\) 073](#) | [arXiv:1511.02138](#) | [DOI:10.1007/JHEP04\(2016\)073](#) | 2015-11-06

1087. Search for Excited Leptons in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2016\) 125](#) | [arXiv:1511.01407](#) | [DOI:10.1007/JHEP03\(2016\)125](#) | 2015-11-04

1088. High Energy Physics Forum for Computational Excellence: Working Group Reports (I. Applications Software II. Software Libraries and Tools III. Systems)

Habib, Salman et al.

[arXiv:1510.08545](#) | 2015-10-28

1089. Reconstruction and identification of τ lepton decays to hadrons and ν_τ at CMS

Khachatryan, Vardan et al.

[JINST 11 \(2016\) P01019](#) | [arXiv:1510.07488](#) | [DOI:10.1088/1748-0221/11/01/P01019](#) | 2015-10-26

1090. Search for a very light NMSSM Higgs boson produced in decays of the 125 GeV scalar boson and decaying into τ leptons in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2016\) 079](#) | [arXiv:1510.06534](#) | [DOI:10.1007/JHEP01\(2016\)079](#) | 2015-10-22

1091. Measurement of the top quark pair production cross section in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 052002](#) | [arXiv:1510.05302](#) | [DOI:10.1103/PhysRevLett.116.052002](#) | 2015-10-18

1092. Search for a light charged Higgs boson decaying to $c\bar{s}$ in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 12 \(2015\) 178](#) | [arXiv:1510.04252](#) | [DOI:10.1007/JHEP12\(2015\)178](#) | 2015-10-14

1093. Transverse momentum spectra of inclusive b jets in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 754 \(2016\) 59](#) | [arXiv:1510.03373](#) | [DOI:10.1016/j.physletb.2016.01.010](#) | 2015-10-12

1094. Measurement of long-range near-side two-particle angular correlations in pp collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 172302](#) | [arXiv:1510.03068](#) | [DOI:10.1103/PhysRevLett.116.172302](#) | 2015-10-11

1095. Measurement of $t\bar{t}$ production with additional jet activity, including b quark jets, in the dilepton decay channel using pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 379](#) | [arXiv:1510.03072](#) | [DOI:10.1140/epjc/s10052-016-4105-x](#) | 2015-10-11

1096. Observation of top quark pairs produced in association with a vector boson in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2016\) 096](#) | [arXiv:1510.01131](#) | [DOI:10.1007/JHEP01\(2016\)096](#) | 2015-10-05

1097. Searches for a heavy scalar boson H decaying to a pair of 125 GeV Higgs bosons hh or for a heavy pseudoscalar boson A decaying to Zh , in the final states with $h \rightarrow \tau\tau$

Khachatryan, Vardan et al.

[Phys.Lett.B 755 \(2016\) 217](#) | [arXiv:1510.01181](#) | [DOI:10.1016/j.physletb.2016.01.056](#) | 2015-10-05

1098. Measurement of transverse momentum relative to dijet systems in PbPb and pp collisions at $\sqrt{s_{NN}}=2.76$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2016\) 006](#) | [arXiv:1509.09029](#) | [DOI:10.1007/JHEP01\(2016\)006](#) | 2015-09-30

1099. Search for the production of an excited bottom quark decaying to tW in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2016\) 166](#) | [arXiv:1509.08141](#) | [DOI:10.1007/JHEP01\(2016\)166](#) | 2015-09-27

1100. Search for the associated production of a Higgs boson with a single top quark in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 06 \(2016\) 177](#) | [arXiv:1509.08159](#) | [DOI:10.1007/JHEP06\(2016\)177](#) | 2015-09-27

1101. Search for $W' \rightarrow tb$ in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2016\) 122](#) | [arXiv:1509.06051](#) | [DOI:10.1007/JHEP02\(2016\)122](#) | 2015-09-20

1102. Measurement of the $t\bar{t}$ production cross section in the all-jets final state in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 128](#) | [arXiv:1509.06076](#) | [DOI:10.1140/epjc/s10052-016-3956-5](#) | 2015-09-20

1103. Measurement of the top quark mass using proton-proton data at $\sqrt{s} = 7$ and 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 072004](#) | [arXiv:1509.04044](#) | [DOI:10.1103/PhysRevD.93.072004](#) | 2015-09-14

1104. Search for vector-like charge $2/3$ T quarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 012003](#) | [arXiv:1509.04177](#) | [DOI:10.1103/PhysRevD.93.012003](#) | 2015-09-14

1105. Measurement of the inelastic cross section in proton-lead collisions at $\sqrt{s_{NN}}= 5.02$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 759 \(2016\) 641](#) | [arXiv:1509.03893](#) | [DOI:10.1016/j.physletb.2016.06.027](#) | 2015-09-13

1106. Search for pair production of first and second generation leptoquarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 032004](#) | [arXiv:1509.03744](#) | [DOI:10.1103/PhysRevD.93.032004](#) | 2015-09-12

1107. Search for single production of scalar leptoquarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 032005](#) | [arXiv:1509.03750](#) | [DOI:10.1103/PhysRevD.93.032005](#) | 2015-09-12

1108. Search for a charged Higgs boson in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 11 \(2015\) 018](#) | [arXiv:1508.07774](#) | [DOI:10.1007/JHEP11\(2015\)018](#) | 2015-08-31

1109. Measurement of differential cross sections for Higgs boson production in the diphoton decay channel in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 13](#) | [arXiv:1508.07819](#) | [DOI:10.1140/epjc/s10052-015-3853-3](#) | 2015-08-31

1110. Search for supersymmetry in the vector-boson fusion topology in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 11 \(2015\) 189](#) | [arXiv:1508.07628](#) | [DOI:10.1007/JHEP11\(2015\)189](#) | 2015-08-30

1111. Study of B Meson Production in p+Pb Collisions at $\sqrt{s_{NN}}=5.02$ TeV Using Exclusive Hadronic Decays

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 116 \(2016\) 032301](#) | [arXiv:1508.06678](#) | [DOI:10.1103/PhysRevLett.116.032301](#) | 2015-08-26

1112. Search for W' decaying to tau lepton and neutrino in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 755 \(2016\) 196](#) | [arXiv:1508.04308](#) | [DOI:10.1016/j.physletb.2016.02.002](#) | 2015-08-18

1113. Measurement of the charge asymmetry in top quark pair production in pp collisions at $\sqrt{s} = 8$ TeV using a template method

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 034014](#) | [arXiv:1508.03862](#) | [DOI:10.1103/PhysRevD.93.034014](#) | 2015-08-16

1114. Search for neutral MSSM Higgs bosons decaying to $\mu^+ \mu^-$ in pp collisions at $\sqrt{s} = 7$ and 8 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 752 \(2016\) 221](#) | [arXiv:1508.01437](#) | [DOI:10.1016/j.physletb.2015.11.042](#) | 2015-08-06

1115. Search for supersymmetry in events with a photon, a lepton, and missing transverse momentum in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 757 \(2016\) 6](#) | [arXiv:1508.01218](#) | [DOI:10.1016/j.physletb.2016.03.039](#) | 2015-08-05

1116. Angular analysis of the decay $B^0 \rightarrow K^{*0} \mu^+ \mu^-$ from pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 753 \(2016\) 424](#) | [arXiv:1507.08126](#) | [DOI:10.1016/j.physletb.2015.12.020](#) | 2015-07-29

1117. Measurement of the CP-violating weak phase ϕ_s and the decay width difference $\Delta\Gamma_s$ using the $B_s^0 \rightarrow J/\psi\phi(1020)$ decay channel in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 757 \(2016\) 97](#) | [arXiv:1507.07527](#) | [DOI:10.1016/j.physletb.2016.03.046](#) | 2015-07-27

1118. Measurement of the underlying event activity using charged-particle jets in proton-proton collisions at $\sqrt{s} = 2.76$ TeV

Khachatryan, Vardan et al.

[JHEP 09 \(2015\) 137](#) | [arXiv:1507.07229](#) | [DOI:10.1007/JHEP09\(2015\)137](#) | 2015-07-26

1119. Search for pair-produced vectorlike B quarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 112009](#) | [arXiv:1507.07129](#) | [DOI:10.1103/PhysRevD.93.112009](#) | 2015-07-25

1120. Limits on the Higgs boson lifetime and width from its decay to four charged leptons

Khachatryan, Vardan et al.

[Phys.Rev.D 92 \(2015\) 072010](#) | [arXiv:1507.06656](#) | [DOI:10.1103/PhysRevD.92.072010](#) | 2015-07-23

1121. Pseudorapidity distribution of charged hadrons in proton-proton collisions at $\sqrt{s} = 13$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 751 \(2015\) 143](#) | [arXiv:1507.05915](#) | [DOI:10.1016/j.physletb.2015.10.004](#) | 2015-07-21

1122. Measurement of the W^+W^- cross section in pp collisions at $\sqrt{s} = 8$ TeV and limits on anomalous gauge couplings

Khachatryan, Vardan et al.

[Eur.Phys.J.C 76 \(2016\) 401](#) | [arXiv:1507.03268](#) | [DOI:10.1140/epjc/s10052-016-4219-1](#) | 2015-07-12

1123. Inclusive and differential measurements of the $t\bar{t}$ charge asymmetry in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 757 \(2016\) 154](#) | [arXiv:1507.03119](#) | [DOI:10.1016/j.physletb.2016.03.060](#) | 2015-07-11

1124. Search for supersymmetry with photons in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 92 \(2015\) 072006](#) | [arXiv:1507.02898](#) | [DOI:10.1103/PhysRevD.92.072006](#) | 2015-07-10

1125. Search for a Higgs boson decaying into $\gamma^* \gamma \rightarrow \ell\ell\gamma$ with low dilepton mass in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 753 \(2016\) 341](#) | [arXiv:1507.03031](#) | [DOI:10.1016/j.physletb.2015.12.039](#) | 2015-07-10

1126. Production of leading charged particles and leading charged-particle jets at small transverse momenta in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 92 \(2015\) 112001](#) | [arXiv:1507.00233](#) | [DOI:10.1103/PhysRevD.92.112001](#) | 2015-07-01

1127. Search for exotic decays of a Higgs boson into undetectable particles and one or more photons

Khachatryan, Vardan et al.

[Phys.Lett.B 753 \(2016\) 363](#) | [arXiv:1507.00359](#) | [DOI:10.1016/j.physletb.2015.12.017](#) | 2015-07-01

1128. Search for neutral MSSM Higgs bosons decaying into a pair of bottom quarks

Khachatryan, Vardan et al.

[JHEP 11 \(2015\) 071](#) | [arXiv:1506.08329](#) | [DOI:10.1007/JHEP11\(2015\)071](#) | 2015-06-27

1129. Search for resonant $t\bar{t}$ production in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 93 \(2016\) 012001](#) | [arXiv:1506.03062](#) | [DOI:10.1103/PhysRevD.93.012001](#) | 2015-06-09

1130. Search for diphoton resonances in the mass range from 150 to 850 GeV in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 750 \(2015\) 494](#) | [arXiv:1506.02301](#) | [DOI:10.1016/j.physletb.2015.09.062](#) | 2015-06-07

1131. Search for a massive resonance decaying into a Higgs boson and a W or Z boson in hadronic final states in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2016\) 145](#) | [arXiv:1506.01443](#) | [DOI:10.1007/JHEP02\(2016\)145](#) | 2015-06-03

1132. Search for the standard model Higgs boson produced through vector boson fusion and decaying to $b\bar{b}$

Khachatryan, Vardan et al.

[Phys.Rev.D 92 \(2015\) 032008](#) | [arXiv:1506.01010](#) | [DOI:10.1103/PhysRevD.92.032008](#) | 2015-06-02

1133. A search for pair production of new light bosons decaying into muons

Khachatryan, V. et al.

[Phys.Lett.B 752 \(2016\) 146](#) | [arXiv:1506.00424](#) | [DOI:10.1016/j.physletb.2015.10.067](#) | 2015-06-01

1134. Search for neutral color-octet weak-triplet scalar particles in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 09 \(2015\) 201](#) | [arXiv:1505.08118](#) | [DOI:10.1007/JHEP09\(2015\)201](#) | 2015-05-29

1135. Comparison of the $Z/\gamma^* + \text{jets}$ to $\gamma + \text{jets}$ cross sections in pp collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 10 \(2015\) 128](#) | [arXiv:1505.06520](#) | [DOI:10.1007/JHEP04\(2016\)010](#) | 2015-05-24

1136. Measurement of the differential cross section for top quark pair production in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 542](#) | [arXiv:1505.04480](#) | [DOI:10.1140/epjc/s10052-015-3709-x](#) | 2015-05-17

1137. Search for a pseudoscalar boson decaying into a Z boson and the 125 GeV Higgs boson in $\ell^+\ell^-bb$ final states

Khachatryan, Vardan et al.

[Phys.Lett.B 748 \(2015\) 221](#) | [arXiv:1504.04710](#) | [DOI:10.1016/j.physletb.2015.07.010](#) | 2015-04-18

1138. Measurement of the Z boson differential cross section in transverse momentum and rapidity in proton-proton collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 749 \(2015\) 187](#) | [arXiv:1504.03511](#) | [DOI:10.1016/j.physletb.2015.07.065](#) | 2015-04-14

1139. Angular coefficients of Z bosons produced in pp collisions at $\sqrt{s} = 8$ TeV and decaying to $\mu^+\mu^-$ as a function of transverse momentum and rapidity

Khachatryan, Vardan et al.

[Phys.Lett.B 750 \(2015\) 154](#) | [arXiv:1504.03512](#) | [DOI:10.1016/j.physletb.2015.08.061](#) | 2015-04-14

1140. Search for the production of dark matter in association with top-quark pairs in the single-lepton final state in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 06 \(2015\) 121](#) | [arXiv:1504.03198](#) | [DOI:10.1007/JHEP06\(2015\)121](#) | 2015-04-13

1141. Search for a Higgs boson in the mass range from 145 to 1000 GeV decaying to a pair of W or Z bosons

Khachatryan, Vardan et al.

[JHEP 10 \(2015\) 144](#) | [arXiv:1504.00936](#) | [DOI:10.1007/JHEP10\(2015\)144](#) | 2015-04-03

1142. Search for Third-Generation Scalar Leptoquarks in the $t\tau$ Channel in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 07 \(2015\) 042](#) | [arXiv:1503.09049](#) | [DOI:10.1007/JHEP11\(2016\)056](#) | 2015-03-31

1143. Measurement of diffraction dissociation cross sections in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 92 \(2015\) 012003](#) | [arXiv:1503.08689](#) | [DOI:10.1103/PhysRevD.92.012003](#) | 2015-03-30

1144. Searches for third-generation squark production in fully hadronic final states in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 06 \(2015\) 116](#) | [arXiv:1503.08037](#) | [DOI:10.1007/JHEP06\(2015\)116](#) | 2015-03-27

1145. Combined Measurement of the Higgs Boson Mass in pp Collisions at $\sqrt{s}=7$ and 8 TeV with the ATLAS and CMS Experiments

Aad, Georges et al.

[Phys.Rev.Lett. 114 \(2015\) 191803](#) | [arXiv:1503.07589](#) | [DOI:10.1103/PhysRevLett.114.191803](#) | 2015-03-26

1146. Study of W boson production in pPb collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 750 \(2015\) 565](#) | [arXiv:1503.05825](#) | [DOI:10.1016/j.physletb.2015.09.057](#) | 2015-03-19

1147. Measurements of the ZZ production cross sections in the $2\ell 2\nu$ channel in proton-proton collisions at $\sqrt{s} = 7$ and 8 TeV and combined constraints on triple gauge

couplings

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 511](#) | [arXiv:1503.05467](#) | [DOI:10.1140/epjc/s10052-015-3706-0](#) | 2015-03-18

1148. Search for resonant pair production of Higgs bosons decaying to two bottom quark-antiquark pairs in proton-proton collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 749 \(2015\) 560](#) | [arXiv:1503.04114](#) | [DOI:10.1016/j.physletb.2015.08.047](#) | 2015-03-13

1149. Search for vector-like T quarks decaying to top quarks and Higgs bosons in the all-hadronic channel using jet substructure

Khachatryan, Vardan et al.

[JHEP 06 \(2015\) 080](#) | [arXiv:1503.01952](#) | [DOI:10.1007/JHEP06\(2015\)080](#) | 2015-03-06

1150. Evidence for transverse momentum and pseudorapidity dependent event plane fluctuations in PbPb and pPb collisions

Khachatryan, Vardan et al.

[Phys.Rev.C 92 \(2015\) 034911](#) | [arXiv:1503.01692](#) | [DOI:10.1103/PhysRevC.92.034911](#) | 2015-03-05

1151. Study of Final-State Radiation in Decays of Z Bosons Produced in pp Collisions at 7 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 092012](#) | [arXiv:1502.07940](#) | [DOI:10.1103/PhysRevD.91.092012](#) | 2015-02-27

1152. Search for Lepton-Flavour-Violating Decays of the Higgs Boson

Khachatryan, Vardan et al.

[Phys.Lett.B 749 \(2015\) 337](#) | [arXiv:1502.07400](#) | [DOI:10.1016/j.physletb.2015.07.053](#) | 2015-02-25

1153. Search for Physics Beyond the Standard Model in Events with Two Leptons, Jets, and Missing Transverse Momentum in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2015\) 124](#) | [arXiv:1502.06031](#) | [DOI:10.1007/JHEP04\(2015\)124](#) | 2015-02-20

1154. Measurement of the $Z\gamma$ Production Cross Section in pp Collisions at 8 TeV and Search for Anomalous Triple Gauge Boson Couplings

Khachatryan, Vardan et al.

[JHEP 04 \(2015\) 164](#) | [arXiv:1502.05664](#) | [DOI:10.1007/JHEP04\(2015\)164](#) | 2015-02-19

1155. Evidence for Collective Multiparticle Correlations in p-Pb Collisions

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 115 \(2015\) 012301](#) | [arXiv:1502.05382](#) | [DOI:10.1103/PhysRevLett.115.012301](#) | 2015-02-18

1156. Nuclear Effects on the Transverse Momentum Spectra of Charged Particles in pPb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 237](#) | [arXiv:1502.05387](#) | [DOI:10.1140/epjc/s10052-015-3435-4](#) | 2015-02-18

1157. Search for Narrow High-Mass Resonances in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV Decaying to a Z and a Higgs Boson

Khachatryan, Vardan et al.

[Phys.Lett.B 748 \(2015\) 255](#) | [arXiv:1502.04994](#) | [DOI:10.1016/j.physletb.2015.07.011](#) | 2015-02-17

1158. Distributions of Topological Observables in Inclusive Three- and Four-Jet Events in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 302](#) | [arXiv:1502.04785](#) | [DOI:10.1140/epjc/s10052-015-3491-9](#) | 2015-02-16

1159. Searches for Supersymmetry using the M_{T2} Variable in Hadronic Events Produced in pp Collisions at 8 TeV

Khachatryan, Vardan et al.

[JHEP 05 \(2015\) 078](#) | [arXiv:1502.04358](#) | [DOI:10.1007/JHEP05\(2015\)078](#) | 2015-02-15

1160. Measurement of J/ψ and $\psi(2S)$ Prompt Double-Differential Cross Sections in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 191802](#) | [arXiv:1502.04155](#) | [DOI:10.1103/PhysRevLett.114.191802](#) | 2015-02-13

1161. Search for a Standard Model Higgs Boson Produced in Association with a Top-Quark Pair and Decaying to Bottom Quarks Using a Matrix Element Method

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 251](#) | [arXiv:1502.02485](#) | [DOI:10.1140/epjc/s10052-015-3454-1](#) | 2015-02-09

1162. Constraints on the pMSSM, AMSB model and on other models from the search for long-lived charged particles in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 325](#) | [arXiv:1502.02522](#) | [DOI:10.1140/epjc/s10052-015-3533-3](#) | 2015-02-09

1163. Performance of Electron Reconstruction and Selection with the CMS Detector in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JINST 10 \(2015\) P06005](#) | [arXiv:1502.02701](#) | [DOI:10.1088/1748-0221/10/06/P06005](#) | 2015-02-09

1164. Performance of Photon Reconstruction and Identification with the CMS Detector in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JINST 10 \(2015\) P08010](#) | [arXiv:1502.02702](#) | [DOI:10.1088/1748-0221/10/08/P08010](#) | 2015-02-09

1165. Search for Supersymmetry Using Razor Variables in Events with b-Tagged Jets in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 052018](#) | [arXiv:1502.00300](#) | [DOI:10.1103/PhysRevD.91.052018](#) | 2015-02-01

1166. Measurements of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ differential cross sections in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 749 \(2015\) 14](#) | [arXiv:1501.07750](#) | [DOI:10.1016/j.physletb.2015.07.037](#) | 2015-01-30

1167. Measurement of the ratio $B(B^0_s \rightarrow J/\psi f_0(980)) / B(B^0_s \rightarrow J/\psi \phi(1020))$ in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 756 \(2016\) 84](#) | [arXiv:1501.06089](#) | [DOI:10.1016/j.physletb.2016.02.047](#) | 2015-01-24

1168. Search for heavy Majorana neutrinos in $\mu^+ \mu^- +$ jets events in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 748 \(2015\) 144](#) | [arXiv:1501.05566](#) | [DOI:10.1016/j.physletb.2015.06.070](#) | 2015-01-22

1169. Search for Decays of Stopped Long-Lived Particles Produced in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 151](#) | [arXiv:1501.05603](#) | [DOI:10.1140/epjc/s10052-015-3367-z](#) | 2015-01-22

1170. Search for resonances and quantum black holes using dijet mass spectra in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 052009](#) | [arXiv:1501.04198](#) | [DOI:10.1103/PhysRevD.91.052009](#) | 2015-01-17

1171. Precise determination of the mass of the Higgs boson and tests of compatibility of its couplings with the standard model predictions using proton collisions at 7 and 8 TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 212](#) | [arXiv:1412.8662](#) | [DOI:10.1140/epjc/s10052-015-3351-7](#) | 2014-12-30

1172. Search for pair-produced resonances decaying to jet pairs in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 747 \(2015\) 98](#) | [arXiv:1412.7706](#) | [DOI:10.1016/j.physletb.2015.04.045](#) | 2014-12-24

1173. Search for physics beyond the standard model in dilepton mass spectra in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, Vardan et al.

[JHEP 04 \(2015\) 025](#) | [arXiv:1412.6302](#) | [DOI:10.1007/JHEP04\(2015\)025](#) | 2014-12-19

1174. Searches for supersymmetry based on events with b jets and four W bosons in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 745 \(2015\) 5](#) | [arXiv:1412.4109](#) | [DOI:10.1016/j.physletb.2015.04.002](#) | 2014-12-12

1175. Measurement of the inclusive 3-jet production differential cross section in proton-proton collisions at 7 TeV and determination of the strong coupling constant in the TeV range

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 186](#) | [arXiv:1412.1633](#) | [DOI:10.1140/epjc/s10052-015-3376-y](#) | 2014-12-04

1176. Measurements of differential and double-differential Drell-Yan cross sections in proton-proton collisions at 8 TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 147](#) | [arXiv:1412.1115](#) | [DOI:10.1140/epjc/s10052-015-3364-2](#) | 2014-12-02

1177. Search for stealth supersymmetry in events with jets, either photons or leptons, and low missing transverse momentum in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 743 \(2015\) 503](#) | [arXiv:1411.7255](#) | [DOI:10.1016/j.physletb.2015.03.017](#) | 2014-11-26

1178. Search for long-lived particles that decay into final states containing two electrons or two muons in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 052012](#) | [arXiv:1411.6977](#) | [DOI:10.1103/PhysRevD.91.052012](#) | 2014-11-25

1179. Search for Long-Lived Neutral Particles Decaying to Quark-Antiquark Pairs in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 012007](#) | [arXiv:1411.6530](#) | [DOI:10.1103/PhysRevD.91.012007](#) | 2014-11-24

1180. Search for disappearing tracks in proton-proton collisions at $\sqrt{s}=8$ TeV

Khachatryan, V. et al.

[JHEP 01 \(2015\) 096](#) | [arXiv:1411.6006](#) | [DOI:10.1007/JHEP01\(2015\)096](#) | 2014-11-21

1181. Measurement of the cross section ratio $\sigma_{t\bar{t}b\bar{b}} / \sigma_{t\bar{t}jj}$ in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 746 \(2015\) 132](#) | [arXiv:1411.5621](#) | [DOI:10.1016/j.physletb.2015.04.060](#) | 2014-11-20

1182. Observation of the rare $B^0_s \rightarrow \mu^+ \mu^-$ decay from the combined analysis of CMS and LHCb data

Khachatryan, Vardan et al.

[Nature 522 \(2015\) 68](#) | [arXiv:1411.4413](#) | [DOI:10.1038/nature14474](#) | 2014-11-17

1183. Constraints on the spin-parity and anomalous HVV couplings of the Higgs boson in proton collisions at 7 and 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 92 \(2015\) 012004](#) | [arXiv:1411.3441](#) | [DOI:10.1103/PhysRevD.92.012004](#) | 2014-11-12

1184. Search for quark contact interactions and extra spatial dimensions using dijet angular distributions in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 746 \(2015\) 79](#) | [arXiv:1411.2646](#) | [DOI:10.1016/j.physletb.2015.04.042](#) | 2014-11-10

1185. Performance of the CMS missing transverse momentum reconstruction in pp data at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JINST 10 \(2015\) P02006](#) | [arXiv:1411.0511](#) | [DOI:10.1088/1748-0221/10/02/P02006](#) | 2014-11-03

1186. Search for new phenomena in monophoton final states in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 755 \(2016\) 102](#) | [arXiv:1410.8812](#) | [DOI:10.1016/j.physletb.2016.01.057](#) | 2014-10-31

1187. Search for a standard model-like Higgs boson in the $\mu^+ \mu^-$ and $e^+ e^-$ decay channels at the LHC

Khachatryan, Vardan et al.

[Phys.Lett.B 744 \(2015\) 184](#) | [arXiv:1410.6679](#) | [DOI:10.1016/j.physletb.2015.03.048](#) | 2014-10-24

1188. Constraints on parton distribution functions and extraction of the strong coupling constant from the inclusive jet cross section in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 288](#) | [arXiv:1410.6765](#) | [DOI:10.1140/epjc/s10052-015-3499-1](#) | 2014-10-24

1189. Study of vector boson scattering and search for new physics in events with two same-sign leptons and two jets

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 051801](#) | [arXiv:1410.6315](#) | [DOI:10.1103/PhysRevLett.114.051801](#) | 2014-10-23

1190. Measurement of the ratio of the production cross sections times branching fractions of $B_c^{\pm} \rightarrow J/\psi \pi^{\pm}$ and $B^{\pm} \rightarrow J/\psi K^{\pm}$ and $\mathcal{B}(B_c^{\pm} \rightarrow J/\psi \pi^{\pm} \pi^{\pm} \pi^{\mp}) / \mathcal{B}(B_c^{\pm} \rightarrow J/\psi \pi^{\pm})$ in pp collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2015\) 063](#) | [arXiv:1410.5729](#) | [DOI:10.1007/JHEP01\(2015\)063](#) | 2014-10-21

1191. Study of Z production in PbPb and pp collisions at $\sqrt{s_{NN}} = 2.76$ TeV in the dimuon and dielectron decay channels

Chatrchyan, Serguei et al.

[JHEP 03 \(2015\) 022](#) | [arXiv:1410.4825](#) | [DOI:10.1007/JHEP03\(2015\)022](#) | 2014-10-17

1192. Identification techniques for highly boosted W bosons that decay into hadrons

Khachatryan, Vardan et al.

[JHEP 12 \(2014\) 017](#) | [arXiv:1410.4227](#) | [DOI:10.1007/JHEP12\(2014\)017](#) | 2014-10-15

1193. Measurement of electroweak production of two jets in association with a Z boson in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 66](#) | [arXiv:1410.3153](#) | [DOI:10.1140/epjc/s10052-014-3232-5](#) | 2014-10-12

1194. Searches for heavy Higgs bosons in two-Higgs-doublet models and for t decay using multilepton and diphoton final states in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 90 \(2014\) 112013](#) | [arXiv:1410.2751](#) | [DOI:10.1103/PhysRevD.90.112013](#) | 2014-10-10

1195. Measurement of Prompt $\psi(2S) \rightarrow J/\psi$ Yield Ratios in Pb-Pb and p-p Collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 113 \(2014\) 262301](#) | [arXiv:1410.1804](#) | [DOI:10.1103/PhysRevLett.113.262301](#) | 2014-10-07

1196. Search for Monotop Signatures in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 101801](#) | [arXiv:1410.1149](#) | [DOI:10.1103/PhysRevLett.114.101801](#) | 2014-10-05

1197. Measurement of the W boson helicity in events with a single reconstructed top quark in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2015\) 053](#) | [arXiv:1410.1154](#) | [DOI:10.1007/JHEP01\(2015\)053](#) | 2014-10-05

1198. Search for Standard Model Production of Four Top Quarks in the Lepton + Jets Channel in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 11 \(2014\) 154](#) | [arXiv:1409.7339](#) | [DOI:10.1007/JHEP11\(2014\)154](#) | 2014-09-25

1199. Measurement of the production cross section ratio $\sigma(Xb2(1P)) / \sigma(Xb1(1P))$ in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 743 \(2015\) 383](#) | [arXiv:1409.5761](#) | [DOI:10.1016/j.physletb.2015.02.048](#) | 2014-09-19

1200. Search for Displaced Supersymmetry in events with an electron and a muon with large impact parameters

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 114 \(2015\) 061801](#) | [arXiv:1409.4789](#) | [DOI:10.1103/PhysRevLett.114.061801](#) | 2014-09-16

1201. Long-range two-particle correlations of strange hadrons with charged particles in pPb and PbPb collisions at LHC energies

Khachatryan, Vardan et al.

[Phys.Lett.B 742 \(2015\) 200](#) | [arXiv:1409.3392](#) | [DOI:10.1016/j.physletb.2015.01.034](#) | 2014-09-11

1202. Searches for electroweak neutralino and chargino production in channels with Higgs, Z, and W bosons in pp collisions at 8 TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 90 \(2014\) 092007](#) | [arXiv:1409.3168](#) | [DOI:10.1103/PhysRevD.90.092007](#) | 2014-09-10

1203. Search for dark matter, extra dimensions, and unparticles in monojet events in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 75 \(2015\) 235](#) | [arXiv:1408.3583](#) | [DOI:10.1140/epjc/s10052-015-3451-4](#) | 2014-08-15

1204. Search for neutral MSSM Higgs bosons decaying to a pair of tau leptons in pp collisions

Khachatryan, Vardan et al.

[JHEP 10 \(2014\) 160](#) | [arXiv:1408.3316](#) | [DOI:10.1007/JHEP10\(2014\)160](#) | 2014-08-14

1205. Measurements of jet multiplicity and differential production cross sections of Z + jets events in proton-proton collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 052008](#) | [arXiv:1408.3104](#) | [DOI:10.1103/PhysRevD.91.052008](#) | 2014-08-13

1206. Search for physics beyond the standard model in final states with a lepton and missing transverse energy in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 91 \(2015\) 092005](#) | [arXiv:1408.2745](#) | [DOI:10.1103/PhysRevD.91.092005](#) | 2014-08-12

1207. Search for the associated production of the Higgs boson with a top-quark pair

Khachatryan, Vardan et al.

[JHEP 09 \(2014\) 087](#) | [arXiv:1408.1682](#) | [DOI:10.1007/JHEP09\(2014\)087](#) | 2014-08-07

1208. Search for pair production of third-generation scalar leptoquarks and top squarks in proton-proton collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 739 \(2014\) 229](#) | [arXiv:1408.0806](#) | [DOI:10.1016/j.physletb.2014.10.063](#) | 2014-08-04

1209. Measurement of the $t\bar{t}$ Production Cross Section in pp Collisions at $\sqrt{s} = 8$ TeV in Dilepton Final States Containing One τ Lepton

Khachatryan, Vardan et al.

[Phys.Lett.B 739 \(2014\) 23](#) | [arXiv:1407.6643](#) | [DOI:10.1016/j.physletb.2014.10.032](#) | 2014-07-24

1210. Search for Heavy Neutrinos and W Bosons with Right-Handed Couplings in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 74 \(2014\) 3149](#) | [arXiv:1407.3683](#) | [DOI:10.1140/epjc/s10052-014-3149-z](#) | 2014-07-14

1211. Search for New Resonances Decaying via WZ to Leptons in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 740 \(2015\) 83](#) | [arXiv:1407.3476](#) | [DOI:10.1016/j.physletb.2014.11.026](#) | 2014-07-13

1212. Study of Hadronic Event-Shape Variables in Multijet Final States in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[JHEP 10 \(2014\) 087](#) | [arXiv:1407.2856](#) | [DOI:10.1007/JHEP10\(2014\)087](#) | 2014-07-10

1213. Observation of the Diphoton Decay of the Higgs Boson and Measurement of Its Properties

Khachatryan, Vardan et al.

[Eur.Phys.J.C 74 \(2014\) 3076](#) | [arXiv:1407.0558](#) | [DOI:10.1140/epjc/s10052-014-3076-z](#) | 2014-07-02

1214. Measurement of Top Quark-Antiquark Pair Production in Association with a W or Z Boson in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 74 \(2014\) 3060](#) | [arXiv:1406.7830](#) | [DOI:10.1140/epjc/s10052-014-3060-7](#) | 2014-06-30

1215. Differential Cross Section Measurements for the Production of a W Boson in Association with Jets in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 741 \(2015\) 12](#) | [arXiv:1406.7533](#) | [DOI:10.1016/j.physletb.2014.12.003](#) | 2014-06-29

1216. Search for Excited Quarks in the γ + Jet Final State in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 738 \(2014\) 274](#) | [arXiv:1406.5171](#) | [DOI:10.1016/j.physletb.2014.09.048](#) | 2014-06-19

1217. Measurement of Jet Fragmentation in PbPb and pp Collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.C 90 \(2014\) 024908](#) | [arXiv:1406.0932](#) | [DOI:10.1103/PhysRevC.90.024908](#) | 2014-06-03

1218. Measurement of the Ratio of Inclusive Jet Cross Sections using the Anti-k_T Algorithm with Radius Parameters R=0.5 and 0.7 in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 072006](#) | [arXiv:1406.0324](#) | [DOI:10.1103/PhysRevD.90.072006](#) | 2014-06-02

1219. Measurement of Prompt J/ψ Pair Production in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[JHEP 09 \(2014\) 094](#) | [arXiv:1406.0484](#) | [DOI:10.1007/JHEP09\(2014\)094](#) | 2014-06-02

1220. Measurement of the pp → ZZ Production Cross Section and Constraints on Anomalous Triple Gauge Couplings in Four-Lepton Final States at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 740 \(2015\) 250](#) | [arXiv:1406.0113](#) | [DOI:10.1016/j.physletb.2016.04.010](#) | 2014-05-31

1221. Searches for Electroweak Production of Charginos, Neutralinos, and Staleptons Decaying to Leptons and W, Z, and Higgs Bosons in pp Collisions at 8 TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 74 \(2014\) 3036](#) | [arXiv:1405.7570](#) | [DOI:10.1140/epjc/s10052-014-3036-7](#) | 2014-05-29

1222. Search for Jet Extinction in the Inclusive Jet-p_t Spectrum from Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 90 \(2014\) 032005](#) | [arXiv:1405.7653](#) | [DOI:10.1103/PhysRevD.90.032005](#) | 2014-05-29

1223. Measurement of Differential Cross Sections for the Production of a Pair of Isolated Photons in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 3129](#) | [arXiv:1405.7225](#) | [DOI:10.1140/epjc/s10052-014-3129-3](#) | 2014-05-28

1224. Description and Performance of Track and Primary-Vertex Reconstruction with the CMS Tracker

Chatrchyan, Serguei et al.

[JINST 9 \(2014\) P10009](#) | [arXiv:1405.6569](#) | [DOI:10.1088/1748-0221/9/10/P10009](#) | 2014-05-26

1225. Search for Top-Squark Pairs Decaying into Higgs or Z Bosons in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 736 \(2014\) 371](#) | [arXiv:1405.3886](#) | [DOI:10.1016/j.physletb.2014.07.053](#) | 2014-05-15

1226. Search for Supersymmetry with Razor Variables in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 112001](#) | [arXiv:1405.3961](#) | [DOI:10.1103/PhysRevD.90.112001](#) | 2014-05-15

1227. Search for Massive Resonances Decaying into Pairs of Boosted Bosons in Semi-Leptonic Final States at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 08 \(2014\) 174](#) | [arXiv:1405.3447](#) | [DOI:10.1007/JHEP08\(2014\)174](#) | 2014-05-14

1228. Constraints on the Higgs Boson Width from Off-Shell Production and Decay to Z-Boson Pairs

Khachatryan, Vardan et al.

[Phys.Lett.B 736 \(2014\) 64](#) | [arXiv:1405.3455](#) | [DOI:10.1016/j.physletb.2014.06.077](#) | 2014-05-14

1229. Search for Massive Resonances in Dijet Systems Containing Jets Tagged as W or Z Boson Decays in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 08 \(2014\) 173](#) | [arXiv:1405.1994](#) | [DOI:10.1007/JHEP08\(2014\)173](#) | 2014-05-08

1230. Measurement of Pseudorapidity Distributions of Charged Particles in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV by the CMS and TOTEM Experiments

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 3053](#) | [arXiv:1405.0722](#) | [DOI:10.1140/epjc/s10052-014-3053-6](#) | 2014-05-04

1231. Search for Anomalous Production of Events with Three or More Leptons in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 032006](#) | [arXiv:1404.5801](#) | [DOI:10.1103/PhysRevD.90.032006](#) | 2014-04-23

1232. Search for WW γ and WZ γ Production and Constraints on Anomalous Quartic Gauge Couplings in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 032008](#) | [arXiv:1404.4619](#) | [DOI:10.1103/PhysRevD.90.032008](#) | 2014-04-17

1233. Measurement of Jet Multiplicity Distributions in tt Production in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2015\) 3014](#) | [arXiv:1404.3171](#) | [DOI:10.1140/epjc/s10052-014-3014-0](#) | 2014-04-11

1234. Measurement of the Ratio $\mathcal{B}(t \rightarrow Wb)/\mathcal{B}(t \rightarrow Wq)$ in pp Collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 736 \(2014\) 33](#) | [arXiv:1404.2292](#) | [DOI:10.1016/j.physletb.2014.06.076](#) | 2014-04-08

1235. Search for Invisible Decays of Higgs Bosons in the Vector Boson Fusion and Associated ZH Production Modes

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2980](#) | [arXiv:1404.1344](#) | [DOI:10.1140/epjc/s10052-014-2980-6](#) | 2014-04-04

1236. Measurement of the t-channel single-top-quark production cross section and of the $|V_{tb}|$ CKM matrix element in pp collisions at $\sqrt{s} = 8$ TeV

Khachatryan, Vardan et al.

[JHEP 06 \(2014\) 090](#) | [arXiv:1403.7366](#) | [DOI:10.1007/JHEP06\(2014\)090](#) | 2014-03-28

1237. Measurement of WZ and ZZ Production in pp Collisions at $\sqrt{s} = 8$ TeV in Final States with b-Tagged Jets

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2973](#) | [arXiv:1403.3047](#) | [DOI:10.1140/epjc/s10052-014-2973-5](#) | 2014-03-12

1238. Alignment of the CMS Tracker with LHC and Cosmic Ray Data

Chatrchyan, Serguei et al.

[JINST 9 \(2014\) P06009](#) | [arXiv:1403.2286](#) | [DOI:10.1088/1748-0221/9/06/P06009](#) | 2014-03-10

1239. Search for New Physics in the Multijet and Missing Transverse Momentum Final State in Proton-Proton Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2014\) 055](#) | [arXiv:1402.4770](#) | [DOI:10.1007/JHEP06\(2014\)055](#) | 2014-02-19

1240. Measurements of the $t\bar{t}$ Charge Asymmetry Using the Dilepton Decay Channel in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2014\) 191](#) | [arXiv:1402.3803](#) | [DOI:10.1007/JHEP04\(2014\)191](#) | 2014-02-16

1241. Search for $W' \rightarrow tb$ Decays in the Lepton + Jets Final State in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2014\) 108](#) | [arXiv:1402.2176](#) | [DOI:10.1007/JHEP05\(2014\)108](#) | 2014-02-10

1242. Measurement of the Production Cross Sections for a Z Boson and One or More b Jets in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2014\) 120](#) | [arXiv:1402.1521](#) | [DOI:10.1007/JHEP06\(2014\)120](#) | 2014-02-06

1243. Measurement of Inclusive W and Z Boson Production Cross Sections in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 191802](#) | [arXiv:1402.0923](#) | [DOI:10.1103/PhysRevLett.112.191802](#) | 2014-02-04

1244. Evidence for the Direct Decay of the 125 GeV Higgs Boson to Fermions

Chatrchyan, Serguei et al.

[Nature Phys. 10 \(2014\) 557](#) | [arXiv:1401.6527](#) | [DOI:10.1038/nphys3005](#) | 2014-01-25

1245. Determination of the Top-Quark Pole Mass and Strong Coupling Constant from the $t\bar{t}$ Production Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 728 \(2014\) 496](#) | [arXiv:1307.1907](#) | [DOI:10.1016/j.physletb.2013.12.009](#) | 2014-01-20

1246. Evidence for the 125 GeV Higgs Boson Decaying to a Pair of τ Leptons

Chatrchyan, Serguei et al.

[JHEP 05 \(2014\) 104](#) | [arXiv:1401.5041](#) | [DOI:10.1007/JHEP05\(2014\)104](#) | 2014-01-20

1247. Studies of Dijet Transverse Momentum Balance and Pseudorapidity Distributions in pPb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2951](#) | [arXiv:1401.4433](#) | [DOI:10.1140/epjc/s10052-014-2951-y](#) | 2014-01-17

1248. Observation of the Associated Production of a Single Top Quark and a W Boson in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 231802](#) | [arXiv:1401.2942](#) | [DOI:10.1103/PhysRevLett.112.231802](#) | 2014-01-13

1249. Measurement of the $t\bar{t}$ production cross section in the dilepton channel in pp collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[JHEP 02 \(2014\) 024](#) | [arXiv:1312.7582](#) | [DOI:10.1007/JHEP02\(2014\)024](#) | 2013-12-29

1250. Measurement of the Production Cross Section for a W Boson and Two b Jets in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 735 \(2014\) 204](#) | [arXiv:1312.6608](#) | [DOI:10.1016/j.physletb.2014.06.041](#) | 2013-12-23

1251. Measurement of Four-Jet Production in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 89 \(2014\) 092010](#) | [arXiv:1312.6440](#) | [DOI:10.1103/PhysRevD.89.092010](#) | 2013-12-22

1252. Measurement of the Muon Charge Asymmetry in Inclusive $pp \rightarrow W+X$ Production at $\sqrt{s} = 7$ TeV and an Improved Determination of Light Parton Distribution Functions

Chatrchyan, Serguei et al.

[Phys.Rev.D 90 \(2014\) 032004](#) | [arXiv:1312.6283](#) | [DOI:10.1103/PhysRevD.90.032004](#) | 2013-12-21

1253. Event Activity Dependence of $Y(nS)$ Production in $\sqrt{s_{NN}}=5.02$ TeV pPb and $\sqrt{s}=2.76$ TeV pp Collisions

Chatrchyan, Serguei et al.

JHEP 04 (2014) 103 | [arXiv:1312.6300](#) | [DOI:10.1007/JHEP04\(2014\)103](#) | 2013-12-21

1254. Study of Double Parton Scattering Using $W + 2$ -Jet Events in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

JHEP 03 (2014) 032 | [arXiv:1312.5729](#) | [DOI:10.1007/JHEP03\(2014\)032](#) | 2013-12-19

1255. Measurement of the Properties of a Higgs Boson in the Four-Lepton Final State

Chatrchyan, Serguei et al.

Phys.Rev.D 89 (2014) 092007 | [arXiv:1312.5353](#) | [DOI:10.1103/PhysRevD.89.092007](#) | 2013-12-18

1256. Search for Flavor-Changing Neutral Currents in Top-Quark Decays $t \rightarrow Zq$ in pp Collisions at $\sqrt{s}=8$ TeV

Chatrchyan, Serguei et al.

Phys.Rev.Lett. 112 (2014) 171802 | [arXiv:1312.4194](#) | [DOI:10.1103/PhysRevLett.112.171802](#) | 2013-12-15

1257. Evidence of b-Jet Quenching in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

Phys.Rev.Lett. 113 (2014) 132301 | [arXiv:1312.4198](#) | [DOI:10.1103/PhysRevLett.113.132301](#) | 2013-12-15

1258. Search for Top Squark and Higgsino Production using Diphoton Higgs Boson Decays

Chatrchyan, Serguei et al.

Phys.Rev.Lett. 112 (2014) 161802 | [arXiv:1312.3310](#) | [DOI:10.1103/PhysRevLett.112.161802](#) | 2013-12-11

1259. Search for Top-Quark Partners with Charge $5/3$ in the Same-Sign Dilepton Final State

Chatrchyan, Serguei et al.

Phys.Rev.Lett. 112 (2014) 171801 | [arXiv:1312.2391](#) | [DOI:10.1103/PhysRevLett.112.171801](#) | 2013-12-09

1260. Studies of Azimuthal Dihadron Correlations in Ultra-Central PbPb Collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

JHEP 02 (2014) 088 | [arXiv:1312.1845](#) | [DOI:10.1007/JHEP02\(2014\)088](#) | 2013-12-06

1261. Measurement of Higgs Boson Production and Properties in the WW Decay Channel with Leptonic Final States

Chatrchyan, Serguei et al.

JHEP 01 (2014) 096 | [arXiv:1312.1129](#) | [DOI:10.1007/JHEP01\(2014\)096](#) | 2013-12-04

1262. Inclusive Search for a Vector-Like T Quark with Charge $2/3$ in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 729 (2014) 149 | [arXiv:1311.7667](#) | [DOI:10.1016/j.physletb.2014.01.006](#) | 2013-11-29

1263. Search for New Physics in Events with Same-Sign Dileptons and Jets in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

JHEP 01 (2014) 163 | [arXiv:1311.6736](#) | [DOI:10.1007/JHEP01\(2014\)163](#) | 2013-11-26

1264. Measurement of the Triple-Differential Cross Section for Photon+Jets Production in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 06 (2014) 009 | [arXiv:1311.6141](#) | [DOI:10.1007/JHEP06\(2014\)009](#) | 2013-11-24

1265. Probing Color Coherence Effects in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2901](#) | [arXiv:1311.5815](#) | [DOI:10.1140/epjc/s10052-014-2901-8](#) | 2013-11-22

1266. Search for Pair Production of Excited Top Quarks in the Lepton + Jets Final State

Chatrchyan, Serguei et al.

[JHEP 06 \(2014\) 125](#) | [arXiv:1311.5357](#) | [DOI:10.1007/JHEP06\(2014\)125](#) | 2013-11-21

1267. Search for Supersymmetry in pp Collisions at $\sqrt{s}=8$ TeV in Events with a Single Lepton, Large Jet Multiplicity, and Multiple b Jets

Chatrchyan, Serguei et al.

[Phys.Lett.B 733 \(2014\) 328](#) | [arXiv:1311.4937](#) | [DOI:10.1016/j.physletb.2014.04.023](#) | 2013-11-19

1268. Measurements of $t\bar{t}$ Spin Correlations and Top-Quark Polarization Using Dilepton Final States in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 112 \(2014\) 182001](#) | [arXiv:1311.3924](#) | [DOI:10.1103/PhysRevLett.112.182001](#) | 2013-11-15

1269. Searches for Light- and Heavy-Flavour Three-Jet Resonances in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 730 \(2014\) 193](#) | [arXiv:1311.1799](#) | [DOI:10.1016/j.physletb.2014.01.049](#) | 2013-11-07

1270. Measurement of Higher-Order Harmonic Azimuthal Anisotropy in PbPb Collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.C 89 \(2014\) 044906](#) | [arXiv:1310.8651](#) | [DOI:10.1103/PhysRevC.89.044906](#) | 2013-10-31

1271. CMS Computing Operations During Run 1

Adelman, J. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 032040](#) | [DOI:10.1088/1742-6596/513/3/032040](#) | 2013-10-30

1272. Measurement of the Differential and Double-Differential Drell-Yan Cross Sections in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2013\) 030](#) | [arXiv:1310.7291](#) | [DOI:10.1007/JHEP12\(2013\)030](#) | 2013-10-27

1273. Jet and Underlying Event Properties as a Function of Charged-Particle Multiplicity in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2674](#) | [arXiv:1310.4554](#) | [DOI:10.1140/epjc/s10052-013-2674-5](#) | 2013-10-16

1274. Towards a Centralized Grid Speedometer

Dzhunov, I. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 032028](#) | [DOI:10.1088/1742-6596/513/3/032028](#) | 2014

1275. Evolution of the Pilot Infrastructure of CMS: Towards a Single GlideinWMS Pool

Belforte, S. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 032041](#) | [DOI:10.1088/1742-6596/513/3/032041](#) | 2014

1276. CMS experience of running glideinWMS in High Availability mode

Sfiligoi, I. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 032086](#) | [DOI:10.1088/1742-6596/513/3/032086](#) | 2014

1277. Deployment of a WLCG network monitoring infrastructure based on the perfSONAR-PS technology

Campana, S. et al.

[J.Phys.Conf.Ser. 513 \(2014\) 062008](#) | [DOI:10.1088/1742-6596/513/6/062008](#) | 2014

1278. Opportunistic Resource Usage in CMS

Kreuzer, Peter et al.

[J.Phys.Conf.Ser. 513 \(2014\) 062028](#) | [DOI:10.1088/1742-6596/513/6/062028](#) | 2014

1279. Search for the Standard Model Higgs Boson Produced in Association with a W or a Z Boson and Decaying to Bottom Quarks

Chatrchyan, Serguei et al.

[Phys.Rev.D 89 \(2014\) 012003](#) | [arXiv:1310.3687](#) | [DOI:10.1103/PhysRevD.89.012003](#) | 2013-10-14

1280. Rapidity Distributions in Exclusive Z + Jet and γ + Jet Events in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 88 \(2013\) 112009](#) | [arXiv:1310.3082](#) | [DOI:10.1103/PhysRevD.88.112009](#) | 2013-10-11

1281. Search for Baryon Number Violation in Top-Quark Decays

Chatrchyan, Serguei et al.

[Phys.Lett.B 731 \(2014\) 173](#) | [arXiv:1310.1618](#) | [DOI:10.1016/j.physletb.2014.02.033](#) | 2013-10-06

1282. Measurement of the Cross Section and Angular Correlations for Associated Production of a Z Boson with b Hadrons in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2013\) 039](#) | [arXiv:1310.1349](#) | [DOI:10.1007/JHEP12\(2013\)039](#) | 2013-10-04

1283. Measurement of Associated W + Charm Production in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 02 \(2014\) 013](#) | [arXiv:1310.1138](#) | [DOI:10.1007/JHEP02\(2014\)013](#) | 2013-10-03

1284. Modification of Jet Shapes in PbPb Collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 730 \(2014\) 243](#) | [arXiv:1310.0878](#) | [DOI:10.1016/j.physletb.2014.01.042](#) | 2013-10-02

1285. Observation of a Peaking Structure in the $J/\psi\phi$ Mass Spectrum from $B^{\pm} \rightarrow J/\psi\phi K^{\pm}$ Decays

Chatrchyan, Serguei et al.

[Phys.Lett.B 734 \(2014\) 261](#) | [arXiv:1309.6920](#) | [DOI:10.1016/j.physletb.2014.05.055](#) | 2013-09-26

1286. Searches for New Physics using the $t\bar{t}$ Invariant Mass Distribution in pp Collisions at $\sqrt{s}=8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 111 \(2013\) 211804](#) | [arXiv:1309.2030](#) | [DOI:10.1103/PhysRevLett.111.211804](#) | 2013-09-08

1287. Measurement of the Production Cross Section for $Z\gamma \rightarrow \nu\bar{\nu}\gamma$ in pp Collisions at $\sqrt{s} = 7$ TeV and Limits on $ZZ\gamma$ and $Z\gamma\gamma$ Triple Gauge Boson Couplings

Chatrchyan, Serguei et al.

[JHEP 10 \(2013\) 164](#) | [arXiv:1309.1117](#) | [DOI:10.1007/JHEP10\(2013\)164](#) | 2013-09-04

1288. Search for a New Bottomonium State Decaying to $\Upsilon(1S)\pi^+\pi^-$ in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 727 \(2013\) 57](#) | [arXiv:1309.0250](#) | [DOI:10.1016/j.physletb.2013.10.016](#) | 2013-09-01

1289. Measurement of the $W\gamma$ and $Z\gamma$ Inclusive Cross Sections in pp Collisions at $\sqrt{s}=7$ TeV and Limits on Anomalous Triple Gauge Boson Couplings

Chatrchyan, Serguei et al.

[Phys.Rev.D 89 \(2014\) 092005](#) | [arXiv:1308.6832](#) | [DOI:10.1103/PhysRevD.89.092005](#) | 2013-08-30

1290. Measurement of the W-Boson Helicity in Top-Quark decays from $t\bar{t}$ Production in Lepton + Jets Events in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

JHEP 10 (2013) 167 | [arXiv:1308.3879](#) | [DOI:10.1007/JHEP10\(2013\)167](#) | 2013-08-18

1291. Angular Analysis and Branching Fraction Measurement of the Decay $B^0 \rightarrow K^{*0} \mu^+ \mu^-$

Chatrchyan, Serguei et al.

[Phys.Lett.B 727 \(2013\) 77](#) | [arXiv:1308.3409](#) | [DOI:10.1016/j.physletb.2013.10.017](#) | 2013-08-15

1292. Search for Top-Squark Pair Production in the Single-Lepton Final State in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2677](#) | [arXiv:1308.1586](#) | [DOI:10.1140/epjc/s10052-013-2677-2](#) | 2013-08-07

1293. Measurement of the Prompt J/ψ and $\psi(2S)$ Polarizations in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 727 \(2013\) 381](#) | [arXiv:1307.6070](#) | [DOI:10.1016/j.physletb.2013.10.055](#) | 2013-07-23

1294. Search for a Higgs Boson Decaying into a Z and a Photon in pp Collisions at $\sqrt{s} = 7$ and 8 TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 726 \(2013\) 587](#) | [arXiv:1307.5515](#) | [DOI:10.1016/j.physletb.2013.09.057](#) | 2013-07-21

1295. Measurement of the $B^0_s \rightarrow \mu^+ \mu^-$ Branching Fraction and Search for $B^0 \rightarrow \mu^+ \mu^-$ with the CMS Experiment

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 111 \(2013\) 101804](#) | [arXiv:1307.5025](#) | [DOI:10.1103/PhysRevLett.111.101804](#) | 2013-07-18

1296. Measurement of the Top-Quark Mass in All-Jets $t\bar{t}$ Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2758](#) | [arXiv:1307.4617](#) | [DOI:10.1140/epjc/s10052-014-2758-x](#) | 2013-07-17

1297. Study of the Production of Charged Pions, Kaons, and Protons in pPb Collisions at $\sqrt{s_{NN}} = 5.02$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 74 \(2014\) 2847](#) | [arXiv:1307.3442](#) | [DOI:10.1140/epjc/s10052-014-2847-x](#) | 2013-07-12

1298. The Performance of the CMS Muon Detector in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV at the LHC

Chatrchyan, Serguei et al.

[JINST 8 \(2013\) P11002](#) | [arXiv:1306.6905](#) | [DOI:10.1088/1748-0221/8/11/P11002](#) | 2013-06-28

1299. Search for Top Squarks in R-Parity-Violating Supersymmetry using Three or More Leptons and B-Tagged Jets

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 111 \(2013\) 221801](#) | [arXiv:1306.6643](#) | [DOI:10.1103/PhysRevLett.111.221801](#) | 2013-06-27

1300. Energy Calibration and Resolution of the CMS Electromagnetic Calorimeter in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JINST 8 \(2013\) P09009](#) | [arXiv:1306.2016](#) | [DOI:10.1088/1748-0221/8/09/P09009](#) | 2013-06-09

1301. Measurement of the W^+W^- Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV and Limits on Anomalous $WW\gamma$ and WWZ Couplings

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2610](#) | [arXiv:1306.1126](#) | [DOI:10.1140/epjc/s10052-013-2610-8](#) | 2013-06-05

1302. Measurement of the Hadronic Activity in Events with a Z and Two Jets and Extraction of the Cross Section for the Electroweak Production of a Z with Two Jets in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

JHEP 10 (2013) 062 | [arXiv:1305.7389](#) | DOI:10.1007/JHEP10(2013)062 | 2013-05-31

1303. Measurement of Neutral Strange Particle Production in the Underlying Event in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

Phys.Rev.D 88 (2013) 052001 | [arXiv:1305.6016](#) | DOI:10.1103/PhysRevD.88.052001 | 2013-05-26

1304. Study of Exclusive Two-Photon Production of W^+W^- in pp Collisions at $\sqrt{s} = 7$ TeV and Constraints on Anomalous Quartic Gauge Couplings

Chatrchyan, Serguei et al.

JHEP 07 (2013) 116 | [arXiv:1305.5596](#) | DOI:10.1007/JHEP07(2013)116 | 2013-05-23

1305. Search for Gluino Mediated Bottom- and Top-Squark Production in Multijet Final States in pp Collisions at 8 TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 725 (2013) 243 | [arXiv:1305.2390](#) | DOI:10.1016/j.physletb.2013.06.058 | 2013-05-10

1306. Searches for Long-Lived Charged Particles in pp Collisions at $\sqrt{s}=7$ and 8 TeV

Chatrchyan, Serguei et al.

JHEP 07 (2013) 122 | [arXiv:1305.0491](#) | DOI:10.1007/JHEP07(2013)122 | 2013-05-02

1307. Multiplicity and Transverse Momentum Dependence of Two- and Four-Particle Correlations in pPb and PbPb Collisions

Chatrchyan, Serguei et al.

Phys.Lett.B 724 (2013) 213 | [arXiv:1305.0609](#) | DOI:10.1016/j.physletb.2013.06.028 | 2013-05-02

1308. Measurement of the Λ_b^0 Lifetime in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

JHEP 07 (2013) 163 | [arXiv:1304.7495](#) | DOI:10.1007/JHEP07(2013)163 | 2013-04-28

1309. Measurement of the Ratio of the Inclusive 3-Jet Cross Section to the Inclusive 2-Jet Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV and First Determination of the Strong Coupling Constant in the TeV Range

Chatrchyan, Serguei et al.

Eur.Phys.J.C 73 (2013) 2604 | [arXiv:1304.7498](#) | DOI:10.1140/epjc/s10052-013-2604-6 | 2013-04-28

1310. Measurement of Masses in the $t\bar{t}$ System by Kinematic Endpoints in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

Eur.Phys.J.C 73 (2013) 2494 | [arXiv:1304.5783](#) | DOI:10.1140/epjc/s10052-013-2494-7 | 2013-04-21

1311. Search for a Standard-Model-Like Higgs Boson with a Mass in the Range 145 to 1000 GeV at the LHC

Chatrchyan, Serguei et al.

Eur.Phys.J.C 73 (2013) 2469 | [arXiv:1304.0213](#) | DOI:10.1140/epjc/s10052-013-2469-8 | 2013-03-31

1312. Measurement of the $\Upsilon(1S)$, $\Upsilon(2S)$, and $\Upsilon(3S)$ Cross Sections in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 727 (2013) 101 | [arXiv:1303.5900](#) | DOI:10.1016/j.physletb.2013.10.033 | 2013-03-23

1313. Search for Microscopic Black Holes in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2013\) 178](#) | [arXiv:1303.5338](#) | [DOI:10.1007/JHEP07\(2013\)178](#) | 2013-03-21

1314. Observation of a New Boson with Mass Near 125 GeV in pp Collisions at $\sqrt{s} = 7$ and 8 TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2013\) 081](#) | [arXiv:1303.4571](#) | [DOI:10.1007/JHEP06\(2013\)081](#) | 2013-03-19

1315. Studies of Jet Mass in Dijet and W/Z + Jet Events

Chatrchyan, Serguei et al.

[JHEP 05 \(2013\) 090](#) | [arXiv:1303.4811](#) | [DOI:10.1007/JHEP05\(2013\)090](#) | 2013-03-19

1316. Measurement of associated production of vector bosons and top quark-antiquark pairs at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 172002](#) | [arXiv:1303.3239](#) | [DOI:10.1103/PhysRevLett.110.172002](#) | 2013-03-13

1317. Search for Supersymmetry in Hadronic Final States with Missing Transverse Energy Using the Variables α_T and b-Quark Multiplicity in pp collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2568](#) | [arXiv:1303.2985](#) | [DOI:10.1140/epjc/s10052-013-2568-6](#) | 2013-03-12

1318. Search for the Standard Model Higgs Boson Produced in Association with a Top-Quark Pair in pp Collisions at the LHC

Chatrchyan, Serguei et al.

[JHEP 05 \(2013\) 145](#) | [arXiv:1303.0763](#) | [DOI:10.1007/JHEP05\(2013\)145](#) | 2013-03-04

1319. Search for Narrow Resonances Using the Dijet Mass Spectrum in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 114015](#) | [arXiv:1302.4794](#) | [DOI:10.1103/PhysRevD.87.114015](#) | 2013-02-19

1320. Measurement of the $X(3872)$ Production Cross Section Via Decays to $J/\psi \pi^+ \pi^-$ in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2013\) 154](#) | [arXiv:1302.3968](#) | [DOI:10.1007/JHEP04\(2013\)154](#) | 2013-02-16

1321. Search for New Physics in Final States with a Lepton and Missing Transverse Energy in pp Collisions at the LHC

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 072005](#) | [arXiv:1302.2812](#) | [DOI:10.1103/PhysRevD.87.072005](#) | 2013-02-12

1322. Search for a Higgs Boson Decaying into a b-Quark Pair and Produced in Association with b Quarks in Proton-Proton Collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 207](#) | [arXiv:1302.2892](#) | [DOI:10.1016/j.physletb.2013.04.017](#) | 2013-02-12

1323. Study of the Underlying Event at Forward Rapidity in pp Collisions at $\sqrt{s} = 0.9, 2.76$, and 7 TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2013\) 072](#) | [arXiv:1302.2394](#) | [DOI:10.1007/JHEP04\(2013\)072](#) | 2013-02-10

1324. Searches for Higgs Bosons in pp Collisions at $\sqrt{s} = 7$ and 8 TeV in the Context of Four-Generation and Fermiophobic Models

Chatrchyan, Serguei et al.

[Phys.Lett.B 725 \(2013\) 36](#) | [arXiv:1302.1764](#) | [DOI:10.1016/j.physletb.2013.06.043](#) | 2013-02-07

1325. Measurement of the $t\bar{t}$ Production Cross Section in the All-Jet Final State in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2013\) 065](#) | [arXiv:1302.0508](#) | [DOI:10.1007/JHEP05\(2013\)065](#) | 2013-02-03

1326. Search for Pair-Produced Dijet Resonances in Four-Jet Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 141802](#) | [arXiv:1302.0531](#) | [DOI:10.1103/PhysRevLett.110.141802](#) | 2013-02-03

1327. Measurement of the $t\bar{t}$ Production Cross Section in the τ +Jets Channel in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2386](#) | [arXiv:1301.5755](#) | [DOI:10.1140/epjc/s10052-013-2386-x](#) | 2013-01-24

1328. Search for Contact Interactions Using the Inclusive Jet p_T Spectrum in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 052017](#) | [arXiv:1301.5023](#) | [DOI:10.1103/PhysRevD.87.052017](#) | 2013-01-21

1329. Measurement of $W^+ W^-$ and ZZ Production Cross Sections in pp Collisions at $\sqrt{s} = 8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 721 \(2013\) 190](#) | [arXiv:1301.4698](#) | [DOI:10.1016/j.physletb.2013.03.027](#) | 2013-01-20

1330. Search for Supersymmetry in Events with Opposite-Sign Dileptons and Missing Transverse Energy Using an Artificial Neural Network

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 072001](#) | [arXiv:1301.0916](#) | [DOI:10.1103/PhysRevD.87.072001](#) | 2013-01

1331. Event Shapes and Azimuthal Correlations in Z + Jets Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 238](#) | [arXiv:1301.1646](#) | [DOI:10.1016/j.physletb.2013.04.025](#) | 2013-01

1332. Interpretation of Searches for Supersymmetry with Simplified Models

Chatrchyan, Serguei et al.

[Phys.Rev.D 88 \(2013\) 052017](#) | [arXiv:1301.2175](#) | [DOI:10.1103/PhysRevD.88.052017](#) | 2013-01

1333. Search for Physics Beyond the Standard Model in Events with τ Leptons, Jets, and Large Transverse Momentum Imbalance in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2493](#) | [arXiv:1301.3792](#) | [DOI:10.1140/epjc/s10052-013-2493-8](#) | 2013-01

1334. Search for Long-Lived Particles Decaying to Photons and Missing Energy in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 273](#) | [arXiv:1212.1838](#) | [DOI:10.1016/j.physletb.2013.04.027](#) | 2012-12

1335. Search for Heavy Resonances in the W/Z-Tagged Dijet Mass Spectrum in pp Collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 723 \(2013\) 280](#) | [arXiv:1212.1910](#) | [DOI:10.1016/j.physletb.2013.05.040](#) | 2012-12

1336. Search for Contact Interactions in $\mu^+ \mu^-$ Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 032001](#) | [arXiv:1212.4563](#) | [DOI:10.1103/PhysRevD.87.032001](#) | 2012-12

1337. Search for Heavy Narrow Dilepton Resonances in pp Collisions at $\sqrt{s}=7$ TeV and $\sqrt{s}=8$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 720 \(2013\) 63](#) | [arXiv:1212.6175](#) | [DOI:10.1016/j.physletb.2013.02.003](#) | 2012-12

1338. Search for New Physics in Events with Same-Sign Dileptons and b Jets in pp Collisions at $\sqrt{s}=8$ TeV

Chatrchyan, Serguei et al.

[JHEP 03 \(2013\) 037](#) | [arXiv:1212.6194](#) | [DOI:10.1007/JHEP03\(2013\)037](#) | 2012-12

1339. Inclusive Search for Supersymmetry Using the Razor Variables in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 111 \(2013\) 081802](#) | [arXiv:1212.6961](#) | [DOI:10.1103/PhysRevLett.111.081802](#) | 2012-12

1340. Measurement of the $t\bar{t}$ Production Cross Section in pp Collisions at $\sqrt{s}=7$ TeV with Lepton + Jets Final States

Chatrchyan, Serguei et al.

[Phys.Lett.B 720 \(2013\) 83](#) | [arXiv:1212.6682](#) | [DOI:10.1016/j.physletb.2013.02.021](#) | 2012-12

1341. Measurements of Differential Jet Cross Sections in Proton-Proton Collisions at $\sqrt{s}=7$ TeV with the CMS Detector

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 112002](#) | [arXiv:1212.6660](#) | [DOI:10.1103/PhysRevD.87.112002](#) | 2012-12

1342. Study of the Mass and Spin-Parity of the Higgs Boson Candidate Via Its Decays to Z Boson Pairs

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 081803](#) | [arXiv:1212.6639](#) | [DOI:10.1103/PhysRevLett.110.081803](#) | 2012-12

1343. Search for Supersymmetry in pp Collisions at $\sqrt{s}=7$ TeV in Events with a Single Lepton, Jets, and Missing Transverse Momentum

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2404](#) | [arXiv:1212.6428](#) | [DOI:10.1140/epjc/s10052-013-2404-z](#) | 2012-12

1344. Measurement of Differential Top-Quark Pair Production Cross Sections in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2339](#) | [arXiv:1211.2220](#) | [DOI:10.1140/epjc/s10052-013-2339-4](#) | 2012-11

1345. Search in Leptonic Channels for Heavy Resonances Decaying to Long-Lived Neutral Particles

Chatrchyan, Serguei et al.

[JHEP 02 \(2013\) 085](#) | [arXiv:1211.2472](#) | [DOI:10.1007/JHEP02\(2013\)085](#) | 2012-11

1346. Search for Supersymmetry in Final States with a Single Lepton, b-Quark Jets, and Missing Transverse Energy in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 052006](#) | [arXiv:1211.3143](#) | [DOI:10.1103/PhysRevD.87.052006](#) | 2012-11

1347. Search for Z' Resonances Decaying to $t\bar{t}$ in Dilepton + Jets Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 072002](#) | [arXiv:1211.3338](#) | [DOI:10.1103/PhysRevD.87.072002](#) | 2012-11

1348. Identification of b-Quark Jets with the CMS Experiment

Chatrchyan, Serguei et al.

[JINST 8 \(2013\) P04013](#) | [arXiv:1211.4462](#) | [DOI:10.1088/1748-0221/8/04/P04013](#) | 2012-11

1349. Search for New Physics in Events with Photons, Jets, and Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 03 \(2013\) 111](#) | [arXiv:1211.4784](#) | [DOI:10.1007/JHEP03\(2013\)111](#) | 2012-11

1350. Measurement of the ZZ Production Cross Section and Search for Anomalous Couplings in 2 l2l ' Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2013\) 063](#) | [arXiv:1211.4890](#) | [DOI:10.1007/JHEP01\(2013\)063](#) | 2012-11

1351. Search for Exotic Resonances Decaying into WZ/ZZ in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 02 \(2013\) 036](#) | [arXiv:1211.5779](#) | [DOI:10.1007/JHEP02\(2013\)036](#) | 2012-11

1352. Search for Anomalous Production of Highly Boosted Z Bosons Decaying to $\mu^+ + \mu^-$ in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 28](#) | [arXiv:1210.0867](#) | [DOI:10.1016/j.physletb.2013.03.037](#) | 2012-10

1353. Measurement of the Relative Prompt Production Rate of χ_{c2} and χ_{c1} in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2251](#) | [arXiv:1210.0875](#) | [DOI:10.1140/epjc/s10052-012-2251-3](#) | 2012-10

1354. Search for Heavy Lepton Partners of Neutrinos in Proton-Proton Collisions in the Context of the Type III Seesaw Mechanism

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2012\) 348](#) | [arXiv:1210.1797](#) | [DOI:10.1016/j.physletb.2012.10.070](#) | 2012-10

1355. Search for Supersymmetry in Events with Photons and Low Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 719 \(2013\) 42](#) | [arXiv:1210.2052](#) | [DOI:10.1016/j.physletb.2012.12.055](#) | 2012-10

1356. Search for Fractionally Charged Particles in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, S. et al.

[Phys.Rev.D 87 \(2013\) 092008](#) | [arXiv:1210.2311](#) | [DOI:10.1103/PhysRevD.87.092008](#) | 2012-10

1357. Search for Narrow Resonances and Quantum Black Holes in Inclusive and b-Tagged Dijet Mass Spectra from pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2013\) 013](#) | [arXiv:1210.2387](#) | [DOI:10.1007/JHEP01\(2013\)013](#) | 2012-10

1358. Search for Heavy Neutrinos and W_R Bosons with Right-Handed Couplings in a Left-Right Symmetric Model in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 261802](#) | [arXiv:1210.2402](#) | [DOI:10.1103/PhysRevLett.109.261802](#) | 2012-10

1359. Search for Excited Leptons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 720 \(2013\) 309](#) | [arXiv:1210.2422](#) | [DOI:10.1016/j.physletb.2013.02.031](#) | 2012-10

1360. Observation of Z Decays to Four Leptons with the CMS Detector at the LHC

Chatrchyan, Serguei et al.

[JHEP 12 \(2012\) 034](#) | [arXiv:1210.3844](#) | [DOI:10.1007/JHEP12\(2012\)034](#) | 2012-10

1361. Observation of Long-Range Near-Side Angular Correlations in Proton-Lead Collisions at the LHC

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 795](#) | [arXiv:1210.5482](#) | [DOI:10.1016/j.physletb.2012.11.025](#) | 2012-10

1362. Search for Third-Generation Leptoquarks and Scalar Bottom Quarks in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2012\) 055](#) | [arXiv:1210.5627](#) | [DOI:10.1007/JHEP12\(2012\)055](#) | 2012-10

1363. Search for Pair Production of Third-Generation Leptoquarks and Top Squarks in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 081801](#) | [arXiv:1210.5629](#) | [DOI:10.1103/PhysRevLett.110.081801](#) | 2012-10

1364. Measurement of the Inelastic Proton-Proton Cross Section at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 722 \(2013\) 5](#) | [arXiv:1210.6718](#) | [DOI:10.1016/j.physletb.2013.03.024](#) | 2012-10

1365. Search for Heavy Quarks Decaying into a Top Quark and a W or Z Boson using Lepton + Jets Events in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2013\) 154](#) | [arXiv:1210.7471](#) | [DOI:10.1007/JHEP01\(2013\)154](#) | 2012-10

1366. Measurement of the Sum of W W and WZ Production with W+Dijet Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 73 \(2013\) 2283](#) | [arXiv:1210.7544](#) | [DOI:10.1140/epjc/s10052-013-2283-3](#) | 2012-10

1367. Search for a Non-Standard-Model Higgs Boson Decaying to a Pair of New Light Bosons in Four-Muon Final States

Chatrchyan, Serguei et al.

[Phys.Lett.B 726 \(2013\) 564](#) | [arXiv:1210.7619](#) | [DOI:10.1016/j.physletb.2013.09.009](#) | 2012-10

1368. Search for Supersymmetry in Final States with Missing Transverse Energy and 0, 1, 2, or at Least 3 b-Quark Jets in 7 TeV pp Collisions using the Variable α_T

Chatrchyan, Serguei et al.

[JHEP 01 \(2013\) 077](#) | [arXiv:1210.8115](#) | [DOI:10.1007/JHEP01\(2013\)077](#) | 2012-10

1369. Search for Pair Produced Fourth-Generation Up-Type Quarks in pp Collisions at $\sqrt{s}=7$ TeV with a Lepton in the Final State

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2012\) 307](#) | [arXiv:1209.0471](#) | [DOI:10.1016/j.physletb.2012.10.038](#) | 2012-09

1370. Combined Search for the Quarks of a Sequential Fourth Generation

Chatrchyan, Serguei et al.

[Phys.Rev.D 86 \(2012\) 112003](#) | [arXiv:1209.1062](#) | [DOI:10.1103/PhysRevD.86.112003](#) | 2012-09

1371. Search for Exclusive or Semi-Exclusive Photon Pair Production and Observation of Exclusive and Semi-Exclusive Electron Pair Production in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2012\) 080](#) | [arXiv:1209.1666](#) | [DOI:10.1007/JHEP11\(2012\)080](#) | 2012-09

1372. Observation of a Diffractive Contribution to Dijet Production in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 87 \(2013\) 012006](#) | [arXiv:1209.1805](#) | [DOI:10.1103/PhysRevD.87.012006](#) | 2012-09

1373. Measurement of the Top-Quark Mass in $t\bar{t}$ Events with Lepton+Jets Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2012\) 105](#) | [arXiv:1209.2319](#) | [DOI:10.1007/JHEP12\(2012\)105](#) | 2012-09

1374. Measurement of the Top-Quark Mass in $t\bar{t}$ Events with Dilepton Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2202](#) | [arXiv:1209.2393](#) | [DOI:10.1140/epjc/s10052-012-2202-z](#) | 2012-09

1375. Measurement of the $Y(1S)$, $Y(2S)$ and $Y(3S)$ Polarizations in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 081802](#) | [arXiv:1209.2922](#) | [DOI:10.1103/PhysRevLett.110.081802](#) | 2012-09

1376. Evidence for Associated Production of a Single Top Quark and W Boson in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 022003](#) | [arXiv:1209.3489](#) | [DOI:10.1103/PhysRevLett.110.022003](#) | 2012-09

1377. Search for a Narrow Spin-2 Resonance Decaying to a Pair of Z Vector Bosons in the Semileptonic Final State

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 1208](#) | [arXiv:1209.3807](#) | [DOI:10.1016/j.physletb.2012.11.063](#) | 2012-09

1378. Search for the Standard Model Higgs Boson Produced in Association with W and Z Bosons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2012\) 088](#) | [arXiv:1209.3937](#) | [DOI:10.1007/JHEP11\(2012\)088](#) | 2012-09

1379. Search for Resonant $t\bar{t}$ Production in Lepton+Jets Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2012\) 015](#) | [arXiv:1209.4397](#) | [DOI:10.1007/JHEP12\(2012\)015](#) | 2012-09

1380. Measurement of the Single-Top-Quark t-Channel Cross Section in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 12 \(2012\) 035](#) | [arXiv:1209.4533](#) | [DOI:10.1007/JHEP12\(2012\)035](#) | 2012-09

1381. Search for Electroweak Production of Charginos and Neutralinos using Leptonic Final States in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2012\) 147](#) | [arXiv:1209.6620](#) | [DOI:10.1007/JHEP11\(2012\)147](#) | 2012-09

1382. Search for a W' Boson Decaying to a Bottom Quark and a Top Quark in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 1229](#) | [arXiv:1208.0956](#) | [DOI:10.1016/j.physletb.2012.12.008](#) | 2012-08

1383. Search for Flavor Changing Neutral Currents in Top Quark Decays in pp Collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 1252](#) | [arXiv:1208.0957](#) | [DOI:10.1016/j.physletb.2012.12.045](#) | 2012-08

1384. Measurement of the Azimuthal Anisotropy of Neutral Pions in PbPb cCollisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 110 \(2013\) 042301](#) | [arXiv:1208.2470](#) | [DOI:10.1103/PhysRevLett.110.042301](#) | 2012-08

1385. Measurement of the $t\bar{t}$ Production Cross Section in the Dilepton Channel in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2012\) 067](#) | [arXiv:1208.2671](#) | [DOI:10.1007/JHEP11\(2012\)067](#) | 2012-08

1386. Observation of Sequential Upsilon Suppression in PbPb Collisions

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 222301](#) | [arXiv:1208.2826](#) | [DOI:10.1103/PhysRevLett.109.222301](#) | 2012-08

1387. Search for Three-Jet Resonances in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2012\) 329](#) | [arXiv:1208.2931](#) | [DOI:10.1016/j.physletb.2012.10.048](#) | 2012-08

1388. Study of the Dijet Mass Spectrum in $pp \rightarrow W + \text{Jets}$ Events at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 251801](#) | [arXiv:1208.3477](#) | [DOI:10.1103/PhysRevLett.109.251801](#) | 2012-08

1389. Search for Supersymmetry in Events with b-Quark Jets and Missing Transverse Energy in pp Collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 86 \(2012\) 072010](#) | [arXiv:1208.4859](#) | [DOI:10.1103/PhysRevD.86.072010](#) | 2012-08

1390. A New Boson with a Mass of 125 GeV Observed with the CMS Experiment at the Large Hadron Collider

Chatrchyan, Serguei et al.

[Science 338 \(2012\) 1569](#) | [DOI:10.1126/science.1230816](#) | 2012

1391. Inclusive and Differential Measurements of the $t\bar{t}$ Charge Asymmetry in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 717 \(2012\) 129](#) | [arXiv:1207.0065](#) | [DOI:10.1016/j.physletb.2012.09.028](#) | 2012-07

1392. Search for Stopped Long-Lived Particles Produced in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2012\) 026](#) | [arXiv:1207.0106](#) | [DOI:10.1007/JHEP08\(2012\)026](#) | 2012-07

1393. Search for New Physics with Long-Lived Particles Decaying to Photons and Missing Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2012\) 172](#) | [arXiv:1207.0627](#) | [DOI:10.1007/JHEP11\(2012\)172](#) | 2012-07

1394. Search for a Fermiophobic Higgs Boson in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 09 \(2012\) 111](#) | [arXiv:1207.1130](#) | [DOI:10.1007/JHEP09\(2012\)111](#) | 2012-07

1395. Search for Supersymmetry in Hadronic Final States using MT_2 in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 10 \(2012\) 018](#) | [arXiv:1207.1798](#) | [DOI:10.1007/JHEP10\(2012\)018](#) | 2012-07

1396. Search for New Physics in the Multijet and Missing Transverse Momentum Final State in Proton-Proton Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 171803](#) | [arXiv:1207.1898](#) | [DOI:10.1103/PhysRevLett.109.171803](#) | 2012-07

1397. Measurement of the Underlying Event Activity in pp Collisions at $\sqrt{s} = 0.9$ and 7 TeV with the Novel Jet-Area/Median Approach

Chatrchyan, Serguei et al.

1398. A Search for a Doubly-Charged Higgs Boson in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2189](#) | [arXiv:1207.2666](#) | [DOI:10.1140/epjc/s10052-012-2189-5](#) | 2012-07

1399. Forward-Backward Asymmetry of Drell-Yan Lepton Pairs in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 752](#) | [arXiv:1207.3973](#) | [DOI:10.1016/j.physletb.2012.10.082](#) | 2012-07

1400. Study of the Inclusive Production of Charged Pions, Kaons, and Protons in pp Collisions at $\sqrt{s}=0.9, 2.76$, and 7 TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2164](#) | [arXiv:1207.4724](#) | [DOI:10.1140/epjc/s10052-012-2164-1](#) | 2012-07

1401. Search for Pair Production of First- and Second-Generation Scalar Leptoquarks in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 86 \(2012\) 052013](#) | [arXiv:1207.5406](#) | [DOI:10.1103/PhysRevD.86.052013](#) | 2012-07

1402. Search for heavy Majorana Neutrinos in $\mu^{\pm}\mu^{\pm} + \text{Jets}$ and $e^{\pm}e^{\pm} + \text{Jets}$ Events in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 717 \(2012\) 109](#) | [arXiv:1207.6079](#) | [DOI:10.1016/j.physletb.2012.09.012](#) | 2012-07

1403. Observation of a New Boson at a Mass of 125 GeV with the CMS Experiment at the LHC

Chatrchyan, Serguei et al.

[Phys.Lett.B 716 \(2012\) 30](#) | [arXiv:1207.7235](#) | [DOI:10.1016/j.physletb.2012.08.021](#) | 2012-07

1404. A New Era for Central Processing and Production in CMS

Fajardo, E. et al.

[J.Phys.Conf.Ser. 396 \(2012\) 042018](#) | [DOI:10.1088/1742-6596/396/4/042018](#) | 2012-06-27

1405. Search for a $W'^{\pm}$ or ρ_{TC} Decaying into WZ in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 141801](#) | [arXiv:1206.0433](#) | [DOI:10.1103/PhysRevLett.109.141801](#) | 2012-06

1406. Search for High-Mass Resonances Decaying into τ -Lepton Pairs in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 716 \(2012\) 82](#) | [arXiv:1206.1725](#) | [DOI:10.1016/j.physletb.2012.07.062](#) | 2012-06

1407. Search for Narrow Resonances in Dilepton Mass Spectra in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 714 \(2012\) 158](#) | [arXiv:1206.1849](#) | [DOI:10.1016/j.physletb.2012.06.051](#) | 2012-06

1408. Measurement of the Electron Charge Asymmetry in Inclusive W Production in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 111806](#) | [arXiv:1206.2598](#) | [DOI:10.1103/PhysRevLett.109.111806](#) | 2012-06

1409. Search for Charge-Asymmetric Production of W' Bosons in $t\bar{t} + \text{Jet}$ Events from pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 717 \(2012\) 351](#) | [arXiv:1206.3921](#) | [DOI:10.1016/j.physletb.2012.09.048](#) | 2012-06

1410. Search for New Physics in Events with Opposite-Sign Leptons, Jets, and Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 815](#) | [arXiv:1206.3949](#) | [DOI:10.1016/j.physletb.2012.11.036](#) | 2012-06

1411. Performance of CMS Muon Reconstruction in pp Collision Events at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JINST 7 \(2012\) P10002](#) | [arXiv:1206.4071](#) | [DOI:10.1088/1748-0221/7/10/P10002](#) | 2012-06

1412. Search for Dark Matter and Large Extra Dimensions in Monojet Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 09 \(2012\) 094](#) | [arXiv:1206.5663](#) | [DOI:10.1007/JHEP09\(2012\)094](#) | 2012-06

1413. Search for a Light Pseudoscalar Higgs Boson in the Dimuon Decay Channel in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 121801](#) | [arXiv:1206.6326](#) | [DOI:10.1103/PhysRevLett.109.121801](#) | 2012-06

1414. Monitoring Techniques and Alarm Procedures for CMS Services and Sites in WLCG

Molina-Perez, J. et al.

[J.Phys.Conf.Ser. 396 \(2012\) 042041](#) | [DOI:10.1088/1742-6596/396/4/042041](#) | 2012

1415. CMS Data Transfer Operations After the First Years of LHC Collisions

Kaselis, Rapolas et al.

[J.Phys.Conf.Ser. 396 \(2012\) 042033](#) | [DOI:10.1088/1742-6596/396/4/042033](#) | 2012

1416. No File Left Behind - Monitoring Transfer Latencies in PhEDEx

Chwalek, T. et al.

[J.Phys.Conf.Ser. 396 \(2012\) 032089](#) | [DOI:10.1088/1742-6596/396/3/032089](#) | 2012

1417. Studies of Jet Quenching using Isolated-Photon + Jet Correlations in PbPb and pp Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 718 \(2013\) 773](#) | [arXiv:1205.0206](#) | [DOI:10.1016/j.physletb.2012.11.003](#) | 2012-05

1418. Search for Heavy Long-Lived Charged Particles in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 713 \(2012\) 408](#) | [arXiv:1205.0272](#) | [DOI:10.1016/j.physletb.2012.06.023](#) | 2012-05

1419. Measurement of the Λ_b Cross Section and the Λ_b to Λ_b Ratio with $J/\Psi \Lambda$ Decays in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 714 \(2012\) 136](#) | [arXiv:1205.0594](#) | [DOI:10.1016/j.physletb.2012.05.063](#) | 2012-05

1420. Measurement of the Pseudorapidity and Centrality Dependence of the Transverse Energy Density in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 152303](#) | [arXiv:1205.2488](#) | [DOI:10.1103/PhysRevLett.109.152303](#) | 2012-05

1421. Search for New Physics in Events with Same-Sign Dileptons and b-Tagged Jets in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2012\) 110](#) | [arXiv:1205.3933](#) | [DOI:10.1007/JHEP08\(2012\)110](#) | 2012-05

1422. Search for a Light Charged Higgs Boson in Top Quark Decays in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2012\) 143](#) | [arXiv:1205.5736](#) | [DOI:10.1007/JHEP07\(2012\)143](#) | 2012-05

1423. Measurement of Jet Fragmentation into Charged Particles in pp and PbPb Collisions at $\sqrt{s_{\text{NN}}}=2.76$ TeV

Chatrchyan, Serguei et al.

[JHEP 10 \(2012\) 087](#) | [arXiv:1205.5872](#) | [DOI:10.1007/JHEP10\(2012\)087](#) | 2012-05

1424. Study of W Boson Production in PbPb and pp Collisions at $\sqrt{s_{\text{NN}}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 715 \(2012\) 66](#) | [arXiv:1205.6334](#) | [DOI:10.1016/j.physletb.2012.07.025](#) | 2012-05

1425. Search for New Physics with Same-Sign Isolated Dilepton Events with Jets and Missing Transverse Energy

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 071803](#) | [arXiv:1205.6615](#) | [DOI:10.1103/PhysRevLett.109.071803](#) | 2012-05

1426. Ratios of Dijet Production Cross Sections as a Function of the Absolute Difference in Rapidity between Jets in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2216](#) | [arXiv:1204.0696](#) | [DOI:10.1140/epjc/s10052-012-2216-6](#) | 2012-04

1427. Search for Dark Matter and Large Extra Dimensions in pp Collisions Yielding a Photon and Missing Transverse Energy

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 108 \(2012\) 261803](#) | [arXiv:1204.0821](#) | [DOI:10.1103/PhysRevLett.108.261803](#) | 2012-04

1428. Search for Heavy Bottom-like Quarks in 4.9 fb^{-1} of pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2012\) 123](#) | [arXiv:1204.1088](#) | [DOI:10.1007/JHEP05\(2012\)123](#) | 2012-04

1429. Measurement of the Underlying Event in the Drell-Yan Process in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2080](#) | [arXiv:1204.1411](#) | [DOI:10.1140/epjc/s10052-012-2080-4](#) | 2012-04

1430. Measurement of the Elliptic Anisotropy of Charged Particles Produced in PbPb Collisions at $\sqrt{s_{\text{NN}}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.C 87 \(2013\) 014902](#) | [arXiv:1204.1409](#) | [DOI:10.1103/PhysRevC.87.014902](#) | 2012-04

1431. Measurement of the $Z/\gamma^*+b\text{-jet}$ Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 126](#) | [arXiv:1204.1643](#) | [DOI:10.1007/JHEP06\(2012\)126](#) | 2012-04

1432. Azimuthal Anisotropy of Charged Particles at High Transverse Momenta in PbPb Collisions at $\sqrt{s_{\text{NN}}}= 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 109 \(2012\) 022301](#) | [arXiv:1204.1850](#) | [DOI:10.1103/PhysRevLett.109.022301](#) | 2012-04

1433. Search for Anomalous $t\bar{t}$ Production in the Highly-Boosted All-Hadronic Final State

Chatrchyan, Serguei et al.

[JHEP 09 \(2012\) 029](#) | [arXiv:1204.2488](#) | [DOI:10.1007/JHEP09\(2012\)029](#) | 2012-04

1434. Measurement of the Mass Difference between Top and Antitop Quarks

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 109](#) | [arXiv:1204.2807](#) | [DOI:10.1007/JHEP06\(2012\)109](#) | 2012-04

1435. Shape, Transverse Size, and Charged Hadron Multiplicity of Jets in pp Collisions at 7 TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 160](#) | [arXiv:1204.3170](#) | [DOI:10.1007/JHEP06\(2012\)160](#) | 2012-04

1436. Search for Physics Beyond the Standard Model in Events with a Z Boson, Jets, and Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 716 \(2012\) 260](#) | [arXiv:1204.3774](#) | [DOI:10.1016/j.physletb.2012.08.026](#) | 2012-04

1437. Search for Leptonic Decays of $W^\pm$ Bosons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2012\) 023](#) | [arXiv:1204.4764](#) | [DOI:10.1007/JHEP08\(2012\)023](#) | 2012-04

1438. Search for Anomalous Production of Multilepton Events in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 169](#) | [arXiv:1204.5341](#) | [DOI:10.1007/JHEP06\(2012\)169](#) | 2012-04

1439. Observation of a new Ξ_b Baryon

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 108 \(2012\) 252002](#) | [arXiv:1204.5955](#) | [DOI:10.1103/PhysRevLett.108.252002](#) | 2012-04

1440. Measurement of the Inclusive Cross Section $\sigma(pp \rightarrow b\bar{b} X \rightarrow \mu\mu X^\pm)$ at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 110](#) | [arXiv:1203.3458](#) | [DOI:10.1007/JHEP06\(2012\)110](#) | 2012-03

1441. Search for $B^0_s \rightarrow \mu^+ \mu^-$ and $B^0 \rightarrow \mu^+ \mu^-$ decays

Chatrchyan, Serguei et al.

[JHEP 04 \(2012\) 033](#) | [arXiv:1203.3976](#) | [DOI:10.1007/JHEP04\(2012\)033](#) | 2012-03

1442. Search for Heavy, Top-Like Quark Pair Production in the Dilepton Final State in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 716 \(2012\) 103](#) | [arXiv:1203.5410](#) | [DOI:10.1016/j.physletb.2012.07.059](#) | 2012-03

1443. Measurement of the Top Quark Pair Production Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV in Dilepton Final States Containing a τ

Chatrchyan, Serguei et al.

[Phys.Rev.D 85 \(2012\) 112007](#) | [arXiv:1203.6810](#) | [DOI:10.1103/PhysRevD.85.112007](#) | 2012-03

1444. Measurement of the Inclusive Production Cross Sections for Forward Jets and for Dijet Events with One Forward and One Central Jet in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2012\) 036](#) | [arXiv:1202.0704](#) | [DOI:10.1007/JHEP06\(2012\)036](#) | 2012-02

1445. Search for a Higgs Boson in the Decay Channel $H \rightarrow ZZ^* \rightarrow q\bar{q}\ell^-\ell^+$ in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2012\) 036](#) | [arXiv:1202.1416](#) | [DOI:10.1007/JHEP04\(2012\)036](#) | 2012-02

1446. Search for the Standard Model Higgs Boson Decaying into Two Photons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 710 \(2012\) 403](#) | [arXiv:1202.1487](#) | [DOI:10.1016/j.physletb.2012.03.003](#) | 2012-02

1447. Combined Results of Searches for the Standard Model Higgs Boson in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 710 (2012) 26 | [arXiv:1202.1488](#) | [DOI:10.1016/j.physletb.2012.02.064](#) | 2012-02

1448. Search for the Standard Model Higgs Boson Decaying to W^+W^- in the Fully Leptonic Final State in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 710 (2012) 91 | [arXiv:1202.1489](#) | [DOI:10.1016/j.physletb.2012.02.076](#) | 2012-02

1449. Search for the Standard Model Higgs Boson in the Decay Channel $H \rightarrow ZZ \rightarrow 4$ Leptons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Rev.Lett. 108 (2012) 111804 | [arXiv:1202.1997](#) | [DOI:10.1103/PhysRevLett.108.111804](#) | 2012-02

1450. Study of High-pT Charged Particle Suppression in PbPb Compared to pp Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

Eur.Phys.J.C 72 (2012) 1945 | [arXiv:1202.2554](#) | [DOI:10.1140/epjc/s10052-012-1945-x](#) | 2012-02

1451. Search for the Standard Model Higgs Boson in the $H \rightarrow ZZ \rightarrow \ell^+ \ell^- \tau^+ \tau^-$ Decay Channel in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 03 (2012) 081 | [arXiv:1202.3617](#) | [DOI:10.1007/JHEP03\(2012\)081](#) | 2012-02

1452. Search for the Standard Model Higgs Boson in the $H \rightarrow ZZ \rightarrow 2\ell 2\nu$ Channel in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 03 (2012) 040 | [arXiv:1202.3478](#) | [DOI:10.1007/JHEP03\(2012\)040](#) | 2012-02

1453. Search for Large Extra Dimensions in Dimuon and Dielectron Events in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 711 (2012) 15 | [arXiv:1202.3827](#) | [DOI:10.1016/j.physletb.2012.03.029](#) | 2012-02

1454. Search for Neutral Higgs Bosons Decaying to Tau Pairs in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 713 (2012) 68 | [arXiv:1202.4083](#) | [DOI:10.1016/j.physletb.2012.05.028](#) | 2012-02

1455. Search for the Standard Model Higgs Boson Decaying to Bottom Quarks in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 710 (2012) 284 | [arXiv:1202.4195](#) | [DOI:10.1016/j.physletb.2012.02.085](#) | 2012-02

1456. Inclusive b-Jet Production in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 04 (2012) 084 | [arXiv:1202.4617](#) | [DOI:10.1007/JHEP04\(2012\)084](#) | 2012-02

1457. Jet Momentum Dependence of Jet Quenching in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

Phys.Lett.B 712 (2012) 176 | [arXiv:1202.5022](#) | [DOI:10.1016/j.physletb.2012.04.058](#) | 2012-02

1458. Search for Quark Compositeness in Dijet Angular Distributions from pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 05 (2012) 055 | [arXiv:1202.5535](#) | [DOI:10.1007/JHEP05\(2012\)055](#) | 2012-02

1459. Search for Microscopic Black Holes in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

JHEP 04 (2012) 061 | [arXiv:1202.6396](#) | [DOI:10.1007/JHEP04\(2012\)061](#) | 2012-02

1460. Measurement of Isolated Photon Production in pp and PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 710 \(2012\) 256](#) | [arXiv:1201.3093](#) | [DOI:10.1016/j.physletb.2012.02.077](#) | 2012-01

1461. Centrality Dependence of Dihadron Correlations and Azimuthal anisotropy Harmonics in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 2012](#) | [arXiv:1201.3158](#) | [DOI:10.1140/epjc/s10052-012-2012-3](#) | 2012-01

1462. Suppression of Non-Prompt J/ψ , Prompt J/ψ , and $Y(1S)$ in PbPb Collisions at $\sqrt{s_{NN}}=2.76$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2012\) 063](#) | [arXiv:1201.5069](#) | [DOI:10.1007/JHEP05\(2012\)063](#) | 2012-01

1463. Search for signatures of extra dimensions in the diphoton mass spectrum at the Large Hadron Collider

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 108 \(2012\) 111801](#) | [arXiv:1112.0688](#) | [DOI:10.1103/PhysRevLett.108.111801](#) | 2011-12

1464. Measurement of the charge asymmetry in top-quark pair production in proton-proton collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 709 \(2012\) 28](#) | [arXiv:1112.5100](#) | [DOI:10.1016/j.physletb.2012.01.078](#) | 2011-12

1465. J/ψ and ψ_{2S} production in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 02 \(2012\) 011](#) | [arXiv:1111.1557](#) | [DOI:10.1007/JHEP02\(2012\)011](#) | 2011-11

1466. Exclusive photon-photon production of muon pairs in proton-proton collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2012\) 052](#) | [arXiv:1111.5536](#) | [DOI:10.1007/JHEP01\(2012\)052](#) | 2011-11

1467. Forward Energy Flow, Central Charged-Particle Multiplicities, and Pseudorapidity Gaps in W and Z Boson Events from pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 72 \(2012\) 1839](#) | [arXiv:1110.0181](#) | [DOI:10.1140/epjc/s10052-011-1839-3](#) | 2011-10

1468. Measurement of energy flow at large pseudorapidities in pp collisions at $\sqrt{s} = 0.9$ and 7 TeV

Chatrchyan, Serguei et al.

[JHEP 11 \(2011\) 148](#) | [arXiv:1110.0211](#) | [DOI:10.1007/JHEP11\(2011\)148](#) | 2011-10

1469. Measurement of the weak mixing angle with the Drell-Yan process in proton-proton collisions at the LHC

Chatrchyan, Serguei et al.

[Phys.Rev.D 84 \(2011\) 112002](#) | [arXiv:1110.2682](#) | [DOI:10.1103/PhysRevD.84.112002](#) | 2011-10

1470. Jet Production Rates in Association with W and Z Bosons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2012\) 010](#) | [arXiv:1110.3226](#) | [DOI:10.1007/JHEP01\(2012\)010](#) | 2011-10

1471. Measurement of the Rapidity and Transverse Momentum Distributions of Z Bosons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 85 \(2012\) 032002](#) | [arXiv:1110.4973](#) | [DOI:10.1103/PhysRevD.85.032002](#) | 2011-10

1472. Measurement of the Production Cross Section for Pairs of Isolated Photons in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 01 \(2012\) 133](#) | [arXiv:1110.6461](#) | [DOI:10.1007/JHEP01\(2012\)133](#) | 2011-10

1473. Search for Supersymmetry at the LHC in Events with Jets and Missing Transverse Energy

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 221804](#) | [arXiv:1109.2352](#) | [DOI:10.1103/PhysRevLett.107.221804](#) | 2011-09

1474. Search for a Vector-like Quark with Charge $2/3$ in $t + Z$ Events from pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 271802](#) | [arXiv:1109.4985](#) | [DOI:10.1103/PhysRevLett.107.271802](#) | 2011-09

1475. Performance of tau-lepton reconstruction and identification in CMS

Chatrchyan, Serguei et al.

[JINST 7 \(2012\) P01001](#) | [arXiv:1109.6034](#) | [DOI:10.1088/1748-0221/7/01/P01001](#) | 2011-09

1476. Measurement of the Drell-Yan Cross Section in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 10 \(2011\) 007](#) | [arXiv:1108.0566](#) | [DOI:10.1007/JHEP10\(2011\)007](#) | 2011-08

1477. Measurement of the Differential Cross Section for Isolated Prompt Photon Production in pp Collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 84 \(2011\) 052011](#) | [arXiv:1108.2044](#) | [DOI:10.1103/PhysRevD.84.052011](#) | 2011-08

1478. Measurement of the $t\bar{t}$ Production Cross Section in pp Collisions at 7 TeV in Lepton + Jets Events Using b-quark Jet Identification

Chatrchyan, Serguei et al.

[Phys.Rev.D 84 \(2011\) 092004](#) | [arXiv:1108.3773](#) | [DOI:10.1103/PhysRevD.84.092004](#) | 2011-08

1479. Measurement of the Underlying Event Activity at the LHC with $\sqrt{s}=7$ TeV and Comparison with $\sqrt{s}=0.9$ TeV

Chatrchyan, Serguei et al.

[JHEP 09 \(2011\) 109](#) | [arXiv:1107.0330](#) | [DOI:10.1007/JHEP09\(2011\)109](#) | 2011-07

1480. Inclusive search for squarks and gluinos in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 85 \(2012\) 012004](#) | [arXiv:1107.1279](#) | [DOI:10.1103/PhysRevD.85.012004](#) | 2011-07

1481. A search for excited leptons in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 704 \(2011\) 143](#) | [arXiv:1107.1773](#) | [DOI:10.1016/j.physletb.2011.09.021](#) | 2011-07

1482. Search for supersymmetry in pp collisions at $\sqrt{s}=7$ TeV in events with a single lepton, jets, and missing transverse momentum

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 156](#) | [arXiv:1107.1870](#) | [DOI:10.1007/JHEP08\(2011\)156](#) | 2011-07

1483. Search for Three-Jet Resonances in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 101801](#) | [arXiv:1107.3084](#) | [DOI:10.1103/PhysRevLett.107.101801](#) | 2011-07

1484. Determination of Jet Energy Calibration and Transverse Momentum Resolution in CMS

Chatrchyan, Serguei et al.

[JINST 6 \(2011\) P11002](#) | [arXiv:1107.4277](#) | [DOI:10.1088/1748-0221/6/11/P11002](#) | 2011-07

1485. Dependence on pseudorapidity and centrality of charged hadron production in PbPb collisions at a nucleon-nucleon centre-of-mass energy of 2.76 TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 141](#) | [arXiv:1107.4800](#) | [DOI:10.1007/JHEP08\(2011\)141](#) | 2011-07

1486. Measurement of the Inclusive W and Z Production Cross Sections in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 10 \(2011\) 132](#) | [arXiv:1107.4789](#) | [DOI:10.1007/JHEP10\(2011\)132](#) | 2011-07

1487. Search for Resonances in the Dijet Mass Spectrum from 7 TeV pp Collisions at CMS

Chatrchyan, Serguei et al.

[Phys.Lett.B 704 \(2011\) 123](#) | [arXiv:1107.4771](#) | [DOI:10.1016/j.physletb.2011.09.015](#) | 2011-07

1488. Search for B(s) and B to dimuon decays in pp collisions at 7 TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 191802](#) | [arXiv:1107.5834](#) | [DOI:10.1103/PhysRevLett.107.191802](#) | 2011-07

1489. Measurement of the Inclusive Jet Cross Section in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 132001](#) | [arXiv:1106.0208](#) | [DOI:10.1103/PhysRevLett.107.132001](#) | 2011-06

1490. Measurement of the ratio of the 3-jet to 2-jet cross sections in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 702 \(2011\) 336](#) | [arXiv:1106.0647](#) | [DOI:10.1016/j.physletb.2011.07.067](#) | 2011-06

1491. Search for physics beyond the standard model using multilepton signatures in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 704 \(2011\) 411](#) | [arXiv:1106.0933](#) | [DOI:10.1016/j.physletb.2011.09.047](#) | 2011-06

1492. Measurement of the Top-antitop Production Cross Section in pp Collisions at $\sqrt{s}=7$ TeV using the Kinematic Properties of Events with Leptons and Jets

Chatrchyan, Serguei et al.

[Eur.Phys.J.C 71 \(2011\) 1721](#) | [arXiv:1106.0902](#) | [DOI:10.1140/epjc/s10052-011-1721-3](#) | 2011-06

1493. Search for Same-Sign Top-Quark Pair Production at $\sqrt{s}=7$ TeV and Limits on Flavour Changing Neutral Currents in the Top Sector

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 005](#) | [arXiv:1106.2142](#) | [DOI:10.1007/JHEP08\(2011\)005](#) | 2011-06

1494. Search for Light Resonances Decaying into Pairs of Muons as a Signal of New Physics

Chatrchyan, Serguei et al.

[JHEP 07 \(2011\) 098](#) | [arXiv:1106.2375](#) | [DOI:10.1007/JHEP07\(2011\)098](#) | 2011-06

1495. Measurement of the t-channel single top quark production cross section in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 091802](#) | [arXiv:1106.3052](#) | [DOI:10.1103/PhysRevLett.107.091802](#) | 2011-06

1496. Search for Supersymmetry in Events with b Jets and Missing Transverse Momentum at the LHC

Chatrchyan, Serguei et al.

[JHEP 07 \(2011\) 113](#) | [arXiv:1106.3272](#) | [DOI:10.1007/JHEP07\(2011\)113](#) | 2011-06

1497. Measurement of the Strange B Meson Production Cross Section with J/Psi ϕ Decays in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.D 84 \(2011\) 052008](#) | [arXiv:1106.4048](#) | [DOI:10.1103/PhysRevD.84.052008](#) | 2011-06

1498. Search for New Physics with Jets and Missing Transverse Momentum in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 155](#) | [arXiv:1106.4503](#) | [DOI:10.1007/JHEP08\(2011\)155](#) | 2011-06

1499. Search for New Physics with a Mono-Jet and Missing Transverse Energy in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 201804](#) | [arXiv:1106.4775](#) | [DOI:10.1103/PhysRevLett.107.201804](#) | 2011-06

1500. Missing transverse energy performance of the CMS detector

Chatrchyan, Serguei et al.

[JINST 6 \(2011\) P09001](#) | [arXiv:1106.5048](#) | [DOI:10.1088/1748-0221/6/09/P09001](#) | 2011-06

1501. Long-range and short-range dihadron angular correlations in central PbPb collisions at a nucleon-nucleon center of mass energy of 2.76 TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2011\) 076](#) | [arXiv:1105.2438](#) | [DOI:10.1007/JHEP07\(2011\)076](#) | 2011-05

1502. Measurement of $W\gamma$ and $Z\gamma$ production in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 701 \(2011\) 535](#) | [arXiv:1105.2758](#) | [DOI:10.1016/j.physletb.2011.06.034](#) | 2011-05

1503. Search for supersymmetry in events with a lepton, a photon, and large missing transverse energy in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2011\) 093](#) | [arXiv:1105.3152](#) | [DOI:10.1007/JHEP06\(2011\)093](#) | 2011-05

1504. Indications of suppression of excited Υ states in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 052302](#) | [arXiv:1105.4894](#) | [DOI:10.1103/PhysRevLett.107.052302](#) | 2011-05

1505. Search for First Generation Scalar Leptoquarks in the $e\nu jj$ channel in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 703 \(2011\) 246](#) | [arXiv:1105.5237](#) | [DOI:10.1016/j.physletb.2011.07.089](#) | 2011-05

1506. Measurement of the $t\bar{t}$ production cross section and the top quark mass in the dilepton channel in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 07 \(2011\) 049](#) | [arXiv:1105.5661](#) | [DOI:10.1007/JHEP07\(2011\)049](#) | 2011-05

1507. Search for Neutral MSSM Higgs Bosons Decaying to Tau Pairs in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 106 \(2011\) 231801](#) | [arXiv:1104.1619](#) | [DOI:10.1103/PhysRevLett.106.231801](#) | 2011-04

1508. Measurement of the Inclusive Z Cross Section via Decays to Tau Pairs in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 117](#) | [arXiv:1104.1617](#) | [DOI:10.1007/JHEP08\(2011\)117](#) | 2011-04

1509. Measurement of the differential dijet production cross section in proton-proton collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 700 \(2011\) 187](#) | [arXiv:1104.1693](#) | [DOI:10.1016/j.physletb.2011.05.027](#) | 2011-04

1510. Measurement of the B^0 production cross section in pp Collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 106 \(2011\) 252001](#) | [arXiv:1104.2892](#) | [DOI:10.1103/PhysRevLett.106.252001](#) | 2011-04

1511. Search for new physics with same-sign isolated dilepton events with jets and missing transverse energy at the LHC

Chatrchyan, Serguei et al.

[JHEP 06 \(2011\) 077](#) | [arXiv:1104.3168](#) | [DOI:10.1007/JHEP06\(2011\)077](#) | 2011-04

1512. Measurement of the Polarization of W Bosons with Large Transverse Momenta in W+Jets Events at the LHC

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 107 \(2011\) 021802](#) | [arXiv:1104.3829](#) | [DOI:10.1103/PhysRevLett.107.021802](#) | 2011-04

1513. Charged particle transverse momentum spectra in pp collisions at $\sqrt{s} = 0.9$ and 7 TeV

Chatrchyan, Serguei et al.

[JHEP 08 \(2011\) 086](#) | [arXiv:1104.3547](#) | [DOI:10.1007/JHEP08\(2011\)086](#) | 2011-04

1514. Search for a $W^\pm$ boson decaying to a muon and a neutrino in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 701 \(2011\) 160](#) | [arXiv:1103.0030](#) | [DOI:10.1016/j.physletb.2011.05.048](#) | 2011-03

1515. Search for Resonances in the Dilepton Mass Distribution in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 05 \(2011\) 093](#) | [arXiv:1103.0981](#) | [DOI:10.1007/JHEP05\(2011\)093](#) | 2011-03

1516. Search for Supersymmetry in pp Collisions at $\sqrt{s} = 7$ TeV in Events with Two Photons and Missing Transverse Energy

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 106 \(2011\) 211802](#) | [arXiv:1103.0953](#) | [DOI:10.1103/PhysRevLett.106.211802](#) | 2011-03

1517. Search for Physics Beyond the Standard Model in Opposite-Sign Dilepton Events at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 06 \(2011\) 026](#) | [arXiv:1103.1348](#) | [DOI:10.1007/JHEP06\(2011\)026](#) | 2011-03

1518. Measurement of the lepton charge asymmetry in inclusive W production in pp collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[JHEP 04 \(2011\) 050](#) | [arXiv:1103.3470](#) | [DOI:10.1007/JHEP04\(2011\)050](#) | 2011-03

1519. Search for Large Extra Dimensions in the Diphoton Final State at the Large Hadron Collider

Chatrchyan, Serguei et al.

[JHEP 05 \(2011\) 085](#) | [arXiv:1103.4279](#) | [DOI:10.1007/JHEP05\(2011\)085](#) | 2011-03

1520. First Measurement of Hadronic Event Shapes in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 699 \(2011\) 48](#) | [arXiv:1102.0068](#) | [DOI:10.1016/j.physletb.2011.03.060](#) | 2011-02

1521. Observation and studies of jet quenching in PbPb collisions at nucleon-nucleon center-of-mass energy = 2.76 TeV

Chatrchyan, Serguei et al.

[Phys.Rev.C 84 \(2011\) 024906](#) | [arXiv:1102.1957](#) | [DOI:10.1103/PhysRevC.84.024906](#) | 2011-02

1522. Measurement of Dijet Angular Distributions and Search for Quark Compositeness in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 201804](#) | [arXiv:1102.2020](#) | [DOI:10.1103/PhysRevLett.106.201804](#) | 2011-02

1523. Measurement of BB Angular Correlations based on Secondary Vertex Reconstruction at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2011\) 136](#) | [arXiv:1102.3194](#) | [DOI:10.1007/JHEP03\(2011\)136](#) | 2011-02

1524. Strange Particle Production in pp Collisions at $\sqrt{s}=0.9$ and 7 TeV

Khachatryan, Vardan et al.

[JHEP 05 \(2011\) 064](#) | [arXiv:1102.4282](#) | [DOI:10.1007/JHEP05\(2011\)064](#) | 2011-02

1525. Search for a Heavy Bottom-like Quark in pp Collisions at $\sqrt{s} = 7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 701 \(2011\) 204](#) | [arXiv:1102.4746](#) | [DOI:10.1016/j.physletb.2011.05.074](#) | 2011-02

1526. Study of Z boson production in PbPb collisions at $\sqrt{s_{NN}} = 2.76$ TeV

Chatrchyan, Serguei et al.

[Phys.Rev.Lett. 106 \(2011\) 212301](#) | [arXiv:1102.5435](#) | [DOI:10.1103/PhysRevLett.106.212301](#) | 2011-02

1527. Measurement of $W^+ W^-$ production and search for the Higgs boson in pp collisions at $\sqrt{s}=7$ TeV

Chatrchyan, Serguei et al.

[Phys.Lett.B 699 \(2011\) 25](#) | [arXiv:1102.5429](#) | [DOI:10.1016/j.physletb.2011.03.056](#) | 2011-02

1528. Measurement of the B^+ Production Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 112001](#) | [arXiv:1101.0131](#) | [DOI:10.1103/PhysRevLett.106.112001](#) | 2011-01

1529. Search for Heavy Stable Charged Particles in pp collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2011\) 024](#) | [arXiv:1101.1645](#) | [DOI:10.1007/JHEP03\(2011\)024](#) | 2011-01

1530. Search for Supersymmetry in pp Collisions at 7 TeV in Events with Jets and Missing Transverse Energy

Khachatryan, Vardan et al.

[Phys.Lett.B 698 \(2011\) 196](#) | [arXiv:1101.1628](#) | [DOI:10.1016/j.physletb.2011.03.021](#) | 2011-01

1531. Measurement of Bose-Einstein Correlations in pp Collisions at $\sqrt{s}=0.9$ and 7 TeV

Khachatryan, Vardan et al.

[JHEP 05 \(2011\) 029](#) | [arXiv:1101.3518](#) | [DOI:10.1007/JHEP05\(2011\)029](#) | 2011-01

1532. Inclusive b-Hadron Production Cross Section with Muons in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[JHEP 03 \(2011\) 090](#) | [arXiv:1101.3512](#) | [DOI:10.1007/JHEP03\(2011\)090](#) | 2011-01

1533. Dijet Azimuthal Decorrelations in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 122003](#) | [arXiv:1101.5029](#) | [DOI:10.1103/PhysRevLett.106.122003](#) | 2011-01

1534. Measurement of the Isolated Prompt Photon Production Cross Section in pp Collisions at $\sqrt{s} = 7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 082001](#) | [arXiv:1012.0799](#) | [DOI:10.1103/PhysRevLett.106.082001](#) | 2010-12

1535. Measurements of Inclusive W and Z Cross Sections in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2011\) 080](#) | [arXiv:1012.2466](#) | [DOI:10.1007/JHEP01\(2011\)080](#) | 2010-12

1536. Search for Microscopic Black Hole Signatures at the Large Hadron Collider

Khachatryan, Vardan et al.

[Phys.Lett.B 697 \(2011\) 434](#) | [arXiv:1012.3375](#) | [DOI:10.1016/j.physletb.2011.02.032](#) | 2010-12

1537. Search for Pair Production of Second-Generation Scalar Leptoquarks in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 201803](#) | [arXiv:1012.4033](#) | [DOI:10.1103/PhysRevLett.106.201803](#) | 2010-12

1538. Search for Pair Production of First-Generation Scalar Leptoquarks in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 201802](#) | [arXiv:1012.4031](#) | [DOI:10.1103/PhysRevLett.106.201802](#) | 2010-12

1539. Upsilon Production Cross-Section in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.D 83 \(2011\) 112004](#) | [arXiv:1012.5545](#) | [DOI:10.1103/PhysRevD.83.112004](#) | 2010-12

1540. Search for a Heavy Gauge Boson W' in the Final State with an Electron and Large Missing Transverse Energy in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 698 \(2011\) 21](#) | [arXiv:1012.5945](#) | [DOI:10.1016/j.physletb.2011.02.048](#) | 2010-12

1541. Prompt and Non-Prompt J/ψ Production in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 71 \(2011\) 1575](#) | [arXiv:1011.4193](#) | [DOI:10.1140/epjc/s10052-011-1575-8](#) | 2010-11

1542. Search for Stopped Gluinos in pp collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 106 \(2011\) 011801](#) | [arXiv:1011.5861](#) | [DOI:10.1103/PhysRevLett.106.011801](#) | 2010-11

1543. Charged Particle Multiplicities in pp Interactions at $\sqrt{s}=0.9, 2.36$, and 7 TeV

Khachatryan, Vardan et al.

[JHEP 01 \(2011\) 079](#) | [arXiv:1011.5531](#) | [DOI:10.1007/JHEP01\(2011\)079](#) | 2010-11

1544. CMS distributed computing workflow experience

Adelman-McCarthy, Jennifer et al.

[J.Phys.Conf.Ser. 331 \(2011\) 072019](#) | [DOI:10.1088/1742-6596/331/7/072019](#) | 2011

1545. Search for Dijet Resonances in 7 TeV pp Collisions at CMS

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 105 \(2010\) 211801](#) | [arXiv:1010.0203](#) | [DOI:10.1103/PhysRevLett.105.211801](#) | 2010-10

1546. Search for Quark Compositeness with the Dijet Centrality Ratio in pp Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 105 \(2010\) 262001](#) | [arXiv:1010.4439](#) | [DOI:10.1103/PhysRevLett.105.262001](#) | 2010-10

1547. First Measurement of the Cross Section for Top-Quark Pair Production in Proton-Proton Collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Lett.B 695 \(2011\) 424](#) | [arXiv:1010.5994](#) | [DOI:10.1016/j.physletb.2010.11.058](#) | 2010-10

1548. Observation of Long-Range Near-Side Angular Correlations in Proton-Proton Collisions at the LHC

Khachatryan, Vardan et al.

[JHEP 09 \(2010\) 091](#) | [arXiv:1009.4122](#) | [DOI:10.1007/JHEP09\(2010\)091](#) | 2010-09

1549. First Measurement of the Underlying Event Activity at the LHC with $\sqrt{s} = 0.9$ TeV

Khachatryan, Vardan et al.

[Eur.Phys.J.C 70 \(2010\) 555](#) | [arXiv:1006.2083](#) | [DOI:10.1140/epjc/s10052-010-1453-9](#) | 2010

1550. CMS Tracking Performance Results from Early LHC Operation

Khachatryan, Vardan et al.

[Eur.Phys.J.C 70 \(2010\) 1165](#) | [arXiv:1007.1988](#) | [DOI:10.1140/epjc/s10052-010-1491-3](#) | 2010

1551. Transverse-momentum and pseudorapidity distributions of charged hadrons in pp collisions at $\sqrt{s}=7$ TeV

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 105 \(2010\) 022002](#) | [arXiv:1005.3299](#) | [DOI:10.1103/PhysRevLett.105.022002](#) | 2010-05

1552. First Measurement of Bose-Einstein Correlations in Proton-Proton Collisions at $\sqrt{s}=0.9$ and 2.36 TeV at the LHC

Khachatryan, Vardan et al.

[Phys.Rev.Lett. 105 \(2010\) 032001](#) | [arXiv:1005.3294](#) | [DOI:10.1103/PhysRevLett.105.032001](#) | 2010-05

1553. Measurement of the Charge Ratio of Atmospheric Muons with the CMS Detector

Khachatryan, Vardan et al.

[Phys.Lett.B 692 \(2010\) 83](#) | [arXiv:1005.5332](#) | [DOI:10.1016/j.physletb.2010.07.033](#) | 2010-05

1554. Transverse Momentum and Pseudorapidity Distributions of Charged Hadrons in pp Collisions at $\sqrt{s} = 0.9$ and 2.36 TeV

Khachatryan, Vardan et al.

[JHEP 02 \(2010\) 041](#) | [arXiv:1002.0621](#) | [DOI:10.1007/JHEP02\(2010\)041](#) | 2010-02

1555. Alignment of the CMS Muon System with Cosmic-Ray and Beam-Halo Muons

Chatrchyan, S et al.

[JINST 5 \(2010\) T03020](#) | [arXiv:0911.4022](#) | [DOI:10.1088/1748-0221/5/03/T03020](#) | 2009-11

1556. Time Reconstruction and Performance of the CMS Electromagnetic Calorimeter

Chatrchyan, S et al.

[JINST 5 \(2010\) T03011](#) | [arXiv:0911.4044](#) | [DOI:10.1088/1748-0221/5/03/T03011](#) | 2009-11

1557. Performance Study of the CMS Barrel Resistive Plate Chambers with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03017](#) | [arXiv:0911.4045](#) | [DOI:10.1088/1748-0221/5/03/T03017](#) | 2009-11

1558. Aligning the CMS Muon Chambers with the Muon Alignment System during an Extended Cosmic Ray Run

Chatrchyan, S et al.

[JINST 5 \(2010\) T03019](#) | [arXiv:0911.4770](#) | [DOI:10.1088/1748-0221/5/03/T03019](#) | 2009-11

1559. CMS Data Processing Workflows during an Extended Cosmic Ray Run

Chatrchyan, S et al.

[JINST 5 \(2010\) T03006](#) | [arXiv:0911.4842](#) | [DOI:10.1088/1748-0221/5/03/T03006](#) | 2009-11

1560. Commissioning of the CMS Experiment and the Cosmic Run at Four Tesla

Chatrchyan, S et al.

[JINST 5 \(2010\) T03001](#) | [arXiv:0911.4845](#) | [DOI:10.1088/1748-0221/5/03/T03001](#) | 2009-11

1561. Performance of the CMS Drift Tube Chambers with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03015](#) | [arXiv:0911.4855](#) | [DOI:10.1088/1748-0221/5/03/T03015](#) | 2009-11

1562. Performance of CMS Hadron Calorimeter Timing and Synchronization using Test Beam, Cosmic Ray, and LHC Beam Data

Chatrchyan, S et al.

[JINST 5 \(2010\) T03013](#) | [arXiv:0911.4877](#) | [DOI:10.1088/1748-0221/5/03/T03013](#) | 2009-11

1563. Identification and Filtering of Uncharacteristic Noise in the CMS Hadron Calorimeter

Chatrchyan, S et al.

[JINST 5 \(2010\) T03014](#) | [arXiv:0911.4881](#) | [DOI:10.1088/1748-0221/5/03/T03014](#) | 2009-11

1564. Commissioning of the CMS High-Level Trigger with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03005](#) | [arXiv:0911.4889](#) | [DOI:10.1088/1748-0221/5/03/T03005](#) | 2009-11

1565. Performance of the CMS Drift-Tube Local Trigger with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03003](#) | [arXiv:0911.4893](#) | [DOI:10.1088/1748-0221/5/03/T03003](#) | 2009-11

1566. Calibration of the CMS Drift Tube Chambers and Measurement of the Drift Velocity with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03016](#) | [arXiv:0911.4895](#) | [DOI:10.1088/1748-0221/5/03/T03016](#) | 2009-11

1567. Fine Synchronization of the CMS Muon Drift-Tube Local Trigger using Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03004](#) | [arXiv:0911.4904](#) | [DOI:10.1088/1748-0221/5/03/T03004](#) | 2009-11

1568. Performance of the CMS Hadron Calorimeter with Cosmic Ray Muons and LHC Beam Data

Chatrchyan, S et al.

[JINST 5 \(2010\) T03012](#) | [arXiv:0911.4991](#) | [DOI:10.1088/1748-0221/5/03/T03012](#) | 2009-11

1569. Performance of the CMS Cathode Strip Chambers with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03018](#) | [arXiv:0911.4992](#) | [DOI:10.1088/1748-0221/5/03/T03018](#) | 2009-11

1570. Performance of CMS Muon Reconstruction in Cosmic-Ray Events

Chatrchyan, S et al.

[JINST 5 \(2010\) T03022](#) | [arXiv:0911.4994](#) | [DOI:10.1088/1748-0221/5/03/T03022](#) | 2009-11

1571. Commissioning and Performance of the CMS Silicon Strip Tracker with Cosmic Ray Muons

Chatrchyan, S et al.

[JINST 5 \(2010\) T03008](#) | [arXiv:0911.4996](#) | [DOI:10.1088/1748-0221/5/03/T03008](#) | 2009-11

1572. Measurement of the Muon Stopping Power in Lead Tungstate

Chatrchyan, S et al.

[JINST 5 \(2010\) P03007](#) | [arXiv:0911.5397](#) | [DOI:10.1088/1748-0221/5/03/P03007](#) | 2009-11

1573. Performance of the CMS Level-1 Trigger during Commissioning with Cosmic Ray Muons

Chatrchyan, S et al.

[JINST 5 \(2010\) T03002](#) | [arXiv:0911.5422](#) | [DOI:10.1088/1748-0221/5/03/T03002](#) | 2009-11

1574. Commissioning and Performance of the CMS Pixel Tracker with Cosmic Ray Muons

Chatrchyan, S et al.

[JINST 5 \(2010\) T03007](#) | [arXiv:0911.5434](#) | [DOI:10.1088/1748-0221/5/03/T03007](#) | 2009-11

1575. Alignment of the CMS Silicon Tracker during Commissioning with Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03009](#) | [arXiv:0910.2505](#) | [DOI:10.1088/1748-0221/5/03/T03009](#) | 2009-10

1576. Performance and Operation of the CMS Electromagnetic Calorimeter

Chatrchyan, S et al.

[JINST 5 \(2010\) T03010](#) | [arXiv:0910.3423](#) | [DOI:10.1088/1748-0221/5/03/T03010](#) | 2009-10

1577. Precise Mapping of the Magnetic Field in the CMS Barrel Yoke using Cosmic Rays

Chatrchyan, S et al.

[JINST 5 \(2010\) T03021](#) | [arXiv:0910.5530](#) | [DOI:10.1088/1748-0221/5/03/T03021](#) | 2009-10

1578. Alignment of the CMS Silicon Strip Tracker during stand-alone Commissioning

Adam, W. et al.

[JINST 4 \(2009\) T07001](#) | [arXiv:0904.1220](#) | [DOI:10.1088/1748-0221/4/07/T07001](#) | 2009-04

1579. Experience Building and Operating the CMS Tier-1 Computing Centres

Albert, M. et al.

[J.Phys.Conf.Ser. 219 \(2010\) 072035](#) | [DOI:10.1088/1742-6596/219/7/072035](#) | 2010

1580. Use of glide-ins in CMS for production and analysis

Bradley, D. et al.

[J.Phys.Conf.Ser. 219 \(2010\) 072013](#) | [DOI:10.1088/1742-6596/219/7/072013](#) | 2010

1581. Validation of Software Releases for CMS

Gutsche, Oliver et al.

[J.Phys.Conf.Ser. 219 \(2010\) 042040](#) | [DOI:10.1088/1742-6596/219/4/042040](#) | 2009-03

1582. Stand-alone Cosmic Muon Reconstruction Before Installation of the CMS Silicon Strip Tracker

Adam, W. et al.

[JINST 4 \(2009\) P05004](#) | [arXiv:0902.1860](#) | [DOI:10.1088/1748-0221/4/05/P05004](#) | 2009-02

1583. Performance studies of the CMS Strip Tracker before installation

Adam, W. et al.

[JINST 4 \(2009\) P06009](#) | [arXiv:0901.4316](#) | [DOI:10.1088/1748-0221/4/06/P06009](#) | 2009-01

1584. Large scale job management and experience in recent data challenges within the LHC CMS experiment

Evans, D. et al.

[PoS ACAT08 \(2008\) 032](#) | [DOI:10.22323/1.070.0032](#) | 2008

1585. Track Reconstruction with Cosmic Ray Data at the Tracker Integration Facility

Adam, Wolfgang et al.

2008-09-17

1586. Measurement of beauty production from dimuon events at HERA

Chekanov, S. et al.

[JHEP 02 \(2009\) 032](#) | [arXiv:0811.0894](#) | [DOI:10.1088/1126-6708/2009/02/032](#) | 2008-09

1587. The CMS Experiment at the CERN LHC

Chatrchyan, S. et al.

[JINST 3 \(2008\) S08004](#) | [DOI:10.1088/1748-0221/3/08/S08004](#) | 2008-08

1588. The CMS tracker operation and performance at the Magnet Test and Cosmic Challenge

Adam, W. et al.

[JINST 3 \(2008\) P07006](#) | [DOI:10.1088/1748-0221/3/07/P07006](#) | 2008

1589. The CMS Remote Analysis Builder (CRAB)

Spiga, D. et al.

[Lect.Notes Comput.Sci. 4873 \(2007\) 580](#) | [DOI:10.1007/978-3-540-77220-0_52](#) | 2007

1590. CMS data and workflow management system

Fanfani, A. et al.

[\(2008\) 441](#) | [DOI:10.1142/9789812819093_0076](#) | 2008

1591. WLCG scale testing during CMS data challenges

Gutsche, Oliver et al.

[J.Phys.Conf.Ser. 119 \(2008\) 062033](#) | [DOI:10.1088/1742-6596/119/6/062033](#) | 2008

1592. CMS technical design report, volume II: Physics performance

Bayatian, G.L. et al.

[J.Phys.G 34 \(2007\) 995](#) | [DOI:10.1088/0954-3899/34/6/S01](#) | 2007

1593. Tracker operation and performance at the Magnet Test and Cosmic Challenge

Abbateo, D. et al.

2007-06

1594. CRAB: The CMS distributed analysis tool development and design

Spiga, D. et al.

[Nucl.Phys.B Proc.Suppl. 177-178 \(2008\) 267](#) | [DOI:10.1016/j.nuclphysbps.2007.11.124](#) | 2008

1595. Status and evolution of CRAB

Farina, Fabio et al.

[PoS ACAT \(2007\) 020](#) | [DOI:10.22323/1.050.0020](#) | 2007

1596. CMS physics technical design report: Addendum on high density QCD with heavy ions

d'Enterria, David G. et al.

[J.Phys.G 34 \(2007\) 2307](#) | [DOI:10.1088/0954-3899/34/11/008](#) | 2007-03

1597. CMS expression of interest in the SLHC

Nash, J. et al.

2007-03

1598. The CMS Simulation Software

Abdouline, S. et al.

[\(2006\) 1655](#) | [DOI:10.1109/NSSMIC.2006.354216](#) | 2006

1599. Measurement of neutral current cross sections at high Bjorken-x with the ZEUS detector at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 49 \(2007\) 523](#) | [arXiv:hep-ex/0608014](#) | [DOI:10.1140/epjc/s10052-006-0164-8](#) | 2006-08

1600. Measurement of prompt photons with associated jets in photoproduction at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 49 \(2007\) 511](#) | [arXiv:hep-ex/0608028](#) | [DOI:10.1140/epjc/s10052-006-0134-1](#) | 2006-08

1601. Inclusive-jet and dijet cross-sections in deep inelastic scattering at HERA

Chekanov, S. et al.

[Nucl.Phys.B 765 \(2007\) 1](#) | [arXiv:hep-ex/0608048](#) | [DOI:10.1016/j.nuclphysb.2006.09.018](#) | 2006-08

1602. Measurement of azimuthal asymmetries in neutral current deep inelastic scattering at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 51 \(2007\) 289](#) | [arXiv:hep-ex/0608053](#) | [DOI:10.1140/epjc/s10052-007-0310-y](#) | 2006-08

1603. CMS Physics

Bayatian, G. L. et al.

[DOI:10.2172/2510878](#) | 2006

1604. Event shapes in deep inelastic scattering at HERA

Chekanov, S. et al.

[Nucl.Phys.B 767 \(2007\) 1](#) | [arXiv:hep-ex/0604032](#) | [DOI:10.1016/j.nuclphysb.2006.05.016](#) | 2006-04

1605. Measurement of high- Q^2 deep inelastic scattering cross sections with a longitudinally polarised positron beam at HERA

Chekanov, S. et al.

[Phys.Lett.B 637 \(2006\) 210](#) | [arXiv:hep-ex/0602026](#) | [DOI:10.1016/j.physletb.2006.04.047](#) | 2006-02

1606. The CMS object-oriented simulation

Abdouline, S. et al.

[\(2005\) 995](#) | [DOI:10.1109/NSSMIC.2005.1596421](#) | 2005

1607. Measurement of charm fragmentation ratios and fractions in photoproduction at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 44 \(2005\) 351](#) | [arXiv:hep-ex/0508019](#) | [DOI:10.1140/epjc/s2005-02397-3](#) | 2005-08

1608. Inclusive jet cross sections and dijet correlations in D^{*+} photoproduction at HERA

Chekanov, S. et al.

[Nucl.Phys.B 729 \(2005\) 492](#) | [arXiv:hep-ex/0507089](#) | [DOI:10.1016/j.nuclphysb.2005.09.021](#) | 2005-07

1609. Measurement of inelastic J/ψ production in deep inelastic scattering at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 44 \(2005\) 13](#) | [arXiv:hep-ex/0505008](#) | [DOI:10.1140/epjc/s2005-02346-2](#) | 2005-05

1610. Exclusive electroproduction of phi mesons at HERA

Chekanov, S. et al.

[Nucl.Phys.B 718 \(2005\) 3](#) | [arXiv:hep-ex/0504010](#) | [DOI:10.1016/j.nuclphysb.2005.04.009](#) | 2005-04

1611. An NLO QCD analysis of inclusive cross-section and jet-production data from the zeus experiment

Chekanov, S. et al.

[Eur.Phys.J.C 42 \(2005\) 1](#) | [arXiv:hep-ph/0503274](#) | [DOI:10.1140/epjc/s2005-02293-x](#) | 2005-03

1612. Measurement of Beauty Quark Cross Sections in Photoproduction with the ZEUS Experiment at the Electron Proton Collider HERA

Gutsche, Oliver et al.

[DOI:10.3204/DESY-THESIS-2005-010](#) | 2005-03

1613. Multijet production in neutral current deep inelastic scattering at HERA and determination of $\alpha(s)$

Chekanov, S. et al.

[Eur.Phys.J.C 44 \(2005\) 183](#) | [arXiv:hep-ex/0502007](#) | [DOI:10.1140/epjc/s2005-02347-1](#) | 2005-02

1614. Study of deep inelastic inclusive and diffractive scattering with the ZEUS forward plug calorimeter

Chekanov, S. et al.

[Nucl.Phys.B 713 \(2005\) 3](#) | [arXiv:hep-ex/0501060](#) | [DOI:10.1016/j.nuclphysb.2005.02.001](#) | 2005-01

1615. Search for pentaquarks decaying to Xi-pi in deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 610 \(2005\) 212](#) | [arXiv:hep-ex/0501069](#) | [DOI:10.1016/j.physletb.2005.02.016](#) | 2005-01

1616. Search for lepton-flavor violation at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 44 \(2005\) 463](#) | [arXiv:hep-ex/0501070](#) | [DOI:10.1140/epjc/s2005-02399-1](#) | 2005-01

1617. Forward jet production in deep inelastic ep scattering and low-x parton dynamics at HERA

Chekanov, S. et al.

[Phys.Lett.B 632 \(2006\) 13](#) | [arXiv:hep-ex/0502029](#) | [DOI:10.1016/j.physletb.2005.09.066](#) | 2005-01

1618. Search for a narrow charmed baryonic state decaying to $D^{*+} p$ in ep collisions at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 38 \(2004\) 29](#) | [arXiv:hep-ex/0409033](#) | [DOI:10.1140/epjc/s2004-02042-9](#) | 2004-09

1619. Dissociation of virtual photons in events with a leading proton at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 38 \(2004\) 43](#) | [arXiv:hep-ex/0408009](#) | [DOI:10.1140/epjc/s2004-02047-4](#) | 2004-08

1620. Diffractive charm and jet production

Gutsche, O. et al.

728 | 2004-08

1621. Substructure dependence of jet cross sections at HERA and determination of $\alpha(s)$

Chekanov, S. et al.

[Nucl.Phys.B 700 \(2004\) 3](#) | [arXiv:hep-ex/0405065](#) | [DOI:10.1016/j.nuclphysb.2004.08.049](#) | 2004-05

1622. Measurement of beauty production in deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 599 \(2004\) 173](#) | [arXiv:hep-ex/0405069](#) | [DOI:10.1016/j.physletb.2004.08.048](#) | 2004-05

1623. Evidence for a narrow baryonic state decaying to $K^0(S) p$ and $K^0(S) \bar{p}$ in deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 591 \(2004\) 7](#) | [arXiv:hep-ex/0403051](#) | [DOI:10.1016/j.physletb.2004.04.024](#) | 2004-03

1624. Study of the pion trajectory in the photoproduction of leading neutrons at HERA

Chekanov, S. et al.

[Phys.Lett.B 610 \(2005\) 199](#) | [arXiv:hep-ex/0404002](#) | [DOI:10.1016/j.physletb.2005.01.101](#) | 2004-03

1625. Exclusive electroproduction of J/ψ mesons at HERA

Chekanov, S. et al.

[Nucl.Phys.B 695 \(2004\) 3](#) | [arXiv:hep-ex/0404008](#) | [DOI:10.1016/j.nuclphysb.2004.06.034](#) | 2004-03

1626. The Dependence of dijet production on photon virtuality in ep collisions at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 35 \(2004\) 487](#) | [arXiv:hep-ex/0404033](#) | [DOI:10.1140/epjc/s2004-01885-2](#) | 2004-03

1627. Observation of isolated high $E(T)$ photons in deep inelastic scattering

Chekanov, S. et al.

- Phys.Lett.B 595 (2004) 86 | [arXiv:hep-ex/0402019](#) | [DOI:10.1016/j.physletb.2004.05.033](#) | 2004-02
1628. Photoproduction of D^{*+-} mesons associated with a leading neutron
Chekanov, S. et al.
[Phys.Lett.B 590 \(2004\) 143](#) | [arXiv:hep-ex/0401017](#) | [DOI:10.1016/j.physletb.2004.03.076](#) | 2004-01
1629. Search for QCD instanton induced events in deep inelastic ep scattering at HERA
Chekanov, S. et al.
[Eur.Phys.J.C 34 \(2004\) 255](#) | [arXiv:hep-ex/0312048](#) | [DOI:10.1140/epjc/s2004-01735-3](#) | 2003-12
1630. Beauty photoproduction measured using decays into muons in dijet events in e p collisions at $s^{1/2} = 318$ -GeV
Chekanov, S. et al.
[Phys.Rev.D 70 \(2004\) 012008](#) | [arXiv:hep-ex/0312057](#) | [DOI:10.1103/PhysRevD.70.012008](#) | 2003-12
1631. High Q^2 neutral current cross-sections in e+ p deep inelastic scattering at $s^{1/2} = 318$ -GeV
Chekanov, S. et al.
[Phys.Rev.D 70 \(2004\) 052001](#) | [arXiv:hep-ex/0401003](#) | [DOI:10.1103/PhysRevD.70.052001](#) | 2003-12
1632. Search for contact interactions, large extra dimensions and finite quark radius in e p collisions at HERA
Chekanov, S. et al.
[Phys.Lett.B 591 \(2004\) 23](#) | [arXiv:hep-ex/0401009](#) | [DOI:10.1016/j.physletb.2004.03.081](#) | 2003-12
1633. Isolated tau leptons in events with large missing transverse momentum at HERA
Chekanov, S. et al.
[Phys.Lett.B 583 \(2004\) 41](#) | [arXiv:hep-ex/0311028](#) | [DOI:10.1016/j.physletb.2003.12.054](#) | 2003-11
1634. Search for single-top production in e p collisions at HERA. (Addendum)
Chekanov, S. et al.
2003-11
1635. Bose-Einstein correlations in one and two-dimensions in deep inelastic scattering
Chekanov, S. et al.
[Phys.Lett.B 583 \(2004\) 231](#) | [arXiv:hep-ex/0311030](#) | [DOI:10.1016/j.physletb.2003.12.068](#) | 2003-10
1636. Observation of $K_0^*(800)$ resonances in deep inelastic scattering at HERA
Chekanov, S. et al.
[Phys.Lett.B 578 \(2004\) 33](#) | [arXiv:hep-ex/0308006](#) | [DOI:10.1016/j.physletb.2003.10.049](#) | 2003-08
1637. Measurement of D^{*+-} production in deep inelastic e+- p scattering at HERA
Chekanov, S. et al.
[Phys.Rev.D 69 \(2004\) 012004](#) | [arXiv:hep-ex/0308068](#) | [DOI:10.1103/PhysRevD.69.012004](#) | 2003-08
1638. Measurement of high Q^2 charged current cross-sections in e+ p deep inelastic scattering at HERA
Chekanov, S. et al.
[Eur.Phys.J.C 32 \(2003\) 1](#) | [arXiv:hep-ex/0307043](#) | [DOI:10.1140/epjc/s2003-01363-5](#) | 2003-07
1639. Measurement of the open charm contribution to the diffractive proton structure function
Chekanov, S. et al.
[Nucl.Phys.B 672 \(2003\) 3](#) | [arXiv:hep-ex/0307068](#) | [DOI:10.1016/j.nuclphysb.2003.09.001](#) | 2003-07
1640. Jet production in charged current deep inelastic e+ p scattering at HERA
Chekanov, S. et al.
[Eur.Phys.J.C 31 \(2003\) 149](#) | [arXiv:hep-ex/0306018](#) | [DOI:10.1140/epjc/s2003-01358-2](#) | 2003-06

1641. A ROOT based client server event display for the ZEUS experiment

Kind, O. et al.

eConf C0303241 (2003) MOLT002 | [arXiv:hep-ex/0305095](#) | 2003-05

1642. Measurement of deeply virtual Compton scattering at HERA

Chekanov, S. et al.

Phys.Lett.B 573 (2003) 46 | [arXiv:hep-ex/0305028](#) | DOI:10.1016/j.physletb.2003.08.048 | 2003-04

1643. A Search for resonance decays to lepton + jet at HERA and limits on leptoquarks

Chekanov, S. et al.

Phys.Rev.D 68 (2003) 052004 | [arXiv:hep-ex/0304008](#) | DOI:10.1103/PhysRevD.68.052004 | 2003-03

1644. Search for single top production in ep collisions at HERA

Chekanov, S. et al.

Phys.Lett.B 559 (2003) 153 | [arXiv:hep-ex/0302010](#) | DOI:10.1016/S0370-2693(03)00333-2 | 2003-02

1645. Dijet angular distributions in photoproduction of charm at HERA

Chekanov, S. et al.

Phys.Lett.B 565 (2003) 87 | [arXiv:hep-ex/0302025](#) | DOI:10.1016/S0370-2693(03)00752-4 | 2003-02

1646. Measurement of subjet multiplicities in neutral current deep inelastic scattering at HERA and determination of $\alpha(s)$

Chekanov, S. et al.

Phys.Lett.B 558 (2003) 41 | [arXiv:hep-ex/0212030](#) | DOI:10.1016/S0370-2693(03)00216-8 | 2002-12

1647. Scaling violations and determination of $\alpha(s)$ from jet production in gamma p interactions at HERA

Chekanov, S. et al.

Phys.Lett.B 560 (2003) 7 | [arXiv:hep-ex/0212064](#) | DOI:10.1016/S0370-2693(03)00380-0 | 2002-12

1648. Observation of the strange sea in the proton via inclusive phi meson production in neutral current deep inelastic scattering at HERA

Chekanov, S. et al.

Phys.Lett.B 553 (2003) 141 | [arXiv:hep-ex/0211025](#) | DOI:10.1016/S0370-2693(02)03206-9 | 2002-11

1649. Measurement of event shapes in deep inelastic scattering at HERA

Chekanov, S. et al.

Eur.Phys.J.C 27 (2003) 531 | [arXiv:hep-ex/0211040](#) | DOI:10.1140/epjc/s2003-01148-x | 2002-11

1650. Study of the azimuthal asymmetry of jets in neutral current deep inelastic scattering at HERA

Chekanov, S. et al.

Phys.Lett.B 551 (2003) 226 | [arXiv:hep-ex/0210064](#) | DOI:10.1016/S0370-2693(02)03072-1 | 2002-10

1651. Measurements of inelastic J / psi and psi-prime photoproduction at HERA

Chekanov, S. et al.

Eur.Phys.J.C 27 (2003) 173 | [arXiv:hep-ex/0211011](#) | DOI:10.1140/epjc/s2002-01130-2 | 2002-10

1652. Leading proton production in e+ p collisions at HERA

Chekanov, S. et al.

Nucl.Phys.B 658 (2003) 3 | [arXiv:hep-ex/0210029](#) | DOI:10.1016/S0550-3213(03)00152-4 | 2002-09

1653. A ZEUS next-to-leading-order QCD analysis of data on deep inelastic scattering

Chekanov, S. et al.

Phys.Rev.D 67 (2003) 012007 | [arXiv:hep-ex/0208023](#) | DOI:10.1103/PhysRevD.67.012007 | 2002-08

1654. Inclusive jet cross-sections in the Breit frame in neutral current deep inelastic scattering at HERA and determination of $\alpha(s)$

Chekanov, S. et al.

[Phys.Lett.B 547 \(2002\) 164](#) | [arXiv:hep-ex/0208037](#) | [DOI:10.1016/S0370-2693\(02\)02763-6](#) | 2002-08

1655. Measurement of high Q^2 e- p neutral current cross-sections at HERA and the extraction of $x_F(3)$

Chekanov, S. et al.

[Eur.Phys.J.C 28 \(2003\) 175](#) | [arXiv:hep-ex/0208040](#) | [DOI:10.1140/epjc/s2003-01163-y](#) | 2002-08

1656. Measurement of diffractive production of D^{*+} -(2010) mesons in deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 545 \(2002\) 244](#) | [arXiv:hep-ex/0206020](#) | [DOI:10.1016/S0370-2693\(02\)02595-9](#) | 2002-06

1657. Leading neutron production in e+ p collisions at HERA

Chekanov, S. et al.

[Nucl.Phys.B 637 \(2002\) 3](#) | [arXiv:hep-ex/0205076](#) | [DOI:10.1016/S0550-3213\(02\)00439-X](#) | 2002-05

1658. Measurement of proton dissociative diffractive photoproduction of vector mesons at large momentum transfer at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 26 \(2003\) 389](#) | [arXiv:hep-ex/0205081](#) | [DOI:10.1140/epjc/s2002-01079-0](#) | 2002-05

1659. Measurement of high Q^2 charged current cross-sections in e^+ -p deep inelastic scattering at HERA

Chekanov, S. et al.

[Phys.Lett.B 539 \(2002\) 197](#) | [arXiv:hep-ex/0205091](#) | [DOI:10.1016/S0370-2693\(02\)02093-2](#) | 2002-05

1660. Measurement of the Q^2 and energy dependence of diffractive interactions at HERA

Chekanov, S. et al.

[Eur.Phys.J.C 25 \(2002\) 169](#) | [arXiv:hep-ex/0203039](#) | [DOI:10.1007/s10052-002-1003-1](#) | 2002-03

1661. Development of the Trigger Algorithm for the MONOLITH Experiment

Gutsche, Oliver et al.

2001-05

1662. MONOLITH: A massive magnetized iron detector for neutrino oscillation studies

Agafonova, N.Y. et al.

2000-08